TJA1121

Rev. 1.0 — 3 December 2024 Product data sheet

1 General description

The TJA1121 is a 1000BASE-T1 compliant MacSec-enabled Ethernet PHY. The device provides 1000 Mbit/s transmit and receive capability over shielded or unshielded twisted-pair cables, supporting a cable length of at least 15 m.

It is available in two variants. The TJA1121A offers RGMII(-ID) MAC interfaces. The variant TJA1121B provides SGMII.

The product integrates an IEEE 802.1AE-2018 compliant MACsec subsystem, which is configurable by the host controller via MDIO. MACsec ensures the authenticity, integrity, and also (optional) confidentiality of any communication over the link. The TJA1121 supports both, point-to-point and end-to-end secure associations. The implementation of MACsec is fully AVB and TSN compliant and maintains predictable latency for both, controlled and uncontrolled, data paths. Also, the MACsec congestion avoidance handling techniques supported by TJA1121 retain full compliance with PTP. A rich set of advanced frame-filtering options allow bypassing selected non-MACsec frame types.

TJA1121 remains fully drop-in compatible to TJA1120, with extended register space and feature set. If MACsec is disabled, the TJA1121 behaves similarly to TJA1120 as a traditional 1000BASE-T1 compliant Ethernet PHY.

For sustainable system efficiency and flexibility, the device is optimized for lowest power consumption and highest robustness, while maintaining a small package footprint. The TJA1121, including the MACsec subsystem, is developed in compliance with ISO 26262, achieving ASIL-B.

Adequate safety features have been implemented on-chip, to support functional safety concepts on system level. Additional documentation, including a safety manual [8], is available on request. In combination with state of the art diagnostic features, the device is perfectly suited to support high availability of communication paths in upcoming automotive electrical/electronic (E/E) architectures.

The TJA1121 is designed to meet AEC-Q100 Grade 1 to support the harsh requirements of automotive applications and is typically used in electronic control units (ECUs), such as Zonal Aggregators, Gateways, Telematics and Head Units, as well as typical end nodes, such as Radar and LiDAR sensors for driver assistance systems (ADAS), as well as a variety of electronic control units (ECUs) such as zonal gateways, telematics and head units.

The device supports OPEN Alliance TC-10 wake-up forwarding over data-line for 1000BASE-T1. By eliminating the need for additional cabling, the feature saves cost and weight.

Due to its efficient IEEE1588v2 and 802.1AS-rev compliant time stamping capability, TJA1121 provides high-resolution accuracy for global time synchronization, which is required to support the timing critical data fusion functions of self-driving cars.

2 Features and benefits

2.1 General

- IEEE 802.3 Clause 97 compliant 1000BASE-T1 PHY [3]
- AEC-Q100 Grade 1 (–40 °C to +125 °C ambient)
- Compact HVQFN 36-pin package (6 × 6 mm) for PCB space-constraint applications
- Standard RGMII- (TJA1121A)
  - Integrated delay (RGMII-ID) in receive and transmit clock path
  - 3.3 V, 2.5 V, or 1.8 V LVCMOS I/O standard for RGMII
- SGMII-compliant (TJA1121B) MAC interface
- OPEN Alliance TC-10 compatible sleep / wake-forwarding over data line [7]
- Low-power Sleep mode
- ISO 26262, ASIL-B compliant
- IEEE1588v2 and 802.1AS-rev 2-step timestamping support[1]
  - Synchronization of timestamp clock with other switch or PHY devices
  - Timestamp delivery via In-band (RX-only) or SMI communication
  - RX and TX SOF (Start of Frame) indication via pins
- IEEE1149.1 and IEEE1149.6 compliant JTAG interface for boundary scan
- Clause 45 MDIO register access
- MDIO register access
  - IEEE 802.3 Clause 45
  - IEEE 802.3 Clause 22 (indirect access)

2.2 Secure communication

- IEEE 802.1AE-2018 MACsec at full wire line performance
- Support for integrity protection and confidentiality protection
- Support for two concurrent security associations per secure channel for re-keying
- Support for 4 end-to-end secure channels in TX and RX direction
- Cipher suites:
  - AES-GCM-128
  - AES-GCM-256
- Extended packet number (XPN) for reduced rekeying frequency
- AVB, TSN compatible flow-control:[2]
  - Cut-through, pre-gapped IPG[3]
  - Cut-through, standard IPG and vendor-specific Ethernet tag in TX direction
- Uncontrolled / controlled port and secure channel selection:
  - Internal lookup table with ternary match-based frame filtering
    - 8 uncontrolled / controlled port filter rules
    - 4 secure-channel selection rules
  - In-band host-side selection through vendor-specific Ethernet tag in TX and RX direction
- Replay protection

1 The PTP frame parser cannot recognize PTP frames if an optional, vendor specific MACsec TX/RX control tag is present.
2 PAUSE frame flow-control is not supported. The PAUSE mechnism would interfere with real-time traffic and introduce undesired packet delay variation.
3 IPG must be extended large enough to allow for adding SecTag and ICV.

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- Jumbo-frame support
- Clear-tag support
  - Up to two VLAN tags and one custom tag (for example 802.1CB R-tag)
- Diagnostics and statistics (MIB) counters

2.3 Optimized for automotive use cases

- Supports IEEE 802.3bu PoDL
- Support for shielded and unshielded twisted-pair cable
- Supports cable length of at least 15 m (type-A link segment as defined in clause 97 of IEEE Std 802.3-2022. [3])
- Enhanced integrated pulse shaping for low RF emissions
- EMC-optimized output driver configuration for RGMII and SGMII
- MDI pins protected against ESD to ± 6 kV (unpowered) and ± 3 kV (powered)
- MDI pins protected against transients in automotive environment IEC62228-5
- MDI pins do not need external filtering or ESD protection

2.4 Low-power mode

- Inhibit output for voltage regulator control
- Robust remote wake-up detection via bus lines
- OPEN Alliance TC-10 compliant sleep/wake-up
  - PHY-level wakeup-forwarding via dedicated pin
- Wake-up via SMI-access

2.5 Diagnosis

- Open Alliance TC-12 Advanced PHY Features
- Built-in self-test (BIST)
- Latent fault detection
- Real-time monitoring of link stability and data quality
- Signal Quality Indicator (SQI)
- Mean Square Error value
- Diagnosis of cable errors (shorts and opens) and error location (TDR method)
- Link training time measurement
- Gap-free supply undervoltage detection
- Temperature monitor with overtemperature detection
- Internal, external and remote loopback modes for diagnostics
- Diagnostic counters
  - Link training time
  - Link losses because reset/power on
  - Symbol errors because power on

2.6 Miscellaneous

- 3.3 V or 2.5 V, 1.8 V, 1.1 V direct supply operation
- 3.3 V or 2.5 V I/O for RST_N, INH and wake interface

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- GPIO, LED, PPS_SYNC and reference clock outputs
- Clock inputs
  - 25 MHz Crystal
  - Two single-ended clock modes are supported:
    - VDD_CORE referenced (1.1 V)
    - VDDIO referenced (1.8, 2.5, 3.3 – depending on VDDIO setting)
  - LVDS differential clock input
- Pin-strapping to configure
  - Leader / Follower mode
  - xMII modes
  - PHY address
  - CLK Select
  - JTAG Mode
  - AUTO/MANAGED Mode
- On-chip termination resistors
- Jumbo frame support up to 16 kB

3 Ordering information

Table 1. Ordering information

Type number	Package		
	Name	Description	Version
TJA1121AHN	HVQFN36	plastic thermal enhanced very thin quad flat package; no leads; 36 terminals; body 6 × 6 × 0.85 mm	SOT1092-2
TJA1121BHN			

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4 Block diagram

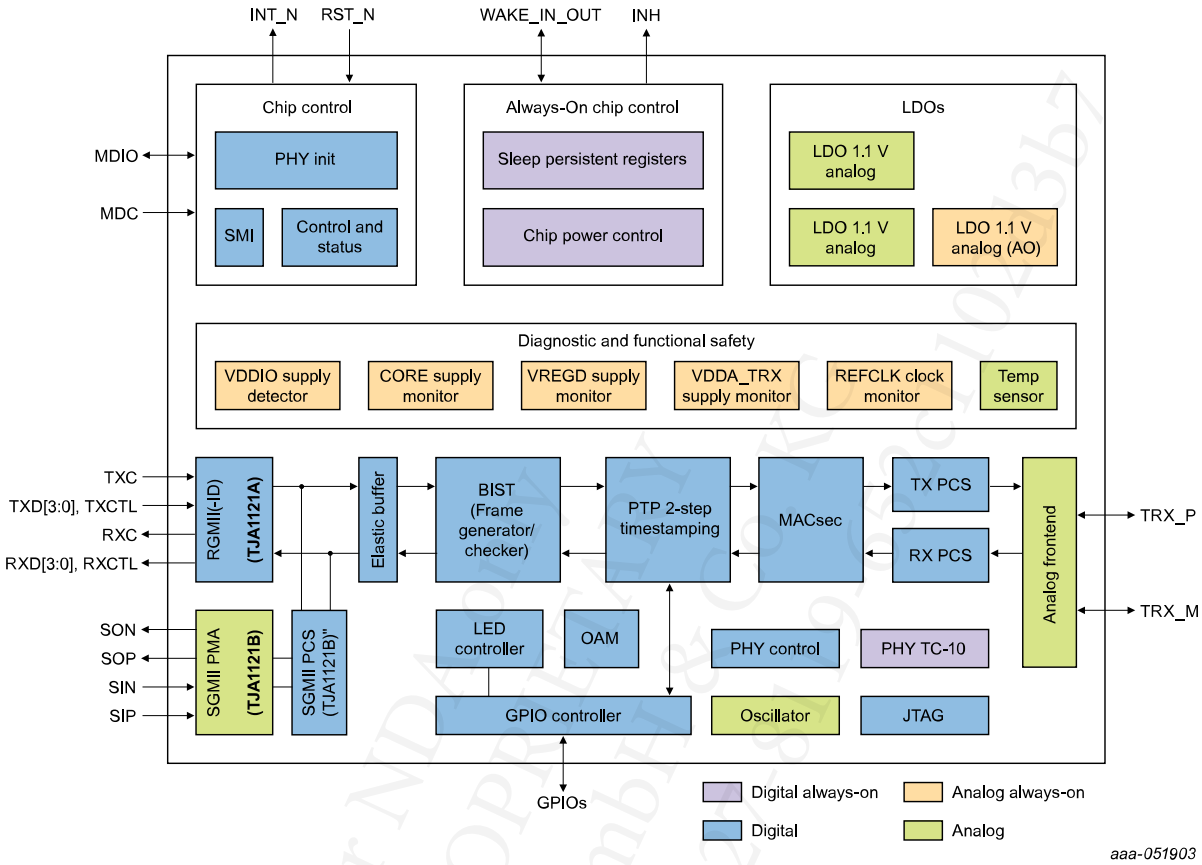


Figure 1. Block diagram

5 Pinning information

5.1 Pinning

The pin configuration of the TJA1121 is shown in Figure 2 and Figure 3.

The following standard interfaces are provided by the TJA1121: MDI, MDO, RGMII (TJA121A) and SGMII (TJA1121B). TJA1121A/B is fully drop-in compatible with TJA1120A/B and pin compatible with TJA1103A/B and TJA1104A/B 100BASE-T1 devices. Refer to the TJA1121 application note [6] for migration guidelines.

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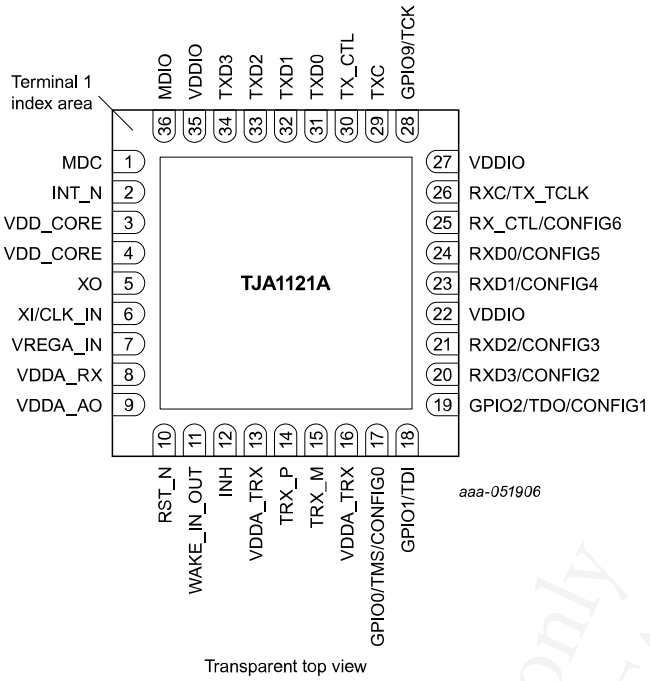


Figure 2. Pinning TJA1121A

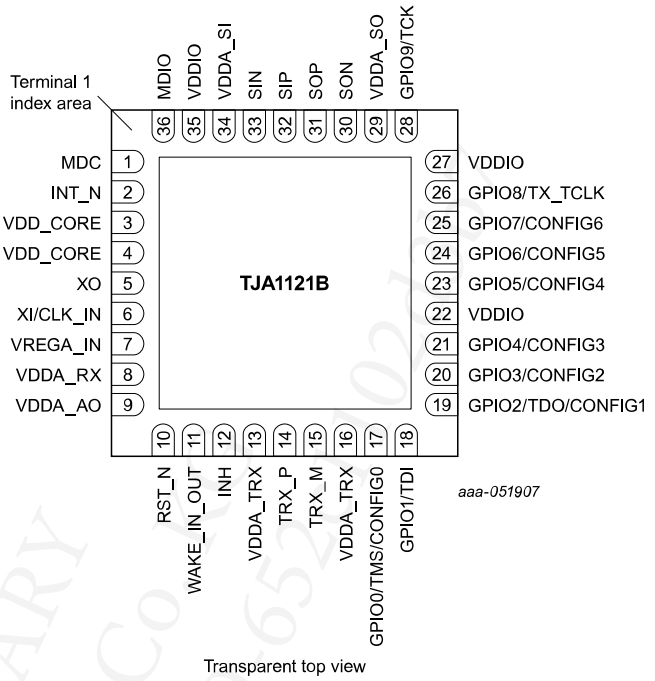


Figure 3. Pinning TJA1121B

5.2 Pin description

Table 2. Pinning TJA1121A

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
MDC	1	I	VDDIO	SMI Clock Input
INT_N	2	O, PU	VDDIO	Interrupt Output
VDD_CORE	3, 4	P	-	Digital supply (1.1V)
XO	5	AO	-	Crystal feedback
XI/CLK_IN	6	AI	-	Crystal input / Digital clock input
VREGA_IN	7	P	-	Analog Regulator input (1,8V, 2.5V, 3.3V)
VDDA_RX	8	P	-	Analog receiver supply (1.8V)
VDDA_AO	9	P	-	Always on supply (2.5 - 3.3V)
RST_N	10	I, PU	VDDA_AO	Reset Input
WAKE_IN_OUT	11	IO	VDDA_AO	Local/forwarding wake-up input/output (configurable)
INH	12	OD	VDDA_AO	Inhibit Output for Vreg control
VDDA_TRX	13	P	-	MDI analog voltage supply (1.8V)
TRX_P	14	AIO	-	Plus terminal for Tx/Rx signal
TRX_M	15	AIO	-	Minus terminal for Tx/Rx signal
VDDA_TRX	16	P	-	MDI analog voltage supply (1.8V)
GPIO0/TMS/CONFIG0	17	IO	VDDIO	GPIO0, JTAG Test mode select, CONFIG0
GPIO1/TDI	18	IO	VDDIO	GPIO1, JTAG Test data in

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Table 2. Pinning TJA1121A*...continued*

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
GPIO2/TDO/CONFIG1	19	IO	VDDIO	GPIO2, JTAG Test data out, CONFIG1
RXD3/CONFIG2	20	IO	VDDIO	RGMII Rx Data 3 output, CONFIG2
RXD2/CONFIG3	21	IO	VDDIO	RGMII Rx Data 2 output, CONFIG3
VDDIO	22	P	-	Digital interface I/O voltage (1.8V, 2.5V, 3.3V)
RXD1/CONFIG4	23	IO	VDDIO	RGMII Rx Data 1 output, CONFIG4
RXD0/CONFIG5	24	IO	VDDIO	RGMII Rx Data 0 output, CONFIG5
RX_CTL/CONFIG6	25	IO	VDDIO	RGMII receive control input, CONFIG6
RXC/TX_TCLK	26	IO	VDDIO	RGMII receive clock, test clock TX_TCLK
VDDIO	27	P	-	Digital interface I/O voltage (1.8V, 2.5V, 3.3V)
GPIO9/TCK	28	IO	VDDIO	GPIO9, JTAG Test clock
TXC	29	IO	VDDIO	RGMII Transmit Clock input
TX_CTL	30	IO	VDDIO	RGMII interface transmit control input
TXD0	31	IO	VDDIO	RGMII Tx Data input
TXD1	32	IO	VDDIO	RGMII Tx Data input
TXD2	33	IO	VDDIO	RGMII Tx Data input
TXD3	34	IO	VDDIO	RGMII Tx Data input
VDDIO	35	P	-	Digital interface I/O voltage (1.8V, 2.5V, 3.3V)
MDIO	36	IO	VDDIO	SMI Data I/O
GNDPAD	37	G	-	Exposed die pad

[1] I: digital input; O: digital output; IO: digital input/output; AO: Analog output; AI: Analog Input; AIO: Analog input/output; PU: weak pull-up; P: power supply; G: ground
[2] Pin type depends on functional mode; consult the functional description for details.
[3] Only applicable to CMOS input/output pins

Table 3. Pinning TJA1121B

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
MDC	1	I	VDDIO	SMI Clock Input
INT_N	2	O, PU	VDDIO	Interrupt Output
VDD_CORE	3, 4	P	-	Digital supply (1.1V)
XO	5	AO	-	Crystal feedback
XI/CLK_IN	6	AI	-	Crystal input / Digital clock input
VREGA_IN	7	P	-	Analog Regulator input (2.5V or 3.3V)
VDDA_RX	8	P	-	Analog receiver supply (1.8V)
VDDA_AO	9	P	-	Always on supply (2.5 or 3.3V)
RST_N	10	I, PU	VDDA_AO	Reset Input
WAKE_IN_OUT	11	IO	VDDA_AO	Local/forwarding wake-up input/output (configurable)

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Table 3. Pinning TJA1121B*...continued*

Symbol	Pin	Type[1][2]	IO supply domain[3]	Description
INH	12	OD	VDDA_AO	Inhibit Output for Vreg control
VDDA_TRX	13	P	-	MDI analog voltage supply (1.8V)
TRX_P	14	AIO	-	Plus terminal for Tx/Rx signal
TRX_M	15	AIO	-	Minus terminal for Tx/Rx signal
VDDA_TRX	16	P	-	MDI analog voltage supply (1.8V)
GPIO0/TMS/CONFIG0	17	IO	VDDIO	GPIO0, JTAG Test mode select, CONFIG0
GPIO1/TDI	18	IO	VDDIO	GPIO1, JTAG Test data in
GPIO2/TDO/CONFIG1	19	IO	VDDIO	GPIO2, JTAG Test data out, CONFIG1
GPIO3/CONFIG2	20	IO	VDDIO	GPIO3, CONFIG2
GPIO4/CONFIG3	21	IO	VDDIO	GPIO4, CONFIG3
VDDIO	22	P	-	Digital interface I/O voltage (1.8V, 2.5V, 3.3V)
GPIO5/CONFIG4	23	IO	VDDIO	GPIO5, CONFIG4
GPIO6/CONFIG5	24	IO	VDDIO	GPIO6, CONFIG5
GPIO7/CONFIG6	25	IO	VDDIO	GPIO7, CONFIG6
GPIO8/TX_TCLK	26	IO	VDDIO	GPIO8, test clock TX_TCLK
VDDIO	27	P	-	Digital interface I/O voltage (1.8V, 2.5V, 3.3V)
GPIO9/TCK	28	IO	VDDIO	GPIO9, JTAG Test clock
VDDA_SO	29	P	-	Analog SGMII transmitter supply (1.1V)
SON	30	AO	-	Minus Terminal for SGMII output
SOP	31	AO	-	Plus Terminal for SGMII output
SIP	32	AI	-	Plus Terminal for SGMII input
SIN	33	AI	-	Minus Terminal for SGMII input
VDDA_SI	34	P	-	Analog SGMII receiver supply (1.1V)
VDDIO	35	P	-	Digital interface I/O voltage (1.8V, 2.5V, 3.3V)
MDIO	36	IO	VDDIO	SMI Data I/O
GNDPAD	37	G	-	Exposed die pad

[1] I: digital input; O: digital output; IO: digital input/output; AO: Analog output; AI: Analog Input; AIO: Analog input/output; PU: weak pull-up; P: power supply; G: ground
[2] Pin type depends on functional mode; consult the functional description for details.
[3] Only applicable to CMOS input/output pins

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6 Functional description

6.1 Operating modes

A mode transition diagram showing all device- and port-level state transitions is shown in Figure 4.

Internally, the device architecture, including the register layout and interrupt tree, is scalable and extends to multiport PHY devices.

See AN14517 [6] for a detailed description of application use-cases.

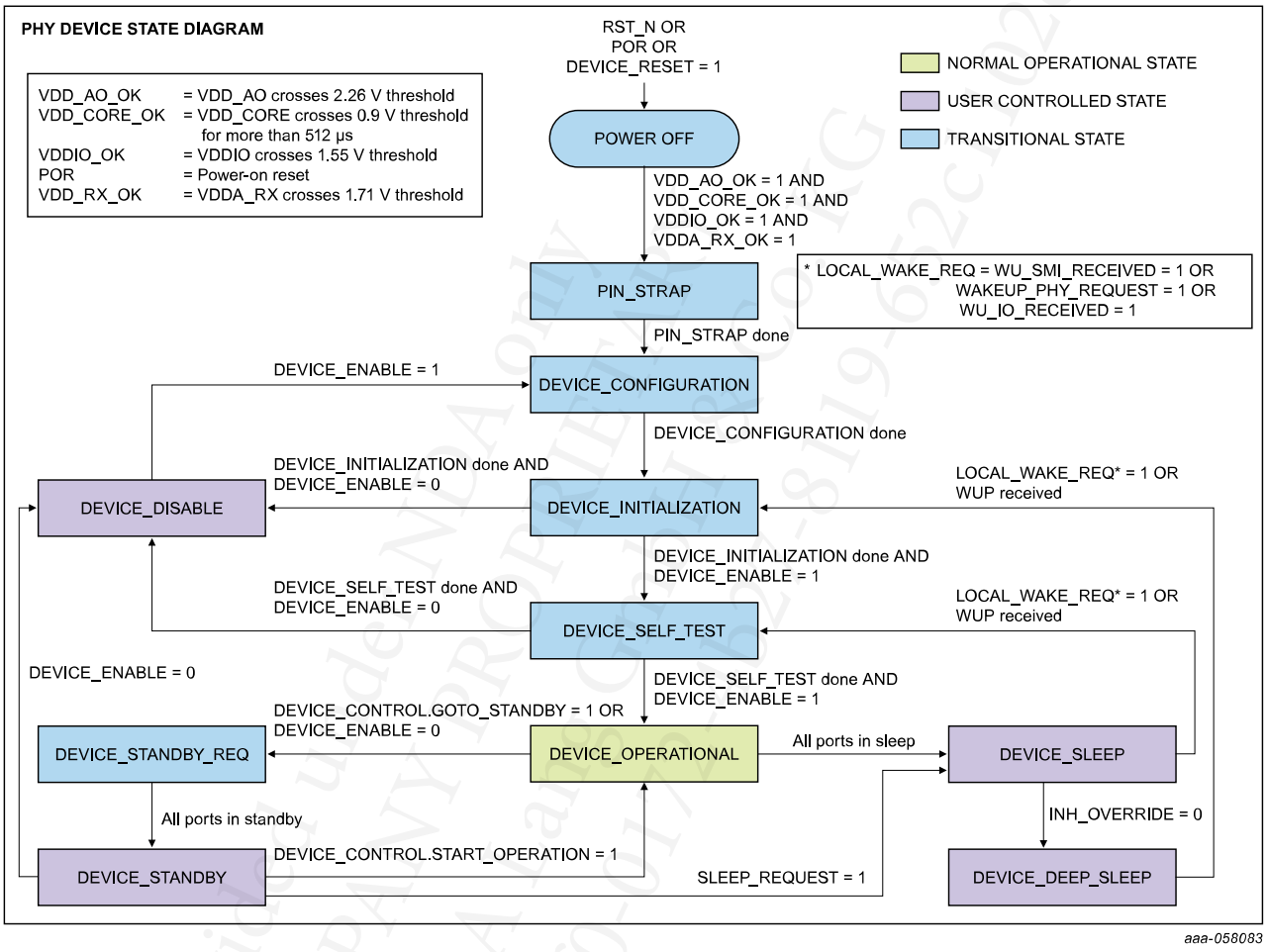


Figure 4. Device-level mode transitions

6.2 Managed/autonomous operation

The device is configured in either managed or autonomous mode, which determines the startup behavior. The mode can be set via pin strapping (see Table 1) or via bit PHY_CONFIG.AUTO_OPERATION (Table 1).

In managed mode, the device transitions to STANDBY state after power-on and device initialization. This allows additional configuration options to be set before bringing the device into OPERATIONAL state by setting EPHY_CONTROL.START_OPERATION. SMI access is required to trigger this transition.

In autonomous mode, the device proceeds to OPERATIONAL state without any interaction needed. In OPERATIONAL state, link-up can begin immediately. This enables very fast link-up after reset and allows

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parallel link-up and host controller boot-up. SMI access is optional in this mode if further configuration or status access is not needed.

6.3 Data path block diagram

The TJA1121 data path consists of multiple connected building blocks: The CMOS IO pads[4] or SGMII SerDes, xMII rate-matching FIFO, the PTP, BIST and MACSec blocks and the 1000BASE-T1 PHY.

Note: *After device reset, PTP and BIST and MACSec blocks are bypassed and xMII and MBT1/ePHY blocks are connected.*

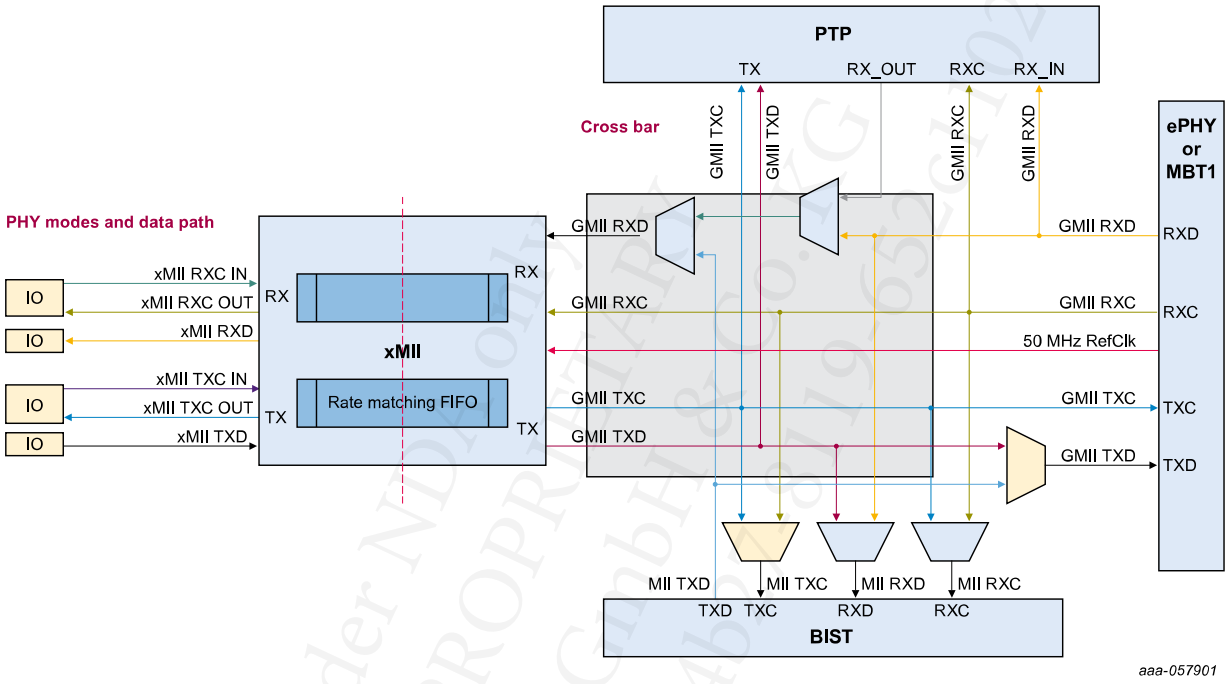


Figure 5. Data path block diagram (except MACsec, refer to 6.12 for details on MACsec)

Table 4. xMII Data path control

MODE	CMOS_OR_SERDES	xMII_MODE	ROLE	TXC dir.	RXC dir.	REFCLK_SRC	TX FIFO	RX FIFO
RGMII	0	111	NA	in	out	XI pin	Yes	No
SGMII	1	001	0/1	-	-	XI pin	Yes	Yes

4 CMOS IO pads (TJA1121A) and SGMII SerDes (TJA1121B)

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Table 5. BIST data path control

Notes:

- *Do not enable XMII_CONTROL.PHY_LOOPBACK(30.AFC0h.1)*
- *Set RGMII_RXC_DELAY_CONFIG.ENABLE(30.AFCDh.15) if external loopback is used.*
- *Refer to Table 18 for possible BIST/loopback combinations*

BIST MODE	BIST_ENABLE	BIST_TX_TO_EPHY	BIST_TX_TO_XMII	BIST_RX_EPHY_OR_XMII	DRIVERMBT1_PORT	CAPTURE_PORT
Mode 1 (EPHY)	1	10	00	1	EPHY_TX	EPHY_RX
Mode 2 (xMII)	1	10	00	0	EPHY_TX	XMII_TX
Mode 3	1	00	10	0	XMII_RX	XMII_TX
Mode 4	1	00	10	1	XMII_RX	EPHY_RX
Disabled (default)	0	-	-	-	DISABLED	

6.4 Startup self-test

The TJA1121 executes a self-test on startup. The test is split into two parts a device-level self-test and a port-level self-test. Both tests are part of the device and port startup sequence (see Section 6.1). The tests consists of several steps including voltage, clock, and latent fault checks.

Voltage supply inputs are tested for overvoltage (OV) and/or undervoltage (UV) conditions. The reference 25 MHz (Crystal) is compared against the internal free-running oscillator.

When both the device and the global self-test checks are completed, the GLOBAL_INFRA_IRQ_SOURCES.DEV_BOOT_DONE is triggered automatically and GLOBAL_FUSA_STATUS.FUSA_CHECK is available to check PASS or FAIL. In case a check has not passed successfully, the issue is reported via the related status register.

In MANAGED mode, Port-level self test will start only if DEVICE_CONTROL.START_OPERATION or PHY_CONTROL.START_OPERATION is set.

In case of Link fail time-out, reading PORT_FUSA_STATUS.FUSA_CHECK needs to be checked for any failures.

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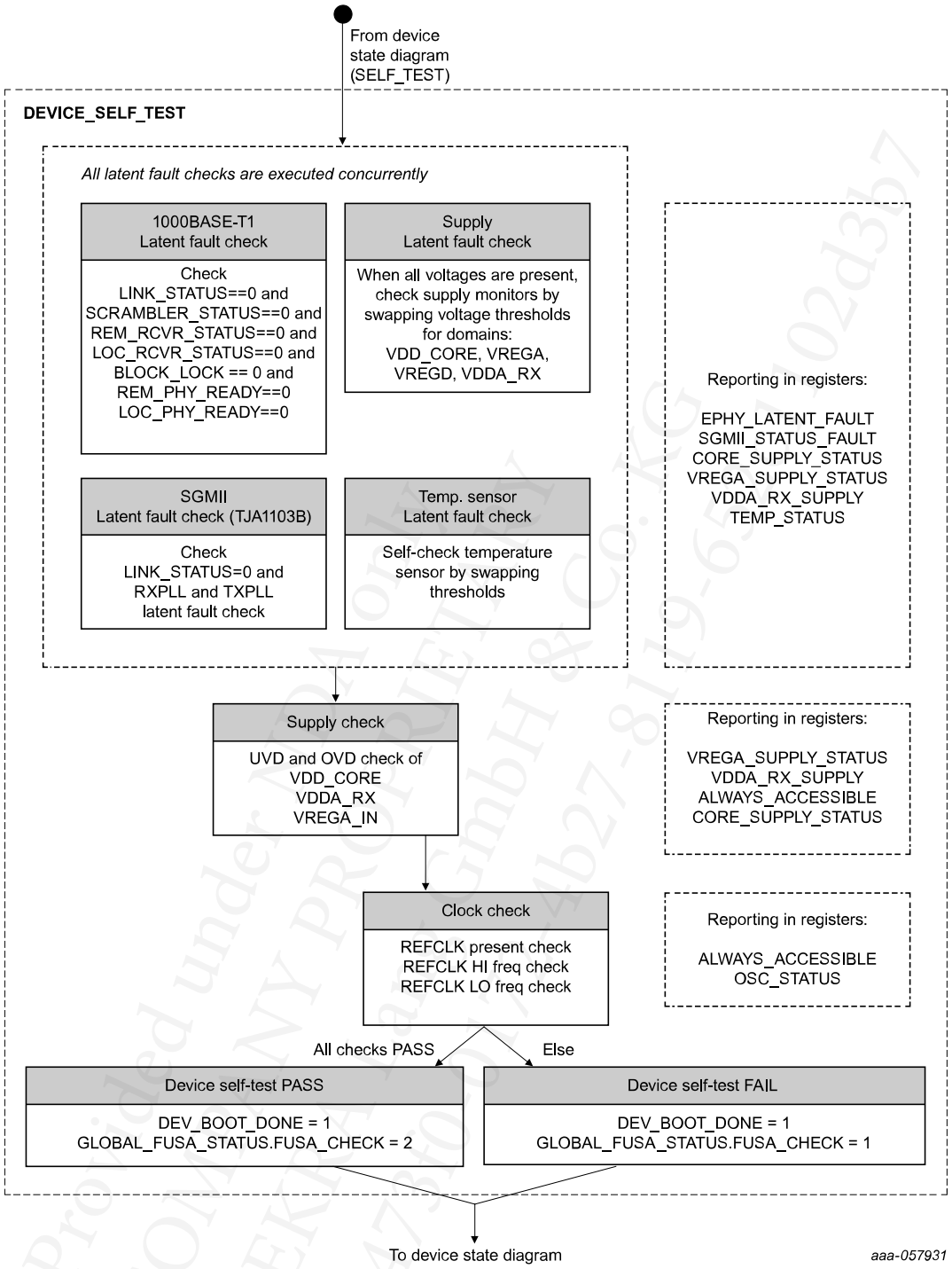


Figure 6. Device self-test sequence

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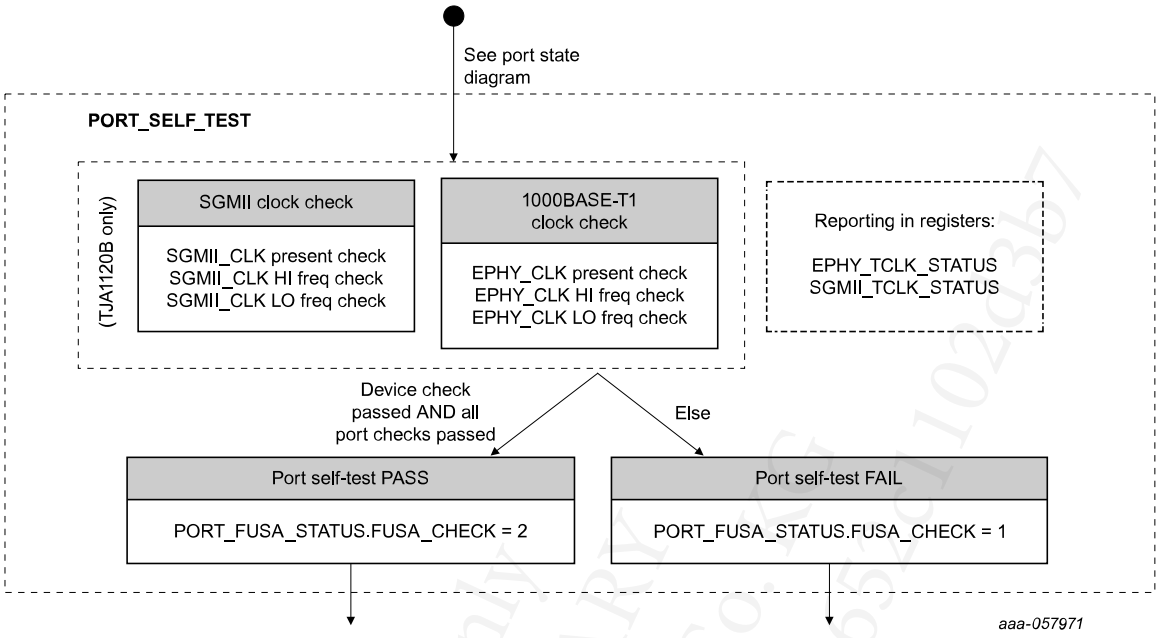


Figure 7. Port self-test sequence

6.5 xMII interfaces

TJA1121 is available in two variants, supporting RGMII (TJA1121A) or SGMII (TJA1121B).

Submodes of RGMII and SGMII can be selected via pinstrapping, as defined in Section 6.7.1. Settings can also be changed through register settings.

6.5.1 RGMII

The RGMII [4] PHY-MAC connection is shown in Figure 8. The RGMII protocol is intended as an alternative to the IEEE 802.3 GMII standard [3], which is not supported. The objective is to reduce the number of pins between MAC and PHY in a cost-effective and technology-independent way.

Control signals are multiplexed and transmitted data is synchronous to both clock edges (double data rate). The interface is intended for 1000 Mbit/s operation.

RGMII is a symmetrical interface and TXC is generated by the MAC and RXC is generated by the PHY. For 1000 Mbit/s operation, the clock operates at 125 MHz (± 50 ppm) and data is driven on both rising and falling edge of the appropriate clock.

RX_CTL and TX_CTL encode the RX_DV and RX_ER as well as TX_EN and TX_ER, respectively. RX_CTL/TX_CTL present the RX_DV/TX_EN on the rising edge and RXERR/TXERR on the falling edge. RXERR and TXERR are encoded to minimize the number of transitions during nominal data transmission.

TXERR <= TX_ER xor TX_EN

RXERR <= RX_ER xor RX_DV

This device implements RGMII v1.3 as well as RGMII 2.0 by supporting the RGMII integrated-delay option (RGMII-ID). Optional delay in the clock path can be added independently in the TXC and the RXC clock path.

Table 6. RGMII modes

RGMII mode[1]	Description
RGMII	In RGMII mode, the interface functions according to RGMII v1.3 (no internal delay is implemented)

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Table 6. RGMII modes...*continued*

RGMII mode[1]	Description
RGMII-ID	In RGMII-ID mode the interface functions as per RGMII v2.0 (internal delay on RXC is enabled, no internal delay on TXC).
RGMII-ID (TX/RX)	In RGMII-ID (TX/RX) mode the interface functions as per RGMII v2.0 (internal delay on both RXC and TXC is enabled).

[1] This table only covers application usecases, which are available as pinstrap options. See Section 6.7. All possible TX/RX combinations for internal delay can be selected via register.

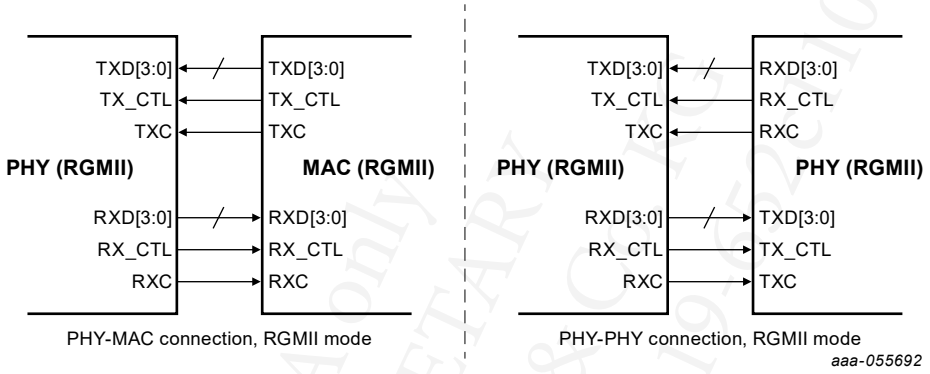


Figure 8. RGMII signaling

6.5.1.1 RGMII pinning

The following table shows the mapping of the RGMII interface pins to device pins. Unused GPIOs can be left floating, consult Section 6.15.2 for details on how to use available GPIOs.

Table 7. Pinning in RGMII-mode

Function	Pin	Symbol	Type[1]	Description
RXD3	20	RXD3/CONFIG2	O	Receive data bit 3 of RXD[3:0] nibble
RXD2	21	RXD2/CONFIG3	O	Receive data bit 2 of RXD[3:0] nibble
RXD1	23	RXD1/CONFIG4	O	Receive data bit 1 of RXD[3:0] nibble
RXD0	24	RXD0/CONFIG5	O	Receive data bit 0 of RXD[3:0] nibble
RX_CTL	25	RX_CTL/CONFIG6	O	Receive data control output
RXC	26	RXC/TX_TCLK	O	Receive clock
TXC	29	TXC	I	Transmit clock
TX_CTL	30	TX_CTL	I, PD	Transmit data control input
TXD0	31	TXD0	I, PD	Transmit data bit 0 of TXD[3:0] nibble
TXD1	32	TXD1	I, PD	Transmit data bit 1 of TXD[3:0] nibble
TXD2	33	TXD2	I, PD	Transmit data bit 2 of TXD[3:0] nibble
TXD3	34	TXD3	I, PD	Transmit data bit 3 of TXD[3:0] nibble

[1] I: digital input; O: digital output; IO: digital input/output; PU: weak pull-up; PD: weak pull-down

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6.5.1.2 RGMII clock delay

The RGMII integrated-delay is controlled by pin strapping and optionally register control. The most common modes can be selected directly from pin strapping, see Section 6.7.1.

The clock delay can be configured via registers RGMII_TXC_DELAY_CONFIG and RGMII_RXC_DELAY_CONFIG.

6.5.2 SGMII

6.5.2.1 Introduction

The Serial-GMII (SGMII) [5] is a SerDes interface for PHY-MAC connection. The SGMII protocol is intended to be an alternative to the RGMII standard. It uses less interface signals and provides better EMC performance by using differential signaling.

The device provides the SGMII 4-wire interface[5], which implements clock data recovery so no additional clock pairs are required.

The transfer rate of SGMII is fixed at 1.25 GBaud. The device automatically enables the SGMII transmitter (SON, SOP) on device startup in order to reduce time to communication.

6.5.2.2 Block diagram

A block diagram of the SGMII subsystem is shown in Figure 9.

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5 SGMII 8-wire interface is not supported.

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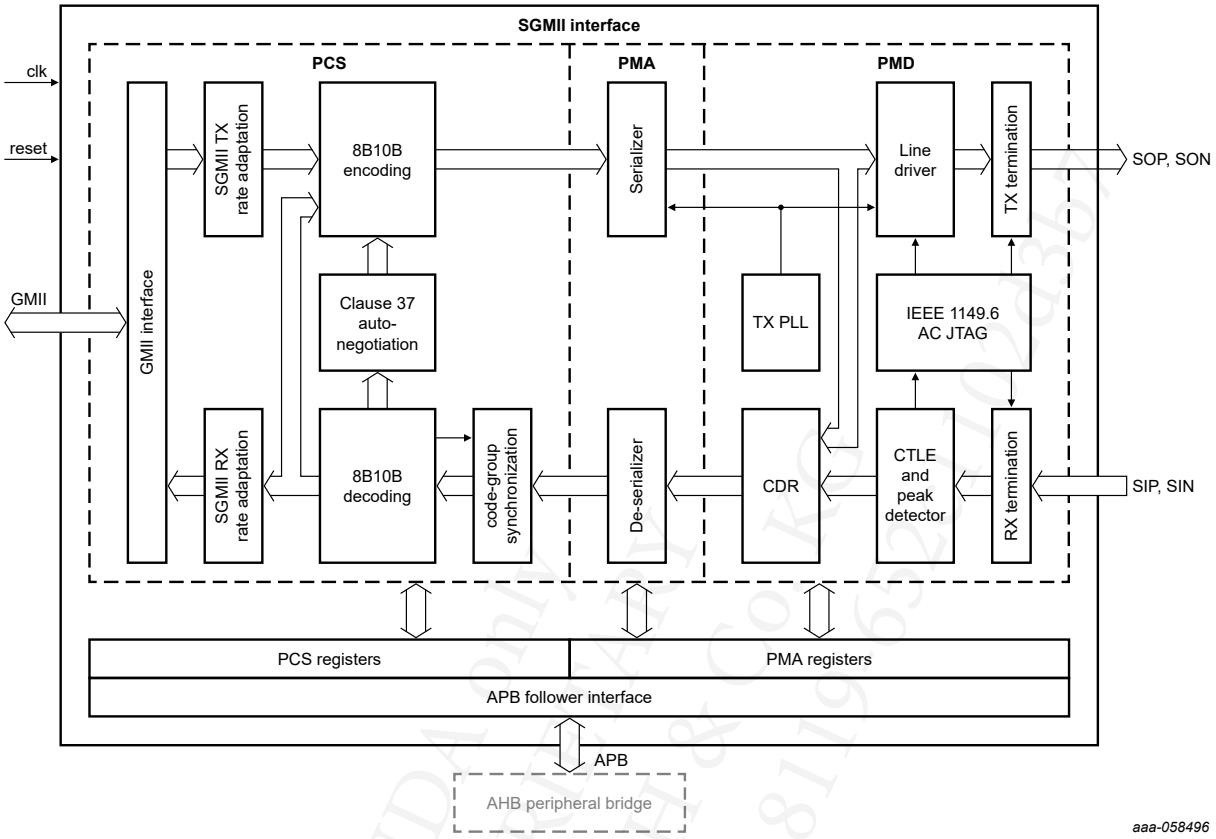


Figure 9. SGMII block diagram

6.5.2.3 Signaling

The SGMII block is AC-compliant to the SGMII 1.8 specification. The interface must always be AC coupled with capacitors of 100 nF.

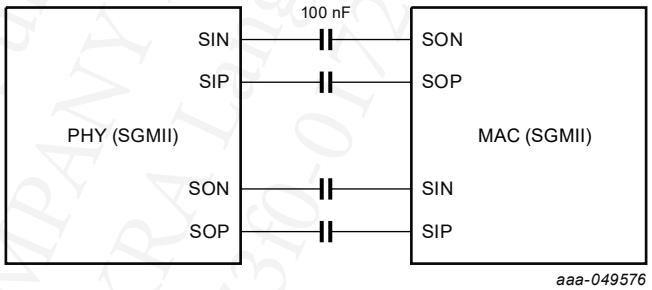


Figure 10. SGMII signalling

6.5.2.4 Polarity

The polarity of the differential pairs SI and SO can be flipped by the register bits SGMII_PCS_CONFIG1.RX_POL_INV_0 and SGMII_PCS_CONFIG1.TX_POL_INV_0, respectively.

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6.5.2.5 Auto-Negotiation

The SGMII interface implements (optional) Auto-Negotation. In this mode, PHY, and MAC handshake the supported interface capabilities to find the optimal operating conditions. The device advertise its speed, duplex mode (full-duplex or half-duplex).

In SGMII-PHY mode, the device communicates link speed as well as 1000BASE-T1 link availability by passing tx_config_Reg from PHY to MAC. If configured in SGMII-MAC mode, the received autonegotiation data is ignored.

Auto-negotiation is automatically enabled and control information automatically exchanged from PHY to MAC via tx_config_Reg[15:0] as shown in Table 13. The PHY advertises the LINK_AVAILABILITY via tx_config_Reg[15]. Duplex mode and speed are fixed to full duplex and 1000 Mbps.

Table 8. Auto-Negotiation control information passed between PHY and MAC

Bit	tx_config_Reg[15:0] sent from the PHY to the MAC	tx_config_Reg[15:0] sent from the MAC to the PHY
15	Link: 1 = link up, 0 = link down	0
14	Reserved for Auto-Negotiation acknowledge as specified in 802.3z	1
13	0	0
12	Duplex mode: 1 = full duplex, 0 = half duplex	0
11 to 10	Speed: Bit 11, 10: 1 1 = Reserved 1 0 = 1000 Mbps 0 1 = 100 Mbps 0 0 = 10 Mbps	0
9 to 1	0h	0h
0	1	1

6.5.2.6 Mode of operation

6.5.2.6.1 Normal operation

During normal operation, the transmit and receive data path of the SGMII interface are active. The transmit path accepts Ethernet data frames from the on-chip GMII interface, applies rate adaptation for 1000 Mbps Ethernet data and applies 1000BASE-X 8B10B encoding. After serialization, the data is driven onto the physical medium by the line driver, which has an on-chip transmitter termination for impedance match to the medium. The TXPLL generates the half bit-rate clock necessary for serializing the data. The chip-internal GMII interface clocks are derived from the clock generated by the TXPLL.

The receive path receives a serial data stream from the physical medium, to which is provides an impedance match with the on-chip receiver termination. The CTLE equalizes the received signals, to compensate for channel losses, before recovering the clock and data from the serial stream by the CDR. The recovered data is deserialized before aligning and decoding 8B10B code groups. Finally, rate adaption is applied for 1000 Mbps Ethernet data before making the data frames available for on-chip GMII interface.

The interface starts autonomously in normal mode. If the interface is in standby state, setting the SGMII_ADVANCED_CONTROL.START_OPERATION bit transitions interface into normal mode.

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6.5.2.6.2 Standby

The SGMII interface can be disabled by setting the bit SGMII_ADVANCED_CONTROL.START_OPERATION.

6.5.2.6.3 PCS loopback

The on-chip PCS RX-to-TX loopback is an operating mode where the received data is not applied by the PCS to the on-chip GMII interface but is applied instead to the transmit path of the PCS. In this operating mode the SGMII interface effectively sends the serial data stream, received from the SGMII link partner, back out to the link partner through the transmitter data path.

6.5.2.6.4 PMA loopback

The on-chip PMA TX-to-RX loopback is an operating mode where the serial data stream of the transmit path is not provided to the line driver but is looped back instead to the CDR, which in turn does not receive the serial data from the SGMII link partner, through the CTLE, but recovers the data from the transmit path's serializer instead.

6.5.3 Alternate xMII pinning

The RGMII standards do not dictate the physical pin arrangement of the xMII bus. Next to the default xMII pinning of the TJA1121A/TJA1121B, which are described in Section 5, the devices have the possibility for alternate mappings. These alternate mappings can be selected through register XMII_DATA_CONFIG. These settings can be applied independently to allow more compact PCB layouts with less or no signal swaps.

If set, the TXD30_SWAP bit will flip the TXD bus from TXD[3:0] to TXD[0:3]. If set, the TXCTL_SELECT bit moves the TX_CTL function on the opposite side of the TXD[3:0] bus.

If set, the RXD30_SWAP bit will flip the RXD bus from RXD[3:0] to RXD[0:3]. If set, the RXCTL_SELECT bit moves the RX_CTL function on the opposite side of the RXD[3:0] bus.

Exemplery alternate pinning configurations are shown in Figure 11.

Any GPIO pin that is made available on RXD[2:3] and TXD[2:3] moves respectively.

Notes:

- *The alternate settings XMII_DATA_CONFIG.TXD30_SWAP and XMII_DATA_CONFIG.RXCTL_SELECT cannot be selected through pin strapping. Thus, on device startup the RGMII pins are as per default configuration.*
- *The alternate pinning described in this section only applies to the TJA1121A variant.*

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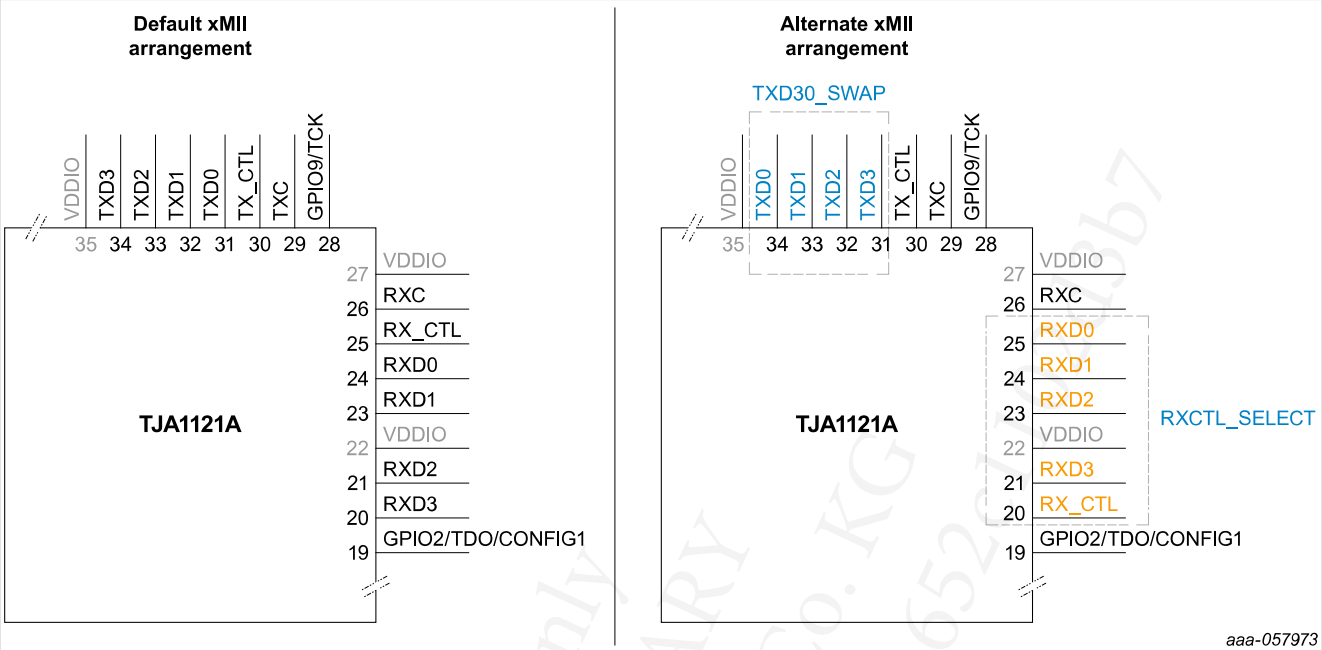


Figure 11. Default and exemplary alternate xMII pin arrangements

6.5.4 xMII elastic buffers

The xMII interface uses elasticity buffers to compensate between clock differences of the 1000BASE-T1 link and the xMII clocks. These buffers are able to compensate rate differences of up to 100 ppm in all xMII modes. The clocking scheme and thus the need of buffering depends on the selected xMII mode.

Note: *When using a crystal that exceed the RGMII specification (50 ppm), the MAC must also be able to operate in such conditions.*

The latencies caused by these buffers are included in the overall xMII to MDI latency numbers reported in the dynamic characteristics. Table 9 shows the mode buffers that are added in the data path and when they are bypassed.

Table 9. Elastic buffer scheme

xMII mode	TX Interface	RX Interface
RGMII	enabled	bypassed
SGMII	enabled	enabled

6.6 MDIO

The MDIO interface is a management interface to control the device through a two-wire shared bus interface. Individual PHYs are selected through a PHYAD identifier, configured through pin strapping. This device supports up to 27 addresses, see Section 6.7.1.

Note: *PHYAD zero is reserved and cannot be assigned. Optionally, the device can listen on PHYAD zero for broadcast read/writes.*

There are two protocol versions of MDIO, IEEE 802.3 Clause 22 and IEEE 802.3 Clause 45. The earlier Clause 22 standard allows to address 32 registers and up to 32 PHY devices. The more recent Clause 45 standard allows to address up to 65536 registers per PHY per MMD. Both MDIO protocols are supported by the TJA1121, but only Clause 45 gives a full view on all registers, see Section 6.6.5 and Section 6.6.4.

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Frame formats of both standards are provided in Section 6.6.2 and Section 6.6.1.

Note: *Do not interleave Clause 22 with Clause 45 accesses. Use Clause 22 or Clause 45 exclusively.*

Figure 12 shows a typical MDIO topology.

Note: *The TJA1121 does not have integrated pull-up or pull-down resistors in MDIO or MDC.*

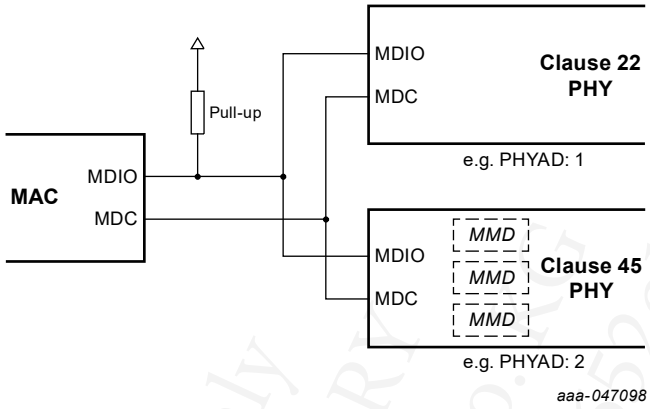
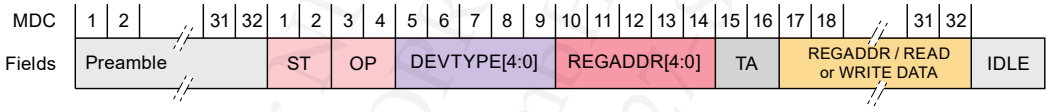


Figure 12. MDIO signaling

6.6.1 IEEE 802.3 Clause 45

Figure 13 shows the IEEE Clause 45 frame format and access types.



	Management Frame Fields							
	PRE	ST	OP	PRTAD	DEVAD	TA	ADDRESS/DATA	IDLE
Address	1...1	00	00	PPPPP	EEEEE	10	AAAAAAAAAAAAAAAA	Z
Write	1...1	00	01	PPPPP	EEEEE	10	DDDDDDDDDDDDDDDD	Z
Read	1...1	00	11	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z
Post-read-increment-address	1...1	00	10	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

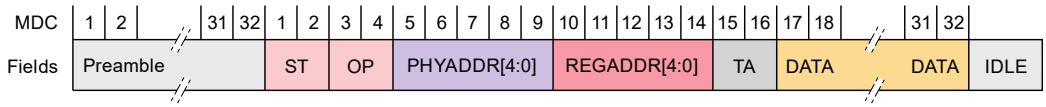
aaa-038231

Figure 13. Clause 45 frame format and access types

6.6.2 IEEE 802.3 Clause 22

Figure 14 shows the IEEE Clause 22 frame format and access types.

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	Management Frame Fields							
	PRE	ST	OP	PHYAD	REGAD	TA	DATA	IDLE
Read	1...1	01	10	AAAAA	RRRRR	Z0	DDDDDDDDDDDDDDDD	Z
Write	1...1	01	01	AAAAA	RRRRR	10	DDDDDDDDDDDDDDDD	Z

aaa-038230

Figure 14. Clause 22 frame format and access types

6.6.3 MDIO Broadcast

The device supports MDIO broadcast functionality, which is disabled by default. If enabled, the TJA1121 responds on a configurable PHY address (by default PHYAD 0) to reads and writes.

This feature can be enabled by setting the bit SMI_CONFIG.BROADCAST_ENABLE.

6.6.4 Clause 45 MMD register organization

The device supports the following MMDs:

- PMA (MMD 01)
- PCS (MMD 03)
- NXP Vendor Specific (MMD 30)
- NXP Vendor Specific (MMD 31)

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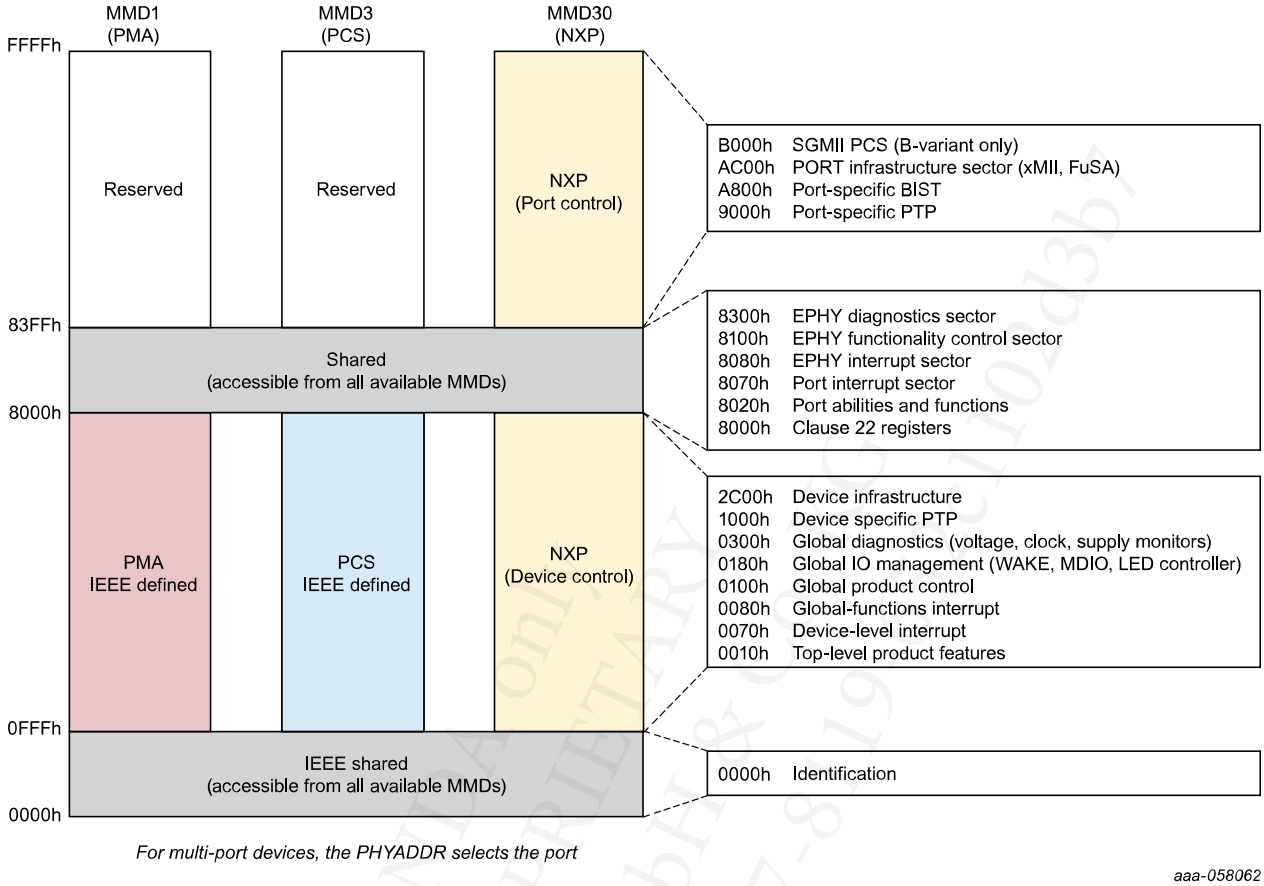


Figure 15. Organization of MMD1, MMD3 and MMD30

Note: *MMD31 is described in the Section 6.12*

The lower part of MMD 1 and MMD 3 following the IEEE 802.3 specification for PCS and PMA devices. The lower part of MMD 30 contains registers that are device or otherwise global control registers (e.g., LED, GPIO,...). The higher part of MMD 30 configures port related functions.

Note: *The register organization separates port from device control in order to support multi-port devices (not used in TJA1121).*

Common functionality is mapped into MMD1/3/30, see Shared MMD. In this sense, the Shared MMD is not a real MMD itself. The shared registers are mapped into the range between 0000h-001fh and 8000h-83ffh.

Standard Clause 22 registers are mapped in the vendor-specific register space of all MMDs from 8000h – 801Fh.

6.6.5 Clause 22 register organization

The device implements limited Clause 22 access as listed in Section 7.1, which are the only registers that are directly accessible through IEEE Clause 22 and IEEE Clause 45 type transactions. It must be noted that the CL22 registers are not exclusive to IEEE Clause 22 accesses.

The CL22 registers are comprised of:

- PHY identification registers.
- FAVORITE_IRQS, which contains common IRQs. This register can be used to shortcut the IRQ tree.
- IEEE Clause 45 indirection registers, used if the MAC does not support Clause 45 MDIO.

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- ALWAYS_ACCESSIBLE register, which is the only register accessible without presence of VDD_CORE. VDDIO and VDDA_AO must be present to access this register.

The CL22 register table lists the register offsets as mapped into Shared MMD Clause 45 space.

Notes:

- *The addresses as listed in the CL22 register table cannot be used directly for a IEEE Clause 22 MDIO access. The CL22 register table lists the register offsets as mapped into Shared MMD Clause 45 space. When accessing these registers through a physical Clause 22 transaction, 8000h must be subtracted from the offset as given in the register table. For example, PHY_ID_1 is located at 2h and ALWAYS_ACCESSIBLE at 1Fh.*
- *There is some limitation in switching between clause 22 and clause 45 access. If both, CL22 and CL45, are used, consult application notes for additional information.*

6.7 Pin strap

6.7.1 Pin strap configuration

The TJA1121 uses pin strap configuration to set critical configuration data. For this, the CONFIG pins of the device support a three-level strapping, which is sensed after device reset (power-on reset, RST_N pin or reset register). All configuration, except the PHYAD, can be changed at any time over MDIO.

During the sampling, it must be ensured that the CONFIG pins are not overruled by the microcontroller pin configuration. This can be achieved if

- The RX interface of the xMII microcontroller pins is high-ohmic during initialization phase (recommended),
- The RX interface of the xMII microcontroller pins is actively driving and are logically consistent with the specified by pin strapping or
- The RST_N pin is not released until the relevant microcontroller pins are configured as inputs (high-ohmic).

Figure 16 shows the application of the pull-up and pull-down configuration. Care must be taken if a CONFIG pin is used to drive loads that are pulling opposite to the configuration resistor. For LEDs, the polarity can be selected to match the direction of the configuration resistor. In case negative LED polarity is used (as shown in the pull-down case), the polarity of the LED driver must be adjusted via software.

The pin strap sequence has successfully latched the configuration latest 2 ms after RST_N is deasserted and power supplies are available.

Table 10. 3-Level Strap Resistor Values

Strap Setting	R_{pu}	R_{pd}
O	OPEN	OPEN
H	10 kΩ	OPEN
L	OPEN	10 kΩ

Notes:

- *The sum of all resistances on a CONFIG pin must be greater than 145 kΩ in order to be reliably detected as O (OPEN).*
- *Consult the application note [6] for layout guidelines.*

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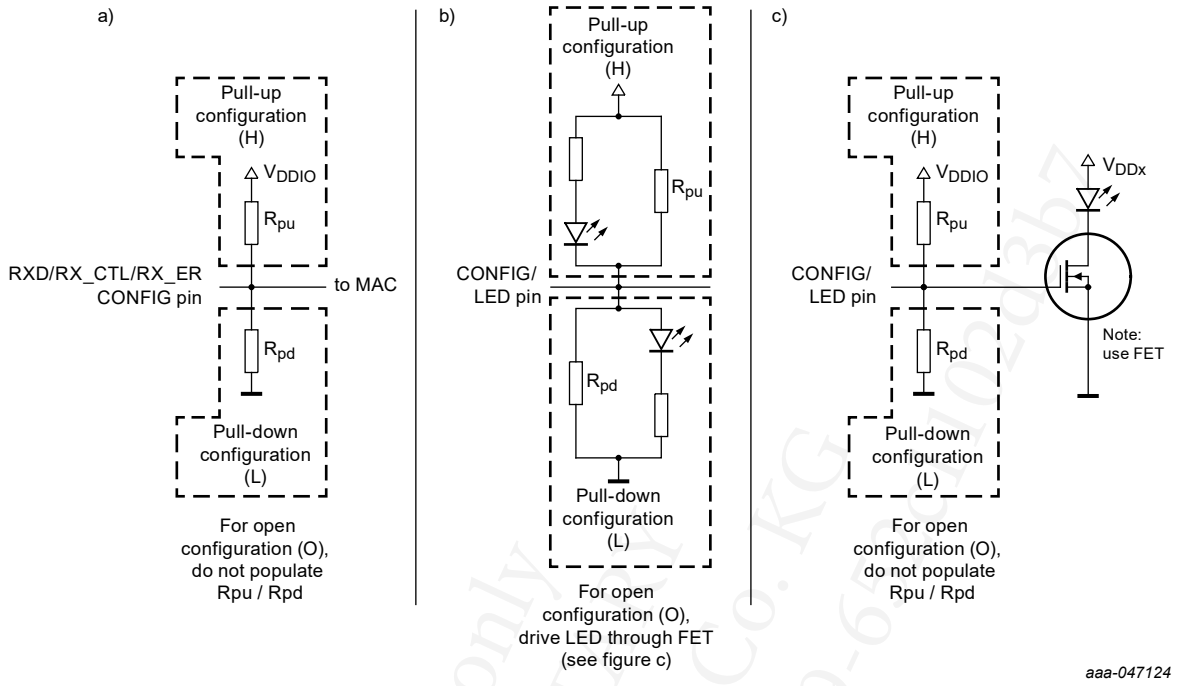


Figure 16. Pin strap configurations

6.7.2 Configuration options

The available CONFIG pins are grouped to configure MODE, PHYAD and miscellaneous settings.

Table 11. MODE group

xMII MODE (TJA1121A)	CONFIG3	CONFIG4
RGMII-ID	O	O
RGMII	H	O
RGMII-ID (TX/RX)	L	O
JTAG	L	L

xMII MODE (TJA1121B)	CONFIG3	CONFIG4
SGMII-PHY	O	O
SGMII-MAC	O	H
JTAG	L	L

Table 12. PHYAD group

PHYAD	CONFIG0	CONFIG1	CONFIG2
LED active-low configuration			
1	H	H	H
2	H	H	L
3	H	H	O

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Table 12. PHYAD group...*continued*

PHYAD	CONFIG0	CONFIG1	CONFIG2
4	H	L	H
5	H	L	L
6	H	L	O
7	H	O	H
8	H	O	L
9	H	O	O
LED active-high configuration			
10	L	H	H
11	L	H	L
12	L	H	O
13	L	L	H
14	L	L	L
15	L	L	O
16	L	O	H
17	L	O	L
18	L	O	O
FET required to drive LED			
19	O	H	H
20	O	H	L
21	O	H	O
22	O	L	H
23	O	L	L
24	O	L	O
25	O	O	H
26	O	O	L
27	O	O	O

Note: *Other CONFIG options are reserved*

Table 13. Clock pinstrapping

CLK select	Pin CONFIG5
LVDS	O
XO mode	H
single ended	L

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Table 14. Miscellaneous pinstrapping

LEADER/FOLLOWER	AUTONOMOUS mode	Pin CONFIG6
LEADER	enabled	O
FOLLOWER	enabled	H
register controlled (default: FOLLOWER)	disabled	L

6.8 Reconfiguration

Writing to the configuration registers is disabled by default to avoid unintended reconfiguration. To change the configuration, configuration register writing must be enabled. There are multiple, hierarchical CONFIG_ENABLE switches. Table 15 gives an overview of the registers and the way each is protected.

There is one device configuration enable bit (SUPER_CONFIG_ENABLE) that enables all reconfiguration. The configuration usually occurs after power-on as part of the software initialization routine. However, PHY reconfiguration is possible at any later time.

All other registers that are not listed in this table are not write protected.

Table 15. Reconfiguration protected registers

Level 1	Level 2	Level 3	Protected registers
MMD30.DEVICE_CONTROL.SUPER_CONFIG_ENABLE			
	MMD30.DEVICE_CONTROL.GLOBAL_CONFIG_ENABLE		
			MMD30.DEVICE_CONFIG
			MMD30.WISE_CONFIG
			MMD30.WISE_PARAMETERS
			MMD30.SMI_CONFIG
			MMD30.DEVICE_CONFIG_EXTENDED
			MMD30.SMI_DATA_IO_CONFIG
			MMD30.COMPL_TEST_CONTROL
	MMD30.GLOBAL_INFRA_CONTROL.CONFIG_ENABLE		
			MMD30.GPIO0_FUNC_CONFIG
			—
			MMD30.GPIO9_FUNC_CONFIG
			MMD30.GPIOx_IO_CONFIG
	MMD30.PORT_CONTROL.CONFIG_ENABLE		
			MMD30.PORT_FUNC_ENABLES
		Shared.PHY_CONTROL.CONFIG_ENABLE	
			MMD1.BASE_T1_PMA_CONTROL
			MMD1.E1000BT1_TEST_MODE_CONTROL
			Shared.PHY_CONFIG
			Shared.PHY_PARAMETERS
			Shared.WAKE_SLEEP_CONFIG
			Shared.WAKE_SLEEP_PARAMETERS
			Shared.COMPL_TEST_CONTROL
		MMD30.PORT_BIST_CONTROL.CONFIG_ENABLE	
			MMD30.BIST_INTERCEPT_CONFIG

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Table 15. Reconfiguration protected registers*...continued*

Level 1	Level 2	Level 3	Protected registers
			MMD30.PORT_BIST_CONFIG
		MMD30.PORT_INFRA_CONTROL.CONFIG_ENABLE	
			MMD30.INFRA_CONFIG
			MMD30.MII_BASIC_CONFIG
			MMD30.XMII_CLK_CONFIG
			MMD30.XMII_CLK_IO_CONFIG
			MMD30.XMII_DATA_CONFIG
			MMD30.XMII_DATA_IO_CONFIG
			MMD30.RGMII_RXC_DELAY_CONFIG
			MMD30.RGMII_TXC_DELAY_CONFIG
			MMD30.MONITOR_CONFIG
		MMD30.SGMII_ADVANCED_CONTROL.CONFIG_ENABLE	
			MMD30.SGMII_ADVANCED_CONFIG
		MMD30.GPTP_CONTROL.CONFIG_ENABLE	
			MMD30.PTP_CLK_NSEC
			MMD30.PTP_SYNC_PERIOD_0
			MMD30.PTP_SYNC_PERIOD_1
			MMD30.LTC_LOOP_CONTROL
		MMD30.PTP_CONTROL.CONFIG_ENABLE	
			MMD30.PTP_MESSAGE_FILTER
			MMD30.PTP_FILTER_ENABLES
			MMD30.TX_DELAY_NSEC
			MMD30.TX_DELAY_XTD
			MMD30.RX_DELAY_NSEC
			MMD30.RX_DELAY_XTD
			MMD30.USER_MAC_DA_0
			MMD30.USER_MAC_DA_1
			MMD30.USER_MAC_DA_2
			MMD30.USER_MAC_DA_MASK_0
			MMD30.USER_MAC_DA_MASK_1
			MMD30.USER_MAC_DA_MASK_2
			MMD30.FILTER_PARAMETERS

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Table 16 gives an overview of the test configuration enable hierarchy.

Table 16. Test configuration register enables

TEST_ENABLE will not affect the configuration but enables the test function.

Level 1	Level 2	Level 3	Affected test functions
MMD30.DEVICE_CONFIG.SUPER_TEST_ENABLE			
	MMD30.DEVICE_CONFIG.TEST_ENABLE		
			MMD30.COMPL_TEST_CONTROL.COMPL_TCLK_EN
	MMD30.PORT_FUNC_ENABLES.TEST_ENABLE		
		Shared.PHY_CONFIG.TEST_ENABLE	
			MMD1.PMA_CONTROL1.REMOTE_LOOPBACK
			MMD1.PMA_CONTROL1.LOCAL_LOOPBACK
			MMD1.E1000BT1_TEST_MODE_CONTROL.TEST_MODE
			MMD3.PCS_CONTROL1.LOOPBACK
			MMD3.E1000BT1_PCS_CONTROL.LOOPBACK
			Shared.TDR_CONTROL.ENABLE
			Shared.TDR_CONTROL.START
		MMD30.PORT_BIST_CONFIG.TEST_ENABLE	
			MMD30.PORT_BIST_CONFIG.PACKET_BIST_ENABLE
		MMD30.INFRA_CONFIG.TEST_ENABLE	
			MMD30.XMII_CONTROL.MAC_LOOPBACK
			MMD30.XMII_CONTROL.PHY_LOOPBACK

6.9 Polarity detection/correction

The 1000BASE-T1 PHY can automatically detect and flip the MDI polarity (polarity detection and correction). Optionally, setting the PHY_CONFIG.POLARITY_CORRECT_DISABLE bit disables automatic polarity correction.

The MDI polarity can be swapped by setting the PHY_CONFIG.POLARITY_SWAP bit. If set, the pins TRX_P and TRX_M are swapped and polarity detection and correction operate on the swapped pinning.

The status of the polarity inversion is reflected by the PHY_STATUS.DETECTED_POLARITY bit. If polarity correction is disabled, the bit PHY_STATUS.DETECTED_POLARITY flags a conflict between actual signal polarity and current MDI polarity. In that case, the pin POL_SWAP should be inverted.

A polarity conflict is also signaled via an IRQ in only PHY_FUNCTIONAL_IRQ_SOURCES group, consult Section 6.11.

6.10 Diagnosis

6.10.1 Link up status

There are multiple flags that capture link up using different semantics.

The PHY_STATUS.LINK_AVAILABLE bit is asserted if all signals, LOC_PHY_READY, REM_PHY_READY, LINK_STATUS, are asserted. Optionally, glitches are filtered with an hysteresis. The hysteresis is configured with PHY_PARAMETERS.LINK_AVAILABLE_HYST_TIME. For LINK_AVAILABLE_HYST_TIME = 0, this corresponds to the compliance 'link up condition'. This signal is then identical to communication ready (comm_ready) as defined in OPEN Alliance.

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The SENDN_OR_DATA status bit indicates if the PHY Control state machine is in SEND_DATA state.

The LINK_MONITOR_STATUS bit indicates the link status from Link Monitor FSM.

The RECEIVE_LINK_STATUS flag shows if the PMA/PMD receive link is up. This flag is latched-low.

Note: *It is recommended to use the LINK_AVAILABILITY flag as a link state indication.*

6.10.2 SQI

The signal quality indicator (SQI) is part of the SIGNAL_QUALITY register. The field SQI determines the link quality between values 0 (worst, no-link) and 7 (best). The SQI value is only valid if the VALID bit in the SIGNAL_QUALITY register is set. Assuming Gaussian noise, an SQI level of 3 and more indicates a BER of 10^{-10} and better.

In addition to the current SQI value, the lowest observed SQI value observed is stored in SQI_WORST. The SQI_WORST is cleared on read and set to the current SQI value. In order to avoid extensive SQI polling, an SQI threshold level can be defined. In case the threshold is exceeded an IRQ (SQI_WARN) is asserted. The threshold can be configured in register SQI_WARN_LIMIT. Setting this register to 0h disables the feature.

6.10.3 MSE

The mean square error is captured after channel equalization which has removed ISI, echo and attenuation effects. The MSE metric is capturing signal distortion caused by uncorrelated noise and interference. The MSE is captured after setting the MSE.ENABLE bit.

The 8-bit MSE value as read from the register is scaled by 1024. To derive the mean square error in units of V2 it must be divided by 1024. The resulting value is between 0.0 V2 (best) and 0.25 V2 (worst). The root mean square error can be derived as follows:

$$\mathrm{RMSE} = \sqrt{\frac{\mathrm{MSE}}{1024}}\, V_{\mathrm{rms}} \tag{1}$$

6.10.4 Test Modes

The 1000BASE-T1 PHY supports six test modes according to IEEE Std 802.3, Clause 96 [3].

These test modes allow testing of the transmitter waveform, transmitter distortion, transmitter jitter, transmitter droop, and BER.

The respective test mode is entered by setting MMD30.EPHY_CONTROL.CONFIG_ENABLE enabling the test mode bit TEST_ENABLE and writing the test mode number to in MMD1.E1000BT1_TEST_MODE_CONTROL.TEST_MODE.

6.10.4.1 Test mode 1

Test mode 1 enables testing of timing jitter on LEADER and FOLLOWER transmitters. LEADER and FOLLOWER 1000BASE-T1 PHYs are connected over a link segment defined in 97.6. When in this mode, the 1000BASE-T1 PHY provides access to TX_TCLK125. This 125 MHz test clock is on sixth frequency divided version of TX_TCLK that times the transmitted symbols.

6.10.4.2 Test mode 2

Test mode 2 is for transmitter jitter testing on MDI when transmitter is in LEADER timing mode. When test mode 2 is enabled, the 1000BASE-T1 PHY transmits a continuous pattern of three {+1} symbols followed by three {–1} symbols with the transmitted symbols timed from its local clock source of 750 MHz. The transmitter output is a 125 MHz signal.

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6.10.4.3 Test mode 3

Test mode 3 is not defined and the corresponding register is reserved for future use.

6.10.4.4 Test mode 4

Test mode 4 is for transmitter linearity testing. When test mode 4 is enabled, 1000BASE-T1 PHY transmits the sequence of symbols generated by the scrambler generator polynomial per the equations below.

The maximum length shift register used to generate the sequences defined by this polynomial is updated once per symbol interval (2/750 MHz). The bits stored in the shift register delay line at a particular time n are denoted by Scr_n[10:0]. At each symbol period the shift register is advanced by one bit and one new bit represented by Scr_n[0] is generated Bits Scr_n[8] and Scr_n[10] are exclusive ORed together to generate the next Scr_n[0] bit. The bit sequences x0n, x1n, and x2n generated from conbinations of the scrambler bits, as shown in the following equations, is used to generate the ternary symbols T0n and T1n. The transmitter times the transmitted symbols from a 750 MHz 0.01 % clock in the LEADER timing mode. The transmit signal level and spectral shaping in this mode is the same as in normal operation mode.

6.10.4.5 Test mode 5

Test mode 5 is for checking whether the transmitter is compliant with the transmit PSD mask and the transmit power level. When test mode 5 is enabled, 1000BASE-T1 PHY transmits as in non-test operation and in LEADER data mode with data set to normal Inter-Frame idle signals.

6.10.4.6 Test mode 6

When test mode 6 is enabled, 1000BASE-T1 PHY transmits a continuous pattern of fifteen {+1} symbols followed by fifteen {-1} symbols with the transmitted symbols timed from its local clock source of 750 MHz. The transmitter output is a 25 MHz signal.

6.10.4.7 Test mode 7

Test mode 7 is for enabling measurement of the bit error ratio of the link including the RS-FEC encoder/decoder, transmit and receive analog front ends of the 1000BASE-T1 PHY, and a cable connecting two 1000BASE-T1 PHYs. This mode reuses the 1000BASE-T1 normal (non-test) mode with zero data pattern. Instead of encoding received data from MAC, continuous zero data pattern is encoded. In the receive side, after FEC and 80/81 decoding, zero data sequence is expected with no error. Any no zero data bit received is counted as an error and calculated in BER.

6.10.5 Recovered clock (TCLK)

The recovered clock can be externalized on Pin 26. To enable TX_TCLK, the following registers must be configured:

- PHY_CONTROL.CONFIG_ENABLE=1 → enable access to PHY_COMPLIANCE_TEST register
- DEVICE_CONTROL.GLOBAL_CONFIG_ENABLE=1 → enable access to DEVICE_CONFIG register
- DEVICE_CONFIG.TEST_ENABLE=1 → allow test functions
- PHY_COMPLIANCE_TEST.TX_TCLK_EN=1 → enable tclk

6.10.6 Loopback

Each PHY supports the following loopback modes:

- PMA local loopback

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- PMA remote loopback
- PCS loopback
- xMII PHY loopback
- xMII MAC loopback

6.10.6.1 PMA local loopback

In this mode, the PMA is placed in local loopback mode. The PMA receive function utilizes the echo signals from the unterminated MDI. It decodes these signals to pass the data back to RGMII Receive Interface as shown in Figure 17.

The MAC can compare the packets sent through RGMII transmit function to the packets received from the RGMII receive function to validate PCS and PMA functions.

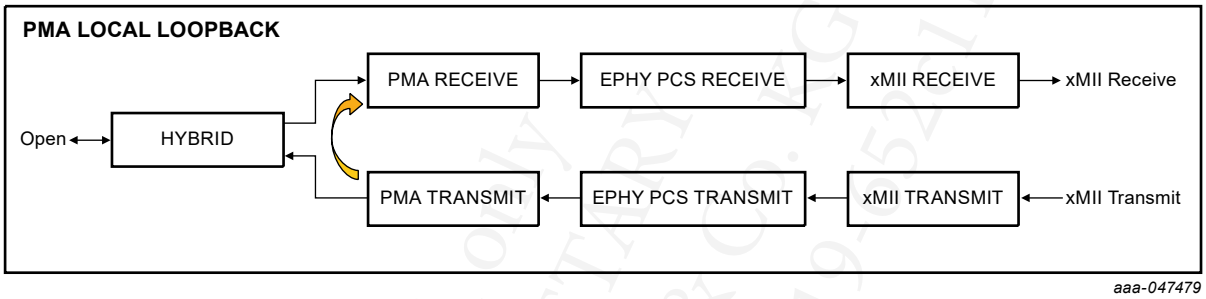


Figure 17. PMA local loopback

To enable PMA local loopback, the LOCAL_LOOPBACK bit in MMD1.PMA_CONTROL1 must be set. The PHY can be in any PHY_MODE. Data can be sent once the following flags in MMD30.PHY_STATUS are set:

- LINK_AVAILABLE
- SENDN_OR_DATA
- REM_RCVR_STATUS
- SCRAMBLER_STATUS
- LINK_STATUS
- LOC_RCVR_STATUS
- LOC_PHY_READY
- REM_PHY_READY
- BLOCK_LOCK

Note: *PCS loopback overrules PMA loopback*

6.10.6.2 PMA remote loopback

In PMA Remote loopback mode, the data transmitted by the link partner on the MDI is mirrored back to the link partner. The received symbols are passed through the PMA receive and PCS receive function, forwarded to the PCS transmit function, which in turn sends it back to the link partner.

PMA Remote loopback allows the MAC to compare the packets sent to the MDI with the packets received back from the MDI and, therefore, to validate the functionality of the physical channel, including both 1000BASE-T1 PHYs.

To enable PMA remote loopback the REMOTE_LOOPBACK bit in MMD1.PMA_CONTROL1 must be set.

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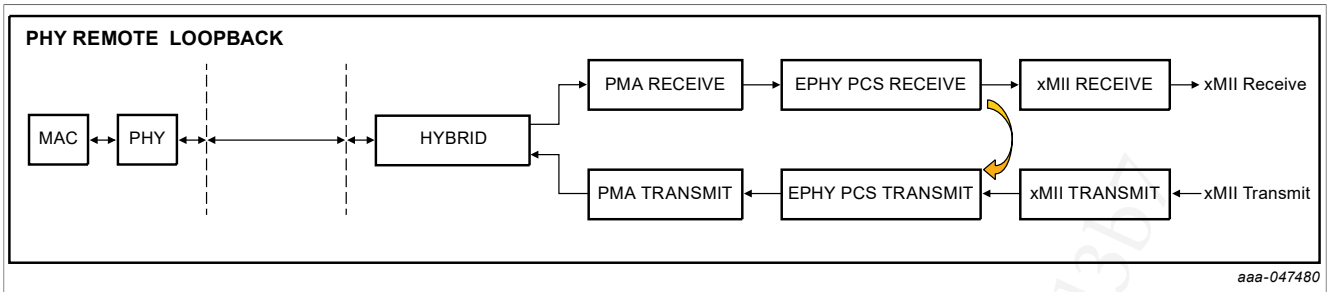


Figure 18. PMA remote loopback

6.10.6.3 PCS loopback

In Internal loopback, the PCS Receive function gets xMII Receive data directly from the PCS Transmit function as shown in Figure 19. The xMII Receive data is transmitted through the PCS subsystem and mirrored to the xMII Transmit interface.

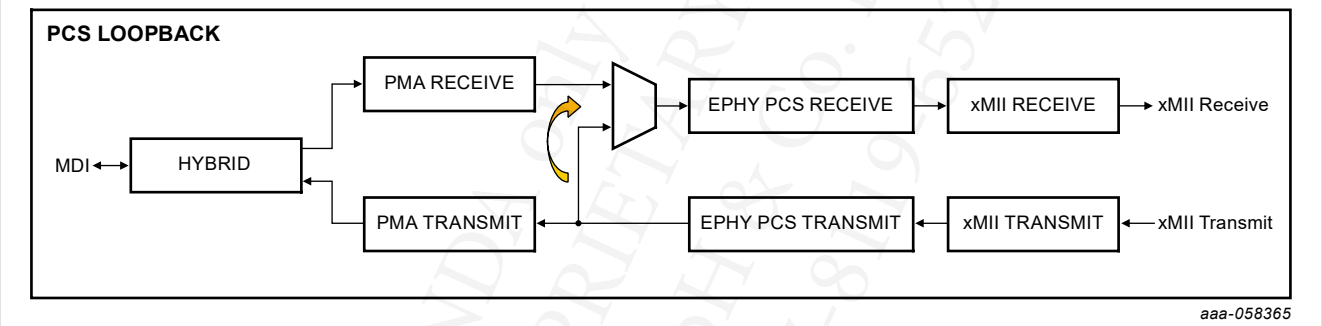


Figure 19. PCS Loopback

PCS loopback is enabled by setting the LOOPBACK bit in MMD3.PCS_CONTROL1.

6.10.6.4 xMII PHY loopback

In xMII PHY loopback mode, the frames received by the link partner at the MDI are passed through the PMA receive and PCS receive functions and forwarded to the xMII transmit function, which in turn sends it back to the link partner from where it came.

xMII PHY loopback allows the MAC to compare the packets sent to the MDI with the packets received back from the MDI and, therefore, to validate the functionality of the physical channel, both 1000BASE-T1 PHYs and the xMII function including rate matching FIFOs.

xMII PHY loopback is enabled by setting the XMII_CONTROL.PHY_LOOPBACK bit.

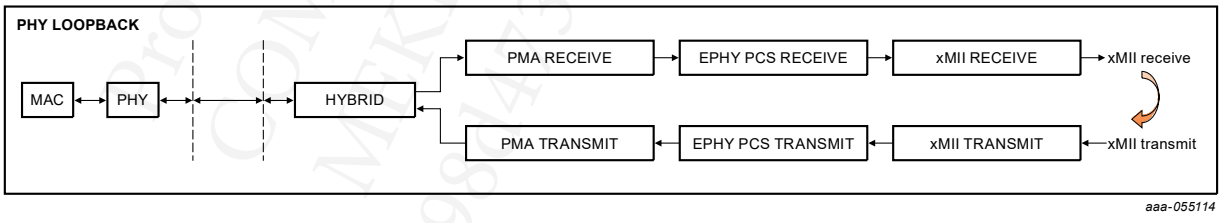


Figure 20. xMII PHY loopback

Note: *BIST mode not to be enabled in this mode of operation.*

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6.10.6.5 xMII MAC loopback

The xMII MAC loopback is similar to PCS loopback but directly loops back at the xMII interface.

xMII PHY loopback is enabled by setting the XMII_CONTROL.MAC_LOOPBACK bit.

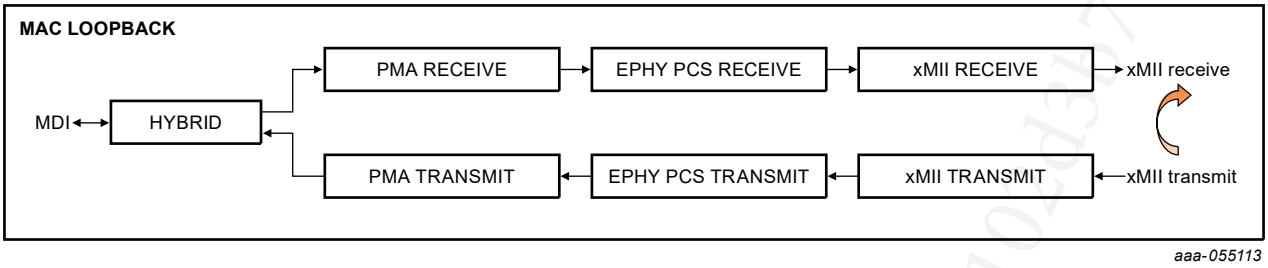


Figure 21. xMII MAC loopback

6.10.7 BIST Generator/Checker

The BIST (built-in self test) is an integrated Ethernet frame generator and checker intended for diagnosis. Both generator and checker can be enabled independently. BIST generator provides full configurability on Ethernet frame header with programmable interpacket gap (IPG).

BIST frame generator can generate mix of both good and bad frames. The BIST checker can also monitor the incoming Ethernet frames and provide statistics about number of good and bad frames received.

Together with loopback modes, TJA1121 supports a number of BIST configuration allowing self-testing of the full communication chain.

6.10.7.1 Introduction

The Built In Self-Test Generator and Checker (BIST) allows to test the data path without MAC connected to a PHY device.

Normally a MAC sends Ethernet frames to the PHY over an xMII interface. These frames are transmitted over the MDI and are eventually received by the remote PHY. The remote PHY transmits the frames through its xMII to the receiving MAC. For debugging purposes, the BIST module is able to replicates this behavior without a MAC being involved. The BIST module consists of a generator and a checker. The generator generates

Ethernet frames and the checker receives these frames and checks for errors. The number of erred frames is captured in statistics counters.

Figure 22 shows the simplified block diagram.

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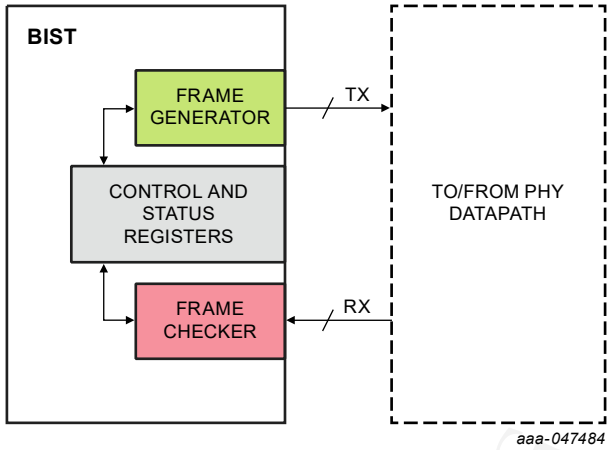


Figure 22. BIST block diagram

6.10.7.2 Functional description

The BIST module is not active by default, and generator and checker need to be enabled (see registers BIST_CHECK_CTRL and BIST_GEN_CTRL).

The generator composes Ethernet frames with configurable length, payload content and other header fields. The generated frames are dispatched either xMII or MDI facing (see BIST_INTERCEPT_CONFIG.TX_INTERCEPT and BIST_INTERCEPT_CONFIG.RX_INTERCEPT.

The BIST checker checks the incoming data based on MDI or xMII (RXD) direction (see CHECK_EPHY_OR_XMII). The checker validates the CRC and RX_ER and logs status and diagnostics information.

There are two modes of operation.

Production mode is intended for production test. In this mode, the total number of frames generated by the checker is configurable. The same number of frames is expected at the checker. The total number of frames is a configurable combination of good frames and error injected frames (erroneous FCS). The BIST checker gathers statistics until a configurable timeout has expired (BIST_WAIT_TIMER) or all expected frames were received. On completion of the test, the bit BIST_CHECK_DONE is set. The test verdict is then stored in register BIST_CHECK_FAIL.

Continuous mode is similar to stress test known from Ethernet test equipment. In this mode, the BIST generator generates continuous Ethernet frames and statistics are continuously collected. This mode can be used to generate Ethernet frames by the BIST module itself or it acts as a sink for frames generated by external test equipment.

6.10.7.3 Generator

The BIST Generator is capable of generating the IEEE 802.3 Ethernet frame format with varying frame length and different types of payload data. Once the BIST generator is set, it starts generating the Ethernet frame according to configuration.

Table 17. Custom BIST generator frame

Preamble 7 bytes	SFD 1 byte	DA 6 bytes	SA 6 bytes	EtherType 2 bytes	Frame Payload Data 46-1500 bytes	FCS 4 bytes	IPG 12 bytes

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The number of preamble octets is configurable (PREAMBLE_LENGTH).

The IPG selection can be configured in one of the following modes:

- Fixed value (IPG_LENGHT_TYPE=0). IPG_LENGTH controls the IPG for all frames.
- Increment-by-one (IPG_LENGHT_TYPE=1) and roll-over to IPG_LENGTH when the current IP reaches 255.
- Random IPG (IPG_LENGHT_TYPE=2). IPG_LENGTH acts as a minimum IPG as well as a LFSR seed.

The EtherType is configurable via the ETHER_TYPE register.

The generator supports various modes for the payload data length generation. It can by FIXED (PAYLOAD_DATA_TYPE=0), INCREMENT (PAYLOAD_DATA_TYPE=1) or RANDOM (PAYLOAD_DATA_TYPE=2).

In FIXED mode the number of octets is set by PAYLOAD_SIZE.

In INCREMENT mode, the data length starts with value PAYLOAD_SIZE[7:0]. The data length increments until a value of of PAYLOAD_SIZE[13:8]×256+255 is reached, then it rolls-over to PAYLOAD_SIZE[7:0].

In RANDOM mode, the payload length is determined by a 14-bit LFSR x14+x13+x12+x2+1. PAYLOAD_SIZE is used as seed. PAYLOAD_SIZE[7:0] sets the minimum payload size. PAYLOAD[13:8] determined the maximum payload size.

If LFSR[13:8] ≤ PAYLOAD_SIZE[13:8], the full 14-bit LFSR[13:0] value determines the payload size.

If LFSR[13:8] > PAYLOAD_SIZE[13:8], the payload size is determined by LFSR[9:0].

If LFSR[13:0] < PAYLOAD_SIZE[7:0], the payload size is set to PAYLOAD_SIZE[7:0].

Frames in the standard range of 64 to 1535 octets are generated if PAYLOAD_SIZE[13:8] is set to 05h and PAYLOAD_SIZE[7:0] is set to 40h.

If PAYLOAD_SIZE[13:8]>05h, jumbo frames will be generated. Jumbo frames up to the max size of 16383 bytes will be generated if PAYLOAD_SIZE[13:8]=3Fh.

The payload data is configurable and the following types are supported:

- FIXED (PAYLOAD_DATA_TYPE = 0). The fixed data octet is controlled via FIXED_PAYLOAD_DATA.
- INCREMENT (PAYLOAD_DATA_TYPE = 1). The data starts with 00h, increments by one and rolls over at ffh.
- PRBS (PAYLOAD_DATA_TYPE = 2). Three different PRBS schemes can be selected via BIST_PRBS_SELECT (PRBS7, PRBS9, PRBS11, PRBS15). The random seed is configured via BIST_LFSR_SEED. It is configurable (LFSR_SEED_USAGE) if the PRBS sequence is reset to the seed for each frame or continues free-running.

The BIST generator calculates the FCS as specified in the Ethernet standard and appends it to the frame. All frames are transmitted with TX_ER=0.

The number of good frames and error frames is programmable. The total number of generated frames is the number of good frames plus number of error injected frames. The generator first produces all good frames, and then produces all bad frames.

In continuous mode, the generator only sends good frames and the good/bad frame plans are ignored. Setting the STOP bit stops the continues generation of frames. To restart frame generation set generation enable.

6.10.7.4 Checker

The BIST checker qualifies all received Ethernet frames and generate the status and statistical results. The checker is enabled by programming BIST_CHECKER_EN. Generator and checker work fully independent of each other. That means that the generator and checker functionality can be distributed across two physical devices.

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The checker classified frames into good and bad frames. Good frames are frames with a correct FCS and bad frames with an incorrect FCS. The checker also counts all frames are are received with RX_ER set (R_RXER_FRAME_CNT).

The statistics (R_GOOD_FRAME_CNT and R_BAD_FRAME_CNT) can be reset by STATISTC_CNT_RST.

In production mode, status informations are available once the BIST test is done.

- Good frame error indicates that the number of good frames does not match the expected number according to the configured plan.
- Bad frame error indicates that the number of bad frames does not match the expected number according to the configured plan.
- RX_ER detected is set, if a frame is received with RX_ER indication asserted.
- RX_DV not detected is set if no data valid indicated is observed.

Table 18 shows the supported combinations of BIST configurations and loopback modes."

Table 18. Supported BIST loopback combinations

No	BIST		MDI Loopbacks			xMII Loopbacks		Supported		Alternate configuration for variant B
	Driver Port	Receiver Port	PMA Local	PCS	PMA Remote	xMII PHY	xMII Ext[1][2]	Variant A	Variant B	
			1.0000h.0	3.0000h.14	1.0000h.1	30.AFC0h.0				
1	xMII RX[3]	xMII TX[4]					enabled	Yes	Yes	
2	xMII RX[3]	MDI RX[5]	enabled			enabled		Yes	Yes	
3	xMII RX[3]	MDI RX[5]		enabled		enabled		Yes	Yes	
4	xMII RX[3]	MDI RX[5]	enabled				enabled	Yes	No	8
5	xMII RX[3]	MDI RX[5]		enabled			enabled	Yes	No	9
6	MDI TX[6]	MDI RX[5]	enabled					Yes	Yes	
7	MDI TX[6]	MDI RX[5]		enabled				Yes	Yes	
8[7]	MDI TX[6]	xMII TX[4]	enabled				enabled	Yes	Yes	
9	MDI TX[6]	xMII TX[4]		enabled			enabled	Yes	Yes	
10	MDI TX[6]	MDI RX[5]			enabled			Yes	Yes	
11	xMII RX[3]	MDI RX[5]			enabled		enabled	Yes	No	12
12[7]	MDI TX[6]	xMII TX[4]			enabled		enabled	Yes	Yes	

[1] Enable RGMII-ID (RX)
[2] External RGMII or SGMII loopback
[3] RX_INTERCEPT = '10'; TX_INTERCEPT = '00'
[4] CHECK_EPHY_OR_XMII = '0'
[5] CHECK_EPHY_OR_XMII = '1'
[6] RX_INTERCEPT = '00'; TX_INTERCEPT = '10'
[7] Recommended for complete path test

6.10.8 Interrupts

The device is able to signal important events, for instance changes in link availability, via its IRQ_N pin. In order to efficiently deduce which of the available events has fired, an interrupt tree is implemented. The IRQ_N pin is driven low when at least one IRQ in the tree is active and enabled. Otherwise, the IRQ_N pin is high-Z.

Multiple devices can share the IRQ_N line using a wired-OR topology.

6.10.8.1 Interrupt tree

The interrupt tree concept is designed to support multi-port PHY devices and differentiates between port, global and device level interrupt sources. All port-level IRQs are present once, because this device has one port. Common functionality that is shared among multiple data ports is captured in the global section.

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On device level the port and global level are joined together. The ALWAYS_ACCESSIBLE register serves a special purpose. This register is available for MDIO read and write access in cases when the VDD_CORE supply or the chip internal 25 MHz REFCLK is not present.

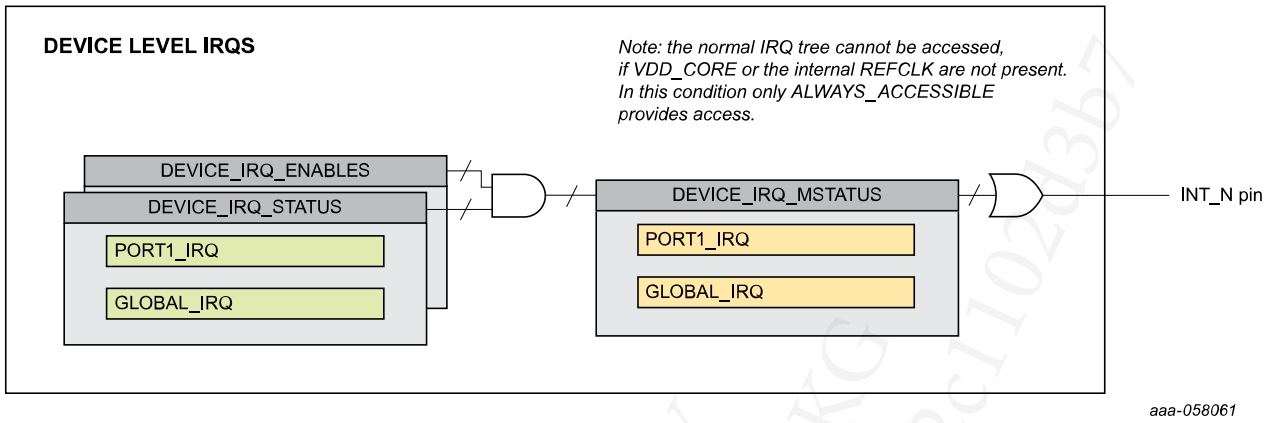


Figure 23. Device IRQ

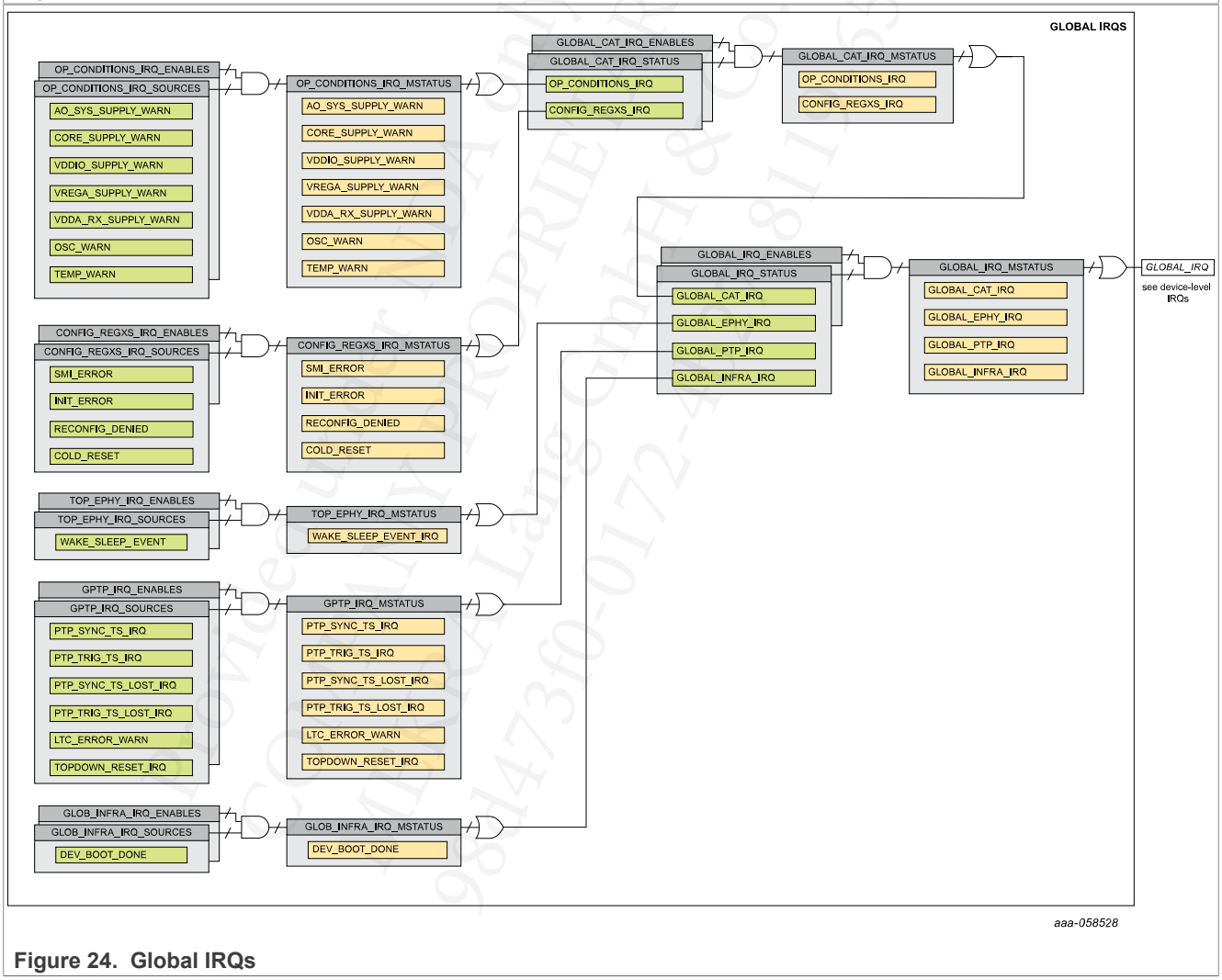


Figure 24. Global IRQs

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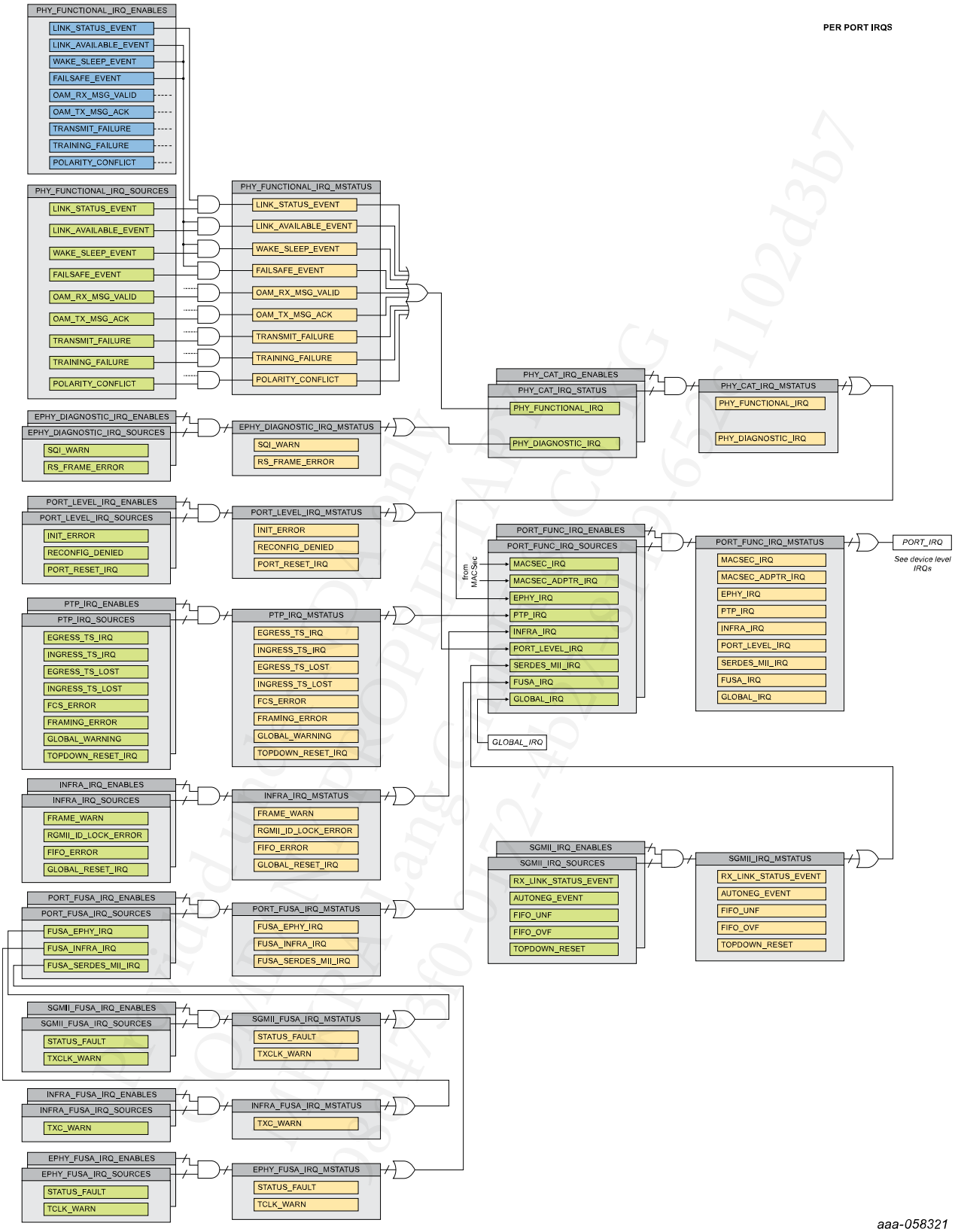


Figure 25. Port IRQs

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6.10.8.2 Non-maskable interrupts

The maskable BOOT_DONE_IRQ signals that all self-checks have passed as described in Section 6.4.

Several IRQs are non-maskable and act as an indication of severe malfunction. The interrupt enable bits that gate these IRQs are ignored.

6.11 Time stamping

The PTP time-stamping module is a generic module which supports both 100/1000 Mbit/s operation. In this device, the block is integrated for the 1000Mbit/s mode and internally connected to the on-chip GMII bus as shown in Section 6.3.

6.11.1 Introduction

IEEE 1588 v2 (PTP) is an industry standard that enables precise time synchronization of nodes in an Ethernet network. The synchronization is achieved by exchanging various protocol-defined PTP event messages. Based on these PTP event messages, the follower can correct offset and drift of its local clock to synchronize with the leader clock.

The PTP module is sandwiched between the MII/GMII interface. The module provides a 1588 digital clock implementation, transmit and receive frame parsing, Start-of-Frame detection, transmit and receive timestamp units, event timestamp unit and a Pulse-Per-Second (PPS) generator.

6.11.1.1 Features

The following features are provided by the IEEE 1588 v2 PTP block:

- Support for 100BASE-T1 and 1000BASE-T1 PHYs
- Supports hardware time stamping features of IEEE 1588 v2 PTP Clock
- Provides timestamping on ingress and egress PTP messages
- Hardware self-locking or software assisted rate correction to an externally provided reference clock
- Timestamping on external pin sync events
- Ingress and Egress timestamps stored in ring buffer for host processing
- Support for two-step operation
- Supports in-band timestamp delivery for ingress path
- Supports detection of PTP frame encoded in Ethernet both untagged and 802.1Q VLAN tagged (double tagged VLAN is not supported)
- Software configurable frame filter to match PTP event frames
- Support MACsec TLV parser in TX and RX direction

6.11.2 Overview

The diagram below depicts the integration of the PTP module at system-level.

The timestamping module uses the PHY's RX_SOF and TX_SOF signals to accurately take a timestamp at the instant when a frame has ingressed/egressed the Ethernet PHY. The MACsec engine is between the PHY and the timestamping module and acts as a deep fifo of more than 64 octets. This implies that an RX_SOF/TX_SOF event cannot be easily correlated with a frame inside the timestamping module. To be able to correspond a frame with an SOF event internal sequence numbers are generated and passed through a sideband channel of the MACsec engine. This makes it possible to correlate SOF events with frames and keep track and frames that were discarded inside the MACsec subsystem.

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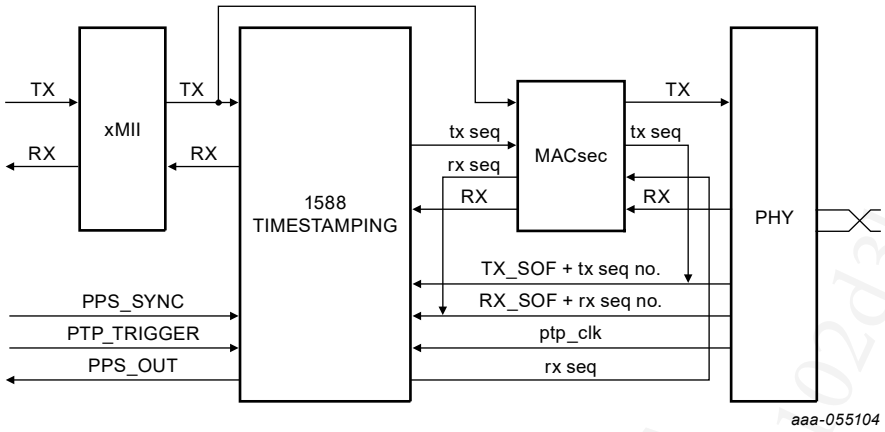


Figure 26. PTP PHY integration

The module consists of the following blocks:

- The actual clock, called *local time counter* (LTC)
- The transmit and receive frame parser and qualifier
- Transmit and receive timestamp FIFOs
- Other miscellaneous functions such as control and status registers, PPS output and syncing

The TX_SOF and RX_SOF indication comes from the PHY sublayer and usually exhibits a pipeline delay. This device-specific delay is constant and can be corrected automatically by configuring the registers MMD30.TX_DELAY_NSEC and MMD30.RX_DELAY_NSEC.

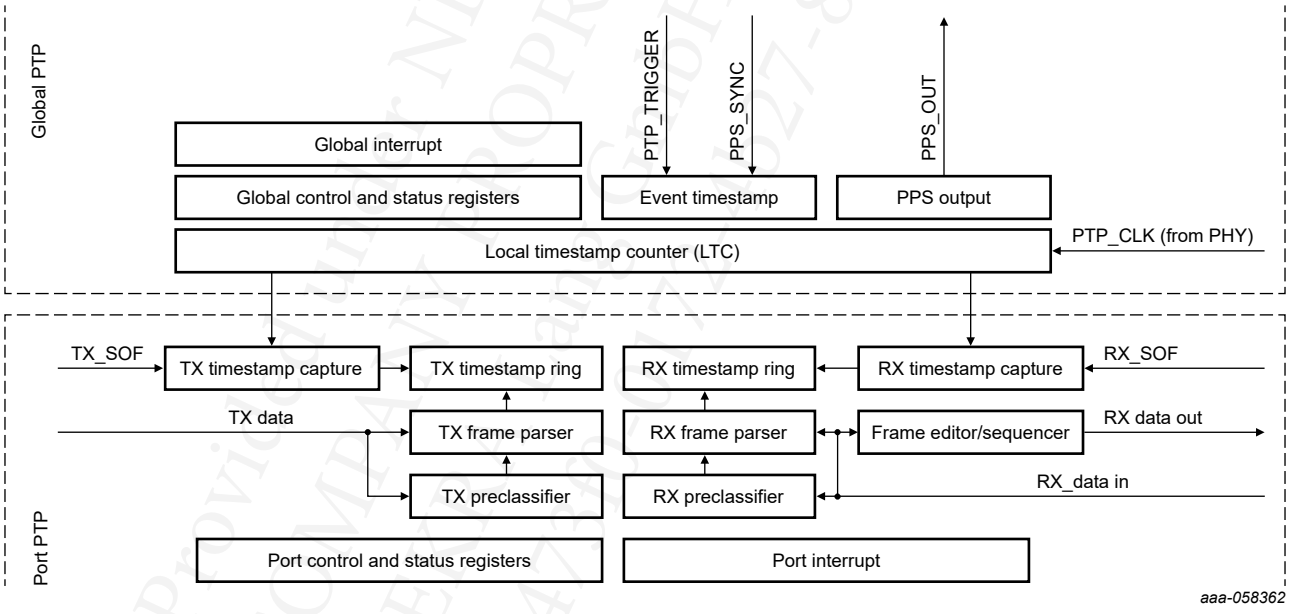


Figure 27. PTP block diagram

6.11.3 Clocking scheme

The IEEE 1588 PTP uses a timestamp clock that is derived from the Ethernet PHY block. The PTP_CLK frequency must be configured in the PTP_CLK_PERIOD register. For 1000BASE-T1, the PTP_CLK frequency is 125 MHz, thus the PTP_CLK_PERIOD must be set to 8 ns.

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6.11.4 Resets

By default, PTP module is inactive. All clocks are disabled and internal logic is under reset. The host can enable the PTP module by writing both GLOBAL_PTP_ENABLE=1 and PORT_PTP_ENABLE=1.

6.11.5 Local time counter

The local time counter (LTC) module is the clock used for timestamping. It is local to the PHY and can be synchronized with other chips via a PPS_SYNC facility. The software can interact with the LTC to control absolute value, the rate as well as change the offset.

The ptp_clk is the digital clock that drives the LTC. Depending on the speed of the PHY, the ptp_clk is either 125 MHz (1000BASE-T1) or 66.66 MHz (100BASE-T1). A block diagram of the LTC structure is shown below:

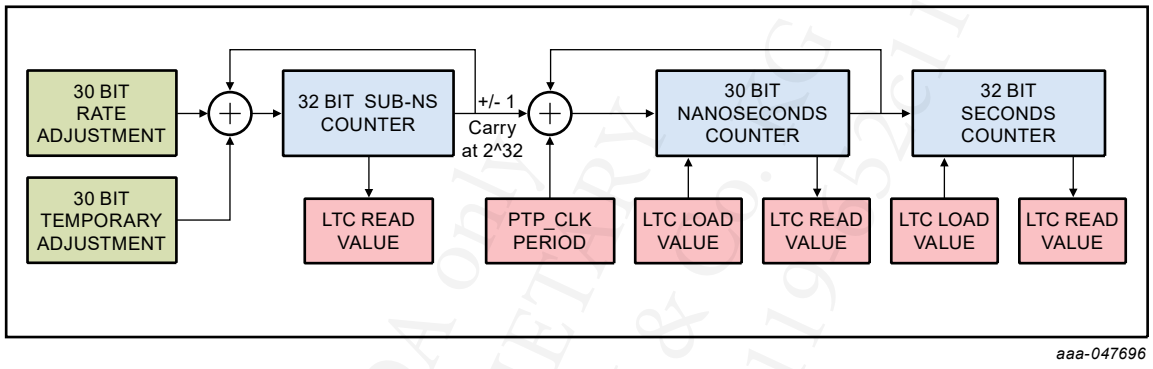


Figure 28. Block diagram of LTC

The local time counter consists of two counters, nanoseconds counter and seconds counter. The nanoseconds counter is 30 bits wide and seconds counter is 32 bits wide.

The ns counter increments by 8 (15) every ptp_clk cycle for 1000 Mbit/s (100 Mbit/s) operation. This translates to 8 ns (15 ns) per ptp_clk tick. Once the nanoseconds counter reaches the rollover value, the seconds counter increments by 1. The 32-bits seconds counter has a rollover time of 136 years.

A sub-ns counter is used for continuous and temporary rate-correction in fractions of a nanosecond per clock. Such ns fractions are accounted for in the 32-bit wide sub-ns counter, which has a granularity of 2^{-32}ns. When the sub-ns counter rolls over, an additional nanosecond is added or subtracted from the nanoseconds counter.

6.11.5.1 Direct read/write

A direct write will set the LTC immediately. The parameters LTC_LOAD_DATA_SEC and LTC_LOAD_DATA_NSEC must be written to load internal seconds counter and nanoseconds counter. By setting the self-clearing load bit, these pre-programmed values are loaded into the LTC.

The seconds counter, nanoseconds counter, sub-nanoseconds counter can be read at any time by asserting the read bit.

6.11.5.1.1 Immediate LTC load

An immediate load causes the values as written to the nanoseconds and seconds register to be transferred to the LTC.

1. Program LTC_LOAD_DATA_0, LTC_LOAD_DATA_1 , LTC_LOAD_DATA_2, LTC_LOAD_DATA_3
2. Load these values into internal counters by setting LOAD_LTC

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6.11.5.1.2 PPS edge triggered load

The PPS edge loading transfers the time as loaded into the nanoseconds and seconds register into the LTC at the rising edge of the PPS_SYNC input.

1. Program LTC_LOAD_DATA_0, LTC_LOAD_DATA_1 , LTC_LOAD_DATA_2, LTC_LOAD_DATA_3
2. Load these values into internal counters by setting LOAD_LTC_ON_SYNC
3. Internally these values are loaded into local counters on event of rising edge of external pps sync

When an external PPS Sync is chosen with a frequency of more than 1 Hz, loading the external PPS sync signal must be staged. The PPS must be paused before loading the LTC. Then the PPS can be started again. This is required in order to associate a particular edge with an absolute time value reliably. Pause the PPS signal, then follow step 1 and 2, and then restart the PPS Sync signal.

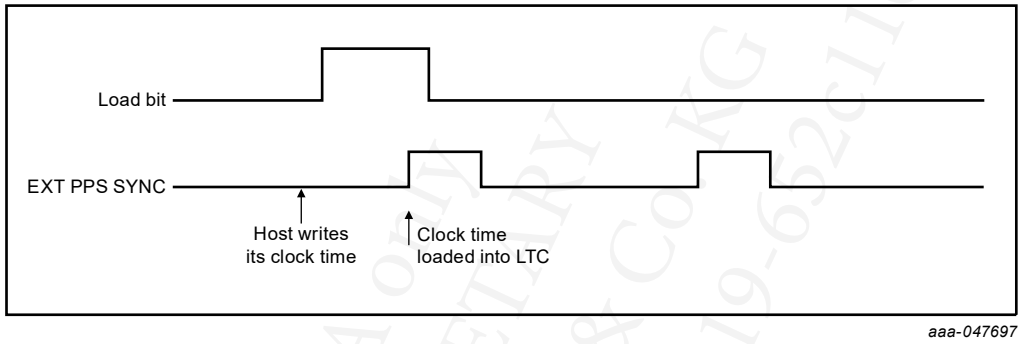


Figure 29. Loading LTC on rising edge of PPS

6.11.5.1.3 Immediate LTC read

1. Program READ_LTC to save the local clock time
2. Read LTC_READ_DATA_0 to LTC_READ_DATA_3
3. Same saved data can be read multiple times as new value is loaded only when READ_LTC is reprogrammed

6.11.5.1.4 PPS edge triggered read

The LTC value is latched at the rising or falling edge of the PPS input. It can then be read from the PTP_SYNC_TS_DATA_x registers.

It is mandatory to read PTP_SYNC_TS_DATA_S first and continue to PTP_SYNC_TS_DATA_3. It is not required to read the full set of registers. The read sequence can be interrupted at any time by writing to any of PTP_SYNC_TS_DATA_x registers. This frees the register for the next PPS edge event.

6.11.5.2 Rate correction

In order to match rate difference between PHY and an external clock source (for instance a microcontroller or switch), a digital rate correction can be applied. Every ptp_clk cycle, a rate correction value RATE_ADJUST_SUBNS parameter is added or subtracted to/from a hidden subns counter. On overflow/ underflow ofthis subns counter, the carry is added/subtracted to/from the nanoseconds counter and integrated in the visible LTC. The RATE_ADJUST_SUBNS parameter is 30 bit wide and in units of 2^{-32} ns. To control the direction (speed up or slow down) the tuning value can be added or subtracted. The direction is set with register RATE_ADJUST_SUBNS_1.SIGN.

As an example, lets assume the clock rate of the LTC is too slow compared to the external PPS. This difference is 0.125 ns in the time interval of one ptp_clk cycle.

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Then, PTP_CLOCK_RATE_SNS = rate * 2^{32} = 30'h2000 0000

6.11.5.3 Temporary rate correction

An LTC offset is corrected by temporarily speeding-up or slowing-down the LTC clock, therefore slowly integrating the offset error over multiple ptp_clk cycles. This avoids abrupt jumps and time discontinuities and guarantees a monotonic clock.

A temporary correction is applied similarly as for rate-correction, except an additional time window is configured during which the rate change is applied. The parameter TEMP_ADJUST_SUBNS holds the temporary rate-correction value and the TEMP_ADJUST_CYCLES parameter is used to configure the number of ptp_clk clock cycles to be adjusted. The register TMP_ADJUST_SUBNS_1.SIGN controls the direction of update (add/subtract). Setting the self-clearing bit TMP_ADJUST_SUBNS_1.EXECUTE triggers the temporary adjustment for one time.

The parameter TEMP_ADJUST_SUBNS is 30-bit wide and in units of 2^{-32} ns.

The temporary rate correction acts additive, on-top of the regular rate-correction as described earlier. For example, lets assume an offset error of 25.5 ns. With an integration rate of 0.125 ns per ptp_cycle,

the temporary correction must be applied for 25.5/0.125 = 204 ptp_cycles. Thus the following registers are programmed:

TEMP_ADJUST_SUBNS = rate * 2^{32} = 30'h2000 0000 TEMP_ADJUST_CYCLES = 25.5/0.125 = 204

6.11.6 LTC synchronization

A typical system including MAC, IEEE 1588 protocol engine and a PHY is depicted in the below diagram.

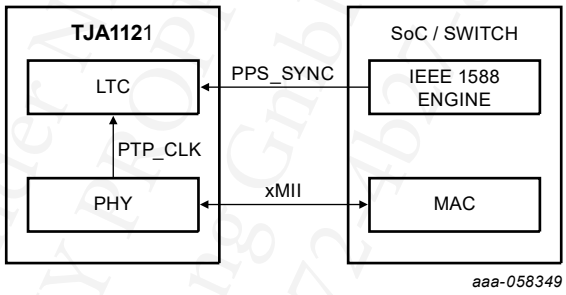


Figure 30. Example clocking scheme

Normally, the clock source used to drive the LTC is an external crystal. This provides a slightly different clock than used inside other timestamping engines. All timestamps taken in multi-port devices such as switches must be taken on the same timebase in order to be consistent.

A time difference between the LTC and any other timestamping clocks can have the following reasons:

1. The clock sources are different and have a relative frequency difference within the 100 ppm limit.
2. The timestamp clock in another device is itself a PTP follower clock, therefore constantly rate and offset corrected.

The PTP module can lock onto an external Pulse Per Second (PPS) signal. The PPS signal is then used to to correct the LTC automatically. The PPS does not need to be a pulse *per second*. The LTC block can track any low frequency signal, such as 1 Hz or 100 kHz. The rising edge or falling edge of external PPS sync is used to correct the timing drift. There are multiple ways to synchronize the LTC with the external reference PPS synchronization signal.

An external software tracking routine can be implemented to synchronize the LTC via a software driven control loop. Alternatively, the PTP block can track an external PPS synchronization pulse fully autonomously.

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6.11.6.1 Software tracking

After initialization, an external host reads the clock time of the PHY at the instance of rising edges of consecutive pps_sync and finds out the rate difference between the PHY and IEEE 1588 protocol engine.

Based on the calculations, the host can update LTC by using rate adjustment method to correct the rate difference and temporary adjustment to correct the offset. The following steps summarize the mechanism for a relatively slow PPS input:

1. The host pre-loads the expected LTC value (LTC_0) at the next PPS sync edge. Once the PPS sync input has fired, the LTC is running.
2. Capture LTC timestamp (LTC_1 at the following active edge of external PPS sync signal. The relative rate difference is calculated by dividing the expected PPS cycle time by the $LTC_1 – LTC_0$. The offset error can be determined through LTC_1.
3. Fix the rate-correction value as well as the offset error as described in Section 6.12.5.
4. Continuously repeat to fix rate-correction and offset based on LTC snapshots LTC_{n+1} and LTC_n.

6.11.6.2 Hardware tracking

The PTP block implements an optional hardware control loop to track an external PPT sync input. The rate difference and offset errors are detected automatically and corrected on-the-fly. Setting GPTP_CONFIG.LTC_AUTO_LOCK enables this feature.

The initialization works identical to the software tracking approach. The LTC value is pre-loaded and transferred into the working registers at the next active PPS edge. The PPS reference frequency must be programmed via registers PTP_SYNC_PERIOD_0 and PTP_SYNC_PERIOD_1.

Once LTC is loaded external PPS sync, the LTC is captured at the next PPY sync event. A phase comparator calculates the phase difference between observed LTC value and expected reference time. A direction signal is generated that tells if the LTC is lagging or leading. With this information, the rate difference and offset errors are computed and adjusted each PPS sync cycle.

If the hardware is not able to track the PPS input and the error exceeds a certain bound set by LTC_ERROR_LIMIT_0 and LTC_ERROR_LIMIT_1, a flag (LTC_ERROR_WARN) is asserted and an IRQ is generated.

6.11.7 PPS output

It is possible to generate a pulse-per-second (PPS) signal based on the internal LTC and externalize it on a GPIO pin. This can be enabled by asserting PPS_OUT_EN. The hardware is able to generate a nominal a 1 Hz signal with a pulse at each one-second transition of time. If enabled, the PPS generator periodically keeps generating the PPS signal.

To change the polarity of the generated PPS, set the PPS_OUT_POLARITY bit.

6.11.8 Event timestamp

External events can be monitored and timestamped. A rising edge on the PTP_TRIGGER input latches the current LTC value into the registers PTP_TRIG_TS_DATA_x.

6.11.9 Egress timestamping unit

The egress timestamping is part of the transmit path. It is responsible for identifying a PTP message inside an Ethernet frame (classify), pull relevant information out of the PTP message (parse) and the actual timestamping.

The captured timestamps are stored into a buffer (dataset capture and TS ring) for retrieval via MDIO. The diagram below depicts the data flow.

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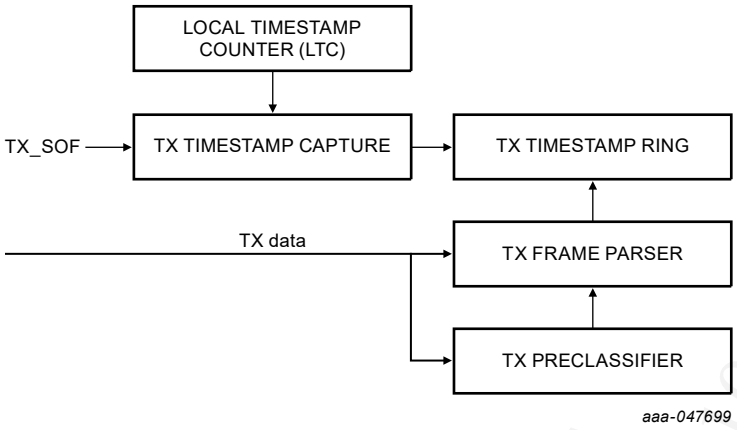


Figure 31. Egress timestamping unit

The TX_SOF is generated by the Ethernet PHY, not the PTP module. The TX_SOF is derived from PCS information and not the exact point of start of frame delimiter (SFD). The parser inside the PTP block detects the actual SFD point and compensates for the delay between TX_SOF and SFD.

6.11.9.1 Pre-classifier and TX parser

The TX parser interfaces with TX GMII/MII interface. It monitors the incoming transmit frames to detect relevant IEEE 1588 PTP event messages. It is able to recognize PTP event messages transported in layer 2 Ethernet frames[6]. The parser supports untagged and 802.1Q VLAN Tagging (single tagged) in Ethernet frames. PTP messages are qualified by the MAC address. Two PTP standard fixed MAC address (01:1B:19:00:00:00 and 01:80:C2:00:00:0E). Additionally, a single entry DA match rule can be used to support user-defined MAC addresses (PTP_FILTER_ENABLES.USER_MAC_DA).

Once TX Enable is asserted on the internal MII/GMII bus the incoming frame is parsed to identify the PTP message. If a frame qualifies, all the necessary PTP message contents are extracted and stored for future use. It also sets a flag bit saying incoming detected message is PTP message.

The following frame filters can be individually enabled/disabled:

- Destination MAC match (two fixed addresses, one configurable. See register PTP_FILTER_ENABLES.USER_MAC_DA)
- VLAN tagged (see register PTP_FILTER_ENABLES.VLAN_TAG)
- Message type: Sync, Delay_Req, Pdelay_Req, Pdelay_Resp event messages (see register PTP_MESSAGE_FILTER)
- Match a specific versionPTP (see register PTP_FILTER_ENABLES.PTP_VERSION)
- Match a specific domainNumber (see register PTP_FILTER_ENABLES.DOMAIN_NUMBER) It is possible to timestamp the following PTP event messages:

Table 19. PTP Event Messages

PTP Message Type	Value
Sync	0
Delay_Req	1
Pdelay_Req	2
PDelay_Resp	3

6 IP encapsulated PTP messages are not recognized.

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6.11.9.2 TX timestamp ring buffer

The TX timestamp (TS) Ring buffer stores the timestamping data in up to 4 available slots. Next to the timestamp additional metadata is stored to correlate the information with an Ethernet frame. This includes PTP message sequenceId, domain number, message type and valid flag to indicate the dataset is not yet read.

The buffer is organized as a ring buffer and the oldest valid entry is exposed to the registers EGRESS_TS_DATA_x for read. Once the timestamp has been read via the MDIO interface, the ring buffer pointer is advanced by writing to any of the EGRESS_TS registers.

If the buffer is full, the oldest entry is flushed out to make room for a new entry. If enabled, an interrupt is generated on overflow.

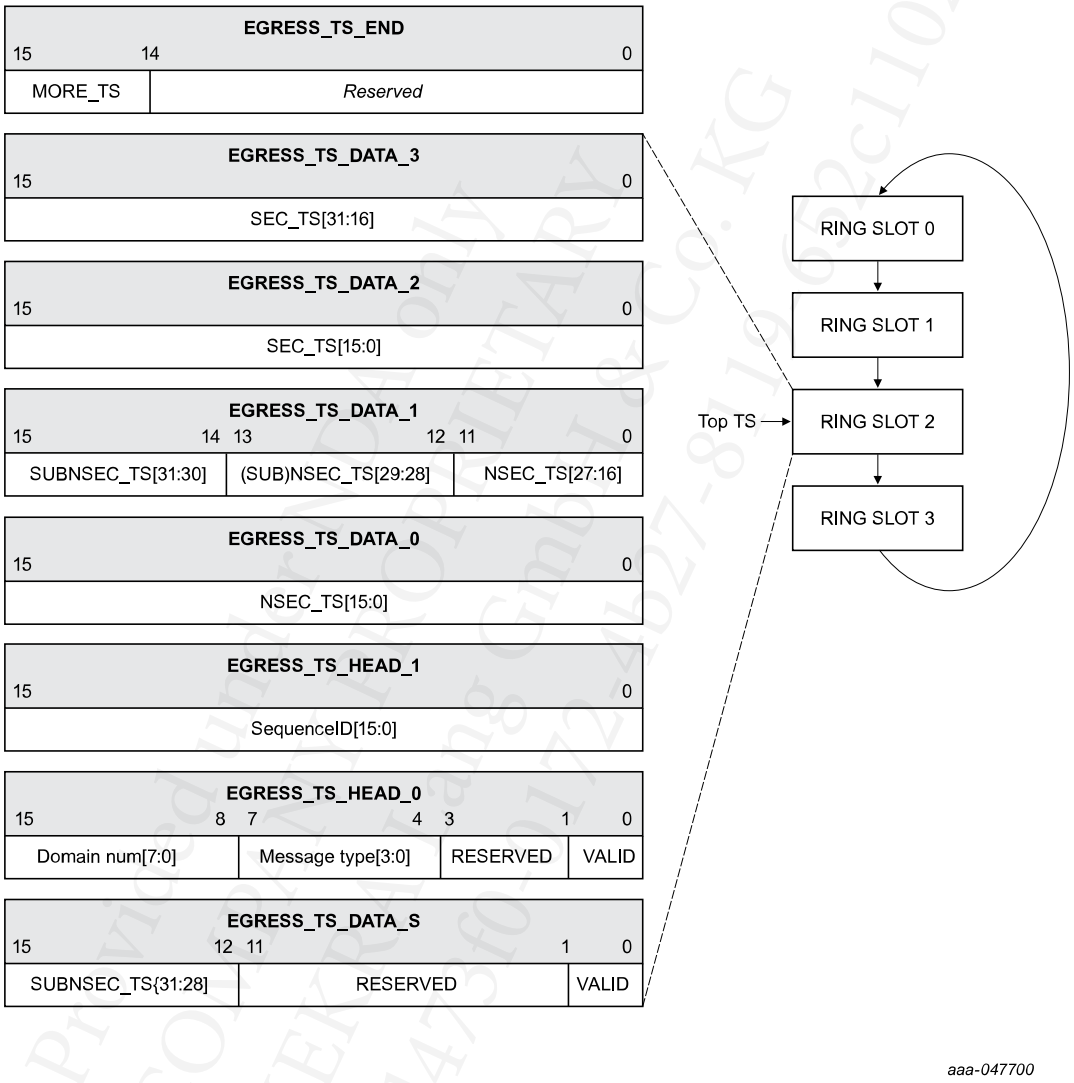


Figure 32. Egress TS ring buffer organization

6.11.10 Ingress timestamping unit

The ingress timestamping is part of the receive path. It is responsible for identifying a PTP message inside an Ethernet frame (classify), pull relevant information out of the PTP message (parse) and for timestamping the frame. The captured timestamps are stored into a buffer (dataset capture TS ring) for retrieval via MDIO.

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Alternatively, the timestamp can be embedded directly into the Ethernet frame (message editor). The diagram Figure 33 depicts the data flow.

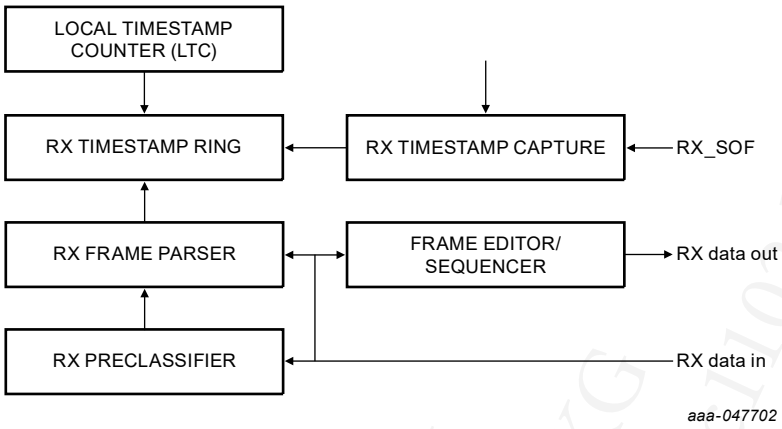


Figure 33. Ingress timestamping unit

The RX_SOF is generated by the Ethernet PHY, not the PTP module. The RX_SOF is derived from PCS information and not the exact point of start of frame delimiter (SFD). The parser inside the PTP block detects the actual SFD point and compensates for the delay between RX_SOF and SFD.

6.11.10.1 Pre-classifier and RX parser

The RX parser interfaces with RX GMII/MII interface. It monitors the incoming received frames to detect relevant IEEE 1588 PTP event messages. The parser and classifier logic is identical to Section 6.12.9.1.

6.11.10.2 RX timestamp ring buffer

The RX timestamp (TS) Ring buffer stores the timestamping data in up to 4 available slots. Next to the timestamp additional metadata is stored to correlate the information with an Ethernet frame. The organization of the ring structure and registers is identical as described in Section 6.12.9.2.

6.11.10.3 Frame message editor

Alternatively to delivering the RX timestamp via the ring buffer exposed via MDIO, it is possible to deliver the timestamp inside the PTP event frame itself. The message editor updates the PTP message header and the FCS in the Ethernet frame.

The engine is able to modify four reserved octets at PTP header offset 16 and overwrite these bits with 32 bits timestamp data. The timestamp format can be selected through register INGRESS_TS_INSERT_CONTROL.

6.11.10.3.1 Checksum regeneration

Two parallel CRC-32 engines are used to check and regenerate a Frame Check Sequence (FCS) in case the frame is modified. Each frame is always checked for corrected by matching the residue with the magic number (32'hC704_DD7B). In parallel a new FCS is computed accounting for the modified 32 bits.

In case a frame qualifies as a PTP event frame and the original FCS is correct the timestamp is patched in and the FCS is regenerated. In case the original FCS was found to be incorrect, the frame will be left corrupted. This ensures that an FCS of an corrupted frame is not accidentally regenerated and the frame appears correct.

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6.11.11 Interrupts

The PTP module is able to signal events via a PTP interrupt signal. This IRQ line is connected on device level to the device interrupt tree. See Section 6.10.8.1.

Following are the global interrupt sources

- PTP Sync timestamp event
- PTP trigger timestamp event
- LTC error exceeds the error limit to achieve LTC auto-lock (LTC_ERROR_WARN)
- PTP Sync TS lost due to overflow
- PTP trigger TS lost due to overflow

Following are the port interrupt sources

- Egress timestamp event
- Ingress timestamp event
- Egress TS lost due to overflow
- Ingress TS lost due to overflow
- Input FCS error detected (in-band signaling mode)
- Ingress frame error (preamble detected but no SFD; Not complete PTP TS data is received)

6.12 MACsec adapter

6.12.1 Introduction

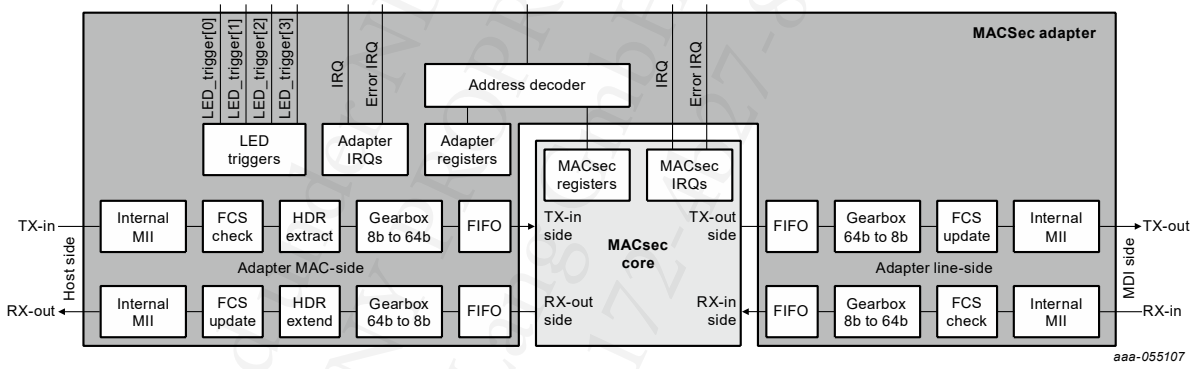


Figure 34. MACSec subsystem block diagram

The MACsec subsystem consists of two sub blocks. The MACsec core which is responsible for the 802.1AE data path and the MACsec adapter which connects the MACsec IP to the on-chip MII datapath.

The adapter implements rudimentary MAC functionality. This includes removal of the Ethernet preamble and SFD (start of frame delimiter) as well as FCS (frame check sequence) checking and recalculation. The MACsec IP is a synchronous IP and decoupled from the MII clock via asynchronous FIFOs.

The MACsec subsystem encompasses a cut-through architecture. This means that for each frame that ingresses the subsystem, after some initial delay, the frame starts to egress while ingress is not yet completed (except for certain error conditions and when policing is used).

The MACsec core processes each frame and, depending on direction appends or removes the 802.1 AE SecTag as well as ICV (integrity check vector). Optionally, the payload data of the Ethernet frame can be encrypted (confidentiality mode, red area below).

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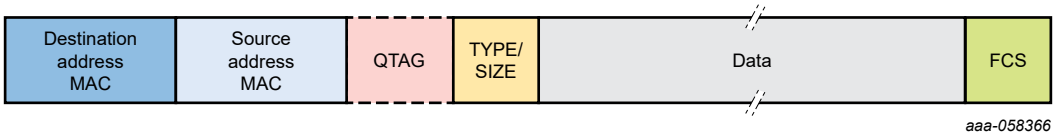


Figure 35. Basic Ethernet frame, without MACsec with VLAN tag

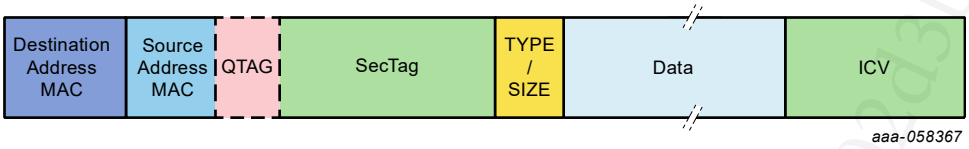


Figure 36. MACsec Ethernet frame with SecTag and VLAN-tag-in-clear

See UM12059 [9] for detailed description of the core MACsec functionality.

Optionally, the HDR extraction block is able to extract additional control information from a proprietary header. Similarly, the HDR extend block is able to annotate received frames with additional meta data.

6.12.2 On-chip MII interface

The MACsec subsystem is connected to the rest of the system through an on-chip MII (Media-independent-Interface) connection. The MII interface implements 802.3 standard behavior. As part of the standard MII processing, the preamble is removed and the TX_ER/RX_ER indications are registered.

Any TX_ER/RX_ER event will be treated internally similar to an FCS error. Thus an TX_ER/RX_ER on one of the input interfaces will cause the frame to be corrupted and transmitted with a corrupted FCS on the respective output interface.

6.12.3 FCS checking and update

The TX_IN and RX_IN ingress stages of the adapter check the Frame check sequence (FCS) of each frame. FCS errors on TX/RX interfaces are counted through RX_STATS_FCS_ERRS and TX_STAT_FCS_ERRS respectively. Because the implementation is cut-through each frame is internally processed and start egressing before it is reliably known that the frame is correct. The egress stages TX_OUT and RX_OUT optionally update the FCS in case a frame was modified.

The FCS of a frame is modified in the following cases:

- A frame entered the device while TX_ERR/RX_ERR signals where asserted. In this case the FCS is corrupted on egress.
- The FCS check during RX/TX ingress did not pass. In this case the FCS is corrupted on egress.
- The frame passed the ingress FCS check and the frame was modified by the MACSec adapter or MACSec core. This can be addition or removal of SecTag/ICV or the addition/removal of a MACsec TX/RX control tag. In this case the FCS is regenerated to match the frame.

In case the FCS check on ingress has passed and a frame was not modified by the MACSec Adapter or MACSec core the FCS is retained as it was.

The FCS corruption on egress is done by calculating the correct FCS and then negating all bits.

6.12.4 Parity protection and ECC

The control and datapath of the MACsec adapter and MACsec core as well as register memories are protected with parity or ECC. Each pipeline stage in the adapter will check the parity and update the parity if needed.

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An adapter internal data parity mismatch as well as a control parity mismatch is indicated via the PARITY_DATA_ERR and PARITY_CNTRL_ERR status flags.

Parity errors inside the MACSec IP are flags via the of MACSEC_HARD_ERR and of MACSEC_SOFT_ERR status flags.

The MACsec subsystem is able to recover from data parity errors. It is is advised to reset the entire subsystem in case of control and hard errors.

6.12.5 Transmit flow control

On data transmission, the principle of the MACsec operation is to add new information (SecTag as well as ICV) to the Ethernet frame. The frame size grows by at most 32 octets. For example: A 64 octets frame *before*

MACsec is extended to 76 bytes *after* MACsec[11]. This process presents a fundamental issue when MACsec is implemented at xMII level (i.e. in a PHY): When the MAC is sending data at a line-rate (e.g., 100 Mbps), the MACSec core cannot add the SecTag/ICV data, because it would exceed the line limit of 100 Mbps.

The following solutions available when using this device

1. **Cut-through pre-gapped:** The MAC must guarantee at least 12+length (SecTag + ICV) IPG octets. This is worst case 12+32 = 44 octets IPG between any two frames.
2. **Cut-through standard IPG with proprietary MACsec TX/RX control tag:** The software network stack can reserve bandwidth by the means of adding a proprietary header to reserve the required bit times. This header is remove by the PHY.
3. Accept frame-loss in case the medium is temporarily over-subscribed.

Without further configuration, the MAC is allowed to burst out frames at minimum IPG of 12 octets. The consequence of a large burst with minimum IPG is that frames are lost in the MACsec subsystem because of FIFO overruns. Such events are counted in the register ABORTED_FRMS. In case such drops occur, the IP_TRACKING register can be used identify critical IPG underrun scenarios.

Note: *The egress IPG of the adapter is actively shaped to be at least TX_OUT_IPG/RX_OUT_IPG octets and set to 12 by default. If the TX IPG enforcement strategy is used, the output IPG must match the input IPG minus MACsec overhead. If the Proprietary MACsec TX/RX control tag strategy is used, the output IPG must match the input IPG. Due to PPM clock differences, the input IPG can be smaller than 12 octets, thus it is recommended to set TX_OUT_IPG/RX_OUT_IPG to 10 to avoid a bottleneck.*

6.12.6 MACsec secure channel

The number of secure channels is fixed to four per direction. The MACsec core IP description [11] shows a higher number of channels, but is limited by the IP usage in the device.

6.12.7 MACsec TX frame classification

There are two distinct mechanism to classify frames on the TX-in interface for MACsec controlled or uncontrolled port as well as steer them to a secure channel.

UM12059, Section 3.3.2 has built-in lookup tables to match Layer-2 Ethernet fields with controlled/uncontrolled port and SCS. For more information, refer to UM12059, Section 3.3.1.

Second, controlled/uncontrolled and SCS can optionally be communicated in-band via a proprietary MACsec TX/RX control tag (MACsec TX/RX control tag). This tag is removed from the frame before sending it to the MACsec core IP and not visible on the medium.

The registers MACSEC_CFG_EXTSCS (see Section 6.13), TX_TAG_ENA, OVERRIDE_TLV_CP and OVERRIDE_TLV_SCI control where and how the frame classification occurs.

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11 The actual growth depends on the configuration of the MACSec core.

Table 20. MACsec controlled/uncontrolled port and secure channel selection overview

MACsec core Register	MACSec adapter Register			TX control tag present	Uncontrolled / controlled port / Secure Channel ID selection
MACSEC_ CFG_ EXTSCS	TX_TAG_ ENA	OVERRIDE_ TLV_CP	OVERRIDE_ TLV_SCI		
0	0	—	—	—	Filtering based on MACsec core settings. MACsec TX/RX control tag parsing is disabled and frames are forward to the MACsec core as is.
0	1	—	—	—	Filtering based on MACsec core settings. If present, MACsec TX/RX control tag is stripped.
1	0	—	—	—	MACsec TX/RX control tag parsing is disabled and frames are forward to the MACsec core IP as is. Fixed CP / SCI settings as per reset values of TX_ TAG_CNTRL1.TLV_CP and TX_TAG_CNTRL1.TLV_ SCI
1	1	0	0	1	MACsec TX/RX control tag is stripped. CP / SCI extracted from MACsec TX/RX control tag
1	1	0	0	0	Fixed CP / SCI as per registers TX_TAG_ CNTRL1. TLV_CP and TX_TAG_CNTRL1.TLV_SCI
1	1	1	0	1	MACsec TX/RX control tag is stripped. SCS as per control information expected via MACsec TX/RX control tag. Fixed CP as per register TX_TAG_CNTRL1.TLV_CP
1	1	1	1	x	Fixed CP / SCS settings as per registers TX_TAG_ CNTRL1.TLV_CP and TX_TAG_CNTRL1.TLV_SCI

6.12.8 MACsec TX/RX control tag

The MACsec TX/RX control tag is an optional proprietary header that serves two purposes:

- Reserve header space that can be later claimed by the MACsec core to add SecTag and ICV.
- Transfer controlled/uncontrolled port and SCS metadata to the MACsec core, thus bypassing its internal filters.

The MACsec TX/RX control tag is parsed by the HDR extract block as part of the TX datapath (see block diagram). This functionality is only enabled when both ADPTR_ENA and TX_TAG_ENA are set.

When TX_TAG_ENA is disabled, the header extraction block is in bypass and all frames including those with MACsec TX/RX control tags are forwarded to the MACsec core for further processing.

When TX_TAG_ENA is enabled, the control tag is parsed and the tag is stripped and the FCS updated. Then, the frame is forwarded to the MACsec IP.

The MACsec TX/RX control tag consists of an configurable Ethertype (see register TLV_ETHTYP) that is used to identify the header. The header is a TLV (tag length value) type and contains the header length as well as a sub-type specification as part of the header information.

The supported sub-type in the TX direction is sub-type=1. This sub-type indicates that the header is a MACsec TX/RX control tag. Other sub-types are not supported and the entire frame is dropped if sub-type is found different than 1.

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The length field of the header contains the number of octets excluding the TLV identification part (4 octets). Practically, the length field is set to 4+P, where P is the number of octets in the padding region.

The version field must be set to zero. A frame with a non-zero version field is dropped.

The SCS and CP fields are optional. See Table 20 for further details. The SCS field encodes the 4-bit logical secure channel identifier that is used to map the frame to a secure channel. The CP field encodes whether the frame is targeted the controlled port (CP=1). If set to zero (CP=0), the frame is send to the uncontrolled port.

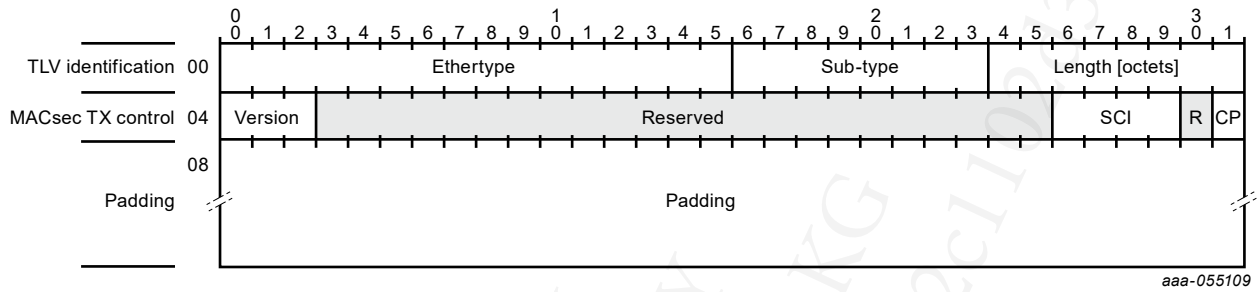


Figure 37. MACsec TX/RX control tag

Note: *If the MACsec TX/RX control tag is used, it is recommended to use it for all (controlled and uncontrolled port) frames to keep latencies constant.*

6.12.8.1 TX control tag padding

The TX control tag can be used to resolve the flow control problem. The purpose of the TX control tag is to make sure that the frame that leaves the MAC is the exact frame size that leaves the MACsec engine. If this is the case, all assumptions that are based on the frame size (such as AVB credit-based shaper, and TSN Qci, Qbv decisions) are still valid. Also no FIFO overflows or rate mismatches occur.

The padding length of the TX control tag must be selected so that the entire TX control tag length matches the added overhead by MACsec:

Length(TX control tag) = Length(Ethertype + SecTag + ICV)

Standard frame MACsec TX control tagged frame

MACsec frame

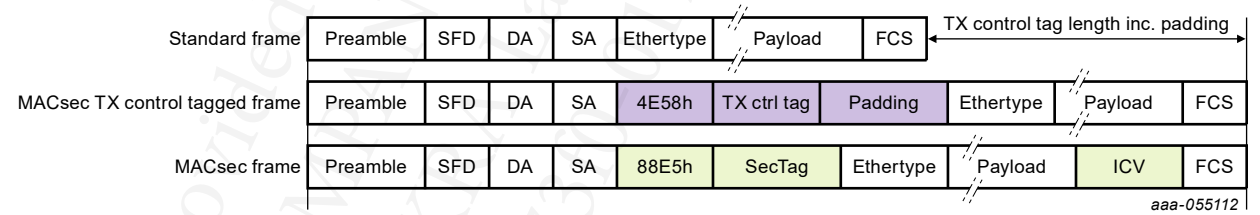


Figure 38. TX control length padding

The Length of the SecTag and ICV depend on the MACsec mode and can be between 16 and 32 octets.

6.12.8.2 TX TLV Policing

The device can be configured to enforce the use of the MACsec TX/RX control tag or reject it. In case the application makes use of this feature and it is expected that all Ethernet frames are MACsec TX/RX control tagged, the POLICE_FRMS register can optionally be set to DROP_STD. Any frame without a tag is then dropped.

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In case the TLV tag is not used, the POLICE_FRMS register can optionally be set to DROP_TLV. Any frames with the MACsec TX/RX control tag present are dropped.

All frames dropped due to this policing setting are reflected in the statistics counter TX_STATS_DISCARDED_FRMS.

6.12.8.3 TX Tag layering

The header extraction block is able to find the MACsec TX/RX control tag in case of a layered header structure. The following tags are recognized and can be skipped:

- VLAN tag (CTAG)
- QinQ tag (STAG)
- 802.1CB tag (RTAG)
- User-defined tag (MISC TAG) of up to 8 octets (see MISC_TAG_LENGTH)

The parser is searching for the first MACsec TX/RX control tag without checking the validity of the header structure itself. An arbitrary sequence of supported tags are accepted. For example, a frame with three VLAN tags is accepted. As soon as the first MACsec TX/RX control tag is found, it is stripped and all remaining octet are treated as payload.

By default, this feature is disabled and individual headers must be enabled (see CTAG_ENA, STAG_ENA, RTAG_ENA and MISC_TAG_ENA).

The header extractor always buffers the frame until the parser can definitely tell that the frame contains a MACsec TX/RX control tag or the maximum depth was reached. Thus, for applications which always place the MACsec TX/RX control tag as the first tag of the frame, keep the detection of additional headers disabled (this is the default setting). This allows the header parser to make decisions as early as possible. Frames that do not have a TX tag but one or multiple other tags are are not delayed because the header parser is waiting for a potential TLV tag after it has seen other supported tags.

The header extractor is able to find the MACsec TX/RX control tag within the first 64 octets (preamble and SFD are not counted).

Note: *If the layering changes from frame to frame, the latency of the MACsec adapter in TX direction is not constant anymore. This is because some frames are buffered longer than others. The latency is always predictable, if the header stack is known.*

6.12.8.4 Error conditions during header modification

If the MACsec TX/RX control tag processing fails, the frame is either fully dropped or invalidated.

Error condition	Manifestation
Unknown version	Frame dropped
Unknown sub-type	Frame dropped
TLV length field > actual frame length	Frame dropped
Frame stops unexpectedly during CTAG/STAG/CTAG/ MISC_TAG parsing	Frame dropped
MACsec TX/RX control tag too deep in the frame	Frame is processed as if no MACsec TX/RX control tag present
FCS error	Frame invalidated (illegal FCS)
MII TX_ERR/RX_ERR set	Frame invalidated (illegal FCS)
Data-parity error	Frame invalidated (illegal FCS)

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Error condition	Manifestation
Control-parity error	Frame either invalidated (illegal FCS) or dropped, depending when and where the error occurs

6.12.9 MACsec RX control tag

The MACsec RX control TAG is conceptually similar to the tag that is used for TX direction. The use of this header is optional and disabled by default. The feature can be enabled by setting RX_TAG_ENA to 1.

The purpose of the tag is to communicate MACsec metadata of the frame with the frame itself. The Ethertype of the MACsec RX control tag is as for the TX control tag set via register TLV_ETHTYP.

For this version of the module, the sub-type is set to 2, length always reports as 4 and version is 0.

The CP field reports the logical port. If CP is set to one, the frame came through the controlled port. If CP is set to zero, the frame came via the uncontrolled port.

The field SCI reports the secure channel index used to process the frame. The field VLD indicates if the SCI is valid. It is set to zero, when a frame came in via the uncontrolled port (CP=0) or a frame came in via the controlled port and the SCI is unknown.

When TAG_UCP_FRMS is set, uncontrolled port frames are tagged with this header. When TAG_CP_FRMS is set, controlled port frames are tagged with this header.

Note: *When TAG_UCP_FRMS is set, the IPG on the rx output side will shrink by 8 octets. Set RX_OUT_IPG to a value lower than 4 to avoid adapter overflow conditions. Consult the application note for details.*

Figure 42. MACsec RX control tag

6.12.9.1 RX Tag layering

By default, the MACsec RX control tag is placed as the first tag, directly after the source MAC address in the Ethernet frame. For applications where the MAC offers hardware offloading and expects certain tags show first, the placement of the MACsec RX Control tag can be tuned.

The header extender block parses the frame on the fly and is able to skip recognized headers before inserting the RX tag. The following tags are recognized and can be skipped:

- VLAN tag (CTAG)
- QinQ tag (STAG)
- 802.1CB tag (RTAG)
- User-defined tag (MISC TAG) of up to 8 octets (see MISC_TAG_LENGTH)

The behavior is controlled via the two register bits SKIP_X and TAG_AFTER_X (where X is either CTAG, STAG, RTAG or MISC_TAG). If SKIP_X is set, header X is skipped and the RX TAG is placed later (but not immediately) and scanning continues. If a header is found that is not recognized, the RX tag is immediately placed.

If TAG_AFTER_X is set, the RX tag is placed immediately after header X. If both TAG_AFTER_X and SKIP_X are set, SKIP_X has precedence.

6.12.10 Latency

The latency introduced by the MACsec subsystem consists of two components: First the MACsec adapter latencies and second the actual MACsec core latencies. The latency of the adapter is constant per-direction, if TX_TAG_ENA=0 and RX_TAG_ENA=0. When header parsing and extension is enabled, the delay can be data-dependent. The tag placement influences the latency. In these cases the latency is always predicable.

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If MACsec TX/RX control tags are used for bandwidth reservation, it is recommended to use them for uncontrolled port and controlled port frames with the same padding settings. That keeps sure that the processing is identical for all frame types. If it is acceptable to have different (but predictable) latencies for UCP/CP this is not required.

The latency of the MACsec core is compensated for uncontrolled and controlled port frames and constant. To summarize, the latency in TX and RX is constant per-direction and independent of:

- Payload size
- Tag structure (assuming TX TLV tag is placed first)
- MACSec action (encrypt, decrypt, verify, clear)

6.12.10.1 TX Latency

The MACSec core matches latency internally, when enabled. The delay is enabled by setting MACSEC_CFG.UFDCD. By default the MACsec core delays uncontrolled port frames by a fixed number of cycles which is not fully matching the latency in this integration scenario. For best matching, the FIFO delay mode must be enabled MACSEC_CFG.UFDCM.

For TX controlled frame processing, the MACSec core will check if the input will create a MACSec short-length frame. For this, the core needs to accumulate up to 80 octets to make the decision if a frame is short-length or not. This will make latency matching impossible.

To disable short-length handling on transmission, MACSEC_CFG.SLD must be set. This makes the internal buffering non-frame size dependent and allows constant latency. For this, the MAC on TX-in must guarantee that the MACsec payload data is more than 64 octets, thus not create incorrect short-length conditions.

The following table assumes UFDCM, UFDCD and SLD are set up as previously explained. TAG can be either S-TAG, C-TAG, R-TAG since they have the same length. See Section 6.13.8.2.

Table 26. TX latency

Table 21.

			MACsec subsystem latency	
TX_TAG_ENA	**TX control tag placement**	**Comment**	**MII-mode [MII cycles]**	**GMII-mode [GMII cycles]**
Disabled	—		176	108
Enabled	TLV always first tag	*Latency depends on number of padding octets*	176	108
Enabled	TAG, TLV		176	112
Enabled	TAG, TAG, TLV	*Order of all TAGs preceding TLV is irrelevant. MISC-TAG is 6 octets*	179	116
Enabled	TAG, TAG, MISC-AG, TLV		195	124
Enabled	TAG, TAG, TAG, MISC-TAG, TLV	*MACSec core only supports 3 tags in clear*	—	—

6.12.10.2 RX Latency

For received frames, the MACsec core does not implement latency matching. Instead active latency compensation is implemented outside the MACsec core in the adapter logic. No additional settings are required.

Table 27. RX latency

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Table 22.

			MACsec subsystem latency	
RX_TAG_ENA	**RX control tag placement**	**Comment**	**MACsec subsystem latency [MII cycles]**	**GMII-mode [GMII cycles]**
Disabled	—		179	122
Enabled	TLV always first tag		179	122
Enabled	TAG, TLV		179	122
Enabled	TAG, TAG, TLV	*Order of all TAGs preceding TLV is irrelevant*	179	122
Enabled	TAG, TAG, TAG, TLV		179	122
Enabled	TAG, TAG, TAG, MISC-TAG, TLV		179	122

6.12.11 Interrupts

The device is able to signal important events, for instance changes in link availability, via its IRQ_N pin. In order to efficiently deduce which of the available events has fired, an interrupt tree is implemented. The IRQ_N pin is driven low when at least one IRQ in the tree is active and enabled. Otherwise, the IRQ_N pin is high-Z.

Multiple devices can share the IRQ_N line using a wired-OR topology.

6.12.12 Adapter Statistics

The adapter keeps track of various events through the means of counters.

- **TX control tag counter** TX_STATS_TLV_FRMS
- **FCS frame errors** TX_STATS_FCS_ERRS, RX_STATS_FCS_ERRS
- **Discarded frames due to policing rules** TX_STATS_DISCARDED_FRMS
- **Aborted frames due to error conditions** TX_STATS_ABORTED_FRMS, RX_STATS_ABORTED_FRMS
- **MII errors** TX_STATS_MII_ERRS , RX_STATS_MII_ERRS

6.12.13 LED interface

The adapter provides four configurable LED triggers to the LED controller. Each LED trigger can be configured individually (ADPTR_LED_TRIGn) to support the following events:

- TX_IN_EN: strobe when a frame is transmitted into the tx_in interface.
- TX_OUT_EN: strobe when a frame is transmitted out of the tx_out interface.
- RX_IN_DV: strobe when a frame is transmitted into the rx_in interface.
- RX_OUT_DV: strobe when a frame is transmitted out of the rx_out interface.
- TX_CP_EVENT: strobe when a frame is transmitted via a controlled port (tx)
- TX_UCP_EVENT: strobe when a frame is transmitted via a uncontrolled port (tx)
- RX_CP_EVENT: strobe when a frame is received via a controlled port (rx)
- RX_UCP_EVENT: strobe when a frame is received via a uncontrolled port (rx) The LED trigger signals are connected to the device's LED controller.

6.12.14 Reset

The adapter reset function accessible via the control register ADPTR_CNTRL.RESET. Setting this bit resets the adapter, the reset is self-clearing.

The MACsec core is reset via the top-level reset triggered by disabling MACsec through PORT_FUNC.MACSEC_ENABLE=0.

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6.12.15 MDIO mapping

The MACsec adapter and core registers are 32-bit wide and mapped into a designated Clause 45 MDIO region in MMD31 between 9000h-9FFFh.

Figure 43. MACsec memory mapping

Since the MDIO access width is limited to 16-bit, each 32-bit register spans two neighboring 16-bit MDIO registers which need to be written/read consecutive from lower to upper 16-bit word. The read/write is only completed after both accesses occurred. The MDIO is resilient against incorrect accesses, but can return wrong data.

Figure 44. 32-bit register arrangement inside MMD31 space

To convert between the relative byte-addresses in the MACsec window and actual MDIO addresses the following equation can be used:

mdio_address = relative_offset / 2 + 9000h

Example: Access the ADPTR_CNTRL register (relative address F00h). First read/write lower 16-bit from MDIO address 9780h (MMD31) and then the upper 16-bit from 9781h (MMD31).

Notes:

- *Lower and upper accesses must be atomic. Never interleave lower and upper accesses with other accesses.*
- *There is some limitation in the SMI access to the MACsec core , consult application notes for additional information.*

6.12.16 Register map

The following addresses are relative byte-addresses inside the MACsec address window.

Table 28.

Table 23.

Relative Byte-address inside 32-bit region	Description
0x000-0xEFF	MACsec core
0xF00-0xFFF	MACSec adapter registers

6.12.17 Configuration and test enable

In order to change the configuration of the MACsec adapter, the CONFIG_ENA bit must be set. This unlocks all configuration bits for change. The change is applied after CONFIG_ENA is deasserted.

Note: *After (re-)configuration is completed, CONFIG_ENA **must** be deasserted. Otherwise, settings are not committed.*

In this integration, TEST_ENA is not used, and test-modes are not available.

When TEST_ENA or CONFIG_ENA is asserted, ADPTR_STATUS.STATE indicates TEST_CONFIG.

6.12.18 Initialization

After device reset or startup, the entire MACsec subsystem is externally bypassed and the MACsec adapter is by default in blocking mode (ADPTR_ENA=0). The following steps are required to enable the adapter.

The adapter is dropping frames until ADPTR_ENA is set to 1. To ensure that no frames accidentally pass through during a partial configuration, set ADPTR_ENA=1 as the last step.

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1. Device specific initialization:

To enable the MACsec function, PORT_FUNC_ENABLES.MACSEC_ENABLE MMD30.8048h.5 must be set.

1. Set CONFIG_ENA=1 to allow adapter reconfiguration.
2. Optionally configure MACsec RX Control Tag and MACsec TX/RX control tag processing
   a. Set TX_TAG_ENA=1 for MACsec TX/RX control tag processing
   b. Set RX_TAG_ENA=1 for MACsec RX Control Tag processing
   c. Configure Ethertype via register TLV_ETHTYP
3. It is recommended to reduce RX_OUT_IPG and TX_OUT_IPG to 10 to achieve highest throughput. ***Note:*** *The TX and RX IPG as normally received by a PHY can already be below 12 due to IPG shrinkage caused by clock-domain-crossing inside the SoC / peer-PHY sides. Setting the RX_OUT_IPG/ TX_OUT_IPG to lower values adds some margin.*

Note: *The settings RX_OUT_IPG=10 together with RX_TAG_ENA=1 can cause issues due to excessive IPG reduction since the RX control tag can increase the frame size.*

1. Enable IRQs, if used

6.12.19 Register Description

Table 24. Register overview: CNTRL_STATUS

Address	Name	Access	Reset	Description
0h	ADPTR_CNTRL	R/W	0h	MACSEC adapter control register
4h	ADPTR_STATUS	R/W	0h	MACSEC adapter status register
Ch	TX_TAG_CNTRL	R/W	0h	TX parser TAG control
10h	TX_TAG_CNTRL1	R/W	0h	TX parser TAG control0
14h	RX_TAG_CNTRL	R/W	0h	RX Extender TAG control
18h	ADPTR_PARAMS	R/W	0h	IPG related adapter parameters
1Ch	TAGS_PARAM	R/W	0h	Ethertype configurations
20h	TAGS_PARAM1	R/W	0h	Ethertype configurations
24h	TAGS_PARAM2	R/W	0h	Ethertype configurations
28h	IPG_TRACKING	R/W	0h	Tracking of IPG
40h	ADPTR_LED_TRIG0	R/W	0h	Configuration of MACSEC led output 0
44h	ADPTR_LED_TRIG1	R/W	0h	Configuration of MACSEC led output 1
48h	ADPTR_LED_TRIG2	R/W	0h	Configuration of MACSEC led output 2
4Ch	ADPTR_LED_TRIG3	R/W	0h	Configuration of MACSEC led output 3
50h	ADPTR_IRQ_STAT	R/W	0h	Adapter Raw Interrupt status
54h	ADPTR_IRQ_ENA	R/W	0h	Adapter interrupt enable
58h	ADPTR_IRQ_INTSTAT	R/W	0h	Adapter interrupt status
80h	TX_STATS_TLV_FRMS	R/W	0h	TX TLV frame counter
84h	TX_STATS_DISCARDED_FRMS	R/W	0h	TX discarded frames counter
88h	TX_STATS_FCS_ERRS	R/W	0h	TX FCS error counter
8Ch	TX_STATS_MII_ERRS	R/W	0h	TX MII error counter

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Table 24. Register overview: CNTRL_STATUS...*continued*

Address	Name	Access	Reset	Description
90h	TX_STATS_ABORTED_FRMS	R/W	0h	TX Aborted frames counter
C0h	RX_STATS_FCS_ERRS	R/W	0h	RX FCS error counter
C4h	RX_STATS_MII_ERRS	R/W	0h	RX MII error counter
C8h	RX_STATS_ABORTED_FRMS	R/W	0h	RX Aborted frames counter
CCh	MACSEC_CFGMOD_SUMMARY	R/W	0h	Register modification status
F8h	MOD_STAT	R/W	0h	Module Configuration Status register
FCh	MOD_ID	R/W	0h	Module Identification register

6.12.19.1 ADPTR_CNTRL register

Table 25. ADPTR_CNTRL register - MACSEC adapter control register (address 0h)

Bit	Symbol	Access	Value	Description
31:16	*reserved*	R	0h	Always write 0, ignore on read.
15	RESET	R/W	0h	Resets the adapter including all registers. Registers can be accessed again at least 10 clock cycles after reset.
14	CONFIG_ENA	R/W	0h	If set, the configuration registers can be changed. Changes in updated registers only take effect after CONFIG_ENA is set back to '0'.
13	TEST_ENA	R/W	0h	If set, test related registers can be changed. This bit is also gating the related test functionality. Thus, if this bit is deasserted, the related test-modes will be deactivated.
12	ADPTR_ENA	R/W	0h	If enabled, the MACSec adapter is operational and will forward frames. If disabled, the MII_RX_IN and MII_RX_OUT stages are disabled and the datapath towards the MACSec IP is gated. Frames which arrive on MII_RX_IN and MII_RX_OUT while the adapter is disabled are counted as aborted frames (see RX/TX_STATS_ABORTED_FRMS).
11:0	*reserved*	R	0h	Always write 0, ignore on read.

6.12.19.2 ADPTR_STATUS register

Table 26. ADPTR_STATUS register - MACSEC adapter status register (address 4h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
31:3	*reserved*	R	0h	Ignore on read.
2:0	STATE	R		Reports the executive state summary of this block.
			0*	Set if MACSEC IP is disabled or in powerdown
			1	MACSEC IP is enabled from top-level, but ADPTR_ENA=0
			2	Set if a test mode or configuration is enabled, here parity test

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Table 26. ADPTR_STATUS register - MACSEC adapter status register (address 4h) *...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
			3	Set if ENA=1 and TEST_ENA=0
			4	Parity error

6.12.19.3 TX_TAG_CNTRL register

Table 27. TX_TAG_CNTRL register - TX parser TAG control (address Ch)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
31	TX_TAG_ENA	R/W	0h	When disabled, the TX tag extraction / frame builder is disabled and operating in bypass. All frames are passing the TX header extraction unmodified
30:20	*reserved*	R	0h	Always write 0, ignore on read.
19	MISC_TAG_ENA	R/W	0h	If enabled, a configurable tag (see MISC_ETHTYP and MISC_TAG_LENGTH) is enabled. If this tag is detected at an arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
18	RTAG_ENA	R/W	0h	If enabled FRER tag parsing is supported. If this tag is detected at an arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
17	STAG_ENA	R/W	0h	If enabled, a QinQ tag parsing is enabled. If this tag is detected at an arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
16	CTAG_ENA	R/W	0h	If enabled, a VLAN tag parsing is enabled. If this tag is detected at an arbitrary position, it will be skipped over and searching for the TLV is continued. If this bit is disabled, the parsing process is stopped once this header is seen.
15:8	TLV_LENGTH	R/W	0h	
7:4	*reserved*	R	0h	Always write 0, ignore on read.
3	POLICE_TLV_ LENGTH	R/W	0h	If set, the TLV parser asserts that the TLV length as seen in the tag has a length that matches TLV_LENGTH, if not the frame is dropped and counted as an ABORTED_FRMS.
2:0	POLICE_FRMS	R/W		If set, the TX parser is able to enforce certain frame types (TLV or STD). This setting controls if and which frames are dropped
			0*	No Policing
			3	Drops TLV frames. If set frames with a MACsec TX/RX control tag are dropped.
			6	If set, all frames without a TLV are dropped.

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6.12.19.4 TX_TAG_CNTRL1 register

Table 28. TX_TAG_CNTRL1 register - TX parser TAG control0 (address 10h)

Bit	Symbol	Access	Value	Description
31:7	*reserved*	R	0h	Always write 0, ignore on read.
6:3	TLV_SCI	RW	0h	Secure channel setting as provided to the MACSec block when OVERRIDE_TLV_SCI is set. This value is also used to drive inputs to MACSec when no TLV is present
2	OVERRIDE_TLV_SCI	RW	0h	If set the Secure-channel identifier (SCI) selection from TLV tag is not used but instead the value of register TLV_SCI is used.
1	TLV_CP	RW	0h	Uncontrolled/Controlled port setting as provided to the MACSec block when OVERRIDE_TLV_CP is set. This value is also used to drive inputs to MACSec when no TLV is present
0	OVERRIDE_TLV_CP	RW	0h	If set, the controlled-port (CP) selection from TLV tag is not used but the value of register TLV_CP is used

6.12.19.5 RX_TAG_CNTRL register

Table 29. RX_TAG_CNTRL register - RX Extender TAG control (address 14h)

Bit	Symbol	Access	Value	Description
31	RX_TAG_ENA	R/W	0h	When disabled, the RX header tag frame builder is disabled and all frames are forwarded unmodified.
30:12	*reserved*	R	0h	Always write 0, ignore on read.
11	SKIP_MISC_TAG	R/W	0h	MISC_TAG. For description see CTAG.
10	SKIP_RTAG	R/W	0h	RTAG. For description see CTAG.
9	SKIP_STAG	R/W	0h	STAG. For description see CTAG.
8	SKIP_CTAG	R/W	0h	If this bit is set, the CTAG will be detected and skipped over during RX header tag generation.
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5	TAG_AFTER_MISC_TAG	R/W	0h	MISC_TAG. For description see CTAG.
4	TAG_AFTER_RTAG	R/W	0h	RTAG. For description see CTAG.
3	TAG_AFTER_STAG	R/W	0h	STAG. For description see CTAG.
2	TAG_AFTER_CTAG	R/W	0h	If this bit is set and a CTAG is found, the RX header tag is added immediately after it, without scanning any further. If SKIP_CTAG is set, skipping has precedence. If multiple TAG_AFTER are set, the TLV tag is placed after the first tag found.
1	TAG_CP_FRMS	R/W	0h	If set, a RX header tag is added to controlled port frames. Attention: this shrinks the IPG!
0	TAG_UCP_FRMS	R/W	0h	If set, a RX header tag is added to uncontrolled port frames. Attention: this shrinks the IPG!

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6.12.19.6 ADPTR_PARAMS register

Table 30. ADPTR_PARAMS register - IPG related adapter parameters (address 18h)

Bit	Symbol	Access	Value	Description
21:16	TX_OUT_IPG	R/W	Ch	Minimum RX IPG maintained on TX_OUT path
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:8	RX_OUT_IPG	R/W	Ch	Minimum RX IPG maintained on RX_OUT path
7:6	*reserved*	R	0h	Always write 0, ignore on read.
5:0	IPG_MON_THR	R/W	Ah	Set the minimum IPG threshold for which an IRQ is flagged. IF an IPG on RX/TX in path is seen with less than this value (in octets) the IRQ is set. Note, due to clock-domain-crossings the expected IPG is less than 12.

6.12.19.7 TAGS_PARAM register

Table 31. TAGS_PARAM register - Ethertype configurations (address 1Ch)

Bit	Symbol	Access	Value	Description
31:16	STAG_ETHTYP	R/W	88A8h	Ethertype used to identify outer VLAN (STAG) tag
15:0	CTAG_ETHTYP	R/W	8100h	Ethertype used to identify VLAN (CTAG) tag

6.12.19.8 TAGS_PARAM1 register

Table 32. TAGS_PARAM1 register - Ethertype configurations (address 20h)

Bit	Symbol	Access	Value	Description
31:16	RTAG_ETHTYP	R/W	F1C1h	Ethertype used to identify RTAG tag
15:0	TLV_ETHTYP	R/W	4E58h	Ethertype used to identify RX/TX MACsec control TLV

6.12.19.9 TAGS_PARAM2 register

Table 33. TAGS_PARAM2 register - Ethertype configurations (address 24h)

Bit	Symbol	Access	Value	Description
18:16	MISC_TAG_LENGTH	R/W	4h	Length used for MISC_ETHTYP
15:0	MISC_ETHTYP	R/W	0h	The MISC tag is catering for any header which is not natively supported. The MISC_ETHTYP is used to identify the tag and the MISC_TAG_LENGTH defines the length in octets not including the Ethertype itself. The supported maximum length is 6 octets. (so the header including the Ethertype is 64bits).

6.12.19.10 IPG_TRACKING register

Table 34. IPG_TRACKING register - Tracking of IPG (address 28h)

Bit	Symbol	Access	Value	Description
13:8	TX_IN_IPG_MIN	R/W	3Fh	Minimum number of IPG octets observed since last clear.
7:6	*reserved*	R	0h	Always write 0, ignore on read.

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Table 34. IPG_TRACKING register - Tracking of IPG (address 28h)*...continued*

Bit	Symbol	Access	Value	Description
5:0	RX_IN_IPG_MIN	R/W	3Fh	Minimum number of IPG octets observed since last clear.

6.12.19.11 ADPTR_LED_TRIG0 register

Table 35. ADPTR_LED_TRIG0 register - Configuration of MACSEC led output 0 (address 40h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

6.12.19.12 ADPTR_LED_TRIG1 register

Table 36. ADPTR_LED_TRIG1 register - Configuration of MACSEC led output 1 (address 44h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected

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Table 36. ADPTR_LED_TRIG1 register - Configuration of MACSEC led output 1 (address 44h)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

6.12.19.13 ADPTR_LED_TRIG2 register

Table 37. ADPTR_LED_TRIG2 register - Configuration of MACSEC led output 2 (address 48h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

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6.12.19.14 ADPTR_LED_TRIG3 register

Table 38. ADPTR_LED_TRIG3 register - Configuration of MACSEC led output 3 (address 4Ch)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	ENA	R/W	0h	Enable LED output
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	R/W		Configures which internal signal to externalize as TRIGGER0
			0*	
			1	If selected, the internal TX_EN signal at the TX_IN interface is selected
			2	If selected, the internal TX_EN signal at the TX_OUT interface is selected
			3	If selected, the internal RX_DV signal at the RX_IN interface is selected
			4	If selected, the internal RX_DV signal at the RX_OUT interface is selected
			5	If selected, the trigger signal is strobed every time a frame is transmitted on the controlled port
			6	If selected, the trigger signal is strobed every time a frame is transmitted on the uncontrolled port
			7	If selected, the trigger signal is strobed every time a frame is received on the controlled port
			8	If selected, the trigger signal is strobed every time a frame is received on the uncontrolled port

6.12.19.15 ADPTR_IRQ_STAT register

Table 39. ADPTR_IRQ_STAT register - Adapter Raw Interrupt status (address 50h)

Bit	Symbol	Access	Value	Description
10	MACSEC_SOFT_ERR	R/W	0h	If set, the MACsec core is in a recoverable error condition which could have caused frame loss Write one to clear.
9	MACSEC_HARD_ERR	R/W	0h	If set, the MACsec core is in an unrecoverable error condition and requires reset. Write one to clear.
8	MACSEC_CFGMOD_EVT	R/W	1h	if set, MACSEC core context register changed due software action or an error caused by internal bit-flips. Write one to clear.
7	DISCARDED_FRM	R/W	0h	If set, a frame was discarded because of the POLICE feature. See TX_STATS_DISCARDED_FRMS register. Write one to clear.
6	ABORTED_FRM	R/W	0h	If set, a frame was aborted. See register TX_STATS_ABORTED_FRMS for the conditions. Write one to clear.

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Table 39. ADPTR_IRQ_STAT register - Adapter Raw Interrupt status (address 50h)*...continued*

Bit	Symbol	Access	Value	Description
5	PARITY_CNTRL_ERR	R/W	0h	If set, a control parity error in the adapter occurred Write one to clear.
4	PARITY_DATA_ERR	R/W	0h	If set, a data parity error in the adapter occurred Write one to clear.
3	TX_IPG_THR_EXCEEDED	R/W	0h	If set, the IPG on the TX_IN interface fell below the threshold Write one to clear.
2	RX_IPG_THR_EXCEEDED	R/W	0h	If set, the IPG on the RX_IN interface fell below the threshold Write one to clear.
1	TX_IN_ERR	R/W	0h	If set, an MII error on the TX_IN interface was observed Write one to clear.
0	RX_IN_ERR	R/W	0h	If set, an MII error on the RX_IN interface was observed Write one to clear.

6.12.19.16 ADPTR_IRQ_ENA register

Table 40. ADPTR_IRQ_ENA register - Adapter interrupt enable (address 54h)

Bit	Symbol	Access	Value	Description
10	MACSEC_SOFT_ERR	R/W	0h	If set, the respective IRQ is enabled
9	MACSEC_HARD_ERR	R/W	0h	If set, the respective IRQ is enabled
8	MACSEC_CFGMOD_EVT	R/W	0h	If set, the respective IRQ is enabled
7	DISCARDED_FRM	R/W	0h	If set, the respective IRQ is enabled
6	ABORTED_FRM	R/W	0h	If set, the respective IRQ is enabled
5	PARITY_CNTRL_ERR	R/W	0h	If set, the respective IRQ is enabled
4	PARITY_DATA_ERR	R/W	0h	If set, the respective IRQ is enabled
3	TX_IPG_THR_EXCEEDED	R/W	0h	If set, the respective IRQ is enabled
2	RX_IPG_THR_EXCEEDED	R/W	0h	If set, the respective IRQ is enabled
1	TX_IN_ERR	R/W	0h	If set, the respective IRQ is enabled
0	RX_IN_ERR	R/W	0h	If set, the respective IRQ is enabled

6.12.19.17 ADPTR_IRQ_INTSTAT register

Table 41. ADPTR_IRQ_INTSTAT register - Adapter interrupt status (address 58h)

Bit	Symbol	Access	Value	Description
10	MACSEC_SOFT_ERR	R	0h	Masked status of the respective interrupt

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Table 41. ADPTR_IRQ_INTSTAT register - Adapter interrupt status (address 58h)*...continued*

Bit	Symbol	Access	Value	Description
9	MACSEC_HARD_ERR	R	0h	Masked status of the respective interrupt
8	MACSEC_CFGMOD_EVT	R	0h	Masked status of the respective interrupt
7	DISCARDED_FRM	R	0h	Masked status of the respective interrupt
6	ABORTED_FRM	R	0h	Masked status of the respective interrupt
5	PARITY_CNTRL_ERR	R	0h	Masked status of the respective interrupt
4	PARITY_DATA_ERR	R	0h	Masked status of the respective interrupt
3	TX_IPG_THR_EXCEEDED	R	0h	Masked status of the respective interrupt
2	RX_IPG_THR_EXCEEDED	R	0h	Masked status of the respective interrupt
1	TX_IN_ERR	R	0h	Masked status of the respective interrupt
0	RX_IN_ERR	R	0h	Masked status of the respective interrupt

6.12.19.18 TX_STATS_TLV_FRMS register

Table 42. TX_STATS_TLV_FRMS register - TX TLV frame counter (address 80h)

Bit	Symbol	Access	Value	Description
31:0	TLV_FRMS	R/W	0h	Counts the number of TLV frames. • This counter is overflowing. • Writing 0x0 clears the counter

6.12.19.19 TX_STATS_DISCARDED_FRMS register

Table 43. TX_STATS_DISCARDED_FRMS register - TX discarded frames counter (address 84h)

Bit	Symbol	Access	Value	Description
31:0	DISCARDED_FRM	R/W	0h	Counts the number of discarded frames because of the POLICE feature. • This counter is overflowing. • Writing 0x0 clears the counter

6.12.19.20 TX_STATS_FCS_ERRS register

Table 44. TX_STATS_FCS_ERRS register - TX FCS error counter (address 88h)

Bit	Symbol	Access	Value	Description
31:0	FCS_ERRS	R/W	0h	Counts the number of frames where the FCS-check on ingress failed. • This counter is overflowing. • Writing 0x0 clears the counter

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6.12.19.21 TX_STATS_MII_ERRS register

Table 45. TX_STATS_MII_ERRS register - TX MII error counter (address 8Ch)

Bit	Symbol	Access	Value	Description
31:0	MII_ERRS	R/W	0h	Counts the number of frames where the TX_ER is set. • This counter is overflowing. • Writing 0x0 clears the counter

6.12.19.22 TX_STATS_ABORTED_FRMS register

Table 46. TX_STATS_ABORTED_FRMS register - TX Aborted frames counter (address 90h)

Bit	Symbol	Access	Value	Description
31:0	ABORTED_FRMS	R/W	0h	Counts the number of frames that are aborted inflight because of an error condition. • This counter is overflowing. • Writing 0x0 clears the counterThere are the following reasons for aborts. 1. Input fifo overflow 2. Output fifo underflow 3. Parity error in data or control 4. Frames that are dropped on TX_IN because ENA = 0

6.12.19.23 RX_STATS_FCS_ERRS register

Table 47. RX_STATS_FCS_ERRS register - RX FCS error counter (address C0h)

Bit	Symbol	Access	Value	Description
31:0	FCS_ERRS	R/W	0h	Counts the number of frames where the FCS-check on ingress failed. • This counter is overflowing. • Writing 0x0 clears the counter.

6.12.19.24 RX_STATS_MII_ERRS register

Table 48. RX_STATS_MII_ERRS register - RX MII error counter (address C4h)

Bit	Symbol	Access	Value	Description
31:0	MII_ERRS	R/W	0h	Counts the number of frames where the RX_ER is set. • This counter is overflowing. • Writing 0x0 clears the counter.

6.12.19.25 RX_STATS_ABORTED_FRMS register

Table 49. RX_STATS_ABORTED_FRMS register - RX Aborted frames counter (address C8h)

Bit	Symbol	Access	Value	Description
31:0	ABORTED_FRMS	R/W	0h	Counts the number of frames that are aborted inflight because of an error condition. • This counter is overflowing. • Writing 0x0 clears the counter

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Table 49. RX_STATS_ABORTED_FRMS register - RX Aborted frames counter (address C8h)

Bit	Symbol	Access	Value	Description
				The following are reasons for aborts. • Input fifo overflow • Output fifo underflow • Parity error in data or control • Frames that are dropped on RX_IN because ENA = 0

6.12.19.26 MACSEC_CFGMOD_SUMMARY register

Table 50. MACSEC_CFGMOD_SUMMARY register - Register modification status (address CCh)

Bit	Symbol	Access	Value	Description
18	TX_ADPTR	R/W	1h	If set, MACSEC TX Adapter settings have changed since this flag was cleared. Write one to clear.
17	RX_ADPTR	R/W	1h	If set, the MACSEC RX Adapter settings have changed since this flag was cleared. Write one to clear.
16	GLOBAL	R/W	0h	If set, the MACSEC global context and uncontrolled port filtering has changed since this flag was cleared. Write one to clear.
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11	TX_SC4	R/W	0h	If set, the MACSEC TX secure channel 4 context has changed since this flag was cleared Write one to clear.
10	TX_SC3	R/W	0h	If set, the MACSEC TX secure channel 3 context has changed since this flag was cleared Write one to clear.
9	TX_SC2	R/W	0h	If set, the MACSEC TX secure channel 2 context has changed since this flag was cleared Write one to clear.
8	TX_SC1	R/W	0h	If set, the MACSEC TX secure channel 1 context has changed since this flag was cleared Write one to clear.
7:4	*reserved*	R	0h	Always write 0, ignore on read.
3	RX_SC4	R/W	0h	If set, the MACSEC RX secure channel 4 context has changed since this flag was cleared Write one to clear.
2	RX_SC3	R/W	0h	If set, the MACSEC RX secure channel 3 context has changed since this flag was cleared Write one to clear.
1	RX_SC2	R/W	0h	If set, the MACSEC RX secure channel 2 context has changed since this flag was cleared Write one to clear.
0	RX_SC1	R/W	0h	If set, the MACSEC RX secure channel 1 context has changed since this flag was cleared.

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Table 50. MACSEC_CFGMOD_SUMMARY register - Register modification status (address CCh)...*continued*

Bit	Symbol	Access	Value	Description
				Note: For all of the following flags in this register, a change can occur due to a deliberate software (re)-configuration or because of an internal error caused by bit-flips. Write one to clear.

6.12.19.27 MOD_STAT register

Table 51. MOD_STAT register - Module Configuration Status register (address F8h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
31:24	N_W	R	0h	Implemented number of write-only registers
23:16	N_R	R	5h	Implemented number of read-only registers
15:8	N_RW	R	12h	Implemented number of read/write registers
7:0	TYPE	R		Indicates type of configuration register that is implemented
			7*	Configuration Register module

6.12.19.28 MOD_ID register

Table 52. MOD_ID register - Module Identification register (address FCh)

Bit	Symbol	Access	Value	Description
31:16	ID	R	A09Bh	Module Identifier: Unique 16-bit code, assigned through NXP internal registration
15:12	MAJOR_REV	R	1h	Major Revision: major revision of module implementation, starting at 0. There may not be software compatibility between major revisions.
11:8	MINOR_REV	R	0h	Minor Revision: minor revision of module implementation, starting at 0. Minor revision of module implementation, starting at 0. Software compatibility is expected between minor revisions.
7:0	APERTURE	R	0h	Aperture Size: Encoded as (aperture size/4K) - 1, so 0 means 4 kByte

6.13 MACsec core

The MACsec memory map and register definition of the MACsec core are described in UM12059. The application notes give additional information on the limitation on MACsec in specific use cases.

6.14 TC10

Figure 39 describes the sleep handshake, which is the intended way to enter sleep when TC10 sleep is configured (WAKE_SLEEP_CONFIG.FUNCTION_SELECT='010'), making sure both PHYs enter sleep mode. Alternatively, e.g., when the handshake is not functioning or for testing purpose, DEVICE_SLEEP can be entered without TC10 hand-shake by SLEEP_REQUEST from DEVICE_STANDBY mode. See Figure 4.

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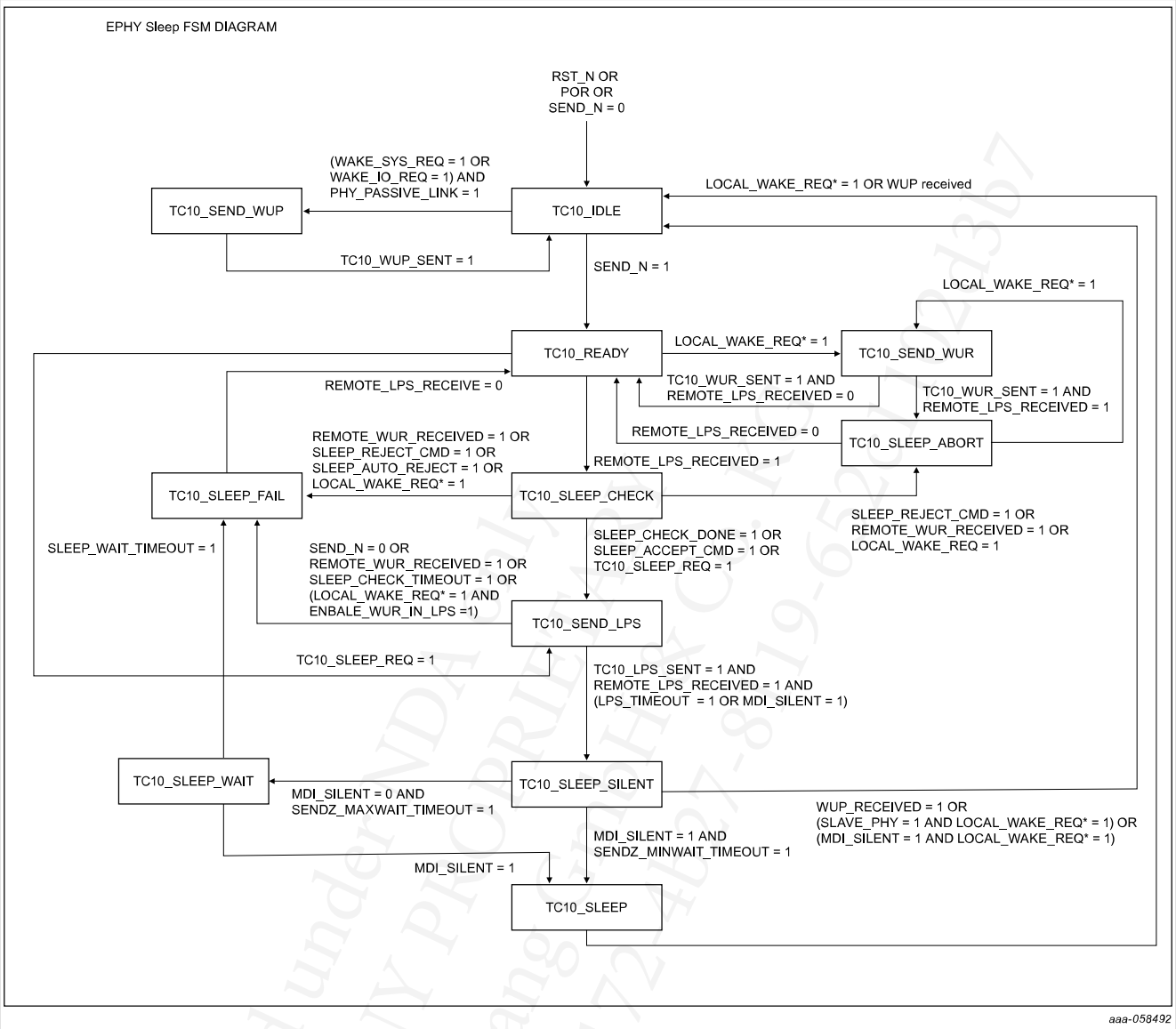


Figure 39. TC10 Ethernet PHY state transitions

6.15 Miscellaneous

6.15.1 JTAG

The TJA1121 supports IEEE1149.1 and IEEE1149.6 compliant JTAG interface for boundary scan. A boundary scan description file (BSDL) is available on request.

JTAG mode is entered via pin-strapping the MODE selection configuration accordingly. In this mode the JTAG TAP interface is accessible on pins TMS, TDI, TDO and TCK as shown in Table 53. Other pins remain in functional mode until a IEEE1149 instruction is executed.

JTAG mode can only be entered by pin strap configuration and left via a device reset though the RST_N pin.

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Table 53. JTAG interface

JTAG Function	Symbol	Pin	Type[1]	Description
TMS	TMS / GPIO0 / CONFIG0	17	IO, PU	JTAG Test mode select (VDDIO domain)
TDI	TDI / GPIO1	18	I, PU	JTAG Test data in (VDDIO domain)
TDO	RX_ER / GPIO2 / TDO / CONFIG1	19	O	JTAG Test data out (VDDIO domain)
TCK	GPIO9 / TCK	28	I	JTAG Test clock (VDDIO domain)

[1] I: digital input; O: digital output; IO: digital input/output; OD: digital open-drain output; AO: Analog output; AI: Analog Input; AIO: Analog input/output; PU: weak pull-up; P: power supply. G: ground

6.15.2 General-purpose I/O

The device has 12 GPIOs multiplexed on functional device pins. Not all pins are available in all modes. The following table gives an overview of the GPIOs that are available in a specified mode of operation.

Table 54. Available GPIOs across the variants.

GPIO	RGMII (TJA1121A)	SGMII (TJA1121B)
GPIO0	Available	Available
GPIO1	Available	Available
GPIO2	Available	Available
GPIO3	—	Available
GPIO4	—	Available
GPIO5	—	Available
GPIO6	—	Available
GPIO7	—	Available
GPIO8	—	Available
GPIO9	Available	Available

Only GPIOs that are free in the configured xMII mode can be used. The configuration settings for GPIOs that are not available are ignored.

Each GPIO can be freely assigned to a function. The available functions are LED, PTP (timestamping) and miscellaneous. The assignment is done through the FUNC_SELECT field of the associated GPIOx_FUNC_CONFIG register.

Some functions, for example the LED function, have multiple internal signals to which a GPIO can be connected, which is configured via SIGNAL_SELECT.

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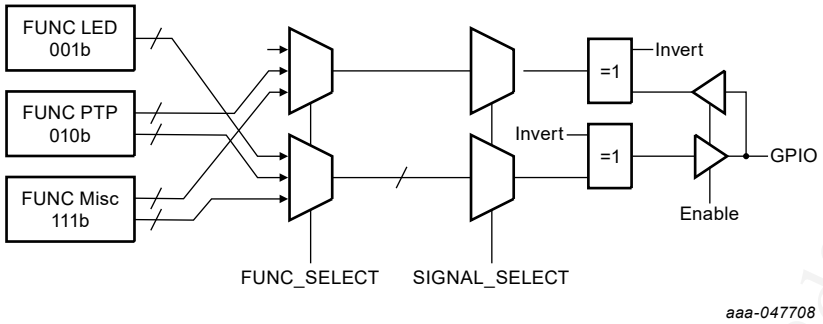


Figure 40. GPIO controller function/signal hierarchy

The table below lists the FUNC_SELECT to function mapping as well as the SIGNAL_SELECT to signal mapping for each function.

Table 55. GPIO function and signal selection (see register GPIOx_FUNC_CONFIG)

Function	Signal	FUNC_ SELECT	SIGNAL_ SELECT	Dir.[1]	Description
LED	LED0_A	001b	00000b	O	LED controller 0, channel A
	LED0_B	001b	00001b	O	LED controller 0, channel B
	LED1_A	001b	00010b	O	LED controller 1, channel A
	LED1_B	001b	00011b	O	LED controller 1, channel B
	LED2_A	001b	00100b	O	LED controller 2, channel A
	LED2_B	001b	00101b	O	LED controller 2, channel B
	LED3_A	001b	00110b	O	LED controller 3, channel A
	LED3_B	001b	00111b	O	LED controller 3, channel B
	reserved	001b	*others*	Z	—
PTP	PTP_SYNC	010b	00000b	I	PTP synchronization input (PPS in)
	PTP_TRIGGER	010b	00001b	I	external PTP timestamp trigger signal
	TX_SOF	010b	10000b	I	External transmit start of frame (SOF) timestamp trigger
	RX_SOF	010b	10001b	I	External receive start of frame (SOF) timestamp trigger
	PPS_OUT	010b	10010b	O	Recovered PPS output
	reserved	010b	*others*		
Misc	REGISTER_OUT	111b	10000b	O	GPIO_OUTPUTS register drives pin
	REF_CLK_OUT	111b	10001b	O	25 MHz REF_CLK_OUT output is enabled. Only available on GPIO9
	TX_CLK_OUT	111b	10010b	O	Test clock output. Only available on GPIO9
	reserved	111b	*others*	Z	—

[1] O: output, I: input, Z: high-Z

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The GPIOx_FUNC_CONFIG.ENABLE acts as a qualifier to and from the GPIO pin. If the selected function is an output function and ENABLE=0, then the GPIO will drive zero. If the selected function is an input function and ENABLE=1, then the GPIO provides a low signal to the selected function.

If GPIOx_FUNC_CONFIG.FUNC_SELECT is set to RESERVED, the respective GPIO is configured to input and the pin is tristated.

If multiple GPIOs are assigned to the same function/signal, the inputs are ORed.

6.15.3 IO driver configuration

MDIO, xMII, and GPIO CMOS drivers are software configurable. The slew rate (SLEW_RATE) and drive strength (DRIVE_STRENGTH) can be adjusted. Table 56 shows an overview of the pins configured by each register.

Table 56. IO driver configuration

Register	Pins / Function
GPIOx_IO_CONFIG	All other GPIO function
GPIO9_IO_CONFIG	GPIO configured for REF_CLK_OUT, TX_CLK_OUT modes
SMI_DATA_IO_CFG	MDIO pin
XMII_CLK_IO_CONFIG	RGMII: RXC
XMII_DATA_IO_CONFIG	RGMII RXD and TXD pins

The device automatically selects the drive characteristics for the selected xMII/VDDIO combination. Using default settings, the device is able to drive 25 pF capacitive load. If the MAC device is physically close to the PHY and trace length is low (< 15 pF capacitive load), an optimized driver configuration can be used. This is configured by asserting XMII_DATA_IO_CONFIG.ALTERNATE_CONFIG. This bit changes the configuration for all data and control xMII pins.

To override the default configuration, MANUAL_CONFIG in the respective register must be asserted. The configuration can be changed by setting SLEW_RATE to a value between 0 (slowest) and 7 (fastest) and DRIVE_STRENGTH for RGMII RX PADS can be configured to value between 0 to 15.

The pins INT_N, INH, WAKE_IN_OUT cannot be configured.

Note: *Changing the slew rate and/or drive strength impacts the xMII/MDIO timing. Custom settings do not guarantee compliance to MDIO/RGMII standards.*

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7 Registers

Reserved register fields must be ignored on read and for writable registers only written with their reset values, unless otherwise indicated. For any register (bit), the single value in the column 'Value' represents that reset value.

Registers are mapped into five groups.

- CL22: MDIO interface registers (IEEE 802.3 Clause 22 and Clause 45)
- PMA registers (MMD 01)
- PCS registers (MMD 03)
- NXP vendor-specific registers (MMD 30)
- MACsec related register (MMD31)
- Shared registers

7.1 CL22

7.1.1 Register Description

Note: *The addresses as listed in the CL22 register table cannot be used directly for a IEEE Clause 22 MDIO access. The CL22 register table lists the register offsets as mapped into Shared MMD Clause 45 space. When accessing these registers through a physical Clause 22 transaction, 8000h must be subtracted from the offset as given in the register table. For example, PHY_ID_1 is located at 2h and ALWAYS_ACCESSIBLE at 1Fh.*

Table 57. Register overview: CL22

Address	Name	Access	Reset	Description
8002h	PHY_ID_1	R	1Bh	PHY identification register 1
8003h	PHY_ID_2	R	B030h	PHY identification register 2
800Dh	CL45_ACCESS_CONTROL	RW	0h	Clause 45 indirect access control register
800Eh	CL45_ADDRESS_DATA	RW	0h	Clause 45 indirect access address/data register
8014h	FAVORITE_IRQS	RW	0h	Favorite interrupts register
801Eh	CL45_ADDRESS	RW	0h	Clause 45 indirect access address register
801Fh	ALWAYS_ACCESSIBLE	RW	0h	Always accessible register (Implemented in AO domain)

7.1.1.1 PHY_ID_1 register

Table 58. PHY_ID_1 register - (address 8002h)

Bit	Symbol	Access	Value	Description
15:0	OUI_BITS_3_TO_18	R	1Bh	OUI = 00-60-37 OUI bits 3:18 = 0000.0000.0001.1011

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7.1.1.2 PHY_ID_2 register

Table 59. PHY_ID_2 register - (address 8003h)

Bit	Symbol	Access	Value	Description
15:10	OUI_BITS_19_TO_24	R	2Ch	OUI = 00-60-37 OUI bits 19:24 = 1011.00
9:4	MODEL	R	3h	6-bit PHY model number
3:0	REVISION	R	0h	4-bit PHY revision number

7.1.1.3 CL45_ACCESS_CONTROL register

Table 60. CL45_ACCESS_CONTROL register - (address 800Dh)

Bit	Symbol	Access	Value	Description
15:14	OP	RW	0h	00= address 01= data, no post increment 10= data, post increment on reads and writes 11= data, post increment on writes only
13:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	MMD	RW	0h	Device address: 0 = Reserved 1 = PMA/PMD 3 = PCS 30 = NXP-specific Other = Reserved

7.1.1.4 CL45_ADDRESS_DATA register

Table 61. CL45_ADDRESS_DATA register - (address 800Eh)

Bit	Symbol	Access	Value	Description
15:0	ADDRESS_DATA	RW	0h	If CL45_ACCESS_CONTROL.OP= '00', MMD DEVAD's address register. Otherwise, MMD DEVAD's data register as indicated by the contents of its address register

7.1.1.5 FAVORITE_IRQS register

Table 62. FAVORITE_IRQS register - (address 8014h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	SGMII_LS_EVENT	RW	0h	1 = SGMII Link_available status bit has changed 0 = SGMII Link_available status bit did not change Write one to clear.
7	SYNC_TS_IRQ	RW	0h	1 = Time-stamps available of detected SYNC signal events in PTP_SYNC_TS buffer. 0 = No time-stamp available in INGRESS buffer Write one to clear.

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Table 62. FAVORITE_IRQS register - (address 8014h)*...continued*

Bit	Symbol	Access	Value	Description
6	INGRESS_TS_IRQ	RW	0h	1 = Time-stamps available of received PTP frames in INGRESS buffer. 0 = No time-stamp available in INGRESS buffer Write one to clear.
5	EGRESS_TS_IRQ	RW	0h	1 = Time-stamps available of transmitted PTP frames in EGRESS buffer. 0 = No time-stamp available in EGRESS buffer Write one to clear.
4	WAKE_SLEEP_EVENT	RW	0h	1 = Change of PHY Wake-Sleep status bits 0 = No change of PHY WakeSleep status bits Write one to clear.
3	LINK_AVAILABLE_EVENT	RW	0h	1 = ePHY Link_available status bit has changed 0 = ePHY Link_available status bit did not change Write one to clear.
2	*reserved*	R	0h	Always write 0, ignore on read.
1	PORT_IRQ	R	0h	1 = Interrupts pending for this PORT 0 = No interrupts pending for this PORT
0	GLOBAL_IRQS	R	0h	1 = Interrupts pending in Global interrupt tree 0 = No interrupts pending in Global interrupt tree

7.1.1.6 CL45_ADDRESS register

Table 63. CL45_ADDRESS register - (address 801Eh)

Bit	Symbol	Access	Value	Description
15:0	ADDRESS	RW	0h	Address pointer of the MMD selected by Dh.4:0

7.1.1.7 ALWAYS_ACCESSIBLE register

Table 64. ALWAYS_ACCESSIBLE register - Implemented in AO Domain (address 801Fh)

Bit	Symbol	Access	Value	Description
15	WU_SMI	RW	0h	1 = Wake-up the device/sub-system by SMI 0 = Default state Write one to clear.
14:5	*reserved*	R	0h	Always write 0, ignore on read.
4	REFCLK_WARN	RW	0h	1 = RefClk frequency does not meet expectations. The frequency is out of range for successful smi trancation for all MDC frequency. 0 = Indicates refclk frequency allows successful smi trancation for all MDC frequency and working range is 12.5 to 37.5 MHz. Write one to clear.
3	CORE_SUPPLY_WARN	RW	0h	1 = Core supply issue detected 0 = No core supply issue detected or a persistant supply fault since last clear.

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Table 64. ALWAYS_ACCESSIBLE register - Implemented in AO Domain (address 801Fh)*...continued*

Bit	Symbol	Access	Value	Description
				Write one to clear.
2	CORE_SUPPLY_ STATUS	R	0h	1 = OKAY all registers readable 0 = NOT OKAY
1	LIMITED_ACCESS	R	0h	0 = All registers accessible read/write If CORE_SUPPLY_STATUS=0: 1 = Only registers 1Fh and 801F are accessible read/write, all other registers not accessible If CORE_SUPPLY_STATUS=1: 1 = All registers readable, only registers 1Fh and 801F are writable.
0	INHIBIT_STATUS	R	0h	1 = This device drives the inhibit pin high 0 = This device does not drive the inhibit pin high

7.2 MMD1

7.2.1 Register Description

Table 65. Register overview: MMD1

Address	Name	Access	Reset	Description
0h	PMA_CONTROL1	RW	40h	PMA/PMD control 1
1h	PMA_STATUS1	R	2h	PMA/PMD status 1 register
4h	PMA_SPEED_ABILITY	R	10h	PMA/PMD speed ability
7h	PMA_CONTROL2	RW	3Dh	PMA/PMD control 2 register
8h	PMA_STATUS2	R	9301h	PMA/PMD status 2 register
9h	PMD_TRANSMIT_DISABLE	RW	0h	PMD transmit disable register
Ah	PMD_RCV_DETECT	R	0h	PMD receive signal detect register
Bh	PMA_EXTENDED_ ABILITIES	R	800h	PMA/PMD extended ability register
12h	BASE_T1_PMA_XTD_ ABILITY	R	2h	BASE-T1 PMA/PMD extended ability
834h	BASE_T1_PMA_CONTROL	RW	8001h	BASE-T1 PMA/PMD control
900h	E1000BT1_PMA_CONTROL	RW	0h	1000BASE-T1 PMA control
901h	E1000BT1_PMA_STATUS	R	B00h	1000BASE-T1 PMA status
902h	E1000BT1_TRAINING	RW	2h	1000BASE-T1 training
903h	E1000BT1_LP_TRAINING	R	0h	1000BASE-T1 link partner training
904h	E1000BT1_TEST_MODE_ CONTROL	RW	0h	1000BASE-T1 PMA/PMD test control

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7.2.1.1 PMA_CONTROL1 register

Table 66. PMA_CONTROL1 register - PMA/PMD control 1 (address 0h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	1=PMA_PMD Reset
14	*reserved*	R	0h	Always write 0, ignore on read.
13	SPEED_SELECT_LSB	RW	0h	Paired with bit 6
12	*reserved*	R	0h	Always write 0, ignore on read.
11	LOW_POWER	RW	0h	1=Enabled low power mode
10:7	*reserved*	R	0h	Always write 0, ignore on read.
6	SPEED_SELECT_MSB	RW	1h	Speed selection bits {SPEED_SELECT_MSB, SPEED_SELECT_LSB} 11 = BITS_5_2_SEL_SPD 10 = 1000MBPS 01 = 100MBPS 00 = 10MBPS
5:2	SPEED_SELECT_XTD	RW	0h	Speed selection 1001=400 Gb/s 1000=200 Gb/s 0111=5 Gb/s 0110=2.5 Gb/s 0101=Reserved 0100=25 Gb/s 0011=100 Gb/s 0010=40 Gb/s 0001=10PASS-TS/2BASE-TL 0000=10 Gb/s
1	REMOTE_LOOPBACK	RW	0h	1=PMA remote loopback enable
0	LOCAL_LOOPBACK	RW	0h	1=PMA local loopback enable

7.2.1.2 PMA_STATUS1 register

Table 67. PMA_STATUS1 register - PMA/PMD status 1 register (address 1h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7	FAULT	R	0h	1=Fault detected indication
6:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	RECEIVE_LINK_STATUS	R	0h	1=Receive link status.
1	LOW_POWER_ABILITY	R	1h	0=PMA/PMD does not support low-power mode

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Table 67. PMA_STATUS1 register - PMA/PMD status 1 register (address 1h)*...continued*

Bit	Symbol	Access	Value	Description
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.2.1.3 PMA_SPEED_ABILITY register

Table 68. PMA_SPEED_ABILITY register - PMA/PMD speed ability (address 4h)

Bit	Symbol	Access	Value	Description
15	ABILITY_400G	R	0h	0=PMA/PMD is not capable of operating at 400 Gb/s
14	ABLITY_5G	R	0h	0=PMA/PMD is not capable of operating as 5 Gb/s
13	ABILITY_2500M	R	0h	0=PMA/PMD is not capable of operating as 2.5 Gb/s
12	ABILITY_200G	R	0h	0=PMA/PMD is not capable of operating at 200 Gb/s
11	ABILITY_25G	R	0h	0=PMA/PMD is not capable of operating at 25 Gb/s
10	ABILITY_10GPASS_XR	R	0h	0=PMA/PMD is not capable of operating as 10GPASS-XR
9	ABILITY_100G	R	0h	0=PMA/PMD is not capable of operating as 100 Gb/s
8	ABILITY_40G	R	0h	0=PMA/PMD is not capable of operating as 40 Gb/s
7	ABILITY_D10G_U1G	R	0h	0=PMA/PMD is not capable of operating at 10 Gb/s downstream and 1 Gb/s upstream.
6	ABILITY_10M	R	0h	0=PMA/PMD is not capable of operating as 10 Mb/s
5	ABILITY_100M	R	0h	0=PMA/PMD is not capable of operating at 100 Mb/s
4	ABILITY_1000M	R	1h	1=PMA/PMD is capable of operating at 1000 Mb/s
3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	ABILITY_10PASS_TS	R	0h	0=PMA/PMD is not capable of operating as 10PASS-TS
1	ABILITY_2BASE_TL	R	0h	0=PMA/PMD is not capable of operating as 2BASE-TL
0	ABILITY_10G	R	0h	0=PMA/PMD is not capable of operating at 10 Gb/s

7.2.1.4 PMA_CONTROL2 register

Table 69. PMA_CONTROL2 register - PMA/PMD control 2 register (address 7h)

Bit	Symbol	Access	Value	Description
15:7	*reserved*	R	0h	Always write 0, ignore on read.
6:0	TYPE_SELECT	RW	3Dh	0 1 1 1 1 0 1 = BASE-T1 PMA/PMD

7.2.1.5 PMA_STATUS2 register

Table 70. PMA_STATUS2 register - PMA/PMD status 2 register (address 8h)

Bit	Symbol	Access	Value	Description
15:14	DEVICE_PRESENT	R	2h	Device Present

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Table 70. PMA_STATUS2 register - PMA/PMD status 2 register (address 8h)*...continued*

Bit	Symbol	Access	Value	Description
				10=Device responding at this address 00=No device responding at this address 01=No device responding at this address 11=No device responding at this address
13	TRANSMIT_FAULT_ABILITY	R	0h	0=PMA/PMD does not have the ability to detect a fault condition on the transmit path
12	RECEIVE_FAULT_ABILITY	R	1h	1=PMA/PMD has the ability to detect a fault condition on the receive path
11	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
10	RECEIVE_FAULT	R	0h	1=Fault condition on receive path
9	EXTENDED_ABILITIES	R	1h	1=PMA/PMD has extended abilities
8	PMD_TRANSMIT_DISABLE	R	1h	1=PMD has the ability to disable the transmit path
7:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	PMA_LOCAL_LOOPBACK_ABILITY	R	1h	1=PMA has the ability to perform a local loopback function

7.2.1.6 PMD_TRANSMIT_DISABLE register

Table 71. PMD_TRANSMIT_DISABLE register - PMD transmit disable register (address 9h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	TX_DISABLE	RW	0h	1=Disable transmitter output

7.2.1.7 PMD_RCV_DETECT register

Table 72. PMD_RCV_DETECT register - PMD receive signal detect register (address Ah)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	RCV_SIGNAL_DETECT	R	0h	1=Signal detected on receive

7.2.1.8 PMA_EXTENDED_ABILITIES register

Table 73. PMA_EXTENDED_ABILITIES register - PMA/PMD extended ability register (address Bh)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11	XTD_ABILITIES_BASE_T1	R	1h	1=PMA/PMD has BASE-T1 extended abilities

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Table 73. PMA_EXTENDED_ABILITIES register - PMA/PMD extended ability register (address Bh)*...continued*

Bit	Symbol	Access	Value	Description
10:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.2.1.9 BASE_T1_PMA_XTD_ABILITY register

Table 74. BASE_T1_PMA_XTD_ABILITY register - BASE-T1 PMA/PMD extended ability (address 12h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	ABILITY_1000 BASE_T1	R	1h	1=PMA/PMD is able to perform 1000BASE-T1
0	ABILITY_100BASE_ T1	R	0h	0=PMA/PMD is not able to perform 100BASE-T1

7.2.1.10 BASE_T1_PMA_CONTROL register

Table 75. BASE_T1_PMA_CONTROL register - BASE-T1 PMA/PMD control (address 834h)

Bit	Symbol	Access	Value	Description
15	MANUAL_LEADER_ FOLLOWER_ CONFIG	R	1h	1=LEADER-FOLLOWER manual config enable
14	LEADER_ FOLLOWER	RW	0h	LEADER-FOLLOWER config value. 1=Configure PHY as LEADER 0=Configure PHY as FOLLOWER
13:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	MODE	RW	1h	Type Selection 0001=1000BASE-T1 0000=100BASE-T1

7.2.1.11 E1000BT1_PMA_CONTROL register

Table 76. E1000BT1_PMA_CONTROL register - 1000BASE-T1 PMA control (address 900h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	1=PMA/PMD reset.
14	TRANSMIT_ DISABLE	RW	0h	1=Transmit disable.
13:12	*reserved*	R	0h	Always write 0, ignore on read.
11	LOW_POWER	RW	0h	1=Low-power mode enable
10:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.2.1.12 E1000BT1_PMA_STATUS register

Table 77. E1000BT1_PMA_STATUS register - 1000BASE-T1 PMA status (address 901h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11	OAM_ABILITY	R	1h	1=PHY has 1000BASE-T1 OAM ability
10	EEE_ABILITY	R	0h	0=PHY does not have EEE ability
9	RECEIVE_FAULT_ ABILITY	R	1h	1=PMA/PMD has the ability to detect a fault condition on the receive path
8	LOW_POWER_ ABILITY	R	1h	1=PMA/PMD has low-power ability
7:3	*reserved*	R	0h	Always write 0, ignore on read. \
2	POLARITY_ DETECT	R	0h	1=Receive polarity is reversed
1	RECEIVE_FAULT	R	0h	1=Fault condition detected
0	RCV_LINK_ STATUS_LL	R	0h	1=PMA/PMD receive link up

7.2.1.13 E1000BT1_TRAINING register

Table 78. E1000BT1_TRAINING register - 1000BASE-T1 training (address 902h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10:4	USER_FIELD	RW	0h	7-bit user defined field to send to the link link partner
3:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	OAM_ ADVERTISEMENT	RW	1h	1=1000BASE-T1 OAM ability advertisement to link partner
0	EEE_ ADVERTISEMENT	RW	0h	1=EEE ability advertised to link partner

7.2.1.14 E1000BT1_LP_TRAINING register

Table 79. E1000BT1_LP_TRAINING register - 1000BASE-T1 link partner training (address 903h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10:4	USER_FIELD	R	0h	7-bit user defined field received from the link partner
3:2	*reserved*	R	0h	Always write 0, ignore on read.
1	OAM_ ADVERTISEMENT	R	0h	1=Link partner 1000BASE-T1 OAM capability advertisement
0	EEE_ ADVERTISEMENT	R	0h	0=Link partner does not have EEE ability

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7.2.1.15 E1000BT1_TEST_MODE_CONTROL register

Table 80. E1000BT1_TEST_MODE_CONTROL register - 1000BASE-T1 PMA/PMD test control (address 904h)

Bit	Symbol	Access	Value	Description
15:13	TEST_MODE	RW	0h	1000BASE-T1 test mode control 111=Test mode 7—Normal operation with zero data pattern. This is for BER monitoring. 110=Test mode 6—Transmitter droop test mode. 101=Test mode 5—Normal operation in Idle mode. This is for the PSD Mask test. 100=Test mode 4—Transmit distortion test. 011=Reserved. 010=Test mode 2—Transmit MDI jitter test in LEADER mode. 001=Test mode 1—Setting LEADER and FOLLOWER PHYs for transmit clock jitter test in linked mode. 000=Normal non-test mode operation.
12:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.3 MMD3

7.3.1 Register Description

Table 81. Register overview: MMD3

Address	Name	Access	Reset	Description
0h	PCS_CONTROL1	RW	40h	PCS control 1
1h	PCS_STATUS1	R	2h	PCS status 1 register
4h	PCS_SPEED_ABILITY	R	0h	PCS speed ability
8h	PCS_STATUS2	R	8000h	PCS status 2 register
900h	E1000BT1_PCS_CONTROL	RW	0h	1000BASE-T1 PCS Control.
901h	E1000BT1_PCS_STATUS1	R	0h	1000BASE-T1 PCS status 1 register
902h	E1000BT1_PCS_STATUS2	R	0h	1000BASE-T1 PCS status 2 register
904h	OAM_TRANSMIT	RW	0h	1000BASE-T1 OAM transmit register
905h	OAM_MESSAGE_1_0	RW	0h	1000BASE-T1 OAM message register 0 and 1
906h	OAM_MESSAGE_3_2	RW	0h	1000BASE-T1 OAM message register 0 and 1
907h	OAM_MESSAGE_5_4	RW	0h	1000BASE-T1 OAM message register 0 and 1
908h	OAM_MESSAGE_7_6	RW	0h	1000BASE-T1 OAM message register 0 and 1
909h	OAM_RECEIVED	R	0h	1000BASE-T1 OAM receive register
90Ah	OAM_LP_MESSAGE_1_0	R	0h	Link partner 1000BASE-T1 OAM message register 0 and 1

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Table 81. Register overview: MMD3...*continued*

Address	Name	Access	Reset	Description
90Bh	OAM_LP_MESSAGE_3_2	R	0h	Link partner 1000BASE-T1 OAM message register 0 and 1
90Ch	OAM_LP_MESSAGE_5_4	R	0h	Link partner 1000BASE-T1 OAM message register 0 and 1
90Dh	OAM_LP_MESSAGE_7_6	R	0h	Link partner 1000BASE-T1 OAM message register 0 and 1

7.3.1.1 PCS_CONTROL1 register

Table 82. PCS_CONTROL1 register - PCS control 1 (address 0h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	1=PCS reset.
14	LOOPBACK	RW	0h	1=PCS Loopback Enable
13	SPEED_SELECT_LSB	RW	0h	Speed selection LSB to be used with SPEED_SELECT_MSB
12	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
11	LOW_POWER	RW	0h	1=Low-power mode
10:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6	SPEED_SELECT_MSB	RW	1h	Speed selection{SPEED_SELECT_MSB, SPEED_SELECT_LSB} 11 = BITS_5_2_SEL_SPD 10 = 1000MBPS 01 = 100MBPS 00 = 10MBPS
5:2	SPEED_SELECT_XTD	RW	0h	speed selection 0001= 10PASS-TS/2BASE-TL 0000= 10 Gb/s
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.3.1.2 PCS_STATUS1 register

Table 83. PCS_STATUS1 register - PCS status 1 register (address 1h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7	FAULT_CONDITION	R	0h	1=Fault condition detected
6:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	RCV_LINK_STATUS_LL	R	0h	1=PCS receive link up

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Table 83. PCS_STATUS1 register - PCS status 1 register (address 1h)*...continued*

Bit	Symbol	Access	Value	Description
1	LOW_POWER_ABILITY	R	1h	1=Low-power ability
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.3.1.3 PCS_SPEED_ABILITY register

Table 84. PCS_SPEED_ABILITY register - PCS speed ability (address 4h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7	ABILITY_5G	R	0h	0=PCS is not capable of operating as 5 Gb/s
6	ABILITY_2500M	R	0h	0=PCS is not capable of operating as 2.5 Gb/s
5:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	ABILITY_10G	R	0h	0=PCS is not capable of operating at 10 Gb/s

7.3.1.4 PCS_STATUS2 register

Table 85. PCS_STATUS2 register - PCS status 2 register (address 8h)

Bit	Symbol	Access	Value	Description
15:14	DEVICE_PRESENT	R	2h	Device present information 10=Device responding at this address 00=No device responding at this address 01=No device responding at this address 11=No device responding at this address
13:12	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
11	TRANSMIT_FAULT	R	0h	0=No fault condition on transmit path
10	RECEIVE_FAULT	R	0h	1=Fault condition on the receive path
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.3.1.5 E1000BT1_PCS_CONTROL register

Table 86. E1000BT1_PCS_CONTROL register - 1000BASE-T1 PCS Control. This bit is a copy of 0x0000.15 and setting or clearing either bit sets or clears the other bit. (address 900h).

Bit	Symbol	Access	Value	Description
15	PCS_RESET	RW	0h	1=PCS Reset
14	LOOPBACK	RW	0h	1=Enable loopback mode
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.3.1.6 E1000BT1_PCS_STATUS1 register

Table 87. E1000BT1_PCS_STATUS1 register - 1000BASE-T1 PCS status 1 register (address 901h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7	FAULT_CONDITION	R	0h	1=Fault condition detected
6:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	RCV_LINK_ STATUS_LL	R	0h	1=PCS receive link up
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.3.1.7 E1000BT1_PCS_STATUS2 register

Table 88. E1000BT1_PCS_STATUS2 register - 1000BASE-T1 PCS status 2 register (address 902h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	RCV_LINK_STATUS	R	0h	1=PCS receive link up
9	HIGH_BER	R	0h	1=PCS reporting a high BER
8	BLOCK_LOCK	R	0h	1=PCS locked to received blocks
7	HIGH_BER_LH	R	0h	1=PCS has reported a latched high BER
6	BLOCK_LOCK_LL	R	0h	Latched latch low block lock
5:0	BER_COUNT	R	0h	BER count

7.3.1.8 OAM_TRANSMIT register

Table 89. OAM_TRANSMIT register - 1000BASE-T1 OAM transmit register (address 904h)

Bit	Symbol	Access	Value	Description
15	MESSAGE_VALID	RW	0h	1=Message data in OAM registers are valid
14	TOGGLE	R	0h	Toggle value Toggle value to be transmitted with message. This bit is set by the state machine and cannot be overridden by the user.
13	MESSAGE_ RECEIVED	R	0h	1=1000BASE-T1 OAM message received by link partner
12	RECEIVED_ TOGGLE	R	0h	Received message toggle value Toggle value of message that was received by link partner as indicated in 3.2308.13.
11:8	MESSAGE_ NUMBER	RW	0h	Message number User-defined message number to send
7:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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Table 89. OAM_TRANSMIT register - 1000BASE-T1 OAM transmit register (address 904h)*...continued*

Bit	Symbol	Access	Value	Description
3	PING_RECEIVED	R	0h	Ping received Received PingTx value from latest good 1000BASE-T1 OAM frame received
2	PING_TRANSMIT	RW	0h	Ping transmit Ping value to send to link partner
1:0	SNR_MARGIN	R	0h	Local SNR 00=PHY link is failing and will drop link and relink within 2 ms to 4 ms after the end of the current 1000BASE-T1 OAM frame. 01=LPI refresh is insufficient to maintain PHY SNR. Request link partner to exit LPI and send idles used only when EEE is enabled." 10=PHY SNR is marginal 11=PHY SNR is good.

7.3.1.9 OAM_MESSAGE_1_0 register

Table 90. OAM_MESSAGE_1_0 register - 1000BASE-T1 OAM message register 0 and 1 (address 905h)

Bit	Symbol	Access	Value	Description
15:8	OAM_MESSAGE_1	RW	0h	1000BASE-T1 OAM message 1. Message octet 1. LSB transmitted first.
7:0	OAM_MESSAGE_0	RW	0h	1000BASE-T1 OAM message 0. Message octet 0. LSB transmitted first.

7.3.1.10 OAM_MESSAGE_3_2 register

Table 91. OAM_MESSAGE_3_2 register - 1000BASE-T1 OAM message register 0 and 1 (address 906h)

Bit	Symbol	Access	Value	Description
15:8	OAM_MESSAGE_3	RW	0h	1000BASE-T1 OAM message 3. Message octet 1. LSB transmitted first.
7:0	OAM_MESSAGE_2	RW	0h	1000BASE-T1 OAM message 2. Message octet 0. LSB transmitted first.

7.3.1.11 OAM_MESSAGE_5_4 register

Table 92. OAM_MESSAGE_5_4 register - 1000BASE-T1 OAM message register 0 and 1 (address 907h)

Bit	Symbol	Access	Value	Description
15:8	OAM_MESSAGE_5	RW	0h	1000BASE-T1 OAM message 5. Message octet 1. LSB transmitted first.
7:0	OAM_MESSAGE_4	RW	0h	1000BASE-T1 OAM message 4. Message octet 0. LSB transmitted first.

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7.3.1.12 OAM_MESSAGE_7_6 register

Table 93. OAM_MESSAGE_7_6 register - 1000BASE-T1 OAM message register 0 and 1 (address 908h)

Bit	Symbol	Access	Value	Description
15:8	OAM_MESSAGE_7	RW	0h	1000BASE-T1 OAM message 7. Message octet 1. LSB transmitted first.
7:0	OAM_MESSAGE_6	RW	0h	1000BASE-T1 OAM message 6. Message octet 0. LSB transmitted first.

7.3.1.13 OAM_RECEIVED register

Table 94. OAM_RECEIVED register - 1000BASE-T1 OAM receive register (address 909h)

Bit	Symbol	Access	Value	Description
15	MESSAGE_VALID	R	0h	1=Link partner 1000BASE-T1 OAM message valid
14	TOGGLE	R	0h	Link partner toggle value. Toggle value received with message.
13:12	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
11:8	MESSAGE_ NUMBER	R	0h	Link partner message number Message number from link partner
7:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1:0	SNR_MARGIN	R	0h	Link partner SNR 00=PHY link is failing and will drop link and relink within 2 ms to 4 ms after the end of the current 1000BASE-T1 OAM frame. 01=LPI refresh is insufficient to maintain PHY SNR. Request link partner to exit LPI and send idles used only when EEE is enabled 10=PHY SNR is marginal 11=PHY SNR is good.

7.3.1.14 OAM_LP_MESSAGE_1_0 register

Table 95. OAM_LP_MESSAGE_1_0 register - Link partner 1000BASE-T1 OAM message register 0 and 1 (address 90Ah)

Bit	Symbol	Access	Value	Description
15:8	OAM_LP_ MESSAGE_1	R	0h	LP 1000BASE-T1 OAM message 1. Message octet 1. LSB transmitted first.
7:0	OAM_LP_ MESSAGE_0	R	0h	LP 1000BASE-T1 OAM message 0. Message octet 0. LSB transmitted first.

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7.3.1.15 OAM_LP_MESSAGE_3_2 register

Table 96. OAM_LP_MESSAGE_3_2 register - Link partner 1000BASE-T1 OAM message register 0 and 1 (address 90Bh)

Bit	Symbol	Access	Value	Description
15:8	OAM_LP_ MESSAGE_3	R	0h	LP 1000BASE-T1 OAM message 3. Message octet 3. LSB transmitted first.
7:0	OAM_LP_ MESSAGE_2	R	0h	LP 1000BASE-T1 OAM message 2. Message octet 2. LSB transmitted first.

7.3.1.16 OAM_LP_MESSAGE_5_4 register

Table 97. OAM_LP_MESSAGE_5_4 register - Link partner 1000BASE-T1 OAM message register 0 and 1 (address 90Ch)

Bit	Symbol	Access	Value	Description
15:8	OAM_LP_ MESSAGE_5	R	0h	LP 1000BASE-T1 OAM message 5. Message octet 5. LSB transmitted first.
7:0	OAM_LP_ MESSAGE_4	R	0h	LP 1000BASE-T1 OAM message 4. Message octet 4. LSB transmitted first.

7.3.1.17 OAM_LP_MESSAGE_7_6 register

Table 98. OAM_LP_MESSAGE_7_6 register - Link partner 1000BASE-T1 OAM message register 0 and 1 (address 90Dh)

Bit	Symbol	Access	Value	Description
15:8	OAM_LP_ MESSAGE_7	R	0h	LP 1000BASE-T1 OAM message 7. Message octet 7. LSB transmitted first.
7:0	OAM_LP_ MESSAGE_6	R	0h	LP 1000BASE-T1 OAM message 6. Message octet 6. LSB transmitted first.

7.4 MMD30

7.4.1 Register Description

Table 99. Register overview: MMD30

Address	Name	Access	Reset	Description
4h	DEVICE_IDENTIFIER3	R	2901h	Device version
8h	MMD30_STATUS	R	8000h	MMD30 status register =NXP-specific MMD
40h	DEVICE_CONTROL	RW	0h	Device/sub-system control register
44h	DEVICE_STATUS_ LATCHED	RW	0h	Device/sub-system latched-status
46h	DEVICE_ABILITIES	R	78h	Device/sub-system abilities
47h	DEVICE_PROPERTIES	R	1h	Device/sub-system abilities extended
48h	DEVICE_CONFIG	RW	8000h	Device function enables

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
49h	DEVICE_CONFIG_ EXTENDED	RW	0h	Device/sub-system configuration
70h	DEVICE_IRQ_STATUS	R	0h	DEVICE interrupt status
72h	DEVICE_IRQ_ENABLES	RW	3h	DEVICE interrupt enables
74h	DEVICE_IRQ_MSTATUS	R	0h	DEVICE interrupt masked-status
78h	GLOBAL_IRQ_STATUS	R	0h	Interrupts from GLOBAL functions. Registers have the same structure as the PORT function interrupt group at 8078:807Fh.
7Ah	GLOBAL_IRQ_ENABLES	RW	100Eh	Interrupts from GLOBAL functions. Registers have the same structure as the PORT function interrupt group at 8078:807Fh.
7Ch	GLOBAL_IRQ_MSTATUS	R	0h	GLOBAL function interrupt masked-status
80h	GLOBAL_CAT_IRQ_ STATUS	R	0h	Global categories interrupt status
81h	GLOBAL_CAT_IRQ_ ENABLES	RW	104h	Global categories interrupt enables
82h	GLOBAL_CAT_IRQ_ MSTATUS	R	0h	Global categories interrupt masked-status
90h	OP_CONDITIONS_IRQ_ SOURCES	RW	0h	Global operating conditions interrupt status
91h	OP_CONDITIONS_IRQ_ ENABLE	RW	0h	Global operating conditions interrupt enables
92h	OP_CONDITIONS_IRQ_ MSTATUS	R	0h	Global operating conditions interrupt masked-status
A0h	TOP_EPHY_IRQ_SOURCES	RW	0h	EPHY-support interrupt sources
A1h	TOP_EPHY_IRQ_ENABLES	RW	0h	EPHY-support interrupt enables
A2h	TOP_EPHY_IRQ_MSTATUS	R	0h	EPHY-support interrupt masked-status
C0h	CONFIG_REGXS_IRQ_ SOURCES	RW	0h	Global configuration interrupt sources
C1h	CONFIG_REGXS_IRQ_ ENABLE	RW	0h	Global config interrupt enables
C2h	CONFIG_REGXS_IRQ_ MSTATUS	R	0h	Global config interrupt masked-status
100h	BROADCAST	RW	0h	Broadcast configuration
101h	PORT1_CONFIG	RW	C000h	PortNr 1 global configuration
180h	WISE_CONTROL	RW	0h	Wake-Inhibit-Sleep-Enable control
181h	WISE_STATUS	RW	0h	Wake-Inhibit-Sleep-Enable status
182h	WISE_CONFIG	RW	3000h	Wake-Inhibit-Sleep-Enable configuration
184h	WISE_PARAMETERS	RW	1h	Wake-Inhibit-Sleep-Enable parameters
189h	SMI_STATUS	RW	0h	Serial Management Interface status
18Ah	SMI_CONFIG	RW	8000h	Serial Management Interface configuration

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
18Ch	SMI_DATA_IO_CONFIG	RW	0h	MDIO pin configuration
190h	GLOBAL_LED_TRIGGER0	RW	0h	LED trigger 0 configuration
191h	GLOBAL_LED_TRIGGER1	RW	0h	LED trigger 1 configuration
192h	GLOBAL_LED_TRIGGER2	RW	0h	LED trigger 2 configuration
193h	GLOBAL_LED_TRIGGER3	RW	0h	LED trigger 3 configuration
1A0h	LED0_CONFIG	RW	C00Ch	LED0 control
1A1h	LED0_TRIG_SOURCE	RW	21h	LED0 trigger source select
1A2h	LED0_TRIG01_CONFIG	RW	E0E0h	LED0 configuration of triggers 0-1
1A3h	LED0_TRIG2_CONFIG	RW	E0h	LED0 configuration of triggers 2-3
1A4h	LED1_CONFIG	RW	0h	LED1 control
1A5h	LED1_TRIG_SOURCE	RW	0h	LED1 trigger source select
1A6h	LED1_TRIG01_CONFIG	RW	0h	LED1 configuration of triggers 0-1
1A7h	LED1_TRIG2_CONFIG	RW	0h	LED1 configuration of triggers 2-3
1A8h	LED2_CONFIG	RW	0h	LED2 control
1A9h	LED2_TRIG_SOURCE	RW	0h	LED2 trigger source select
1AAh	LED2_TRIG01_CONFIG	RW	0h	LED2 configuration of triggers 0-1
1ABh	LED2_TRIG2_CONFIG	RW	0h	LED2 configuration of triggers 2-3
1ACh	LED3_CONFIG	RW	0h	LED3 control
1ADh	LED3_TRIG_SOURCE	RW	0h	LED3 trigger source select
1AEh	LED3_TRIG01_CONFIG	RW	0h	LED3 configuration of triggers 0-1
1AFh	LED3_TRIG2_CONFIG	RW	0h	LED3 configuration of triggers 2-3
311h	AO_SYSTEM_SUPPLY_STATUS	RW	0h	Always-on supply status
313h	CORE_SUPPLY_STATUS	RW	0h	Core supply status
315h	VDDIO_SUPPLY_STATUS	RW	0h	VDDIO supply status
319h	VREGA_SUPPLY_STATUS	RW	0h	Reference supply status
31Ah	VDDA_RX_SUPPLY	RW	0h	VDDA_RX supply status
31Eh	OSC_STATUS	RW	0h	Reference clock status
31Fh	TEMP_STATUS	RW	A1h	Temperature status
3F0h	COMPL_TEST_CONTROL	RW	0h	EPHY compliance test control
1000h	GPTP_CONTROL	RW	0h	General control
1006h	GPTP_CONFIG	RW	4h	General configuration
1008h	GPTP_IRQ_SOURCES	RW	0h	Interrupt status signals
100Ah	GPTP_IRQ_ENABLES	RW	0h	Interrupt enable control
100Ch	GPTP_IRQ_MSTATUS	R	0h	Interrupt status signals
1010h	SYNC_TRIG_FILTER	RW	0h	Edge selection for PTP stnc & PTP trigger timestamping.

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
1012h	REGISTER_TS_CONTROL	RW	4h	Timestamp format control for TS_DATA_ 1 register. This is common for both PTP_ SYNC & PTP_TRIG timestamp.
1020h	PTP_CLK_NSEC	RW	8h	Clock period of ptp_clk. Local LTC increment it's counters by this value for every ptp clock ticks. This register is write protected with config_enable.
1022h	PTP_SYNC_PERIOD_0	RW	CA00h	Period of external PPS sync in nanosecond. Used for HW LTC self locking. This register is write protected with config_enable.
1023h	PTP_SYNC_PERIOD_1	RW	3B9Ah	Period of external PPS sync in nanosecond. Used for HW LTC self locking. This register is write protected with config_enable.
1030h	RATE_ADJUST_SUBNS_0	RW	0h	SW Rate adjust value
1031h	RATE_ADJUST_SUBNS_1	RW	0h	SW Rate adjust value
1032h	TMP_ADJUST_CYCLES_0	RW	0h	Temporary adjustment window
1033h	TMP_ADJUST_CYCLES_1	RW	0h	Temporary adjustment window
1034h	TMP_ADJUST_SUBNS_0	RW	0h	SW temporary rate adjust value
1035h	TMP_ADJUST_SUBNS_1	RW	0h	SW temporary rate adjust value
1036h	LTC_ERROR_LIMIT_0	RW	D090h	Error limitation by HW LTC self locking engine
1037h	LTC_ERROR_LIMIT_1	RW	3h	Error limitation by HW LTC self locking engine
1038h	LTC_LOOP_CONTROL	RW	303h	This register defines control parameters to adapt behavior of the LTC loop if LTC_ AUTO_LOCK=1. This register is write protected with config_enable.
1040h	LTC_LOAD_DATA_0	RW	0h	Loading nanoseconds counter value
1041h	LTC_LOAD_DATA_1	RW	0h	Loading nanoseconds counter value
1042h	LTC_LOAD_DATA_2	RW	0h	Loading seconds counter value
1043h	LTC_LOAD_DATA_3	RW	0h	Loading seconds counter value
1048h	LTC_READ_DATA_0	R	0h	Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set.
1049h	LTC_READ_DATA_1	R	0h	Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set.
104Ah	LTC_READ_DATA_2	R	0h	Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set.
104Bh	LTC_READ_DATA_3	R	0h	Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set.

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
1060h	PTP_SYNC_TS_DATA_S	R	0h	When reading of the first register of this time-stamp PTP_SYNC_TS_DATA_S, the next available PTP_SYNC time-stamp will be latched and locked into the PTP_SYNC_ TS registers, if the previous time-stamp was unlocked. This allows to read a consistent value set from these registers. Any of the PTP_SYNC_TS registers can be read multiple times and in any order. The time-stamp is unlocked by any write transaction to any PTP_SYNC_TS register from DATA_S to END.
1061h	PTP_SYNC_TS_DATA_0	R	0h	The time-stamp is unlocked by any write transaction to any PTP_SYNC_TS register from DATA_S to END.
1062h	PTP_SYNC_TS_DATA_1	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
1063h	PTP_SYNC_TS_DATA_2	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
1064h	PTP_SYNC_TS_DATA_3	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
1065h	PTP_SYNC_TS_END	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock.
1070h	PTP_TRIG_TS_DATA_S	R	0h	When reading of the first register of this time-stamp PTP_TRIG_TS_DATA_S, the next available PTP_TRIG time-stamp will be latched and locked into the PTP_TRIG_ TS registers, if the previous time-stamp was unlocked. This allows to read a consistent value set from these registers. Any of the PTP_TRIG_TS registers can be read multiple times and in any order. The time-stamp is unlocked by any write transaction to any PTP_TRIG_TS register from DATA_S to END.
1071h	PTP_TRIG_TS_DATA_0	R	0h	The time-stamp is unlocked by any write transaction to any PTP_TRIG_TS register from DATA_S to END.
1072h	PTP_TRIG_TS_DATA_1	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
1073h	PTP_TRIG_TS_DATA_2	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
1074h	PTP_TRIG_TS_DATA_3	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
1075h	PTP_TRIG_TS_END	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock.
10FFh	GPTP_IDENTIFIER	R	200h	Identifier which indicates the integrated version of the GLOBAL PTP-functionality that is shared for all PORTs like LTC, SYNC, and TRIGGERs
1100h	LOOP_OTP_CONTROL	RW	0h	3 bit OTP bits limits the programmed value of integral gain.
1101h	PTP_MISC_CONTROL	RW	0h	3 bit OTP bits limits the programmed value of integral gain.
1202h	GLOBAL_FUSA_STATUS	R	0h	Global FUSA status
2C00h	GLOBAL_INFRA_CONTROL	RW	0h	Infrastructure control
2C06h	GLOBAL_INFRA_CONFIG	RW	0h	Infrastructure configuration
2C08h	GLOBAL_INFRA_IRQ_SOURCES	RW	0h	Infrastructure interrupt sources
2C0Ah	GLOBAL_INFRA_IRQ_ENABLES	RW	2h	Infrastructure interrupt enables
2C0Ch	GLOBAL_INFRA_IRQ_MSTATUS	R	0h	Infrastructure interrupt masked-status
2C20h	GPIO_OUTPUTS	RW	0h	GPIO outputs
2C21h	GPIO_STATUS	R	0h	GPIO status
2C40h	GPIO0_FUNC_CONFIG	RW	8020h	GPIO0 functional configuration register
2C41h	GPIO1_FUNC_CONFIG	RW	8000h	GPIO1 functional configuration register
2C42h	GPIO2_FUNC_CONFIG	RW	8000h	GPIO2 functional configuration register
2C43h	GPIO3_FUNC_CONFIG	RW	8000h	GPIO3 functional configuration register
2C44h	GPIO4_FUNC_CONFIG	RW	8000h	GPIO4 functional configuration register
2C45h	GPIO5_FUNC_CONFIG	RW	8000h	GPIO5 functional configuration register
2C46h	GPIO6_FUNC_CONFIG	RW	8000h	GPIO6 functional configuration register
2C47h	GPIO7_FUNC_CONFIG	RW	8000h	GPIO7 functional configuration register
2C48h	GPIO8_FUNC_CONFIG	RW	8000h	GPIO8 functional configuration register
2C49h	GPIO9_FUNC_CONFIG	RW	8000h	GPIO9 functional configuration register
2C80h	GPIO0_IO_CONFIG	RW	0h	IO-cell configuration for GPIO0
2C81h	GPIO1_IO_CONFIG	RW	0h	IO-cell configuration for GPIO1
2C82h	GPIO2_IO_CONFIG	RW	0h	IO-cell configuration for GPIO2
2C83h	GPIO3_IO_CONFIG	RW	0h	IO-cell configuration for GPIO3

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
2C84h	GPIO4_IO_CONFIG	RW	0h	IO-cell configuration for GPIO4
2C85h	GPIO5_IO_CONFIG	RW	0h	IO-cell configuration for GPIO5
2C86h	GPIO6_IO_CONFIG	RW	0h	IO-cell configuration for GPIO6
2C87h	GPIO7_IO_CONFIG	RW	0h	IO-cell configuration for GPIO7
2C88h	GPIO8_IO_CONFIG	RW	0h	IO-cell configuration for GPIO8
2C89h	GPIO9_IO_CONFIG	RW	0h	IO-cell configuration for GPIO9
2F06h	REFCLK_CONFIG	RW	0h	Auxiliary function configuration
9000h	PTP_CONTROL	RW	0h	General control
9004h	PTP_ABILITIES	R	Ch	PTP IP Ability registers
9008h	PTP_IRQ_SOURCES	RW	0h	Interrupt status signals
900Ah	PTP_IRQ_ENABLES	RW	0h	Interrupt enable control
900Ch	PTP_IRQ_MSTATUS	R	0h	Interrupt status signals
9010h	PTP_MESSAGE_FILTER	RW	Dh	PTP event message consideration. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9011h	PTP_FILTER_ENABLES	RW	3h	Ingress or egress packet DA filtering. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9012h	INGRESS_TS_INSERT_CONTROL	RW	4h	Inband timestamp insertion controls at ingress path
9020h	TX_DELAY_NSEC	RW	0h	Egress data path latency from timestamping point of SOF signal to mdi nanoseconds part. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9021h	TX_DELAY_XTD	RW	0h	Egress data path latency from timestamping point of SOF signal to mdi sub-nanoseconds part. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9022h	RX_DELAY_NSEC	RW	0h	Ingress data path latency of SOF signal from MDI line to timestamping point nanoseconds part. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9023h	RX_DELAY_XTD	RW	0h	Ingress data path latency of SOF signal from MDI line to timestamping point sub-nanoseconds part. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9030h	USER_MAC_DA_0	RW	0h	[15:0] of User defined MAC address. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9031h	USER_MAC_DA_1	RW	0h	[31:16] of User defined MAC address. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
9032h	USER_MAC_DA_2	RW	0h	[47:32] of User defined MAC address. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9033h	USER_MAC_DA_MASK_0	RW	0h	User MAC DA Mask control. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9034h	USER_MAC_DA_MASK_1	RW	0h	User MAC DA Mask control. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9035h	USER_MAC_DA_MASK_2	RW	0h	User MAC DA Mask control. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9050h	FILTER_PARAMETERS	RW	0h	Ingress or egress packet filtering parameters. This register is write protected with GPTP_CONTROL.CONFIG_ENABLE.
9060h	EGRESS_TS_DATA_S	R	0h	When reading any of the first two registers of this time-stamp EGRESS_TS_DATA_S or _HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the EGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values.. Any of the EGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any EGRESS_TS register from _DATA_S to _END.
9061h	EGRESS_TS_HEAD_0	R	0h	When reading any of the first two registers of this time-stamp EGRESS_TS_DATA_S or _HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the EGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values.. Any of the EGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any EGRESS_TS register from _DATA_S to _END.
9062h	EGRESS_TS_HEAD_1	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9063h	EGRESS_TS_DATA_0	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9064h	EGRESS_TS_DATA_1	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
9065h	EGRESS_TS_DATA_2	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9066h	EGRESS_TS_DATA_3	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9067h	EGRESS_TS_END	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock.
9070h	INGRESS_TS_DATA_S	R	0h	When reading any of the first two registers of this time-stamp INGRESS_TS_DATA_S or INGRESS_TS_DATA_HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the INGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values.. Any of the INGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any INGRESS_TS register from INGRESS_TS_DATA_S to INGRESS_TS_DATA_END.
9071h	INGRESS_TS_HEAD_0	R	0h	When reading any of the first two registers of this time-stamp INGRESS_TS_DATA_S or INGRESS_TS_DATA_HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the INGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values. Any of the INGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any INGRESS_TS register from INGRESS_TS_DATA_S to _END.
9072h	INGRESS_TS_HEAD_1	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9073h	INGRESS_TS_DATA_0	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9074h	INGRESS_TS_DATA_1	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9075h	INGRESS_TS_DATA_2	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
9076h	INGRESS_TS_DATA_3	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked.
9077h	INGRESS_TS_END	R	0h	Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock.
90FFh	PORT_PTP_IDENTIFIER	R	200h	Identifier which indicates the integrated version of the PORT-specific PTP-functionality like INGRESS and EGRESS time-stamping.
9100h	PTP_DEBUG_CTRL	RW	0h	General ring buffer control
9101h	RING_BUF_0	R	0h	Egress/Ingress ring buffer content for debug purpose to check if FIFO is malfunctioning.
9102h	RING_BUF_1	R	0h	Egress/Ingress ring buffer content for debug purpose to check if FIFO is malfunctioning.
9103h	PTP_PKT_RCV_CNT	R	0h	Number of received ptp event packet
9104h	PTP_FCS_ERR_CNT	R	0h	Number of received ptp event packet has fcs error.
9105h	PTP_MISC_STATUS	RW	0h	Number of received ptp event packet has fcs error.
9202h	PORT_FUSA_STATUS	R	0h	Port FUSA status
9208h	PORT_FUSA_IRQ_SOURCES	R	0h	Port FUSA interrupt sources
920Ah	PORT_FUSA_IRQ_ENABLES	RW	3004h	Port FUSA interrupt enables
920Ch	PORT_FUSA_IRQ_MSTATUS	R	0h	Port FUSA interrupt masked-status
9224h	EPHY_FUSA_IRQ_SOURCES	RW	0h	Functional safety interrupt sources for EPHY
9225h	EPHY_FUSA_IRQ_ENABLES	RW	0h	Functional safety interrupt enables for EPHY
9226h	EPHY_FUSA_IRQ_MSTATUS	R	0h	Functional safety interrupt masked-status for EPHY
9228h	EPHY_LATENT_FAULT	RW	0h	EPHY status latent-fault
922Ah	EPHY_TCLK_STATUS	RW	0h	EPHY transmit clock status
92C4h	INFRA_FUSA_IRQ_SOURCES	RW	0h	Functional safety interrupt sources for INFRAstructure
92C5h	INFRA_FUSA_IRQ_ENABLES	RW	0h	Functional safety interrupt enables for INFRAstructure
92C6h	INFRA_FUSA_IRQ_MSTATUS	R	0h	Functional safety interrupt masked-status for INFRAstructure
92CAh	TXC_STATUS	RW	0h	MII transmit clock status

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
92D4h	SGMII_FUSA_IRQ_SOURCES	RW	0h	Functional safety interrupt sources for Serial-MII SGMII
92D5h	SGMII_FUSA_IRQ_ENABLES	RW	0h	Functional safety interrupt enables for Serial-MII SGMI
92D6h	SGMII_FUSA_IRQ_MSTATUS	R	0h	Functional safety interrupt masked-status for SGMII
92D8h	SGMII_STATUS_FAULT	RW	0h	SGMII status latent-faults
92DAh	SGMII_TCLK_STATUS	RW	0h	SGMII transmit clock status
A800h	PORT_BIST_CONTROL	RW	0h	Port BIST control
A806h	PORT_BIST_CONFIG	RW	0h	Port BIST configuration
A807h	BIST_INTERCEPT_CONFIG	RW	0h	BIST intercept configuration
A880h	BIST_GEN_CTRL	RW	0h	BSIT generator control register
A881h	BIST_GEN_STATUS	R	0h	BIST generator status
A887h	PREAMBL_IPG_SIZE	RW	700Ch	Preamble and IPG lengths
A888h	BIST_DA_0	RW	0h	lower 16 bits of DA value[15:0]
A889h	BIST_DA_1	RW	0h	middle 16 bits of DA value[31:16]
A88Ah	BIST_DA_2	RW	0h	higher 16 bits of DA value[47:32]
A88Bh	BIST_SA_0	RW	0h	lower 16 bits of SA value[15:0]
A88Ch	BIST_SA_1	RW	0h	middle 16 bits of SA value[31:16]
A88Dh	BIST_SA_2	RW	0h	higher 16 bits of SA value[47:32]
A88Fh	BIST_PTP_CONFIG	RW	0h	PTP frame generation configuration
A890h	BIST_ETHER_TYPE	RW	0h	Ethernet type T/L
A891h	PAYLOAD_CONFIG	RW	20A5h	Payload data controls
A892h	PAYLOAD_SIZE	RW	2Eh	Length of frame payload data in bytes
A893h	PRBS_DATA_CONFIG	RW	0h	PRBS data configuration
A894h	BIST_LFSR_SEED	RW	7FFFh	LFSR Seed value
A8A2h	GOOD_FRAMES_PLAN	RW	64h	Number of good frames to be generated by BIST generator
A8A4h	G_GOOD_FRAME_CNT	R	0h	Generated good frames in continuous mode.
A8A6h	BAD_FRAMES_PLAN	RW	0h	Number of bad frames to be generated by BIST generator
A8C0h	BIST_CHECK_CTRL	RW	0h	BIST checker control register
A8C1h	BIST_PROD_STATUS	R	0h	BIST status used when BIST is in production mode and NOT valid for continuous mode
A8C4h	BIST_WAIT_TIMER	RW	FFh	Timer timeout value for checking all status in production mode. This timeout is useful when expected number of generated frames is never received.
A8D0h	R_GOOD_FRAME_CNT	R	0h	Received good frame count

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
A8D2h	R_BAD_FRAME_CNT	R	0h	Received bad frame count
A8D4h	R_RXER_FRAME_CNT	R	0h	RX_ER detected when rx_dv is set.
AC00h	PORT_INFRA_CONTROL	RW	0h	Infrastructure control
AC04h	INFRA_ABILITY	R	8000h	Infrastructure abilities
AC06h	INFRA_CONFIG	RW	0h	Infrastructure configuration
AC08h	INFRA_IRQ_SOURCES	RW	0h	Infrastructure interrupt sources
AC0Ah	INFRA_IRQ_ENABLES	RW	0h	Infrastructure interrupt enables
AC0Ch	INFRA_IRQ_MSTATUS	R	0h	Infrastructure interrupt masked-status
AC80h	MONITOR_CONTROL	RW	0h	Monitor control
AC82h	MONITOR_STATUS	RW	0h	Monitor status
AC86h	MONITOR_CONFIG	RW	0h	Monitor configuration
AC90h	TX_IPG_LENGTH	RW	FFh	Egress Inter-Packet Gap Length
AC91h	RX_IPG_LENGTH	RW	FFh	Ingress Inter-Packet Gap Length
AC92h	TX_PREMBEL_COUNT	RW	Fh	Egress Preamble Length
AC93h	RX_PREMBEL_COUNT	RW	Fh	Ingress Preamble Length
ACA0h	TX_FRAME_COUNT	R	0h	Egress frame count
ACA1h	TX_FRAME_COUNT_XTD	R	0h	Egress frame count extended
ACA2h	RX_FRAME_COUNT	R	0h	Ingress frame count
ACA3h	RX_FRAME_COUNT_XTD	R	0h	Ingress frame count extended
ACA4h	TX_LOST_FRAMES	R	0h	Egress lost frames
ACA5h	TX_LOST_FRAMES_XTD	R	0h	Egress lost frames extended
ACA6h	RX_LOST_FRAMES	R	0h	Ingress lost frames
ACA7h	RX_LOST_FRAMES_XTD	R	0h	Ingress lost frames extended
AD10h	PORT_MIN_TX_DELAY	R	0h	Minimum transmit path delay
AD12h	PORT_MAX_TX_DELAY	R	0h	Maximum transmit path delay
AD14h	PORT_MIN_RX_DELAY	R	0h	Minimum receive path delay
AD16h	PORT_MAX_RX_DELAY	R	0h	Maximum receive path delay
AFC0h	XMII_CONTROL	RW	0h	XMII control
AFC2h	XMII_STATUS	RW	0h	XMII status
AFC4h	XMII_ABILITIES	R	A: c000h B: 1h	XMII abilities
AFC5h	SERDES_MII_ABILITIES	R	A: 0h B: 2h	SERDES MII abilities
AFC6h	MII_BASIC_CONFIG	RW	A: 7h B: 9h	XMII basic configuration
AFC8h	XMII_CLK_CONFIG	RW	0h	XMII Clock Configuration
AFC9h	XMII_CLK_IO_CONFIG	RW	0h	XMII Clock IO configuration

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Table 99. Register overview: MMD30*...continued*

Address	Name	Access	Reset	Description
AFCAh	XMII_DATA_CONFIG	RW	0h	XMII DATA configuration
AFCBh	XMII_DATA_IO_CONFIG	RW	0h	XMII DATA IO configuration
AFCCh	RGMII_TXC_DELAY_CONFIG	RW	12h	RGMII TXC delay configuration
AFCDh	RGMII_RXC_DELAY_CONFIG	RW	12h	RGMII RXC delay configuration
B000h	SGMII_BASIC_CONTROL	R/W	1140h[1] 1040h[2]	SGMII PCS register: SR MII MMD Control Register.
B001h	SGMII_BASIC_STATUS	R/W	0109h	SGMII PCS register: SR MII MMD Status Register.
B002h	SGMII_SR_MII_DEV_ID1	R	0h	SGMII DEVICE ID1 register
B003h	SGMII_SR_MII_DEV_ID1	R	0h	SGMII DEVICE ID2 register
B004h	SGMII_AUTONEG_ADVERTISE	R/W	20h	SGMII PCS register: SR MII MMD AN Advertisement Register.
B005h	SGMII_AUTONEG_LP_ABILITY	R/W	0h	SGMII PCS register: SR MII MMD AN Link Partner Base Ability Register.
B006h	SGMII_AUTONEG_EXPANSION	R/W	0h	SGMII PCS register: SR MII MMD AN Expansion Register.
B00Fh	SGMII_EXTENDED_STATUS	R/W	C000h	SGMII PCS register: SR MII MMD Extended Status Register.
B010h	SGMII_ADVANCED_CONTROL	R/W	0h	Advanced control
B011h	SGMII_ADVANCED_STATUS	R/W	0h	Advanced status
B013h	SGMII_ADVANCED_CONFIG	R/W	0h[1] 1h[3]	Advanced configuration
B014h	SGMII_IRQ_SOURCES	R/W	0h	SGMII interrupt source
B015h	SGMII_IRQ_ENABLES	R/W	0h	SGMII interrupt enables
B016h	SGMII_IRQ_MSTATUS	R/W	0h	SGMII interrupt mask status. This is AND of IRQ_SOURCE and IRQ_ENABLE.
B040h	SGMII_LED_TRIGGER0	R/W	0h	SGMII LED trigger 0
B041h	SGMII_LED_TRIGGER1	R/W	0h	see SGMII_LED_TRIGGER0
B042h	SGMII_LED_TRIGGER2	R/W	0h	see SGMII_LED_TRIGGER0
B043h	SGMII_LED_TRIGGER3	R/W	0h	see SGMII_LED_TRIGGER0
B070h	SGMII_PCS_CONTROL	R/W	2000h[1] 0201h[2]	see SGMII_PCS_CONTROL
B073h	SGMII_PCS_CONFIG1	R/W	0h[1] 10h[2]	see SGMII_PCS_CONFIG1
B074h	SGMII_PCS_CONFIG2	R/W	0h[1] 10Dh[2]	see SGMII_PCS_CONFIG2
B075h	SGMII_AUTONEG_STATUS	R/W	0h	see SGMII_AUTONEG_STATUS

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[1] Default value in managed operation
[2] Default value in autonomous operation and in managed operation once SGMII_ADVANCED_CONTROL.START_OPERATION is set.
[3] Default value in autonomous operation

7.4.1.1 DEVICE_IDENTIFIER3 register

Table 100. DEVICE_IDENTIFIER3 register - Device version (address 4h)

Bit	Symbol	Access	Value	Description
15:12	SILICON_VERSION	R	1h	Silicon update counter for prototype silicon spins and for design changes that allow a drop-in replacement by the customer.
11:8	SAMPLE_TYPE	R	9h	#9 = production
7:0	NVM_VERSION	R	1h	NVM version number

7.4.1.2 MMD30_STATUS register

Table 101. MMD30_STATUS register - MMD30 status register =NXP-specific MMD (address 8h)

Bit	Symbol	Access	Value	Description
15:14	DEVICE_PRESENT	R	2h	15 14 1 0 = Device responding at this address 1 1 = No device responding at this address 0 1 = No device responding at this address 0 0 = No device responding at this address
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.3 DEVICE_CONTROL register

Table 102. DEVICE_CONTROL register - Device/sub-system control register (address 40h)

Bit	Symbol	Access	Value	Description
15	DEVICE_RESET	RW	0h	Write: 1 = PHY reset will be asserted 0 = no action Read: 1 = Phy reset in progress. Other register's value not stable or not valid 0 = Normal operation Note: Normal Operation defination : completed OTP & PS configuration done and accessible to all registersEnabled IPs accessible to all registers indication is available in LIMITED_REG_ACCESS801F.1CL22 register
14	GLOBAL_CONFIG_ENABLE	RW	0h	1 = Configuration registers in the Global address space can be changed 0 = Configuration registers in the Global address space are write protected 0048h DEVICE_CONFIG 0049h DEVICE_CONFIG_EXTENDED 0182h WISE_CONFIG 0184h WISE_PARAMETERS

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Table 102. DEVICE_CONTROL register - Device/sub-system control register (address 40h)*...continued*

Bit	Symbol	Access	Value	Description
				018Ah SMI_CONFIG 018Ch SMI_DATA_IO_CONFIG 03F0h COMPL_TEST_CONTROL
13	SUPER_CONFIG_ ENABLE	RW	0h	1 = All configuration registers and bits can be changed for any PHY and the Global sub-system 0 = Configuration bits of PHYs and Global are protected by individual config_enable bits of the modules
12:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	GOTO_STANDBY	RW	0h	1 = Directs the DEV to standby state for reconfiguration 0 = DEV continues to operate normally
0	START_ OPERATION	RW	0h	1 = Exit standby state and start DEV operation 0 = No effect

7.4.1.4 DEVICE_STATUS_LATCHED register

Table 103. DEVICE_STATUS_LATCHED register - Device/sub-system latched-status (address 44h)

Bit	Symbol	Access	Value	Description
15	PIN_RESET	RW	0h	1 = Device reset by RST_N pin assertion 0 = No reset by RST_N since last clear Write one to clear.
14	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
13	AO_POR_EVENT	RW	0h	1 = Device got reset due to an always-on system supply POR event 0 = No device reset due to an always-on system supply POR event, since last clear Write one to clear.
12:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.5 DEVICE_ABILITIES register

Table 104. DEVICE_ABILITIES register - Device/sub-system abilities (address 46h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11	BIST_ABILITY	R	0h	1 = This device has Global BIST functionality 0 = This device lacks Global BISTfunctionality
10:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6	MACSEC_ADPTR_ ABILITY	R	1h	1 = MACsec adapter functionality is enabled 0 = MACsec adapter functionality is disabled
5	MACSEC_ABILITY	R	1h	1 = MACsec functionality is enabled

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Table 104. DEVICE_ABILITIES register - Device/sub-system abilities (address 46h)*...continued*

Bit	Symbol	Access	Value	Description
				0 = MACsec functionality is disabled
4	FUSA_ABILITY	R	1h	1 = Device supports Functional Safety
3	PTP_ABILITY	R	1h	1 = PTP functionality is enabled 0 = PTP functionality is disabled
2:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.6 DEVICE_PROPERTIES register

Table 105. DEVICE_PROPERTIES register - Device/sub-system abilities extended (address 47h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	NUMBER_OF_PORTS	R	1h	Number of ports in the PHY sub-system

7.4.1.7 DEVICE_CONFIG register

Table 106. DEVICE_CONFIG register - Device function enables (address 48h)

Bit	Symbol	Access	Value	Description
15	DEVICE_ENABLE	RW	1h	1 = Global Device Subsystem Functionality is Enabled 0 = Global Device Subsystem Functionality is disabled
14:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	PTP_ENABLE	RW	0h	1 = PTP functionality is enabled 0 = PTP functionality is disabled
2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	SUPER_TEST_ENABLE	RW	0h	1 = All test features of PORT and GLOBAL functions can be used. 0 = Specific test mode enable & configuration fields of PORTs and GLOBAL are protected by individual test_enable bits inside PORTs and GLOBAL.
0	TEST_ENABLE	RW	0h	0 = Selecting a test mode of a GLOBAL function does not have effect. 1 = Enabled to bring GLOBAL functions into test mode.

7.4.1.8 DEVICE_CONFIG_EXTENDED register

Table 107. DEVICE_CONFIG_EXTENDED register - Device/sub-system configuration (address 49h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	OP_IRQ_RECOVERY	RW	0h	0 = Operating condition interrupts only set on rising edge of trigger signal =assert event

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Table 107. DEVICE_CONFIG_EXTENDED register - Device/sub-system configuration (address 49h)*...continued*

Bit	Symbol	Access	Value	Description
				1 = Operating condition interrupts get asserted when their trigger signal gets either asserted or de-asserted =change event
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.9 DEVICE_IRQ_STATUS register

Table 108. DEVICE_IRQ_STATUS register - DEVICE interrupt status (address 70h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	PORT_IRQ	R	0h	1: PortNr1 has its interrupt line active 0: PortNr1 has no active interrupt
0	GLOBAL_IRQ	R	0h	1: Global has its interrupt line active 0: Global has no active interrupt

7.4.1.10 DEVICE_IRQ_ENABLES register

Table 109. DEVICE_IRQ_ENABLES register - DEVICE interrupt enables (address 72h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	PORT_IRQ	RW	1h	Enable for interrupt from PortNr 1. 1 = Enable, 0=Disable
0	GLOBAL_IRQ	RW	1h	Enable for interrupt from GLOBAL functions 1 = Enable, 0=Disable

7.4.1.11 DEVICE_IRQ_MSTATUS register

Table 110. DEVICE_IRQ_MSTATUS register - if INT_N is set, then this register reads Zero's then read 0x801F (address 74h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	PORT_IRQ	R	0h	Masked-status of PortNr 1 interrupt
0	GLOBAL_IRQ	R	0h	Masked-status of GLOBAL interrupt

7.4.1.12 GLOBAL_IRQ_STATUS register

Table 111. GLOBAL_IRQ_STATUS register - Interrupts from GLOBAL functions. Registers have the same structure as the PORT function interrupt group at 8078:807Fh. (address 78h)

Bit	Symbol	Access	Value	Description
12	GLOBAL_INFRA_IRQ	R	0h	Status bit indicating an interrupt from device infrastructure functionality.

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Table 111. GLOBAL_IRQ_STATUS register - Interrupts from GLOBAL functions. Registers have the same structure as the PORT function interrupt group at 8078:807Fh. (address 78h)*...continued*

Bit	Symbol	Access	Value	Description
11:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	GLOBAL_PTP_IRQ	R	0h	Status bit indicating an interrupt from shared top-level PTP-functionality.
2	GLOBAL_EPHY_IRQ	R	0h	Status bit indicating an interrupt from shared top-level ePHY-related functions.
1	GLOBAL_CAT_IRQ	R	0h	Status bit indicating a device-level categorized interrupt
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.13 GLOBAL_IRQ_ENABLES register

Table 112. GLOBAL_IRQ_ENABLES register - Interrupts from GLOBAL functions. Registers have the same structure as the PORT function interrupt group at 8078:807Fh. (address 7Ah)

Bit	Symbol	Access	Value	Description
12	GLOBAL_INFRA_IRQ	RW	1h	Enable for infrastructure interrupt
11:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	GLOBAL_PTP_IRQ	RW	1h	Enable for PTP interrupt
2	GLOBAL_EPHY_IRQ	RW	1h	Enable for ePHY-related interrupt
1	GLOBAL_CAT_IRQ	RW	1h	Enable for GLOBAL categorized interrupt
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.14 GLOBAL_IRQ_MSTATUS register

Table 113. GLOBAL_IRQ_MSTATUS register - GLOBAL function interrupt masked-status (address 7Ch)

Bit	Symbol	Access	Value	Description
12	GLOBAL_INFRA_IRQ	R	0h	Masked-status of infrastructure interrupt
11:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	GLOBAL_PTP_IRQ	R	0h	Masked-status of PTP interrupt
2	GLOBAL_EPHY_IRQ	R	0h	Masked-status of ePHY-related interrupt
1	GLOBAL_CAT_IRQ	R	0h	Masked-status of global categorized interrupt
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.15 GLOBAL_CAT_IRQ_STATUS register

Table 114. GLOBAL_CAT_IRQ_STATUS register - Global categories interrupt status (address 80h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	CONFIG_REGXS_IRQ	R	0h	1 = Active interrupts in the global 'Config & Register Access' interrupts source register 0 = No active interrupts in the global 'Config & Register Access' interrupts source register
7:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	OP_CONDITIONS_IRQ	R	0h	1 = Active interrupts present in global 'Operating condition interrupts' source register 0 = No active interrupts in global 'Operating condition interrupts' source register
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.16 GLOBAL_CAT_IRQ_ENABLES register

Table 115. GLOBAL_CAT_IRQ_ENABLES register - Global categories interrupt enables (address 81h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	CONFIG_REGXS_IRQ	RW	1h	1= enables the respective source to influence the interrupt output
7:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	OP_CONDITIONS_IRQ	RW	1h	1= enables the respective source to influence the interrupt output
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.17 GLOBAL_CAT_IRQ_MSTATUS register

Table 116. GLOBAL_CAT_IRQ_MSTATUS register - Global categories interrupt masked-status (address 82h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	CONFIG_REGXS_IRQ	R	0h	1=respective interrupt signal AND respective enables are both '1'
7:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	OP_CONDITIONS_IRQ	R	0h	1=respective interrupt signal AND respective enables are both '1'
1:0	*reserved*	R	0h	Always write 0, ignore on read.

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Table 116. GLOBAL_CAT_IRQ_MSTATUS register - Global categories interrupt masked-status (address 82h)*...continued*

Bit	Symbol	Access	Value	Description
				Always write 0, ignore on read.

7.4.1.18 OP_CONDITIONS_IRQ_SOURCES register

Table 117. OP_CONDITIONS_IRQ_SOURCES register - Global operating conditions interrupt status (address 90h)

Bit	Symbol	Access	Value	Description
15	TEMP_WARN	RW	0h	1 = Die temperature has exceeded the limit. Write one to clear.
14	OSC_WARN	RW	0h	1 = One or more clocks out of spec Write one to clear.
13:11	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
10	VDDA_RX_SUPPLY_WARN	RW	0h	1 = Line driver supply issue detected 0 = No line driver supply issue detected or a persistant supply fault since last clear. Write one to clear.
9	VREGA_SUPPLY_WARN	RW	0h	1 = VREGA supply issue detected 0 = No VREGA supply issue detected or a persistant supply fault since last clear. Write one to clear.
8:6	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
5	VDDIO_SUPPLY_WARN	RW	0h	1 = VDDIO supply issue detected 0 = No VDDIO supply issue detected or a persistant supply fault since last clear. Write one to clear.
4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	CORE_SUPPLY_WARN	RW	0h	1 = Core supply issue detected 0 = No core supply issue detected or a persistant supply fault since last clear. Write one to clear.
2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	AO_SYS_SUPPLY_WARN	RW	0h	1 = Always-on system supply issue detected 0 = No always-on system supply issue detected or a persistant supply fault since last clear. This bit is set at '1' whenever a main always-on supply cycle occurs. POR Write one to clear.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.19 OP_CONDITIONS_IRQ_ENABLE register

Table 118. OP_CONDITIONS_IRQ_ENABLE register - Global operating conditions interrupt enables (address 91h)

Bit	Symbol	Access	Value	Description
15	TEMP_WARN	RW	0h	1= enables the respective source to influence the interrupt output
14	OSC_WARN	RW	0h	
13:11	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
10	VDDA_RX_ SUPPLY_WARN	RW	0h	1= enables the respective source to influence the interrupt output
9	VREGA_SUPPLY_ WARN	RW	0h	1= enables the respective source to influence the interrupt output
8:6	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
5	VDDIO_SUPPLY_ WARN	RW	0h	1= enables the respective source to influence the interrupt output
4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	CORE_SUPPLY_ WARN	RW	0h	1= enables the respective source to influence the interrupt output
2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	AO_SYS_SUPPLY_ WARN	RW	0h	1= enables the respective source to influence the interrupt output
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.20 OP_CONDITIONS_IRQ_MSTATUS register

Table 119. OP_CONDITIONS_IRQ_MSTATUS register - Global operating conditions interrupt masked-status (address 92h)

Bit	Symbol	Access	Value	Description
15	TEMP_WARN	R	0h	1=respective interrupt signal AND respective enables are both '1'
14	OSC_WARN	R	0h	1=respective interrupt signal AND respective enables are both '1'
13:11	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
10	VDDA_RX_ SUPPLY_WARN	R	0h	1=respective interrupt signal AND respective enables are both '1'
9	VREGA_SUPPLY_ WARN	R	0h	1=respective interrupt signal AND respective enables are both '1'
8:6	*reserved*	R	0h	Always write 0, ignore on read.

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Table 119. OP_CONDITIONS_IRQ_MSTATUS register - Global operating conditions interrupt masked-status (address 92h)*...continued*

Bit	Symbol	Access	Value	Description
				Always write 0, ignore on read.
5	VDDIO_SUPPLY_WARN	R	0h	1=respective interrupt signal AND respective enables are both '1'
4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	CORE_SUPPLY_WARN	R	0h	1=respective interrupt signal AND respective enables are both '1'
2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	AO_SYS_SUPPLY_WARN	R	0h	1=respective interrupt signal AND respective enables are both '1'
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.21 TOP_EPHY_IRQ_SOURCES register

Table 120. TOP_EPHY_IRQ_SOURCES register - EPHY-support interrupt sources (address A0h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	WAKE_SLEEP_EVENT	RW	0h	1 = Change of global WakeSleep status bits 0 = No change of global WakeSleep status bits Write one to clear.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.22 TOP_EPHY_IRQ_ENABLES register

Table 121. TOP_EPHY_IRQ_ENABLES register - EPHY-support interrupt enables (address A1h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	WAKE_SLEEP_EVENT	RW	0h	Enable for global Wake-Sleep event interrupts
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.23 TOP_EPHY_IRQ_MSTATUS register

Table 122. TOP_EPHY_IRQ_MSTATUS register - EPHY-support interrupt masked-status (address A2h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	WAKE_SLEEP_EVENT_IRQ	R	0h	Masked-status of Wake-Sleep event interrupt

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Table 122. TOP_EPHY_IRQ_MSTATUS register - EPHY-support interrupt masked-status (address A2h)*...continued*

Bit	Symbol	Access	Value	Description
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.24 CONFIG_REGXS_IRQ_SOURCES register

Table 123. CONFIG_REGXS_IRQ_SOURCES register - Global configuration interrupt sources (address C0h)

Bit	Symbol	Access	Value	Description
15	COLD_RESET	RW	0h	Hardware-triggered reset of sub-system. All configuration settings reset to defaults. Write one to clear.
14:13	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
12	RECONFIG_ DENIED	RW	0h	1 = Denied configuration attempt to global register 0 = No denied configuration attempt to global registers Write one to clear.
11	INIT_ERROR	RW	0h	1 = Global initialization of configuration settings not successfully completed Write one to clear.
10:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	SMI_ERROR	RW	0h	1 = Invalid SMI transaction to IC 0 = Only valid SMI transactions to IC Write one to clear.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.25 CONFIG_REGXS_IRQ_ENABLE register

Table 124. CONFIG_REGXS_IRQ_ENABLE register - Global config interrupt enables (address C1h)

Bit	Symbol	Access	Value	Description
15	COLD_RESET	RW	0h	1= enables the respective source to influence the interrupt output
14:13	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
12	RECONFIG_ DENIED	RW	0h	1= enables the respective source to influence the interrupt output
11	INIT_ERROR	RW	0h	1= enables the respective source to influence the interrupt output
10:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	SMI_ERROR	RW	0h	1= enables the respective source to influence the interrupt output

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Table 124. CONFIG_REGXS_IRQ_ENABLE register - Global config interrupt enables (address C1h)*...continued*

Bit	Symbol	Access	Value	Description
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.26 CONFIG_REGXS_IRQ_MSTATUS register

Table 125. CONFIG_REGXS_IRQ_MSTATUS register - Global config interrupt masked-status (address C2h)

Bit	Symbol	Access	Value	Description
15	COLD_RESET	R	0h	1=respective interrupt signal AND respective enables are both '1'
14:13	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
12	RECONFIG_ DENIED	R	0h	1=respective interrupt signal AND respective enables are both '1'
11	INIT_ERROR	R	0h	1=respective interrupt signal AND respective enables are both '1'
10:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	SMI_ERROR	R	0h	1=respective interrupt signal AND respective enables are both '1'
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.27 BROADCAST register

Table 126. BROADCAST register - Broadcast configuration (address 100h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4:0	ADDRESS	RW	0h	5-bit SMI PHY address used for broadcasting

7.4.1.28 PORT1_CONFIG register

Table 127. PORT1_CONFIG register - PortNr 1 global configuration (address 101h)

Bit	Symbol	Access	Value	Description
15	PRESENT	R	1h	1 = PORT is present and can be used. 0 = PORT is not present absent
14	ENABLE	RW	1h	1 = PORT enabled, 0 = PORT disabled
13:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4:0	ADDRESS	R	0h	5-bit PHY address assigned to PortNr

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7.4.1.29 WISE_CONTROL register

Table 128. WISE_CONTROL register - Wake-Inhibit-Sleep-Enable control (address 180h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	INH_OVERRIDE	RW	0h	1 = Drive inhibit signal high at any time as long as the device is powered 0 = Drive of inhibit pin reflects need of this device This control is set HIGH at each time that the actual power has been switched-off by this device the actual power may stay active due to other devices driving the board level wired-or.

7.4.1.30 WISE_STATUS register

Table 129. WISE_STATUS register - Wake-Inhibit-Sleep-Enable status (address 181h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	WU_IO_RECEIVED	RW	0h	1 = Wake-up request received on WU_IO pin Write one to clear.
13	WU_SMI_ RECEIVED	RW	0h	1 = Wake-up received by WU_SMI bit assertion Write one to clear.
12:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	DEV_SLEEP_ STATUS	R	0h	1 = Device in SLEEP 0 = Not Sleep in Sleep Mode

7.4.1.31 WISE_CONFIG register

Table 130. WISE_CONFIG register - Wake-Inhibit-Sleep-Enable configuration (address 182h)

Bit	Symbol	Access	Value	Description
15	WAKE_SLEEP_ ENABLE	RW	0h	1 = Wake-Sleep functionality enabled 0 = Wake-Sleep functionality disabled
14	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
13:12	WU_IO	RW	3h	00 = Wake-Up IO is disabled 01 = Wake-Up IO is input only, output disabled 10 = Wake-Up IO is output only, input disabled 11 = Wake-Up IO is input and output
11	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
10	FWD_WU_SMI_TO_ IO	RW	0h	1 = Forward WU_SYS request to WU_IO pin 0 = Do not forward WU_SYS to WU_IO pin
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.32 WISE_PARAMETERS register

Table 131. WISE_PARAMETERS register - Wake-Inhibit-Sleep-Enable parameters (address 184h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4	WU_IO_DETECT_LIMIT	RW	0h	0 = 10-40us 1 = 10-20 ms
3:0	WU_IO_PULSE_LENGTH	RW	1h	Length of a wake-up pulse on the Wake-Up IO pin at nominal ref clock frequency: 50us*2^n. 0=50us, 1=100us, 2=200us, 3=400us, 4=800us, 5=1.6ms, 6=3.2ms, 7=6.4ms, 8=12.8ms, 9=25ms, A=50ms, B=100ms, C=200ms, D=400ms, E=800ms, F=1.6s

7.4.1.33 SMI_STATUS register

Table 132. SMI_STATUS register - Serial Management Interface status (address 189h)

Bit	Symbol	Access	Value	Description
15	CL45_DETECT	RW	0h	1 = Clause 45 transaction detected on MDIO 0 = No Clause 45 transaction detected on MDIO Write one to clear.
14	CL22_INDIRECT_DETECT	RW	0h	1 = Clause 22 registers 0Dh or 0Eh accessed 0 = Clause 22 registers 0Dh and 0Eh not accessed Write one to clear.
13	CL22_DIRECT_DETECT	RW	0h	1 = Regular Clause 22 transaction detected on MDIO excluding access of registers 0Dh and 0Eh 0 = No regular Clause 22 transactions detected on MDIO excluding access of registers 0Dh and 0Eh Write one to clear.
12	SMI_TIMEOUT	RW	0h	SMI did reset when clock is not proper Write one to clear.
11:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.34 SMI_CONFIG register

Table 133. SMI_CONFIG register - Serial Management Interface configuration (address 18Ah)

Bit	Symbol	Access	Value	Description
15	ACCESS_ENABLE	R	1h	1 = Enable register access via SMI for all individual Phy's 0 = Disable register access via SMI Note: Broadcasting can be enabled even if this field is set at '0'.
14	SIDE_EFFECTS_DISABLE	RW	0h	Side-effects are effects on a register value due to a read action. This include Clear-on-Read for Latch-High, Latch-Low bit and Multi-Word reads. 1 = Disable side-effects for debugging purposes 0 = Enable side-effects according to IEEE standards
13	IDLE_PRECHARGE	RW	0h	1 = Drive MDIO high before return to idle high-Z

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Table 133. SMI_CONFIG register - Serial Management Interface configuration (address 18Ah)*...continued*

Bit	Symbol	Access	Value	Description
				0 = Passive return-to-idle-level by resistive pull-up
12	SYNC_ON_16_ONES	RW	0h	0 = 16-ones requirement for SMI synchronization 1 = SMI synchronizes on first zero
11:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	BROADCAST_READ_FUNC	RW	0h	1 = MRA2 only Broadcast write to all PHYs; Broadcast read of shared address space only MMD30LH, all-zero response for read of PHY address space. 0 = Broadcast write to all PHYs; Broadcast read of full address space. Response on reads to PHY address space is a bit-wise OR-function of that register for all present PHYs.
0	BROADCAST_ENABLE	RW	0h	1 = Broadcast address access enabled 0 = No broadcast address supPHYed Note: When BROADCAST_ENABLE is '1', broadcasting is enabled independent of the value of smi_access_enable.

7.4.1.35 SMI_DATA_IO_CONFIG register

Table 134. SMI_DATA_IO_CONFIG register - MDIO pin configuration (address 18Ch)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of of MDIO cells 0 = Auto-configration of MDIO cells: values in this register are ignored.
14	CAP_LOAD	RW	0h	0 = 25 pF max load load capacitance 1 = 15 pF maximum load capacitance
13:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Select drive strength of MDIO outpit IO cells used for the configured. 0 = weakest 7 = strongest
3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2:0	SLEW_RATE	RW	0h	Select slew rate of MDC output IO cells used for the configured. 0 = slowest 7 = fastest

7.4.1.36 GLOBAL_LED_TRIGGER0 register

Table 135. GLOBAL_LED_TRIGGER0 register - LED trigger 0 configuration (address 190h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = Enabled, 0 = Disabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.

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Table 135. GLOBAL_LED_TRIGGER0 register - LED trigger 0 configuration (address 190h)*...continued*

Bit	Symbol	Access	Value	Description
				Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Safety status 1 = Supplies valid 2 = Reserved 3 = XTAL oscillator running / clock available 4 to 15: Reserved

7.4.1.37 GLOBAL_LED_TRIGGER1 register

Table 136. GLOBAL_LED_TRIGGER1 register - LED trigger 1 configuration (address 191h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = Enabled, 0 = Disabled
14:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Safety status 1 = Supplies valid * 2 = Reserved 3 = XTAL oscillator running / clock available 4-15 = Reserved

7.4.1.38 GLOBAL_LED_TRIGGER2 register

Table 137. GLOBAL_LED_TRIGGER2 register - LED trigger 2 configuration (address 192h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = Enabled, 0 = Disabled
14:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Safety status 1 = Supplies valid * 2 = Reserved 3 = XTAL oscillator running / clock available 4-15 = Reserved

7.4.1.39 GLOBAL_LED_TRIGGER3 register

Table 138. GLOBAL_LED_TRIGGER3 register - LED trigger 3 configuration (address 193h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = Enabled, 0 = Disabled
14:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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Table 138. GLOBAL_LED_TRIGGER3 register - LED trigger 3 configuration (address 193h)*...continued*

Bit	Symbol	Access	Value	Description
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Safety status 1 = Supplies valid * 2 = Reserved 3 = XTAL oscillator running / clock available 4-15 = Reserved

7.4.1.40 LED0_CONFIG register

Table 139. LED0_CONFIG register - LED0 control (address 1A0h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = LED controller is enabled. 0 = LED controller is disabled.
14	INVERT	RW	1h	1 = LED0A becomes high to drive the LED. 0 = LED0A becomes low to drive the LED.
13	SPLIT_UP	RW	0h	0 = Control one LED blink/PWM/multi-color; One output A or two correlated outputs A/B. 1 = Control two LEDs blink only with two synchronous, but uncorrelated outputs A/B.
12:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3:0	BLINKING_PERIOD	RW	Ch	Sets the blinking period. 0000 = Reserved 0001 = 250us 0010 = 500us 0011 = 1ms 0100 = 2ms 0101 = 4ms 0110 = 8ms 0111 = 16ms 1000 = 32ms 1001 = 64ms 1010 = 128ms 1011 = 256ms ~0.25 sec 1100 = 512ms ~0.5 sec 1101 = 1024ms ~1 sec 1110 = 2048ms ~2 sec 1111 = latched reset by disable

7.4.1.41 LED0_TRIG_SOURCE register

Table 140. LED0_TRIG_SOURCE register - LED0 trigger source select (address 1A1h)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	RW	0h	0 = Global LED Trigger independent PORT_SELECT 1 = PHY LED trigger

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Table 140. LED0_TRIG_SOURCE register - LED0 trigger source select (address 1A1h)*...continued*

Bit	Symbol	Access	Value	Description
				2 = PTP LED trigger 3 = SGMII Trigger 4-6: Reserved 7: Reserved For XMII LED trigger
12:8	PORTNR_OR_TOPFUNC_B	RW	0h	If FUNC_SELECT_B≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_B=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved
7:5	FUNC_SELECT_A	RW	1h	0 = Global LED trigger 1 = PHY LED trigger 2 = PTP LED trigger 3 = SGMII LED Trigger 4-6: Reserved 7: Reserved for XMII LED trigger
4:0	PORTNR_OR_TOPFUNC_A	RW	1h	If FUNC_SELECT_A≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_A=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved

7.4.1.42 LED0_TRIG01_CONFIG register

Table 141. LED0_TRIG01_CONFIG register - LED0 configuration of triggers 0-1 (address 1A2h)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	RW	1h	1 = TRIG1 enabled 0 = TRIG1 disabled
14	TRIG1_POLARITY	RW	1h	0 = TRIG1 event when TRIG0=0. 1 = TRIG1 event when TRIG0=1.
13	TRIG1_HILO_PRIORITY	RW	1h	0 = Stretch/prioritize TRIG1=0 occurences 1 = Stretch/prioritize TRIG1=1 occurences
12	TRIG1_SELECT	RW	0h	0 = TRIG1 selected by PORT/FUNC_SELECT_A 1 = TRIG1 selected by PORT/FUNC_SELECT_B
11:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG0_ENABLE	RW	1h	1 = TRIG0 enabled

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Table 141. LED0_TRIG01_CONFIG register - LED0 configuration of triggers 0-1 (address 1A2h)*...continued*

Bit	Symbol	Access	Value	Description
				0 = TRIG0 disabled
6	TRIG0_POLARITY	RW	1h	0 = TRIG0 event when TRIG0=0. 1 = TRIG0 event when TRIG0=1.
5	TRIG0_HILO_PRIORITY	RW	1h	0 = Stretch/prioritize TRIG0=0 occurences 1 = Stretch/prioritize TRIG0=1 occurences
4	TRIG0_SELECT	RW	0h	0 = TRIG0 selected by PORT/FUNC_SELECT_A 1 = TRIG0 selected by PORT/FUNC_SELECT_B
3	TRIG0_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.43 LED0_TRIG2_CONFIG register

Table 142. LED0_TRIG2_CONFIG register - LED0 configuration of triggers 2-3 (address 1A3h)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	RW	0h	1 = TRIG3 enabled 0 = TRIG3 disabled
14	TRIG3_POLARITY	RW	0h	0 = TRIG3 event when TRIG0=0. 1 = TRIG3 event when TRIG0=1.
13	TRIG3_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG3=0 occurences 1 = Stretch/prioritize TRIG3=1 occurences
12	TRIG3_SELECT	RW	0h	0 = TRIG3 selected by PORT/FUNC_SELECT_A 1 = TRIG3 selected by PORT/FUNC_SELECT_B
11:10	TRIG3_FUNCTION	RW	0h	This field has no effect when SPLIT_UP=1 01A0h.13 and TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A. If SPLIT_UP=0: 00 = TRIG3 is AND'ed with TRIG0 event. if TRIG2_MODULATION=1: 01 = Blink color modulation by TRIG2/3 1x = Reserved If TRIG2_MODULATION=0: 01 = ColorB replaces OFF for TRIG3 event 10 = Direct color modulation by TRIG3 11 = Reserved
9:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG2_ENABLE	RW	1h	1 = TRIG2 enabled 0 = TRIG2 disabled
6	TRIG2_POLARITY	RW	1h	0 = TRIG2 event when TRIG0=0. 1 = TRIG2 event when TRIG0=1.

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Table 142. LED0_TRIG2_CONFIG register - LED0 configuration of triggers 2-3 (address 1A3h)*...continued*

Bit	Symbol	Access	Value	Description
5	TRIG2_HILO_ PRIORITY	RW	1h	0 = Stretch/prioritize TRIG2=0 occurences 1 = Stretch/prioritize TRIG2=1 occurences
4	TRIG2_SELECT	RW	0h	0 = TRIG2 selected by PORT/FUNC_SELECT_A 1 = TRIG2 selected by PORT/FUNC_SELECT_B
3	TRIG2_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2	TRIG2_ MODULATION	RW	0h	0 = Blink time modulation 1 = Blink color modulation This bit has no effect when SPLIT_UP=1 01A0h.13 or TRIG2_RAW=1. If SPLIT_UP=1, TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A.
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.44 LED1_CONFIG register

Table 143. LED1_CONFIG register - LED1 control (address 1A4h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = LED controller is enabled. 0 = LED controller is disabled.
14	INVERT	RW	0h	1 = LED0A becomes high to drive the LED. 0 = LED0A becomes low to drive the LED.
13	SPLIT_UP	RW	0h	0 = Control one LED blink/PWM/multi-color; One output A or two correlated outputs A/B. 1 = Control two LEDs blink only with two synchronous, but uncorrelated outputs A/B.
12:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3:0	BLINKING_PERIOD	RW	0h	Sets the blinking period. 0000 = Reserved 0001 = 250us 0010 = 500us 0011 = 1ms 0100 = 2ms 0101 = 4ms 0110 = 8ms 0111 = 16ms 1000 = 32ms 1001 = 64ms 1010 = 128ms 1011 = 256ms ~0.25 sec 1100 = 512ms ~0.5 sec 1101 = 1024ms ~1 sec 1110 = 2048ms ~2 sec 1111 = latched reset by disable

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7.4.1.45 LED1_TRIG_SOURCE register

Table 144. LED1_TRIG_SOURCE register - LED1 trigger source select (address 1A5h)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	RW	0h	0 = Global LED Trigger independent PORT_SELECT 000 = PHY LED trigger 001 = PTP LED trigger 010 = SGMII Trigger 100 to 110: Reserved 111 = Reserved For XMII LED trigger
12:8	PORTNR_OR_TOPFUNC_B	RW	0h	If FUNC_SELECT_B≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_B=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved
7:5	FUNC_SELECT_A	RW	0h	000 = Global LED trigger 001 = PHY LED trigger 010 = PTP LED trigger 011 = SGMII LED Trigger 100 to 110: Reserved 111 = Reserved for XMII LED trigger
4:0	PORTNR_OR_TOPFUNC_A	RW	0h	If FUNC_SELECT_A≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_A=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved

7.4.1.46 LED1_TRIG01_CONFIG register

Table 145. LED1_TRIG01_CONFIG register - LED1 configuration of triggers 0-1 (address 1A6h)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	RW	0h	1 = TRIG1 enabled 0 = TRIG1 disabled
14	TRIG1_POLARITY	RW	0h	0 = TRIG1 event when TRIG0=0. 1 = TRIG1 event when TRIG0=1.
13	TRIG1_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG1=0 occurences 1 = Stretch/prioritize TRIG1=1 occurences
12	TRIG1_SELECT	RW	0h	0 = TRIG1 selected by PORT/FUNC_SELECT_A 1 = TRIG1 selected by PORT/FUNC_SELECT_B

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Table 145. LED1_TRIG01_CONFIG register - LED1 configuration of triggers 0-1 (address 1A6h)*...continued*

Bit	Symbol	Access	Value	Description
11:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG0_ENABLE	RW	0h	1 = TRIG0 enabled 0 = TRIG0 disabled
6	TRIG0_POLARITY	RW	0h	0 = TRIG0 event when TRIG0=0. 1 = TRIG0 event when TRIG0=1.
5	TRIG0_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG0=0 occurences 1 = Stretch/prioritize TRIG0=1 occurences
4	TRIG0_SELECT	RW	0h	0 = TRIG0 selected by PORT/FUNC_SELECT_A 1 = TRIG0 selected by PORT/FUNC_SELECT_B
3	TRIG0_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.47 LED1_TRIG2_CONFIG register

Table 146. LED1_TRIG2_CONFIG register - LED1 configuration of triggers 2-3 (address 1A7h)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	RW	0h	1 = TRIG3 enabled 0 = TRIG3 disabled
14	TRIG3_POLARITY	RW	0h	0 = TRIG3 event when TRIG0=0. 1 = TRIG3 event when TRIG0=1.
13	TRIG3_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG3=0 occurences 1 = Stretch/prioritize TRIG3=1 occurences
12	TRIG3_SELECT	RW	0h	0 = TRIG3 selected by PORT/FUNC_SELECT_A 1 = TRIG3 selected by PORT/FUNC_SELECT_B
11:10	TRIG3_FUNCTION	RW	0h	This field has no effect when SPLIT_UP=1 01A4h.13 and TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A. If SPLIT_UP=0: 00 = TRIG3 is AND'ed with TRIG0 event. if TRIG2_MODULATION=1: 01 = Blink color modulation by TRIG2/3 1x = Reserved If TRIG2_MODULATION=0: 01 = ColorB replaces OFF for TRIG3 event 10 = Direct color modulation by TRIG3 11 = Reserved
9:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG2_ENABLE	RW	0h	1 = TRIG2 enabled 0 = TRIG2 disabled

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Table 146. LED1_TRIG2_CONFIG register - LED1 configuration of triggers 2-3 (address 1A7h)*...continued*

Bit	Symbol	Access	Value	Description
6	TRIG2_POLARITY	RW	0h	0 = TRIG2 event when TRIG0=0. 1 = TRIG2 event when TRIG0=1.
5	TRIG2_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG2=0 occurences 1 = Stretch/prioritize TRIG2=1 occurences
4	TRIG2_SELECT	RW	0h	0 = TRIG2 selected by PORT/FUNC_SELECT_A 1 = TRIG2 selected by PORT/FUNC_SELECT_B
3	TRIG2_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2	TRIG2_MODULATION	RW	0h	0 = Blink time modulation 1 = Blink color modulation This bit has no effect when SPLIT_UP=1 01A0h.13 or TRIG2_RAW=1. If SPLIT_UP=1, TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A.
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.48 LED2_CONFIG register

Table 147. LED2_CONFIG register - LED2 control (address 1A8h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = LED controller is enabled. 0 = LED controller is disabled.
14	INVERT	RW	0h	1 = LED0A becomes high to drive the LED. 0 = LED0A becomes low to drive the LED.
13	SPLIT_UP	RW	0h	0 = Control one LED blink/PWM/multi-color; One output A or two correlated outputs A/B. 1 = Control two LEDs blink only with two synchronous, but uncorrelated outputs A/B.
12:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3:0	BLINKING_PERIOD	RW	0h	Sets the blinking period. 0000 = Reserved 0001 = 250us 0010 = 500us 0011 = 1ms 0100 = 2ms 0101 = 4ms 0110 = 8ms 0111 = 16ms 1000 = 32ms 1001 = 64ms 1010 = 128ms 1011 = 256ms ~0.25 sec 1100 = 512ms ~0.5 sec 1101 = 1024ms ~1 sec

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Table 147. LED2_CONFIG register - LED2 control (address 1A8h)*...continued*

Bit	Symbol	Access	Value	Description
				1110 = 2048ms ~2 sec 1111 = latched reset by disable

7.4.1.49 LED2_TRIG_SOURCE register

Table 148. LED2_TRIG_SOURCE register - LED2 trigger source select (address 1A9h)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	RW	0h	0 = Global LED Trigger independent PORT_SELECT 1 = PHY LED trigger 2 = PTP LED trigger 3 = SGMII Trigger 4-6: Reserved 7: Reserved For XMII LED trigger
12:8	PORTNR_OR_TOPFUNC_B	RW	0h	If FUNC_SELECT_B≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_B=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved
7:5	FUNC_SELECT_A	RW	0h	0 = Global LED trigger 1 = PHY LED trigger 2 = PTP LED trigger 3 = SGMII LED Trigger 4-6: Reserved 7: Reserved for XMII LED trigger
4:0	PORTNR_OR_TOPFUNC_A	RW	0h	If FUNC_SELECT_A≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_A=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved

7.4.1.50 LED2_TRIG01_CONFIG register

Table 149. LED2_TRIG01_CONFIG register - LED2 configuration of triggers 0-1 (address 1AAh)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	RW	0h	1 = TRIG1 enabled 0 = TRIG1 disabled
14	TRIG1_POLARITY	RW	0h	0 = TRIG1 event when TRIG0=0.

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Table 149. LED2_TRIG01_CONFIG register - LED2 configuration of triggers 0-1 (address 1AAh)*...continued*

Bit	Symbol	Access	Value	Description
				1 = TRIG1 event when TRIG0=1.
13	TRIG1_HILO_ PRIORITY	RW	0h	0 = Stretch/prioritize TRIG1=0 occurences 1 = Stretch/prioritize TRIG1=1 occurences
12	TRIG1_SELECT	RW	0h	0 = TRIG1 selected by PORT/FUNC_SELECT_A 1 = TRIG1 selected by PORT/FUNC_SELECT_B
11:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG0_ENABLE	RW	0h	1 = TRIG0 enabled 0 = TRIG0 disabled
6	TRIG0_POLARITY	RW	0h	0 = TRIG0 event when TRIG0=0. 1 = TRIG0 event when TRIG0=1.
5	TRIG0_HILO_ PRIORITY	RW	0h	0 = Stretch/prioritize TRIG0=0 occurences 1 = Stretch/prioritize TRIG0=1 occurences
4	TRIG0_SELECT	RW	0h	0 = TRIG0 selected by PORT/FUNC_SELECT_A 1 = TRIG0 selected by PORT/FUNC_SELECT_B
3	TRIG0_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.51 LED2_TRIG2_CONFIG register

Table 150. LED2_TRIG2_CONFIG register - LED2 configuration of triggers 2-3 (address 1ABh)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	RW	0h	1 = TRIG3 enabled 0 = TRIG3 disabled
14	TRIG3_POLARITY	RW	0h	0 = TRIG3 event when TRIG0=0. 1 = TRIG3 event when TRIG0=1.
13	TRIG3_HILO_ PRIORITY	RW	0h	0 = Stretch/prioritize TRIG3=0 occurences 1 = Stretch/prioritize TRIG3=1 occurences
12	TRIG3_SELECT	RW	0h	0 = TRIG3 selected by PORT/FUNC_SELECT_A 1 = TRIG3 selected by PORT/FUNC_SELECT_B
11:10	TRIG3_FUNCTION	RW	0h	This field has no effect when SPLIT_UP=1 01A8h.13 and TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A. If SPLIT_UP=0: 00 = TRIG3 is AND'ed with TRIG0 event. if TRIG2_MODULATION=1: 01 = Blink color modulation by TRIG2/3 1x = Reserved If TRIG2_MODULATION=0: 01 = ColorB replaces OFF for TRIG3 event 10 = Direct color modulation by TRIG3

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Table 150. LED2_TRIG2_CONFIG register - LED2 configuration of triggers 2-3 (address 1ABh)*...continued*

Bit	Symbol	Access	Value	Description
				11 = Reserved
9:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG2_ENABLE	RW	0h	1 = TRIG2 enabled 0 = TRIG2 disabled
6	TRIG2_POLARITY	RW	0h	0 = TRIG2 event when TRIG0=0. 1 = TRIG2 event when TRIG0=1.
5	TRIG2_HILO_ PRIORITY	RW	0h	0 = Stretch/prioritize TRIG2=0 occurences 1 = Stretch/prioritize TRIG2=1 occurences
4	TRIG2_SELECT	RW	0h	0 = TRIG2 selected by PORT/FUNC_SELECT_A 1 = TRIG2 selected by PORT/FUNC_SELECT_B
3	TRIG2_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2	TRIG2_ MODULATION	RW	0h	0 = Blink time modulation 1 = Blink color modulation This bit has no effect when SPLIT_UP=1 01A0h.13 or TRIG2_RAW=1. If SPLIT_UP=1, TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A.
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.52 LED3_CONFIG register

Table 151. LED3_CONFIG register - LED3 control (address 1ACh)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = LED controller is enabled. 0 = LED controller is disabled.
14	INVERT	RW	0h	1 = LED0A becomes high to drive the LED. 0 = LED0A becomes low to drive the LED.
13	SPLIT_UP	RW	0h	0 = Control one LED blink/PWM/multi-color; One output A or two correlated outputs A/B. 1 = Control two LEDs blink only with two synchronous, but uncorrelated outputs A/B.
12:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3:0	BLINKING_PERIOD	RW	0h	Sets the blinking period. 0000 = Reserved 0001 = 250us 0010 = 500us 0011 = 1ms 0100 = 2ms 0101 = 4ms 0110 = 8ms 0111 = 16ms

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Table 151. LED3_CONFIG register - LED3 control (address 1ACh)*...continued*

Bit	Symbol	Access	Value	Description
				1000 = 32ms 1001 = 64ms 1010 = 128ms 1011 = 256ms ~0.25 sec 1100 = 512ms ~0.5 sec 1101 = 1024ms ~1 sec 1110 = 2048ms ~2 sec 1111 = latched reset by disable

7.4.1.53 LED3_TRIG_SOURCE register

Table 152. LED3_TRIG_SOURCE register - LED3 trigger source select (address 1ADh)

Bit	Symbol	Access	Value	Description
15:13	FUNC_SELECT_B	RW	0h	0 = Global LED Trigger independent PORT_SELECT 1 = PHY LED trigger 2 = PTP LED trigger 3 = SGMII Trigger 4-6: Reserved 7: Reserved For XMII LED trigger
12:8	PORTNR_OR_TOPFUNC_B	RW	0h	If FUNC_SELECT_B≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_B=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved
7:5	FUNC_SELECT_A	RW	0h	0 = Global LED trigger 1 = PHY LED trigger 2 = PTP LED trigger 3 = SGMII LED Trigger 4-6: Reserved 7: Reserved for XMII LED trigger
4:0	PORTNR_OR_TOPFUNC_A	RW	0h	If FUNC_SELECT_A≠0: Select PORT number 1-32 00000=32 If FUNC_SELECT_A=0: Select GLOBAL sub-function: 00000 = Global shared resources 00001 = Reserved for Global ePHY-related 00010 = Reserved for Global PTP 00111 = Reserved for Global XMII-related Otherwise = Reserved

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7.4.1.54 LED3_TRIG01_CONFIG register

Table 153. LED3_TRIG01_CONFIG register - LED3 configuration of triggers 0-1 (address 1AEh)

Bit	Symbol	Access	Value	Description
15	TRIG1_ENABLE	RW	0h	1 = TRIG1 enabled 0 = TRIG1 disabled
14	TRIG1_POLARITY	RW	0h	0 = TRIG1 event when TRIG0=0. 1 = TRIG1 event when TRIG0=1.
13	TRIG1_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG1=0 occurences 1 = Stretch/prioritize TRIG1=1 occurences
12	TRIG1_SELECT	RW	0h	0 = TRIG1 selected by PORT/FUNC_SELECT_A 1 = TRIG1 selected by PORT/FUNC_SELECT_B
11:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG0_ENABLE	RW	0h	1 = TRIG0 enabled 0 = TRIG0 disabled
6	TRIG0_POLARITY	RW	0h	0 = TRIG0 event when TRIG0=0. 1 = TRIG0 event when TRIG0=1.
5	TRIG0_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG0=0 occurences 1 = Stretch/prioritize TRIG0=1 occurences
4	TRIG0_SELECT	RW	0h	0 = TRIG0 selected by PORT/FUNC_SELECT_A 1 = TRIG0 selected by PORT/FUNC_SELECT_B
3	TRIG0_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.55 LED3_TRIG2_CONFIG register

Table 154. LED3_TRIG2_CONFIG register - LED3 configuration of triggers 2-3 (address 1AFh)

Bit	Symbol	Access	Value	Description
15	TRIG3_ENABLE	RW	0h	1 = TRIG3 enabled 0 = TRIG3 disabled
14	TRIG3_POLARITY	RW	0h	0 = TRIG3 event when TRIG0=0. 1 = TRIG3 event when TRIG0=1.
13	TRIG3_HILO_PRIORITY	RW	0h	0 = Stretch/prioritize TRIG3=0 occurences 1 = Stretch/prioritize TRIG3=1 occurences
12	TRIG3_SELECT	RW	0h	0 = TRIG3 selected by PORT/FUNC_SELECT_A 1 = TRIG3 selected by PORT/FUNC_SELECT_B
11:10	TRIG3_FUNCTION	RW	0h	This field has no effect when SPLIT_UP=1 01ACh.13 and TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A. If SPLIT_UP=0: 00 = TRIG3 is AND'ed with TRIG0 event. if TRIG2_MODULATION=1:

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Table 154. LED3_TRIG2_CONFIG register - LED3 configuration of triggers 2-3 (address 1AFh)*...continued*

Bit	Symbol	Access	Value	Description
				01 = Blink color modulation by TRIG2/3 1x = Reserved If TRIG2_MODULATION=0: 01 = ColorB replaces OFF for TRIG3 event 10 = Direct color modulation by TRIG3 11 = Reserved
9:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	TRIG2_ENABLE	RW	0h	1 = TRIG2 enabled 0 = TRIG2 disabled
6	TRIG2_POLARITY	RW	0h	0 = TRIG2 event when TRIG0=0. 1 = TRIG2 event when TRIG0=1.
5	TRIG2_HILO_ PRIORITY	RW	0h	0 = Stretch/prioritize TRIG2=0 occurences 1 = Stretch/prioritize TRIG2=1 occurences
4	TRIG2_SELECT	RW	0h	0 = TRIG2 selected by PORT/FUNC_SELECT_A 1 = TRIG2 selected by PORT/FUNC_SELECT_B
3	TRIG2_RAW	RW	0h	0 = Detected TRIG0 event is stretched 1 = TRIG0 input directly used to drive LED
2	TRIG2_ MODULATION	RW	0h	0 = Blink time modulation 1 = Blink color modulation This bit has no effect when SPLIT_UP=1 01ACh.13 or TRIG2_RAW=1. If SPLIT_UP=1, TRIG2/3 determine output-B with similar behavior as TRIG0/1 for output-A.
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.56 AO_SYSTEM_SUPPLY_STATUS register

Table 155. AO_SYSTEM_SUPPLY_STATUS register - Always-on supply status (address 311h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	UV_EVENT	RW	0h	1 = An under-voltage has occured since last clear 0 = No under-voltage has occured since last clear Write one to clear.
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.57 CORE_SUPPLY_STATUS register

Table 156. CORE_SUPPLY_STATUS register - Core supply status (address 313h)

Bit	Symbol	Access	Value	Description
15	OV_EVENT	RW	0h	1 = An over-voltage has occured since last clear 0 = No over-voltage has occured since last clear

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Table 156. CORE_SUPPLY_STATUS register - Core supply status (address 313h)*...continued*

Bit	Symbol	Access	Value	Description
				Write one to clear.
14	UV_EVENT	RW	0h	1 = An under-voltage has occured since last clear 0 = No under-voltage has occured since last clear Write one to clear.
13	OV_STATUS	R	0h	1 = An over-voltage exists at this moment 0 = No over-voltage exits at this moment
12	UV_STATUS	R	0h	1 = An under-voltage exists at this moment 0 = No under-voltage exits at this moment
11	OV_LATENT_FAULT	RW	0h	1 = FAIL: No over-voltage in latent-fault test 0 = PASS: Over-voltage in latent fault test Write one to clear.
10	UV_LATENT_FAULT	RW	0h	1 = FAIL: no under-voltage in latent-fault test 0 = PASS: Under-voltage in latent-fault test Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.58 VDDIO_SUPPLY_STATUS register

Table 157. VDDIO_SUPPLY_STATUS register - VDDIO supply status (address 315h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	UV_EVENT	RW	0h	1 = An under-voltage has occured since last clear 0 = No under-voltage has occured since last clear Write one to clear.
13	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
12	UV_STATUS	R	0h	1 = An under-voltage exists at this moment 0 = No under-voltage exits at this moment
11:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1:0	VDDIO_LEVEL	R	0h	Supply voltage level 11 = 3.3V 10 = 2.5V 01 = 1.8V 00 = Reserved

7.4.1.59 VREGA_SUPPLY_STATUS register

Table 158. VREGA_SUPPLY_STATUS register - Reference supply status (address 319h)

Bit	Symbol	Access	Value	Description
15	OV_EVENT	RW	0h	1 = An over-voltage has occured since last clear 0 = No over-voltage has occured since last clear

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Table 158. VREGA_SUPPLY_STATUS register - Reference supply status (address 319h)*...continued*

Bit	Symbol	Access	Value	Description
				Write one to clear.
14	UV_EVENT	RW	0h	1 = An under-voltage has occured since last clear 0 = No under-voltage has occured since last clear Write one to clear.
13	OV_STATUS	R	0h	1 = An over-voltage exists at this moment 0 = No over-voltage exits at this moment
12	UV_STATUS	R	0h	1 = An under-voltage exists at this moment 0 = No under-voltage exits at this moment
11	OV_LATENT_FAULT	RW	0h	1 = FAIL: No over-voltage in latent-fault test 0 = PASS: Over-voltage in latent fault test Write one to clear.
10	UV_LATENT_FAULT	RW	0h	1 = FAIL: no under-voltage in latent-fault test 0 = PASS: Under-voltage in latent-fault test Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.60 VDDA_RX_SUPPLY register

Table 159. VDDA_RX_SUPPLY register - VDDA_RX supply status (address 31Ah)

Bit	Symbol	Access	Value	Description
15	OV_EVENT	RW	0h	1 = An over-voltage has occured since last clear 0 = No over-voltage has occured since last clear Write one to clear.
14	UV_EVENT	RW	0h	1 = An under-voltage has occured since last clear 0 = No under-voltage has occured since last clear Write one to clear.
13	OV_STATUS	R	0h	1 = An over-voltage exists at this moment 0 = No over-voltage exits at this moment
12	UV_STATUS	R	0h	1 = An under-voltage exists at this moment 0 = No under-voltage exits at this moment
11	OV_LATENT_FAULT	RW	0h	1 = FAIL: No over-voltage in latent-fault test 0 = PASS: Over-voltage in latent fault test Write one to clear.
10	UV_LATENT_FAULT	RW	0h	1 = FAIL: no under-voltage in latent-fault test 0 = PASS: Under-voltage in latent-fault test Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.4.1.61 OSC_STATUS register

Table 160. OSC_STATUS register - Reference clock status (address 31Eh)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	RW	0h	1 = Ratio between the REFCLK and FRO frequencies was too high 0 = Ratio between the REFCLK and FRO frequencies was not too high Write one to clear.
14	LOW_FQ_EVENT	RW	0h	1 = Ratio between the REFCLK and FRO frequencies was too low 0 = Ratio between the REFCLK and FRO frequencies was not too low Write one to clear.
13	HIGH_FQ_NOW	R	0h	1 = Ratio of REFCLK and FRO frequencies is too high 0 = Ratio of REFCLK and FRO frequencies is not too high
12	LOW_FQ_NOW	R	0h	1 = Ratio of REFCLK and FRO frequencies is too low 0 = Ratio of REFCLK and FRO frequencies is not too low
11	HIGH_FQ_LATENT_FAULT	RW	0h	Latent-fault check for low frequency ratio 1 = Fail 0 = Pass Write one to clear.
10	LOW_FQ_LATENT_FAULT	RW	0h	Latent-fault check for low frequency ratio 1 = Fail 0 = Pass Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.62 TEMP_STATUS register

Table 161. TEMP_STATUS register - Temperature status (address 31Fh)

Bit	Symbol	Access	Value	Description
15	HIGH_TEMP_EVENT	RW	0h	1 = A temperature beyond T_limit has occured since last clear. 0 = Temperature has not exceeded T_limit since last clear. Write one to clear.
14	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
13	HIGH_TEMP_NOW	R	0h	1 = Die temperature is above T_limit now 0 = Die temperature is below T_limit now
12	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
11	HIGH_TEMP_LATENT_FAULT	RW	0h	1 = Failure during temperature sensor latent-fault test Write one to clear.
10:8	*reserved*	R	0h	Always write 0, ignore on read.

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Table 161. TEMP_STATUS register - Temperature status (address 31Fh)*...continued*

Bit	Symbol	Access	Value	Description
				Always write 0, ignore on read.
7:0	TEMP_LIMIT	RW	A1h	T_limit=7.22*x^2+38.8*x+65.3 with x=TEMP_LIMIT/100

7.4.1.63 COMPL_TEST_CONTROL register

Table 162. COMPL_TEST_CONTROL register - EPHY compliance test control (address 3F0h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	COMPL_TCLK_EN	RW	0h	1 = Testing is enabled and configured by this register 0 = Normal operation no test

7.4.1.64 GPTP_CONTROL register

Table 163. GPTP_CONTROL register - General control (address 1000h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	RESET	RW		When this bit is set to one, the whole PTP IP is reset including its registers and including also all PORT specific PTP hardware. During such a reset, only this reset bit remains accessible and other Global and PORT PTP registers are ignored. This is self-clearing bit.
			0*	Normal operation of PTP
			1	PTP function is reset
14	CONFIG_ENABLE	RW		Configuration bit for write protection. This bit is super seeded with SoC level config enable.
			0*	Configuration registers of global PTP functionality are write protected
			1	Configuration registers of global PTP functionality can be changed
13:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	LOAD_LTC_ON_SYNC	RW	0h	When this bit is set and on next rising edge of external pps_sync signal, LTC_LOAD_DATA is loaded into the LTC. This bit is cleared after the loading at PPS edgewith additional 4-5 APB clocks CDC latency This bit should be programmed only after LTC_LOAD_DATA is written.
2	LOAD_LTC	RW	0h	Self-clearing when LTC_LOAD_DATA is loaded into the LTC. This should be set only after LTC_LOAD_DATA is written.
1	READ_LTC	RW	0h	When set, captures the current internal LTC clock time in LTC_READ_DATA register. This bit is cleared on capture.

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Table 163. GPTP_CONTROL register - General control (address 1000h) *...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.65 GPTP_CONFIG register

Table 164. GPTP_CONFIG register - General configuration (address 1006h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:6	*reserved*	R	0h	Always write 0, ignore on read.
5	PPS_OUT_POLARITY	RW		Inverts the polarity of generated output 1 PPS signal.
			0*	No inversion
			1	Invert a 1 PPS out
4	PPS_OUT_ENABLE	RW		Enable the PPS_OUT signal generation
			0*	PPS_OUT is not generated
			1	PPS_OUT is generated
3	LTC_SYNC_EDGE_SELECT	RW		When switching this bit, inverted PPS Sync may be disturbed for first pulse after bit is switched. This bit is not dynamically changing bit.
			0*	Rising edge used to sychronize LTC
			1	Falling edge used for synchronize LTC
2	LTC_AUTO_LOCK	RW		Rate correction and offset correction is handled by HW.
			0	LTC lock based on software interaction.
			1*	After initializing the LTC, it locks to the PTP_SYNC without software support.
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.66 GPTP_IRQ_SOURCES register

Table 165. GPTP_IRQ_SOURCES register - Interrupt status signals (address 1008h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET_IRQ	RW		Top-down propagated PTP reset due to a higher-level reset event. All PTP configuration settings got back to defaults.
			0*	Interrupt is cleared.
			1	higher-level reset event is detected
14	LTC_ERROR_WARN	RW		Asserts if LTC_AUTO_LOCK is selected and Local-Time-Counter LTC error exceeds the limit programmed in registers LTC_ERROR_LIMIT. 250us is limit if 1 PTP_sync period is 1 second time. Interrupt status is cleared when it is written to 1 self-clear bit.

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Table 165. GPTP_IRQ_SOURCES register - Interrupt status signals (address 1008h)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
			0*	Interrupt is cleared.
			1	LTC error exceeds the limit programmed in registers
13:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	PTP_TRIG_TS_LOST_IRQ	RW		New PTP_TRIG time-stamp has overwritten an older unread time-stamp. Interrupt status is cleared when it is written to 1 self-clear bit.
			0*	Interrupt is cleared.
			1	New PTP_SYNC time-stamp has overwritten an older unread time-stamp.
3	PTP_SYNC_TS_LOST_IRQ	RW		New PTP_SYNC time-stamp has overwritten an older unread time-stamp. Interrupt status is cleared when it is written to 1 self-clear bit.
			0*	Interrupt is cleared.
			1	New PTP_SYNC time-stamp has overwritten an older unread time-stamp.
2	PTP_TRIG_TS_IRQ	RW		This interrupt gets asserted on the first PTP_TRIG event time-stamp after enabling the PTP functionality ánd after an MDIO read of this time-stamp type responding 'invalid'. The 'invalid' response informs the controller there were no more time-stamp available, and this interrupt indicates that the buffer is not empty anymore. This bit may be cleared by the controller at any time. This bit does not get set again for a next time-stamp if there was no MDIO read with an 'invalid' response between this and the previous time-stamp event. Interrupt status is cleared when it is written to 1 self-clear bit.
			0*	Interrupt is cleared.
			1	New time-stamp of PTP_TRIG event available after the corresponding time-stamp buffer showed empty.
1	PTP_SYNC_TS_IRQ	RW		This interrupt gets asserted on the first PTP_SYNC event time-stamp after enabling the PTP functionality ánd after an MDIO read of this time-stamp type responding 'invalid'. The 'invalid' response informs the controller there were no more time-stamp available, and this interrupt indicates that the buffer is not empty anymore. This bit may be cleared by the controller at any time. This bit does not get set again for a next time-stamp if there was no MDIO read with an 'invalid' response between this and the previous time-stamp event. Interrupt status is cleared when it is written to 1 self-clear bit.
			0*	Interrupt is cleared.
			1	New time-stamp of PTP_SYNC event available after the corresponding time-stamp buffer showed empty.
0	EXTENDED_IRQ	R	0h	This bit will be one when all bits of this register have been assigned, and another register contains additional interrupt sources.

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7.4.1.67 GPTP_IRQ_ENABLES register

Table 166. GPTP_IRQ_ENABLES register - Interrupt enable control (address 100Ah)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET	RW		Enable for TOPDOWN_RESET interrupt
			0*	Mask the interrupt
			1	Enable the interrupt
14	LTC_ERROR_WARN	RW		Enable for LTC_ERROR_WARN interrupt
			0*	Mask the interrupt
			1	Enable the interrupt
13:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	PTP_TRIG_TS_LOST	RW		Enable for PTP_TRIG_TS_LOST interrupt
			0*	Mask the interrupt
			1	Enable the interrupt
3	PTP_SYNC_TS_LOST	RW		Enable for PTP_SYNC_TS_LOST interrupt
			0*	Mask the interrupt
			1	Enable the interrupt
2	PTP_TRIG_TS_IRQ	RW		Enable for PTP_TRIG_TS interrupt
			0*	Mask the interrupt
			1	Enable the interrupt
1	PTP_SYNC_TS_IRQ	RW		Enable for PTP_SYNC_TS interrupt
			0*	Mask the interrupt
			1	Enable the interrupt
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.68 GPTP_IRQ_MSTATUS register

Table 167. GPTP_IRQ_MSTATUS register - Interrupt status signals (address 100Ch)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET	R	0h	This is masked interrupt status AND of Interrupt status TOPDOWN_RESET and enable bit TOPDOWN_RESET.
14	LTC_ERROR_WARN	R	0h	This is masked interrupt status AND of Interrupt status LTC_ERROR_WARN and enable bit LTC_ERROR_WARN.
13:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	PTP_TRIG_TS_LOST	R	0h	This is masked interrupt status AND of Interrupt status PTP_TRIG_TS_LOST and enable bit PTP_TRIG_TS_LOST.
3	PTP_SYNC_TS_LOST	R	0h	This is masked interrupt status AND of Interrupt status PTP_SYNC_TS_LOST and enable bit PTP_SYNC_TS_LOST.

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Table 167. GPTP_IRQ_MSTATUS register - Interrupt status signals (address 100Ch)*...continued*

Bit	Symbol	Access	Value	Description
2	PTP_TRIG_TS_IRQ	R	0h	This is masked interrupt status AND of Interrupt status PTP_TRIG_TS_IRQ and enable bit PTP_TRIG_TS_IRQ.
1	PTP_SYNC_TS_IRQ	R	0h	This is masked interrupt status AND of Interrupt status PTP_SYNC_TS_IRQ and enable bit PTP_SYNC_TS_IRQ.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.69 SYNC_TRIG_FILTER register

Table 168. SYNC_TRIG_FILTER register - Edge selection for PTP stnc & PTP trigger timestamping. (address 1010h)

Bit	Symbol	Access	Value	Description
15:4	*reserved*	R	0h	Always write 0, ignore on read.
3	PTP_TRIG_RISE_TS	RW	0h	1 = Raising edge used for time-stamping
2	PTP_TRIG_FALL_TS	RW	0h	1 = Falling edge used for time-stamping
1	PTP_SYNC_RISE_TS	RW	0h	1 = Raising edge used for time-stamping
0	PTP_SYNC_FALL_TS	RW	0h	1 = Falling edge used for time-stamping

7.4.1.70 REGISTER_TS_CONTROL register

Table 169. REGISTER_TS_CONTROL register - Timestamp format control for TS_DATA_1 register. This is common for both PTP_SYNC & PTP_TRIG timestamp. (address 1012h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4	TS_FORMAT	RW		Time stamp format for both PTP_SYNC & PTP_TRIG
			0*	Replace TS_DATA_1[15:16-K] with the number of subns bits K= NR_OF_SUBNS_BITS selected by bits[3:0] of this register.
			1	The two least-significant second bits SEC[1:0] are put in bits [31:30] of TS_DATA_1 registers. This is common for both PTP_SYNC & PTP_TRIG timestamp.
3:0	NR_OF_SUBNS_BITS	RW	4h	Number of SUBNS bits in TS_DATA_1 registers. This is common for both PTP_SYNC & PTP_TRIG timestamp.

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7.4.1.71 PTP_CLK_NSEC register

Table 170. PTP_CLK_NSEC register - Clock period of ptp_clk. Local LTC increment it's counters by this value for every ptp clock ticks. This register is write protected with config_enable. (address 1020h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:0	NSEC	RW	8h	Clock period of ptp_clk. Local LTC increment it's counters by this value for every ptp clock ticks. In 100BT1, it is 15 ns 66.66 MHz. In 1000BT1, it is 8ns 125 MHz

7.4.1.72 PTP_SYNC_PERIOD_0 register

Table 171. PTP_SYNC_PERIOD_0 register - Period of external PPS sync in nanosecond. Used for HW LTC self locking. This register is write protected with config_enable. (address 1022h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_LSW	RW	CA00h	[15:0] of period of external PPS sync in terms of nanosecond.

7.4.1.73 PTP_SYNC_PERIOD_1 register

Table 172. PTP_SYNC_PERIOD_1 register - Period of external PPS sync in nanosecond. Used for HW LTC self locking. This register is write protected with config_enable. (address 1023h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	NSEC_MSW	RW	3B9Ah	[29:16] of period of external PPS sync in terms of nanosecond.

7.4.1.74 RATE_ADJUST_SUBNS_0 register

Table 173. RATE_ADJUST_SUBNS_0 register - SW Rate adjust value (address 1030h)

Bit	Symbol	Access	Value	Description
15:0	SUB_NSEC_LSW	RW	0h	[15:0] of RATE_ADJUST_SUBNS. For every clock ticks, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls past zero, then value 1 is added or subtracted to the nanoseconds counter based on direction bit. Note the number is interpreted in 2^{-32} ns units.

7.4.1.75 RATE_ADJUST_SUBNS_1 register

Table 174. RATE_ADJUST_SUBNS_1 register - SW Rate adjust value (address 1031h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	UPDATE	RW	0h	Loads new value of rate correction by SW. For every clock ticks, value in RATE_ADJUST_SUBNS is applied for rate correction. This bit is self cleared after 4-5 APB clocks CDC latency. This bit should be programmed after all required

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Table 174. RATE_ADJUST_SUBNS_1 register - SW Rate adjust value (address 1031h)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
				configuration registers has been programmed including SW adjustment is selected.
14	SIGN	RW	0*	For every clock ticks, value in RATE_ADJUST_SUBNS is added to the sub-nanoseconds counter. Value 1 is subtracted from the nanoseconds counter when sub-nanoseconds counter rolls over past zero.
			1	For every clock ticks, value in RATE_ADJUST_SUBNS is added to the sub-nanoseconds counter. Value 1 is added to the nanoseconds counter when sub-nanoseconds counter rolls over past zero.
13:0	SUB_NSEC_MSW	RW	0h	[29:16] of RATE_ADJUST_SUBNS. For every clock ticks, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls over past zero, then value 1 will be added or subtracted to the nanoseconds counter based on direction bit. Note the number is interpreted in 2^{-32} ns units, so the max value is ~0.25 ns

7.4.1.76 TMP_ADJUST_CYCLES_0 register

Table 175. TMP_ADJUST_CYCLES_0 register - Temporary adjustment window (address 1032h)

Bit	Symbol	Access	Value	Description
15:0	TMP_ADJUST_CYCLES_0	RW	0h	[15:0] of TEMP_ADJUST_CYCLES. Number of PTP clock ticks for which temporary adjustment is applied.

7.4.1.77 TMP_ADJUST_CYCLES_1 register

Table 176. TMP_ADJUST_CYCLES_1 register - Temporary adjustment window (address 1033h)

Bit	Symbol	Access	Value	Description
15:0	TMP_ADJUST_CYCLES_1	RW	0h	[31:16] of TEMP_ADJUST_CYCLES. Number of PTP clock ticks for which temporary adjustment is applied.

7.4.1.78 TMP_ADJUST_SUBNS_0 register

Table 177. TMP_ADJUST_SUBNS_0 register - SW temporary rate adjust value (address 1034h)

Bit	Symbol	Access	Value	Description
15:0	SUB_NSEC_LSW	RW	0h	[15:0] of TEMP_ADJUST_SUBNS. For a programmable clock duration, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls over past zero, then value 1 will be added or subtracted to the nanoseconds counter based on direction bit. Note the number is interpreted in 2^{-32} ns units.

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7.4.1.79 TMP_ADJUST_SUBNS_1 register

Table 178. TMP_ADJUST_SUBNS_1 register - SW temporary rate adjust value (address 1035h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	EXECUTE	RW	0h	Setting a bit triggers the temporary adjustment for a programmed duration TMP_ADJUST_CYCLES_0. This bit is self cleared after the programmed duration is completed with additional 4-5 APB clocks CDC latency. This bit should be programmed after all required configuration registers has been programmed including SW adjustment is selected.
14	SIGN	RW	0*	Value 1 is subtracted from the nanoseconds counter when sub-nanoseconds counter rolls over past zero.
			1	Value 1 is added to the nanoseconds counter when sub-nanoseconds counter rolls over past zero.
13:0	SUB_NSEC_MSW	RW	0h	[29:16] of TEMP_ADJUST_SUBNS. For a programmable clock duration, this value is added to the sub-nanoseconds counter. When the sub-nanoseconds counter rolls over past zero, then value 1 will be added or subtracted to the nanoseconds counter based on direction bit. Note the number is interpreted in 2^{-32} ns units.

7.4.1.80 LTC_ERROR_LIMIT_0 register

Table 179. LTC_ERROR_LIMIT_0 register - Error limitation by HW LTC self locking engine (address 1036h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_LSW	RW	D090h	[15:0] of LTC_ERROR_LIMIT. This register limits the number of nanoseconds beyond which HW can not self-lock with external PPS sync. Interrupt is asserted if the error correction limit is exceeded.

7.4.1.81 LTC_ERROR_LIMIT_1 register

Table 180. LTC_ERROR_LIMIT_1 register - Error limitation by HW LTC self locking engine (address 1037h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:0	NSEC_MSW	RW	3h	[23:16] of LTC_ERROR_LIMIT. This register limits the number of nanoseconds beyond which HW can not self-lock with external PPS sync. Interrupt is asserted if the error correction limit is exceeded.

7.4.1.82 LTC_LOOP_CONTROL register

Table 181. LTC_LOOP_CONTROL register - This register defines control parameters to adapt behavior of the LTC loop if LTC_AUTO_LOCK=1. This register is write protected with config_enable. (address 1038h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.

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Table 181. LTC_LOOP_CONTROL register - This register defines control parameters to adapt behavior of the LTC loop if LTC_AUTO_LOCK=1. This register is write protected with config_enable. (address 1038h)*...continued*

Bit	Symbol	Access	Value	Description
11:8	LOG2_PROP_GAIN	RW	3h	Proportional path gain is 2^-PROP_GAIN
7:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4:0	LOG2_INTEGRAL_GAIN	RW	3h	The integrating path gain is: 2^-LOG2_INTEGRAL_GAIN Note that if LOG2_INTEGRAL_GAIN < LOG2_INTEGRAL_GAIN_LIMIT, the applied gain is: 2^-LOG2_INTEGRAL_GAIN_LIMIT

7.4.1.83 LTC_LOAD_DATA_0 register

Table 182. LTC_LOAD_DATA_0 register - Loading nanoseconds counter value (address 1040h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_0	RW	0h	[15:0] of LTC_LOAD_DATA_NSEC. This is nanoseconds part of clock time loaded into the internal LTC.

7.4.1.84 LTC_LOAD_DATA_1 register

Table 183. LTC_LOAD_DATA_1 register - Loading nanoseconds counter value (address 1041h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	NSEC_1	RW	0h	[29:16] of LTC_LOAD_DATA_NSEC. This is nanoseconds part of clock time loaded into the internal LTC.

7.4.1.85 LTC_LOAD_DATA_2 register

Table 184. LTC_LOAD_DATA_2 register - Loading seconds counter value (address 1042h)

Bit	Symbol	Access	Value	Description
15:0	SEC_0	RW	0h	[15:0] of LTC_LOAD_DATA_SEC. This is seconds part of clock time loaded into the internal LTC.

7.4.1.86 LTC_LOAD_DATA_3 register

Table 185. LTC_LOAD_DATA_3 register - Loading seconds counter value (address 1043h)

Bit	Symbol	Access	Value	Description
15:0	SEC_1	RW	0h	[31:16] of LTC_LOAD_DATA_SEC. This is seconds part of clock time loaded into the internal LTC.

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7.4.1.87 LTC_READ_DATA_0 register

Table 186. LTC_READ_DATA_0 register - Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set. (address 1048h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_0	R	0h	[15:0] of LTC_READ_DATA_NSEC. This reads the value of internal nanoseconds counter.

7.4.1.88 LTC_READ_DATA_1 register

Table 187. LTC_READ_DATA_1 register - Local LTC nanoseconds part. The value read is the saved value of local counter when READ_LTC is set. (address 1049h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	NSEC_1	R	0h	[29:16] of LTC_READ_DATA_NSEC. This reads the value of internal nanoseconds counter.

7.4.1.89 LTC_READ_DATA_2 register

Table 188. LTC_READ_DATA_2 register - Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set. (address 104Ah)

Bit	Symbol	Access	Value	Description
15:0	SEC_0	R	0h	[15:0] of LTC_READ_DATA_SEC. This reads the value of internal seconds counter.

7.4.1.90 LTC_READ_DATA_3 register

Table 189. LTC_READ_DATA_3 register - Local LTC seconds part. The value read is the saved value of local counter when READ_LTC is set. (address 104Bh)

Bit	Symbol	Access	Value	Description
15:0	SEC_1	R	0h	[31:16] of LTC_READ_DATA_SEC. This reads the value of internal seconds counter.

7.4.1.91 PTP_SYNC_TS_DATA_S register

Table 190. PTP_SYNC_TS_DATA_S register - (address 1060h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:12	SUBNS	R	0h	Sub-nano-seconds = binary fractional part of nano-seconds. Bit[15:12] holds SUB-NS[31:28]
11:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	RISE_OR_FALL	R		Edge for which time-stamp is presented
			0*	Rising edge time-stamp is presented
			1	Rising edge time-stamp is presented

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Table 190. PTP_SYNC_TS_DATA_S register - (address 1060h)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
0	VALID	R		This valid bit become one when a new time-stamp is latched into the PTP_SYNC_TS registers. It becomes zero when the time-stamp is unlocked by writing to any of the PTP_SYNC_TS registers. If the previous time-stamp was unlocked, and there is no new time-stamp available, this bit will remain zero when reading PTP_SYNC_TS_DATA_S or PTP_SYNC_TS_DATA_0.
			0*	Latched time-stamp is not valid
			1	Rising edge time-stamp is presented

7.4.1.92 PTP_SYNC_TS_DATA_0 register

Table 191. PTP_SYNC_TS_DATA_0 register - The time-stamp is unlocked by any write transaction to any PTP_SYNC_TS register from DATA_S to END. (address 1061h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_0	R	0h	[15:0] of PTP_SYNC_TS_NSEC. Holds nanoseconds part of clock time at the event of active edge of external pps sync.

7.4.1.93 PTP_SYNC_TS_DATA_1 register

Table 192. PTP_SYNC_TS_DATA_1 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 1062h)

Bit	Symbol	Access	Value	Description
15:14	NSEC_1A_SUBNS_1A_SEC	R	0h	Holds nanoseconds/sub-nanoseconds/seconds part of clock time at the event of active edge of external pps sync. If TS_FORMAT=0: "If NR_OF_SUBNS_BITS>1: PTP_SYNC_LTC_SUBNS[31:30]" "If NR_OF_SUBNS_BITS=1: bit[15]= PTP_SYNC_LTC_SUBNS[31] and bit[14] = zero." "If NR_OF_SUBNS_BITS=0: two zeros" If TS_FORMAT=1: "PTP_SYNC_LTC_SEC[1:0]." independent of NR_OF_SUBNS_BITS.
13:12	NSEC_1B_SUBNS_1B	R	0h	Holds nanoseconds/sub-nanoseconds part of clock time at the event of active edge of external pps sync. If NR_OF_SUBNS_BITS=K>2: K≤4=N PTP_SYNC_LTC_SUBNS[29:32-K] concatenated with PTP_SYNC_LTC_NSEC30[31-K:32-N] otherwise: PTP_SYNC_LTC_NSEC30[29:32-N]
11:0	NSEC_1	R	0h	[27:16] of PTP_SYNC_TS_NSEC. Holds nanoseconds part of clock time at the event of active edge of external pps sync.

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7.4.1.94 PTP_SYNC_TS_DATA_2 register

Table 193. PTP_SYNC_TS_DATA_2 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 1063h)

Bit	Symbol	Access	Value	Description
15:0	SEC_0	R	0h	[15:0] of PTP_SYNC_LTC_SEC. Holds seconds part of clock time at the event of active edge of external pps sync.

7.4.1.95 PTP_SYNC_TS_DATA_3 register

Table 194. PTP_SYNC_TS_DATA_3 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 1064h)

Bit	Symbol	Access	Value	Description
15:0	SEC_1	R	0h	[31:16] of PTP_SYNC_LTC_SEC. Holds seconds part of clock time at the event of active edge of external pps sync.

7.4.1.96 PTP_SYNC_TS_END register

Table 195. PTP_SYNC_TS_END register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock. (address 1065h)

Bit	Symbol	Access	Value	Description
15	MORE_TS	R	0h	1 = another time-stamp available in the buffer behind the currently exposed one. This bit shows if there is another PTP_SYNC time-stamp available in the buffer behind the exposed one. This next time-stamp becomes available in the PTP_SYNC registers when unlocking the current time-stamp, by writing to any of the PTP_SYNC registers.
14:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.97 PTP_TRIG_TS_DATA_S register

Table 196. PTP_TRIG_TS_DATA_S register - (address 1070h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:12	SUBNS	R	0h	Sub-nano-seconds = binary fractional part of nano-seconds. Bit[15:12] holds SUB-NS[31:28]
11:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	RISE_OR_FALL	R		Edge for which time-stamp is presented
			0*	Rising edge time-stamp is presented
			1	Rising edge time-stamp is presented
0	VALID	R		This valid bit become one when a new time-stamp is latched into the PTP_TRIG_TS registers. It becomes zero when the time-stamp is unlocked by writing to any of the PTP_TRIG_TS registers. If the previous time-stamp was unlocked, and there is no new time-stamp available, this bit will remain zero

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Table 196. PTP_TRIG_TS_DATA_S register - (address 1070h)*...continued*

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
				when reading PTP_TRIG_TS_DATA_S or PTP_TRIG_TS_DATA_0.
			0*	Latched time-stamp is not valid
			1	Rising edge time-stamp is presented

7.4.1.98 PTP_TRIG_TS_DATA_0 register

Table 197. PTP_TRIG_TS_DATA_0 register - The time-stamp is unlocked by any write transaction to any PTP_TRIG_TS register from DATA_S to END. (address 1071h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_0	R	0h	[15:0] of PTP_TRIG_TS_NSEC. Holds nanoseconds part of clock time at the event of active edge of external pps sync.

7.4.1.99 PTP_TRIG_TS_DATA_1 register

Table 198. PTP_TRIG_TS_DATA_1 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 1072h)

Bit	Symbol	Access	Value	Description
15:14	NSEC_1A_SUBNS_1A_SEC	R	0h	Holds nanoseconds/sub-nanoseconds/seconds part of clock time at the event of active edge of PTP TRIG. If TS_FORMAT=0: "If NR_OF_SUBNS_BITS>1: PTP_SYNC_LTC_SUBNS[31:30]" "If NR_OF_SUBNS_BITS=1: bit[15]= PTP_SYNC_LTC_SUBNS[31] and bit[14] = zero." "If NR_OF_SUBNS_BITS=0: two zeros" If TS_FORMAT=1: "PTP_SYNC_LTC_SEC[1:0]." independent of NR_OF_SUBNS_BITS.
13:12	NSEC_1B_SUBNS_1B	R	0h	Holds nanoseconds/sub-nanoseconds part of clock time at the event of active edge of external pps sync. If NR_OF_SUBNS_BITS=K>2: K≤4=N PTP_TRIG_LTC_SUBNS[29:32-K] concatenated with PTP_TRIG_LTC_NSEC30[31-K:32-N] otherwise: PTP_TRIG_LTC_NSEC30[29:32-N]
11:0	NSEC_1	R	0h	[27:16] of PTP_TRIG_TS_NSEC. Holds nanoseconds part of clock time at the event of active edge of external pps sync.

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7.4.1.100 PTP_TRIG_TS_DATA_2 register

Table 199. PTP_TRIG_TS_DATA_2 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 1073h)

Bit	Symbol	Access	Value	Description
15:0	SEC_0	R	0h	[15:0] of PTP_TRIG_LTC_SEC. Holds seconds part of clock time at the event of active edge of external pps sync.

7.4.1.101 PTP_TRIG_TS_DATA_3 register

Table 200. PTP_TRIG_TS_DATA_3 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 1074h)

Bit	Symbol	Access	Value	Description
15:0	SEC_1	R	0h	[31:16] of PTP_TRIG_LTC_SEC. Holds seconds part of clock time at the event of active edge of external pps sync.

7.4.1.102 PTP_TRIG_TS_END register

Table 201. PTP_TRIG_TS_END register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock. (address 1075h)

Bit	Symbol	Access	Value	Description
15	MORE_TS	R	0h	1 = another time-stamp available in the buffer behind the currently exposed one. This bit shows if there is another PTP_TRIG time-stamp available in the buffer behind the exposed one. This next time-stamp becomes available in the PTP_TRIG registers when unlocking the current time-stamp, by writing to any of the PTP_TRIG registers.
14:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.103 GPTP_IDENTIFIER register

Table 202. GPTP_IDENTIFIER register - Identifier which indicates the integrated version of the GLOBAL PTP-functionality that is shared for all PORTs like LTC, SYNC, and TRIGGERs (address 10FFh)

Bit	Symbol	Access	Value	Description
15:8	VERSION	R	2h	Versio number non-zero always
7:0	REVISION	R	0h	Revision number

7.4.1.104 LOOP_OTP_CONTROL register

Table 203. LOOP_OTP_CONTROL register - 3 bit OTP bits limits the programmed value of integral gain. (address 1100h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	PTP_OTP_CTRL	RW	0h	3 bit OTP bits limits the programmed value of integral gain.

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7.4.1.105 PTP_MISC_CONTROL register

Table 204. PTP_MISC_CONTROL register - 3 bit OTP bits limits the programmed value of integral gain. (address 1101h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	PPS_OUT_EVENT	RW		Indicates PPS_OUT signal is generated LH signal. Status is cleared when it is written to 1 self-clear bit.
			0*	Status is cleared.
			1	PPS out is generated
0	PPS_SYNC_ RETURN	RW		Return external PPS Sync on PPS_OUT
			0*	Synchronized ptp_clk external pps_sync is not routed back on PPS_OUT
			1	Synchronized ptp_clk external pps_sync is routed back on PPS_OUT

7.4.1.106 GLOBAL_FUSA_STATUS register

Table 205. GLOBAL_FUSA_STATUS register - Global FUSA status (address 1202h)

Bit	Symbol	Access	Value	Description
15:14	FUSA_CHECK	R	0h	00b=NOT_COMPLETED 10b=PASS 01b=FAIL 11b=Reserved
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.107 GLOBAL_INFRA_CONTROL register

Table 206. GLOBAL_INFRA_CONTROL register - Infrastructure control (address 2C00h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	Global Infra Reset
14	CONFIG_ENABLE	RW	0h	Global Infra Config enable Write protected registers from 30.2C40h-30.2C4BhGPIO Config Group
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.108 GLOBAL_INFRA_CONFIG register

Table 207. GLOBAL_INFRA_CONFIG register - Infrastructure configuration (address 2C06h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	TEST_ENABLE	RW	0h	1 = Global INFRA test modes can be used

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Table 207. GLOBAL_INFRA_CONFIG register - Infrastructure configuration (address 2C06h)*...continued*

Bit	Symbol	Access	Value	Description
				0 = Global INFRA test modes are blocked

7.4.1.109 GLOBAL_INFRA_IRQ_SOURCES register

Table 208. GLOBAL_INFRA_IRQ_SOURCES register - Infrastructure interrupt sources (address 2C08h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	DEV_BOOT_DONE	RW	0h	1 = The device has booted, and global FuSa checks are completed when enabled. Write one to clear.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.110 GLOBAL_INFRA_IRQ_ENABLES register

Table 209. GLOBAL_INFRA_IRQ_ENABLES register - Infrastructure interrupt enables (address 2C0Ah)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	DEV_BOOT_DONE	RW	1h	1 = The device has booted, and global FuSa checks are completed when enabled.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.111 GLOBAL_INFRA_IRQ_MSTATUS register

Table 210. GLOBAL_INFRA_IRQ_MSTATUS register - Infrastructure interrupt masked-status (address 2C0Ch)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	DEV_BOOT_DONE	R	0h	Masked-status of DEVICE_BOOT_DONE interrupt
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.112 GPIO_OUTPUTS register

Table 211. GPIO_OUTPUTS register - GPIO outputs (address 2C20h)

Bit	Symbol	Access	Value	Description
15:10	*reserved*	R	0h	Always write 0, ignore on read.
9	GPIO_OUT9	RW	0h	Output on GPIO9
8	GPIO_OUT8	RW	0h	Output on GPIO8
7	GPIO_OUT7	RW	0h	Output on GPIO7

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Table 211. GPIO_OUTPUTS register - GPIO outputs (address 2C20h)*...continued*

Bit	Symbol	Access	Value	Description
6	GPIO_OUT6	RW	0h	Output on GPIO6
5	GPIO_OUT5	RW	0h	Output on GPIO5
4	GPIO_OUT4	RW	0h	Output on GPIO4
3	GPIO_OUT3	RW	0h	Output on GPIO3
2	GPIO_OUT2	RW	0h	Output on GPIO2
1	GPIO_OUT1	RW	0h	Output on GPIO1
0	GPIO_OUT0	RW	0h	Output on GPIO0

7.4.1.113 GPIO_STATUS register

Table 212. GPIO_STATUS register - GPIO status (address 2C21h)

Bit	Symbol	Access	Value	Description
15:10	*reserved*	R	0h	Always write 0, ignore on read.
9	GPIO_OUT9	R	0h	Status of signal on GPIO9
8	GPIO_OUT8	R	0h	Status of signal on GPIO8
7	GPIO_OUT7	R	0h	Status of signal on GPIO7
6	GPIO_OUT6	R	0h	Status of signal on GPIO6
5	GPIO_OUT5	R	0h	Status of signal on GPIO5
4	GPIO_OUT4	R	0h	Status of signal on GPIO4
3	GPIO_OUT3	R	0h	Status of signal on GPIO3
2	GPIO_OUT2	R	0h	Status of signal on GPIO2
1	GPIO_OUT1	R	0h	Status of signal on GPIO1
0	GPIO_OUT0	R	0h	Status of signal on GPIO0

7.4.1.114 GPIO0_FUNC_CONFIG register

Table 213. GPIO0_FUNC_CONFIG register - GPIO0 functional configuration register (address 2C40h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	1h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved

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Table 213. GPIO0_FUNC_CONFIG register - GPIO0 functional configuration register (address 2C40h)*...continued*

Bit	Symbol	Access	Value	Description
				111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

7.4.1.115 GPIO1_FUNC_CONFIG register

Table 214. GPIO1_FUNC_CONFIG register - GPIO1 functional configuration register (address 2C41h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B

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Table 214. GPIO1_FUNC_CONFIG register - GPIO1 functional configuration register (address 2C41h)*...continued*

Bit	Symbol	Access	Value	Description
				xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

7.4.1.116 GPIO2_FUNC_CONFIG register

Table 215. GPIO2_FUNC_CONFIG register - GPIO2 functional configuration register (address 2C42h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A

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Table 215. GPIO2_FUNC_CONFIG register - GPIO2 functional configuration register (address 2C42h)*...continued*

Bit	Symbol	Access	Value	Description
				xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

7.4.1.117 GPIO3_FUNC_CONFIG register

Table 216. GPIO3_FUNC_CONFIG register - GPIO3 functional configuration register (address 2C43h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved

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Table 216. GPIO3_FUNC_CONFIG register - GPIO3 functional configuration register (address 2C43h)*...continued*

Bit	Symbol	Access	Value	Description
				1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

7.4.1.118 GPIO4_FUNC_CONFIG register

Table 217. GPIO4_FUNC_CONFIG register - GPIO4 functional configuration register (address 2C44h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC:

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Table 217. GPIO4_FUNC_CONFIG register - GPIO4 functional configuration register (address 2C44h)*...continued*

Bit	Symbol	Access	Value	Description
				0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

7.4.1.119 GPIO5_FUNC_CONFIG register

Table 218. GPIO5_FUNC_CONFIG register - GPIO5 functional configuration register (address 2C45h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test

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Table 218. GPIO5_FUNC_CONFIG register - GPIO5 functional configuration register (address 2C45h)*...continued*

Bit	Symbol	Access	Value	Description
				Otherwise = Reserved

7.4.1.120 GPIO6_FUNC_CONFIG register

Table 219. GPIO6_FUNC_CONFIG register - GPIO6 functional configuration register (address 2C46h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

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7.4.1.121 GPIO7_FUNC_CONFIG register

Table 220. GPIO7_FUNC_CONFIG register - GPIO7 functional configuration register (address 2C47h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

7.4.1.122 GPIO8_FUNC_CONFIG register

Table 221. GPIO8_FUNC_CONFIG register - GPIO8 functional configuration register (address 2C48h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output,

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Table 221. GPIO8_FUNC_CONFIG register - GPIO8 functional configuration register (address 2C48h)*...continued*

Bit	Symbol	Access	Value	Description
				0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = RESERVED 1xx10 = TX_CLK_OUT test Otherwise = Reserved

7.4.1.123 GPIO9_FUNC_CONFIG register

Table 222. GPIO9_FUNC_CONFIG register - GPIO9 functional configuration register (address 2C49h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1 = Enabled, 0 = Disabled
14	INVERT	RW	0h	1 = Inverted input or output, 0 = Non-inverted input or output
13	PIN_SNIF	RW	0h	1 = Pin status readable from register 2C21h. 0 = Corresponding bit in register 2C21h reads zero independent of pin status.

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Table 222. GPIO9_FUNC_CONFIG register - GPIO9 functional configuration register (address 2C49h) *...continued*

Bit	Symbol	Access	Value	Description
12:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7:5	FUNC_SELECT	RW	0h	000 = According to default PIN assignment 001 = LED output 010 = PTP 011:110 = Reserved 111 = MISC
4:0	SIGNAL_SELECT	RW	0h	If FUNC_SELECT=LED: xx000 = LED0_A xx001 = LED0_B xx010 = LED1_A xx011 = LED1_B xx100 = LED2_A xx101 = LED2_B xx110 = LED3_A xx111 = LED3_B If FUNC_SELECT=PTP: 0xx00 = PTP_SYNC 0xx01 = PTP_TRIGGER 0xx1x = Reserved 1xx00 = TX_SOF 1xx01 = RX_SOF 1xx10 = PPS_OUT 1xx11 = Reserved If FUNC_SELECT=MISC: 0xx00 = SNIF input direction control 1xx00 = REGISTER_OUT 1xx01 = REF_CLK_OUT 1xx10 = TX_CLK_OUT test 1xx11 = CLK_OUT_125M Otherwise = Reserved

7.4.1.124 GPIO0_IO_CONFIG register

Table 223. GPIO0_IO_CONFIG register - IO-cell configuration for GPIO0 (address 2C80h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 3 = fastest

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7.4.1.125 GPIO1_IO_CONFIG register

Table 224. GPIO1_IO_CONFIG register - IO-cell configuration for GPIO1 (address 2C81h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

7.4.1.126 GPIO2_IO_CONFIG register

Table 225. GPIO2_IO_CONFIG register - IO-cell configuration for GPIO2 (address 2C82h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

7.4.1.127 GPIO3_IO_CONFIG register

Table 226. GPIO3_IO_CONFIG register - IO-cell configuration for GPIO3 (address 2C83h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Manual drive-strength setting of GPIO IO-cell. 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

7.4.1.128 GPIO4_IO_CONFIG register

Table 227. GPIO4_IO_CONFIG register - IO-cell configuration for GPIO4 (address 2C84h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell

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Table 227. GPIO4_IO_CONFIG register - IO-cell configuration for GPIO4 (address 2C84h)*...continued*

Bit	Symbol	Access	Value	Description
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Manual drive-strength setting of GPIO IO-cell. 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

7.4.1.129 GPIO5_IO_CONFIG register

Table 228. GPIO5_IO_CONFIG register - IO-cell configuration for GPIO5 (address 2C85h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Manual drive-strength setting of GPIO IO-cell. 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

7.4.1.130 GPIO6_IO_CONFIG register

Table 229. GPIO6_IO_CONFIG register - IO-cell configuration for GPIO6 (address 2C86h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Manual drive-strength setting of GPIO IO-cell. 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

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7.4.1.131 GPIO7_IO_CONFIG register

Table 230. GPIO7_IO_CONFIG register - IO-cell configuration for GPIO7 (address 2C87h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Manual drive-strength setting of GPIO IO-cell. 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

7.4.1.132 GPIO8_IO_CONFIG register

Table 231. GPIO8_IO_CONFIG register - IO-cell configuration for GPIO8 (address 2C88h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Manual drive-strength setting of GPIO IO-cell. 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 7 = fastest

7.4.1.133 GPIO9_IO_CONFIG register

Table 232. GPIO9_IO_CONFIG register - IO-cell configuration for GPIO9 (address 2C89h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of GPIO output IO-cells 0 = Auto-configuration of GPIO output IO-cell
14:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1:0	SLEW_RATE	RW	0h	Manual slew-rate setting of GPIO IO-cell. 0 = slowest 3 = fastest

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7.4.1.134 REFCLK_CONFIG register

Table 233. REFCLK_CONFIG register - Auxiliary function configuration (address 2F06h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1:0	DEV_REFCLK_MODE	RW	0h	Select global reference clock mode: 00b* = VDDIO single-ended input clock provided on XI enforced clock follower mode 01b = VDDCORE single-ended input clock provided on XI enforced 11b* = XTAL-generated clock or single-ended VDDCORE-level input clock provided on XI 10b* = Low-swing differential input clock provided on XI/XO

7.4.1.135 PTP_CONTROL register

Table 234. PTP_CONTROL register - General control (address 9000h)

Bit	Symbol	Access	Value	Description
15	RESET	RW		When this bit is set to one, the PTP part belonging to this PORT is reset, including its registers. During such a reset, only this reset bit remains accessible and other all other bits in this register group should be ignored. This is self-clearing bit.
			0*	Normal operation of PTP
			1	PTP function is reset
14	CONFIG_ENABLE	RW		Configuration bit for write protection. This bit is super seeded with SoC level config enable.
			0*	Configuration registers of the PTP of this PORT are write-protected.
			1	Configuration registers of the PTP of this PORT can be changed.
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.136 PTP_ABILITIES register

Table 235. PTP_ABILITIES register - PTP IP Ability registers (address 9004h)

Bit	Symbol	Access	Value	Description
15:4	*reserved*	R	0h	Always write 0, ignore on read.
3	INGRESS_IN_BAND	R		Indicates PTP IP supports insertion of timestamping information in "rserved" field.
			0	Does not support in-band deliver
			1*	Supports in-band deliver
2	PTP_TWO_STEP	R		Indicates PTP IP supports two-step operation.
			0	Does not support two-step operation

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Table 235. PTP_ABILITIES register - PTP IP Ability registers (address 9004h)*...continued*

Bit	Symbol	Access	Value	Description
			1*	Supports TWO-step operation
1	PTP_ONE_STEP	R		Indicates PTP IP supports one-step operation.
			0*	Does not support one-step operation
			1	Supports one-step operation
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.137 PTP_IRQ_SOURCES register

Table 236. PTP_IRQ_SOURCES register - Interrupt status signals (address 9008h)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_ RESET_IRQ	RW		Top-down propagated PTP reset due to a higher-level reset event. All PTP configuration settings got back to defaults.
			0*	Interrupt is cleared.
			1	higher-level reset event is detected
14	GLOBAL_ WARNING	RW		This interrupt makes PTP driver software aware of a global PTP issue that may make PORT-level time-stamps unreliable. This bit gets asserted when the LTC_ERROR_ LIMIT is exceeded. Interrupt status is cleared when it is written to 1.
			0*	Interrupt is cleared.
			1	PTP time-stamps of this PORT can not be trusted.
13	FRAMING_ERROR	RW		Detected errors in ingress path: 1 Detected pre-amble, but no detection of an SFD. 2 Prematurely ended PTP frame before all required fields for the time-stamp have been received. Interrupt status is cleared when it is written to 1.
			0*	Interrupt is cleared.
			1	Error detected in frame format.
12	FCS_ERROR	RW		Incorrect FCS value observed on a received PTP frame. Note that this is only available if ingress in-band time-stamping is enabled. Interrupt status is cleared when it is written to 1.
			0*	Interrupt is cleared.
			1	Incorrect FCS value observed on a received PTP frame.
11:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	INGRESS_TS_ LOST	RW		New ingress time-stamp has overwritten the oldest unread time-stamp in the buffer. Interrupt status is cleared when it is written to 1.
			0*	Interrupt is cleared.

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Table 236. PTP_IRQ_SOURCES register - Interrupt status signals (address 9008h)*...continued*

Bit	Symbol	Access	Value	Description
			1	New ingress time-stamp has overwritten the oldest unread time-stamp in the buffer.
3	EGRESS_TS_LOST	RW		New egress time-stamp has overwritten the oldest unread time-stamp in the buffer. Interrupt status is cleared when it is written to 1.
			0*	Interrupt is cleared.
			1	New egress time-stamp has overwritten the oldest unread time-stamp in the buffer.
2	INGRESS_TS_IRQ	RW		This interrupt gets asserted on the first INGRESS time-stamp after enabling the PTP PORT- specific functionality ánd after an MDIO read of this time-stamp type responding 'invalid'. The 'invalid' response informs the controller there were no more time-stamp available, and this interrupt indicates that the buffer is not empty anymore. This bit may be cleared by the controller at any time. This bit does not get set again for a next time-stamp if there was no MDIO read with an 'invalid' response between this and the previous time-stamp event. Interrupt status is cleared when it is written to 1.
			0*	Interrupt is cleared.
			1	New time-stamp of INGRESS PTP frame available after the INGRESS time-stamp buffer showed empty.
1	EGRESS_TS_IRQ	RW		This interrupt gets asserted on the first EGRESS time-stamp after enabling the PTP PORT- specific functionality ánd after an MDIO read of this time-stamp type responding 'invalid'. The 'invalid' response informs the controller there were no more time-stamp available, and this interrupt indicates that the buffer is not empty anymore. This bit may be cleared by the controller at any time. This bit does not get set again for a next time-stamp if there was no MDIO read with an 'invalid' response between this and the previous time-stamp event. Interrupt status is cleared when it is written to 1.
			0*	Interrupt is cleared.
			1	New time-stamp of EGRESS PTP frame available after the EGRESS time-stamp buffer showed empty.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.138 PTP_IRQ_ENABLES register

Table 237. PTP_IRQ_ENABLES register - Interrupt enable control (address 900Ah)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET	RW		Enable for TOPDOWN_RESET interrupt
			0*	Mask the interrupt
			1	Enable the interrupt
14	GLOBAL_ WARNING	RW	0*	Mask the interrupt

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Table 237. PTP_IRQ_ENABLES register - Interrupt enable control (address 900Ah)*...continued*

Bit	Symbol	Access	Value	Description
			1	Enable the interrupt
13	FRAMING_ERROR	RW	0*	Mask the interrupt
			1	Enable the interrupt
12	FCS_ERROR	RW	0*	Mask the interrupt
			1	Enable the interrupt
11:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	INGRESS_TS_LOST	RW	0*	Mask the interrupt
			1	Enable the interrupt
3	EGRESS_TS_LOST	RW	0*	Mask the interrupt
			1	Enable the interrupt
2	INGRESS_TS_IRQ	RW		Interrupt - ptp message in ingress path is timestamped
			0*	Mask the interrupt
			1	Enable the interrupt
1	EGRESS_TS_IRQ	RW		Interrupt - ptp message in egress path is timestamped
			0*	Mask the interrupt
			1	Enable the interrupt
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.139 PTP_IRQ_MSTATUS register

Table 238. PTP_IRQ_MSTATUS register - Interrupt status signals (address 900Ch)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET	R	0h	This is masked interrupt status AND of Interrupt status TOPDOWN_RESET and enable bit TOPDOWN_RESET.
14	GLOBAL_WARNING	R	0h	This is masked interrupt status AND of respective Interrupt status and it's enable bit.
13	FRAMING_ERROR	R	0h	This is masked interrupt status AND of respective Interrupt status and it's enable bit.
12	FCS_ERROR	R	0h	This is masked interrupt status AND of respective Interrupt status and it's enable bit.
11:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	INGRESS_TS_LOST	R	0h	This is masked interrupt status AND of respective Interrupt status and it's enable bit.
3	EGRESS_TS_LOST	R	0h	This is masked interrupt status AND of respective Interrupt status and it's enable bit.

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Table 238. PTP_IRQ_MSTATUS register - Interrupt status signals (address 900Ch)*...continued*

Bit	Symbol	Access	Value	Description
2	INGRESS_TS_IRQ	R	0h	This is masked interrupt status AND of respective Interrupt status and it's enable bit.
1	EGRESS_TS_IRQ	R	0h	This is masked interrupt status AND of respective Interrupt status and it's enable bit.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.140 PTP_MESSAGE_FILTER register

Table 239. PTP_MESSAGE_FILTER register - PTP event message consideration. This register is write protected with config_enable. (address 9010h)

Bit	Symbol	Access	Value	Description
15:4	*reserved*	R	0h	Always write 0, ignore on read.
3	PDELAY_RESPONSE	RW		Controls whether Pdelay_Resp event should be considered for timestamping or not.
			0	Do not timestamp Pdelay_Resp event message.
			1*	Consider Pdelay_Resp event message to be timestamped.
2	PDELAY_REQUEST	RW		Controls whether Pdelay_Request event should be considered for timestamping or not.
			0	Do not timestamp Pdelay_Request event message.
			1*	Consider Pdelay_Request event message to be timestamped.
1	DELAY_REQUEST	RW		Controls whether Delay_Request event should be considered for timestamping or not.
			0*	Do not timestamp Delay_Request event message.
			1	Consider Delay_Request event message to be timestamped.
0	SYNC	RW		Controls whether sync event should be considered for timestamping or not.
			0	Do not timestamp Sync event message.
			1*	Consider Sync event message to be timestamped.

7.4.1.141 PTP_FILTER_ENABLES register

Table 240. PTP_FILTER_ENABLES register - Ingress or egress packet DA filtering. This register is write protected with config_enable. (address 9011h)

Bit	Symbol	Access	Value	Description
15	DOMAIN_NUMBER	RW		Controls whether domainNumber to be checked against programmed domain number or not to qualify PTP frame.
			0*	domainNumber is not checked
			1	domainNumber is checked against SW bit
14	PTP_VERSION	RW		Controls whether received versionPTP to be checked against programmed version to qualify PTP frame.

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Table 240. PTP_FILTER_ENABLES register - Ingress or egress packet DA filtering. This register is write protected with config_enable. (address 9011h)*...continued*

Bit	Symbol	Access	Value	Description
			0*	versionPTP is not checked
			1	versionPTP is checked against SW bits
13	VLAN_TAG	RW		Controls whether VLAN tagged PTP frame is allowed to timestamp.
			0*	VLAN tagged ptp frame is not timestamped
			1	Allow vlan tagged ptp frame for timestamping
12:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	USER_MAC_DA			Controls whether user DA can be used for honoring PTP message.
			0*	If any DA filter bits are set, then the received DA should match either one of the specified DA of the respective DA filter bit is set. If none of the DA filter bits are set, then all PTP messages are honored.
			1	Match the received DA with programmed user DA. If matched, then honor the PTP message.
3:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	FIXED_MAC_DA_1			Controls whether fixed DA 01:1B:19:00:00:00 can be used for honoring PTP message.
			0	If any DA filter bits are set, then the received DA should match either one of the specified DA of the respective DA filter bit is set. If none of the DA filter bits are set, then all PTP messages are honored.
			1*	Match the received DA with 01:1B:19:00:00:00. If matched, then honor the PTP message.
0	FIXED_MAC_DA_0			Controls whether fixed DA 01:80:C2:00:00:0E can be used for honoring PTP message.
			0	If any DA filter bits are set, then the received DA should match either one of the specified DA of the respective DA filter bit is set. If none of the DA filter bits are set, then all PTP messages are honored.
			1*	Match the received DA with 01:80:C2:00:00:0E. If matched, then honor the PTP message.

7.4.1.142 INGRESS_TS_INSERT_CONTROL register

Table 241. INGRESS_TS_INSERT_CONTROL register - Inband timestamp insertion controls at ingress path (address 9012h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW		Enables 32 bit ingress TS data insertion. {INGRESS_TS_DATA_1, INGRESS_TS_DATA_0} is inserted.

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Table 241. INGRESS_TS_INSERT_CONTROL register - Inband timestamp insertion controls at ingress path (address 9012h)*...continued*

Bit	Symbol	Access	Value	Description
			0*	Ingress timestamp is not inserted into the “reserved” field of ptp header.
			1	Ingress timestamp is inserted into the “reserved” field of ptp header.
14:8	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
7	ENDIANNESS	RW		Sequence of insertion of calculated timestamp into the “reserved” field
			0*	Little endian: MSByte first, LSByte last
			1	Big endian: LSByte first, MSByte last
6:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	TS_FORMAT	RW		Defines the format of timestamp for both ingress & egress. Note regarding TS_FORMAT=0: if SUBNS_BITS > 2, the most-significant NSEC bits will be lost. These bits will not be available from other registers. This format is allowed in order to enable the same format as the 32-bit in-band time-stamping by reading only two TS_DATA registers. In this case further TS_DATA registers are not supposed to be read.
			0*	Replace the K most-significant bits of the 32-bit inserted time-stamp K subns bits, where K = NR_OF_SUBNS_BITS as selected by bits[3:0] of this register.
			1	The two least-significant second bits SEC[1:0] are put in bits [31:30] of TS_DATA_1 registers.
3:0	NR_OF_SUBNS_BITS	RW	4h	Number of SUBNS bits. The NSEC bits will shift-up accordingly. Note that the same number of most-significatn nsec bits will not be represented in the 32-bit time-stamp. Max is 4.

7.4.1.143 TX_DELAY_NSEC register

Table 242. TX_DELAY_NSEC register - Egress data path latency from timestamping point of SOF signal to mdi nanoseconds part. This register is write protected with config_enable. (address 9020h)

Bit	Symbol	Access	Value	Description
15:0	NSEC	RW	0h	Egress data path latency from timestamping point of SOF signal to mdi. This is nanoseconds part.

7.4.1.144 TX_DELAY_XTD register

Table 243. TX_DELAY_XTD register - Egress data path latency from timestamping point of SOF signal to mdi sub-nanoseconds part. This register is write protected with config_enable. (address 9021h)

Bit	Symbol	Access	Value	Description
15:12	SUB_NSEC	RW	0h	Egress data path latency from timestamping point of SOF signal to mdi.

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Table 243. TX_DELAY_XTD register - Egress data path latency from timestamping point of SOF signal to mdi sub-nanoseconds part. This register is write protected with config_enable. (address 9021h)*...continued*

Bit	Symbol	Access	Value	Description
				This is sub-nanoseconds part
11:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.145 RX_DELAY_NSEC register

Table 244. RX_DELAY_NSEC register - Ingress data path latency of SOF signal from MDI line to timestamping point nanoseconds part. This register is write protected with config_enable. (address 9022h)

Bit	Symbol	Access	Value	Description
15:0	NSEC	RW	0h	Ingress data path latency of SOF signal from MDI line to timestamping point. This is nanoseconds part.

7.4.1.146 RX_DELAY_XTD register

Table 245. RX_DELAY_XTD register - Ingress data path latency of SOF signal from MDI line to timestamping point sub-nanoseconds part. This register is write protected with config_enable. (address 9023h)

Bit	Symbol	Access	Value	Description
15:12	SUB_NSEC	RW	0h	Ingress data path latency of SOF signal from MDI line to timestamping point. This is subnanoseconds part.
11:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.147 USER_MAC_DA_0 register

Table 246. USER_MAC_DA_0 register - [15:0] of User defined MAC address. This register is write protected with config_enable. (address 9030h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_0	RW	0h	[15:0] of User defined MAC address

7.4.1.148 USER_MAC_DA_1 register

Table 247. USER_MAC_DA_1 register - [31:16] of User defined MAC address. This register is write protected with config_enable. (address 9031h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_1	RW	0h	[31:16] of User defined MAC address

7.4.1.149 USER_MAC_DA_2 register

Table 248. USER_MAC_DA_2 register - [47:32] of User defined MAC address. This register is write protected with config_enable. (address 9032h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_2	RW	0h	[47:32] of User defined MAC address

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7.4.1.150 USER_MAC_DA_MASK_0 register

Table 249. USER_MAC_DA_MASK_0 register - User MAC DA Mask control. This register is write protected with config_enable. (address 9033h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_MASK_0	RW		Controls [15:0] of User MAC DA Mask bits
			0*	DA bits are compared for matching.
			1	DA bits are ignored from being matched. Hence DA range comparison is possible.

7.4.1.151 USER_MAC_DA_MASK_1 register

Table 250. USER_MAC_DA_MASK_1 register - User MAC DA Mask control. This register is write protected with config_enable. (address 9034h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_MASK_1	RW		Controls [31:16] of User MAC DA Mask bits
			0*	DA bits are compared for matching.
			1	DA bits are ignored from being matched. Hence DA range comparison is possible.

7.4.1.152 USER_MAC_DA_MASK_2 register

Table 251. USER_MAC_DA_MASK_2 register - User MAC DA Mask control. This register is write protected with config_enable. (address 9035h)

Bit	Symbol	Access	Value	Description
15:0	USER_MAC_DA_MASK_2	RW		Controls [47:32] of User MAC DA Mask bits
			0*	DA bits are compared for matching.
			1	DA bits are ignored from being matched. Hence DA range comparison is possible.

7.4.1.153 FILTER_PARAMETERS register

Table 252. FILTER_PARAMETERS register - Ingress or egress packet filtering parameters. This register is write protected with config_enable. (address 9050h)

Bit	Symbol	Access	Value	Description
15:8	DOMAIN_NUMBER	RW	0h	Domain number to be compared with received versionPTP field to qualify PTP frame.
7:4	PTP_VERSION	RW	0h	This value is compared against received versionPTP field to qualify PTP frame.
3:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.154 EGRESS_TS_DATA_S register

Table 253. EGRESS_TS_DATA_S register - When reading any of the first two registers of this time-stamp EGRESS_TS_DATA_S or _HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the EGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values.. Any of the EGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any EGRESS_TS register from _DATA_S to _END. (address 9060h)

Bit	Symbol	Access	Value	Description
15:12	SUBNS	R	0h	Sub-nano-seconds = binary fractional part of nano-seconds. Bit[15:12] holds SUB-NS[31:28]
11:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	VALID	R		This valid bit becomes one when a new time-stamp is latched into the EGRESS_TS registers. It becomes zero when the time-stamp is unlocked by writing to any of the EGRESS_TS registers. If the previous time-stamp was unlocked, and there is no new time-stamp available, this bit will remain zero when reading EGRESS_TS_DATA_S, or EGRESS_TS_HEAD_0 the first two registers of the time-stamp.
			0*	No new valid time-stamp is present and the last-read time stamp was unlocked.
			1	A valid time-stamp is present and locked.

7.4.1.155 EGRESS_TS_HEAD_0 register

Table 254. EGRESS_TS_HEAD_0 register - When reading any of the first two registers of this time-stamp EGRESS_TS_DATA_S or _HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the EGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values.. Any of the EGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any EGRESS_TS register from _DATA_S to _END. (address 9061h)

Bit	Symbol	Access	Value	Description
15:8	DOMAIN_NUMBER	R	0h	Domain number of time-stamped egress PTP frame.
7:4	MESSAGE_TYPE	R	0h	Message type of time-stamped egress PTP frame.
3:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	VALID	R		This valid bit becomes one when a new time-stamp is latched into the EGRESS_TS registers. It becomes zero when the time-stamp is unlocked by writing to any of the EGRESS_TS registers. If the previous time-stamp was unlocked, and there is no new time-stamp available, this bit will remain zero when reading EGRESS_TS_DATA_S, or EGRESS_TS_HEAD_0 the first two registers of the time-stamp.
			0*	No new valid time-stamp is present and the last-read time stamp was unlocked.
			1	A valid time-stamp is present and locked.

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7.4.1.156 EGRESS_TS_HEAD_1 register

Table 255. EGRESS_TS_HEAD_1 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9062h)

Bit	Symbol	Access	Value	Description
15:0	SEQUENCE_ID	R	0h	Sequence-ID of time-stamped egress PTP frame.

7.4.1.157 EGRESS_TS_DATA_0 register

Table 256. EGRESS_TS_DATA_0 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9063h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_0	R	0h	[15:0] of a nanoseconds counter value of a time-stamp of an egress PTP frame.

7.4.1.158 EGRESS_TS_DATA_1 register

Table 257. EGRESS_TS_DATA_1 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9064h)

Bit	Symbol	Access	Value	Description
15:14	NSEC_1A_SUBNS_1A_SEC	R	0h	Holds nanoseconds/sub-nanoseconds/seconds part of time-stamp of an egress PTP frame. If TS_FORMAT=0: "If NR_OF_SUBNS_BITS>1: TS_LTC_SUBNS[31:30]" "If NR_OF_SUBNS_BITS=1: bit[15]= TS_LTC_SUBNS[31] and bit[14] = zero." "If NR_OF_SUBNS_BITS=0: two zeros." If TS_FORMAT=1: "TS_LTC_SEC[1:0]." independent of NR_OF_SUBNS_BITS.
13:12	NSEC_1B_SUBNS_1B	R	0h	Holds nanoseconds/sub-nanoseconds part of time-stamp of an egress PTP frame. If NR_OF_SUBNS_BITS=K>2: K≤4=N TS_LTC_SUBNS[29:32-K] concatenated with TS_LTC_NSEC30[31-K:32-N] otherwise: TS_LTC_NSEC30[29:32-N]
11:0	NSEC_1	R	0h	[27:16] of Nanosecond part. Holds nanoseconds part of time-stamp of an egress PTP frame.

7.4.1.159 EGRESS_TS_DATA_2 register

Table 258. EGRESS_TS_DATA_2 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9065h)

Bit	Symbol	Access	Value	Description
15:0	SEC_0	R	0h	[15:0] of TS_LTC_SEC. Holds seconds part of time-stamp of an egress PTP frame.

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7.4.1.160 EGRESS_TS_DATA_3 register

Table 259. EGRESS_TS_DATA_3 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9066h)

Bit	Symbol	Access	Value	Description
15:0	SEC_1	R	0h	[31:16] of TS_LTC_SEC. Holds seconds part of time-stamp of an egress PTP frame.

7.4.1.161 EGRESS_TS_END register

Table 260. EGRESS_TS_END register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock. (address 9067h)

Bit	Symbol	Access	Value	Description
15	MORE_TS	R	0h	1 = another time-stamp available in the buffer behind the currently exposed one. This bit shows if there is another EGRESS time-stamp available in the buffer behind the exposed one. This next time-stamp becomes available in the EGRESS_TS registers when unlocking the current time-stamp, by writing to any of the EGRESS_TS registers.
14:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.162 INGRESS_TS_DATA_S register

Table 261. INGRESS_TS_DATA_S register - When reading any of the first two registers of this time-stamp INGRESS_TS_DATA_S or INGRESS_TS_DATA_HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the INGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values.. Any of the INGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any INGRESS_TS register from INGRESS_TS_DATA_S to INGRESS_TS_DATA_END. (address 9070h)

Bit	Symbol	Access	Value	Description
15:12	SUBNS	R	0h	Sub-nano-seconds = binary fractional part of nano-seconds. Bit[15:12] holds SUB-NS[31:28]
11:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	VALID	R		This valid bit becomes one when a new time-stamp is latched into the INGRESS_TS registers. It becomes zero when the time-stamp is unlocked by writing to any of the INGRESS_TS registers. If the previous time-stamp was unlocked, and there is no new time-stamp available, this bit will remain zero when reading INGRESS_TS_DATA_S, or INGRESS_TS_HEAD_0 the first two registers of the time-stamp.
			0*	No new valid time-stamp is present and the last-read time stamp was unlocked.
			1	A valid time-stamp is present and locked.

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7.4.1.163 INGRESS_TS_HEAD_0 register

Table 262. INGRESS_TS_HEAD_0 register - When reading any of the first two registers of this time-stamp INGRESS_TS_DATA_S or INGRESS_TS_DATA_HEAD_0, the next available egress PTP time-stamp will be loaded from the buffer and locked into the INGRESS_TS registers, if the previous time-stamp was unlocked. This allows to read a consistent set of values. Any of the INGRESS_TS registers can be read multiple times and in any order. The time-stamp will be unlocked by any write transaction to any INGRESS_TS register from INGRESS_TS_DATA_S to _END. (address 9071h)

Bit	Symbol	Access	Value	Description
15:8	DOMAIN_NUMBER	R	0h	Domain number of time-stamped ingress PTP frame.
7:4	MESSAGE_TYPE	R	0h	Message type of time-stamped ingress PTP frame.
3:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	VALID	R		This valid bit becomes one when a new time-stamp is latched into the INGRESS_TS registers. It becomes zero when the time-stamp is unlocked by writing to any of the INGRESS_TS registers. If the previous time-stamp was unlocked, and there is no new time-stamp available, this bit will remain zero when reading INGRESS_TS_DATA_S, or INGRESS_TS_HEAD_0 the first two registers of the time-stamp.
			0*	No new valid time-stamp is present and the last-read time stamp was unlocked.
			1	A valid time-stamp is present and locked.

7.4.1.164 INGRESS_TS_HEAD_1 register

Table 263. INGRESS_TS_HEAD_1 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9072h)

Bit	Symbol	Access	Value	Description
15:0	SEQUENCE_ID	R	0h	Sequence-ID of time-stamped ingress PTP frame.

7.4.1.165 INGRESS_TS_DATA_0 register

Table 264. INGRESS_TS_DATA_0 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9073h)

Bit	Symbol	Access	Value	Description
15:0	NSEC_0	R	0h	[15:0] of a nanoseconds counter value of a time-stamp of an ingress PTP frame.

7.4.1.166 INGRESS_TS_DATA_1 register

Table 265. INGRESS_TS_DATA_1 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9074h)

Bit	Symbol	Access	Value	Description
15:14	NSEC_1A_SUBNS_1A_SEC	R	0h	Holds nanoseconds/sub-nanoseconds/seconds part of time-stamp of an ingress PTP frame. If TS_FORMAT=0: "If NR_OF_SUBNS_BITS>1: TS_LTC_SUBNS[31:30]"

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Table 265. INGRESS_TS_DATA_1 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9074h)*...continued*

Bit	Symbol	Access	Value	Description
				"If NR_OF_SUBNS_BITS=1: bit[15]= TS_LTC_SUBNS[31] and bit[14] = zero." "If NR_OF_SUBNS_BITS=0: two zeros." If TS_FORMAT=1: * TS_LTC_SEC[1:0]. independent of NR_OF_SUBNS_BITS.
13:12	NSEC_1B_SUBNS_1B	R	0h	Holds nanoseconds/sub-nanoseconds part of time-stamp of an ingress PTP frame. If NR_OF_SUBNS_BITS=K>2: K≤4=N TS_LTC_SUBNS[29:32-K] concatenated with TS_LTC_NSEC30[31-K:32-N] otherwise: TS_LTC_NSEC30[29:32-N]
11:0	NSEC_1	R	0h	[27:16] of Nanosecond part. Holds nanoseconds part of time-stamp of an ingress PTP frame.

7.4.1.167 INGRESS_TS_DATA_2 register

Table 266. INGRESS_TS_DATA_2 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9075h)

Bit	Symbol	Access	Value	Description
15:0	SEC_0	R	0h	[15:0] of TS_LTC_SEC. Holds seconds part of time-stamp of an ingress PTP frame.

7.4.1.168 INGRESS_TS_DATA_3 register

Table 267. INGRESS_TS_DATA_3 register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. (address 9076h)

Bit	Symbol	Access	Value	Description
15:0	SEC_1	R	0h	[31:16] of TS_LTC_SEC. Holds seconds part of time-stamp of an ingress PTP frame.

7.4.1.169 INGRESS_TS_END register

Table 268. INGRESS_TS_END register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock. (address 9077h)

Bit	Symbol	Access	Value	Description
15	MORE_TS	R	0h	1 = another time-stamp available in the buffer behind the currently exposed one. This bit shows if there is another INGRESS time-stamp available in the buffer behind the exposed one. This next time-stamp becomes available in the INGRESS_TS registers when unlocking the current time-stamp, by writing to any of the INGRESS_TS registers.

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Table 268. INGRESS_TS_END register - Any write transaction unlocks this time-stamp, if locked. Transaction error if not locked. Once TS_DATA is read and no need to further read the same data set, then this bit also can be set to '1' to unlock. (address 9077h)*...continued*

Bit	Symbol	Access	Value	Description
14:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.170 PORT_PTP_IDENTIFIER register

Table 269. PORT_PTP_IDENTIFIER register - Identifier which indicates the integrated version of the PORT-specific PTP-functionality like INGRESS and EGRESS time-stamping. (address 90FFh)

Bit	Symbol	Access	Value	Description
15:8	VERSION	R	2h	Versio number non-zero always
7:0	REVISION	R	0h	Revision number

7.4.1.171 PTP_DEBUG_CTRL register

Table 270. PTP_DEBUG_CTRL register - General ring buffer control (address 9100h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4	CLEAR_STATISTIC	RW	0h	Writing to this register clears the statistic counters PTP_PKT_RCV_CNT and PTP_FCS_ERR_CNT. This is self clearing bits
3	READ_BUFFER	RW	0h	Setting this bit presents the content of selected entry of egress/ingress ring buffer into the RING_BUF_0/1. This bit has to be written to '1' for any new data to be read. This is self-clearing bit.
2	RING_BUF_SEL	RW		Selects either egress or ingress ring buffer entry to be presented to the host. This is for debug purpose.
			0*	Egress ring buffer entry is presented.
			1	Ingress ring buffer entry is presented.
1:0	RING_BUF_LOC_SEL	RW		Selection of egress/ingress ring buffer entry to be presented to the host. This is for debug purpose.
			0*	First location is presented
			1	Second location is presented
			2	Third location is presented
			3	Fourth location is presented

7.4.1.172 RING_BUF_0 register

Table 271. RING_BUF_0 register - Egress/Ingress ring buffer content for debug purpose to check if FIFO is malfunctioning. (address 9101h)

Bit	Symbol	Access	Value	Description
15:0	RING_BUF_0	R	0h	[7:0] - Domain number corresponding to the reading TS ring buffer.

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Table 271. RING_BUF_0 register - Egress/Ingress ring buffer content for debug purpose to check if FIFO is malfunctioning. (address 9101h)

Bit	Symbol	Access	Value	Description
				[11:8] - Message type corresponding to the reading TS ring buffer. [14:12] - 3'b000 [15] - Indicates valid timestamp information is present in reading TS ring buffer.

7.4.1.173 RING_BUF_1 register

Table 272. RING_BUF_1 register - Egress/Ingress ring buffer content for debug purpose to check if FIFO is malfunctioning. (address 9102h)

Bit	Symbol	Access	Value	Description
15:0	RING_BUF_1	R	0h	SequenceId corresponding to the reading TS ring buffer.

7.4.1.174 PTP_PKT_RCV_CNT register

Table 273. PTP_PKT_RCV_CNT register - Number of received ptp event packet (address 9103h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:0	PTP_PKT_RCV_CNT	R	0h	Number of received ptp event packet

7.4.1.175 PTP_FCS_ERR_CNT register

Table 274. PTP_FCS_ERR_CNT register - Number of received ptp event packet has fcs error. (address 9104h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:0	PTP_FCS_ERR_CNT	R	0h	Number of received ptp event packet has fcs error. Note that this is only available if ingress in-band time-stamping is enabled.

7.4.1.176 PTP_MISC_STATUS register

Table 275. PTP_MISC_STATUS register - Number of received ptp event packet has fcs error. (address 9105h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	RX_SOF_LOST	RW		RX SOF is not detected when ptp event packet is received. Status is cleared when it is written to 1.
			0*	Status is cleared.
			1	RX SOF is not detected when ptp event packet is received.
0	TX_SOF_LOST	RW		TX SOF is not detected when ptp event packet is received. Status is cleared when it is written to 1.

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Table 275. PTP_MISC_STATUS register - Number of received ptp event packet has fcs error. (address 9105h)*...continued*

Bit	Symbol	Access	Value	Description
			0*	Status is cleared.
			1	TX SOF is not detected when ptp event packet is received.

7.4.1.177 PORT_FUSA_STATUS register

Table 276. PORT_FUSA_STATUS register - Port FUSA status (address 9202h)

Bit	Symbol	Access	Value	Description
15:14	FUSA_CHECK	R	0h	00b=NOT_COMPLETED 10b=PASS 01b=FAIL 11b=Reserved
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.178 PORT_FUSA_IRQ_SOURCES register

Table 277. PORT_FUSA_IRQ_SOURCES register - Port FUSA interrupt sources (address 9208h)

Bit	Symbol	Access	Value	Description
13	FUSA_SERDES_MII_IRQ	R	0h	Serial MII SGMII only Fusa interrupt source 0x92D4
12	FUSA_INFRA_IRQ	R	0h	Infra Fusa interrupt source 0x92C4
11:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	FUSA_EPHY_IRQ	R	0h	EPHY Fusa interrupt source 0x9224
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.179 PORT_FUSA_IRQ_ENABLES register

Table 278. PORT_FUSA_IRQ_ENABLES register - Port FUSA interrupt enables (address 920Ah)

Bit	Symbol	Access	Value	Description
13	FUSA_SERDES_MII_IRQ	RW	1h	1= enables the respective source to influence the interrupt output
12	FUSA_INFRA_IRQ	RW	1h	1= enables the respective source to influence the interrupt output
11:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	FUSA_EPHY_IRQ	RW	1h	1= enables the respective source to influence the interrupt output
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.180 PORT_FUSA_IRQ_MSTATUS register

Table 279. PORT_FUSA_IRQ_MSTATUS register - Port FUSA interrupt masked-status (address 920Ch)

Bit	Symbol	Access	Value	Description
13	FUSA_SERDES_MII_IRQ	R	0h	1=respective interrupt signal AND respective enables are both '1'
12	FUSA_INFRA_IRQ	R	0h	1=respective interrupt signal AND respective enables are both '1'
11:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	FUSA_EPHY_IRQ	R	0h	1=respective interrupt signal AND respective enables are both '1'
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.181 EPHY_FUSA_IRQ_SOURCES register

Table 280. EPHY_FUSA_IRQ_SOURCES register - Functional safety interrupt sources for EPHY (address 9224h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TCLK_WARN	RW	0h	1 = EPHY clock warning Write one to clear.
9	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
8	STATUS_FAULT	RW	0h	1 = Failing ePHY status bit detected 0 = No failing ePHY status bit detected Note: fail may for example be stuck-at-one Write one to clear.
7:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.182 EPHY_FUSA_IRQ_ENABLES register

Table 281. EPHY_FUSA_IRQ_ENABLES register - Functional safety interrupt enables for EPHY (address 9225h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TCLK_WARN	RW	0h	Enable for TCLOCK_WARN interrupt
9	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
8	STATUS_FAULT	RW	0h	Enable for STATUS_FAULT interrupt
7:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.183 EPHY_FUSA_IRQ_MSTATUS register

Table 282. EPHY_FUSA_IRQ_MSTATUS register - Functional safety interrupt masked-status for EPHY (address 9226h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TCLK_WARN	R	0h	Masked-status of TCLOCK_WARN interrupt
9	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
8	STATUS_FAULT	R	0h	Masked-status of STATUS_FAULT interrupt
7:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.184 EPHY_LATENT_FAULT register

Table 283. EPHY_LATENT_FAULT register - EPHY status latent-fault (address 9228h)

Bit	Symbol	Access	Value	Description
15	PLL_STATUS_ FAULT	RW	0h	Write one to clear.
14:11	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
10	BLOCK_LOCK	RW	0h	Write one to clear.
9	REM_PHY_READY	RW	0h	Write one to clear.
8	LOC_PHY_READY	RW	0h	Write one to clear.
7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6	LINK_AVAILABLE_ FAULT	RW	0h	Write one to clear.
5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	REM_RCVR_STS_ FAULT	RW	0h	Write one to clear.
3	SCR_STATUS_ FAULT	RW	0h	Write one to clear.
2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	LOC_RCVR_STS_ FAULT	RW	0h	Write one to clear.
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.185 EPHY_TCLK_STATUS register

Table 284. EPHY_TCLK_STATUS register - EPHY transmit clock status (address 922Ah)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	RW	0h	1 = Ratio of TCLK and REFCLK frequencies has been too high since last clear 0 = Ratio of TCLK and REFCLK frequencies has not been too high since last clear Write one to clear.
14	LOW_FQ_EVENT	RW	0h	1 = Ratio of TCLK and REFCLK frequencies has been too low 0 = Ratio of TCLK and REFCLK frequencies has not been too low Write one to clear.
13	HIGH_FQ_NOW	R	0h	1 = Ratio of TCLK and REFCLK frequencies is too high 0 = Ratio of TCLK and REFCLK frequencies is not too high
12	LOW_FQ_NOW	R	0h	1 = Ratio of TCLK and REFCLK frequencies is too low 0 = Ratio of TCLK and REFCLK frequencies is not too low
11	HIGH_FQ_LATENT_FAULT	RW	0h	Latent-fault check for high frequency ratio 1 = Fail 0 = Pass Write one to clear.
10	LOW_FQ_LATENT_FAULT	RW	0h	Latent-fault check for low frequency ratio Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.186 INFRA_FUSA_IRQ_SOURCES register

Table 285. INFRA_FUSA_IRQ_SOURCES register - Functional safety interrupt sources for INFRAstructure (address 92C4h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TXC_WARN	RW	0h	1 = TXC clock warning Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.187 INFRA_FUSA_IRQ_ENABLES register

Table 286. INFRA_FUSA_IRQ_ENABLES register - Functional safety interrupt enables for INFRAstructure (address 92C5h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TXC_WARN	RW	0h	Enable for TXC_WARN interrupt
9:0	*reserved*	R	0h	Always write 0, ignore on read.

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Table 286. INFRA_FUSA_IRQ_ENABLES register - Functional safety interrupt enables for INFRAstructure (address 92C5h)*...continued*

Bit	Symbol	Access	Value	Description
				Always write 0, ignore on read.

7.4.1.188 INFRA_FUSA_IRQ_MSTATUS register

Table 287. INFRA_FUSA_IRQ_MSTATUS register - Functional safety interrupt masked-status for INFRAstructure (address 92C6h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TXC_WARN	R	0h	Masked-status of TXC_WARN interrupt
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.189 TXC_STATUS register

Table 288. TXC_STATUS register - MII transmit clock status (address 92CAh)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	RW	0h	1 = Ratio between the TXC and FRO frequencies was too high 0 = Ratio between the TXC and FRO frequencies was not too high Write one to clear.
14	LOW_FQ_EVENT	RW	0h	1 = Ratio between the TXC and FRO frequencies was too low 0 = Ratio between the TXC and FRO frequencies was not too low Write one to clear.
13	HIGH_FQ_NOW	R	0h	1 = Ratio of TXC and FRO frequencies is too high 0 = Ratio of TXC and FRO frequencies is not too high
12	LOW_FQ_NOW	R	0h	1 = Ratio of TXC and FRO frequencies is too low 0 = Ratio of TXC and FRO frequencies is not too low
11	HIGH_FQ_LATENT_FAULT	RW	0h	Latent-fault check for high frequency ratio 1 = Fail 0 = Pass Write one to clear.
10	LOW_FQ_LATENT_FAULT	RW	0h	Latent-fault check for low frequency ratio 1 = Fail 0 = Pass Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.190 SGMII_FUSA_IRQ_SOURCES register

Table 289. SGMII_FUSA_IRQ_SOURCES register - Functional safety interrupt sources for Serial-MII SGMII (address 92D4h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TXCLK_WARN	RW	0h	1 = SGMII clock warning Write one to clear.
9	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
8	STATUS_FAULT	RW	0h	1 = SGMII status failure Write one to clear.
7:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.191 SGMII_FUSA_IRQ_ENABLES register

Table 290. SGMII_FUSA_IRQ_ENABLES register - Functional safety interrupt enables for Serial-MII SGMI (address 92D5h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TXCLK_WARN	RW	0h	Enable for SGMII_CLOCK_WARN interrupt
9	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
8	STATUS_FAULT	RW	0h	Enable for STATUS_FAULT interrupt
7:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.192 SGMII_FUSA_IRQ_MSTATUS register

Table 291. SGMII_FUSA_IRQ_MSTATUS register - Functional safety interrupt masked-status for SGMII (address 92D6h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	TXCLK_WARN	R	0h	Masked-status of PLLCLOCK_WARN interrupt
9	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
8	STATUS_FAULT	R	0h	Masked-status of SGMII STATUS_FAULT interrupt
7:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.193 SGMII_STATUS_FAULT register

Table 292. SGMII_STATUS_FAULT register - SGMII status latent-faults (address 92D8h)

Bit	Symbol	Access	Value	Description
15	RXPLL_STATUS_FAULT	RW	0h	SGMII RX PLL Latent fault event Write one to clear.
14	TXPLL_STATUS_FAULT	RW	0h	SGMII TX PLL Latent Fault Event Write one to clear.
13:3	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
2	LINK_STATUS_FAULT	RW	0h	SGMII LINK Latent fault Write one to clear.
1:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.194 SGMII_TCLK_STATUS register

Table 293. SGMII_TCLK_STATUS register - SGMII transmit clock status (address 92DAh)

Bit	Symbol	Access	Value	Description
15	HIGH_FQ_EVENT	RW	0h	1 = Ratio of TCLK and REFCLK frequencies has been too high since last clear 0 = Ratio of TCLK and REFCLK frequencies has not been too high since last clear Write one to clear.
14	LOW_FQ_EVENT	RW	0h	1 = Ratio of TCLK and REFCLK frequencies has been too low 0 = Ratio of TCLK and REFCLK frequencies has not been too low Write one to clear.
13	HIGH_FQ_NOW	R	0h	1 = Ratio of TCLK and REFCLK frequencies is too high 0 = Ratio of TCLK and REFCLK frequencies is not too high
12	LOW_FQ_NOW	R	0h	1 = Ratio of TCLK and REFCLK frequencies is too low 0 = Ratio of TCLK and REFCLK frequencies is not too low
11	HIGH_FQ_LATENT_FAULT	RW	0h	Latent-fault check for high frequency ratio 1 = Fail 0 = Pass Write one to clear.
10	LOW_FQ_LATENT_FAULT	RW	0h	Latent-fault check for high frequency ratio 1 = Fail 0 = Pass Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.195 PORT_BIST_CONTROL register

Table 294. PORT_BIST_CONTROL register - Port BIST control (address A800h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	Reset
14	CONFIG_ENABLE	RW	0h	1 = All bist configuration registers can be changed 0 = Config-protected bist configuration registers are write-protected register list 30.A807h port_bist_intercept_config
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.196 PORT_BIST_CONFIG register

Table 295. PORT_BIST_CONFIG register - Port BIST configuration (address A806h)

Bit	Symbol	Access	Value	Description
1	PACKET_BIST_ENABLE	RW	0h	1 = Packet BIST is enabled, 0 = Disabled
0	TEST_ENABLE	RW	0h	1 = PORT-specific BIST test modes can be used 0 = PORT-specific BIST test modes are blocked

7.4.1.197 BIST_INTERCEPT_CONFIG register

Table 296. BIST_INTERCEPT_CONFIG register - BIST intercept configuration (address A807h)

Bit	Symbol	Access	Value	Description
11	CHECK_EPHY_OR_XMII	RW	0h	1 = EPHY received data to PORT BIST RXD 0 = XMII received data to PORT BIST RXD
10:6	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
5:4	TX_INTERCEPT	RW	0h	Based on Bits 5:4 00 = Normal operation 01 =Reserved output zeros to ePHY 10 =Intercept ePHY TXD-path with PHY BIST TXD 11 = Reserved output zeros to ePHY
3:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1:0	RX_INTERCEPT	RW	0h	Based on Bits 1:0 00 = Normal operation 01 = Reserved 10 = Intercept XMII RXD-path with PHY BIST TXD 11 = Reserved

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7.4.1.198 BIST_GEN_CTRL register

Table 297. BIST_GEN_CTRL register - BSIT generator control register (address A880h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	BIST_GEN_EN	RW		BIST generator enable
			0*	BIST generator is disabled
			1	BIST generator is enabled
13	BIST_GEN_MODE	RW		Controls whether BIST is operating in continuous mode or production mode.
			0*	Production mode. BIST generator generates number of programmed good frame and number of programmed bad frames.
			1	BIST generator continuously generate good frames only. Hence, GOOD_FRAMES_PLAN and BAD_FRAMES_PLAN are ignored.
12	STOP	RW	0h	STOP control. Stop generation new frames after the currently generated one has completed. This is self-clearing bit.
11:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.199 BIST_GEN_STATUS register

Table 298. BIST_GEN_STATUS register - BIST generator status (address A881h)

Bit	Symbol	Access	Value	Description
15	BIST_GEN_DONE	R	0h	This status flag is set when programmed number of good frame and bad frame are completely generated and sent. This bit will not get asserted when continuous mode is enabled.
14:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.200 PREAMBL_IPG_SIZE register

Table 299. PREAMBL_IPG_SIZE register - Preamble and IPG lengths (address A887h)

Bit	Symbol	Access	Value	Description
15:12	PREAMBLE_ LENGTH	RW	7h	Length of preamble in bytes.
11:10	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
9:8	IPG_LENGTH_ TYPE	RW		Selection of type of IPG length
			0*	IPG length is fixed to IPG_LENGTH
			1	IPG length starts with IPG_LENGTH bytes, incrementing by one every next IPG until 255 is reached. The IPG length rolls-over to IPG_LENGTH bytes after reaching an IPG

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Table 299. PREAMBL_IPG_SIZE register - Preamble and IPG lengths (address A887h)*...continued*

Bit	Symbol	Access	Value	Description
				length of 255 bytes, and starts incrementing from there again.
			2	IPG length is determined by 8bit LFSR. If LFSR value is less than IPG_LENGTH register then IPG length would be IPG_ LENGTH. If LFSR value is greater than or equal to IPG_ LENGTH register then IPG length would be LFSR value. IPG_LENGTH also acts as seed for 8 bit LFSR.
7:0	IPG_LENGTH	RW	Ch	Defines minimum length of IPG in bytes and maximum length is decided by 8 bit LFSR. If LFSR value is less than IPG_ LENGTH register then IPG length would be IPG_LENGTH. If LFSR value is greater than or equal to IPG_LENGTH register then IPG length would be LFSR value. IPG_LENGTH also acts as seed for LFSR.

7.4.1.201 BIST_DA_0 register

Table 300. BIST_DA_0 register - lower 16 bits of DA value[15:0] (address A888h)

Bit	Symbol	Access	Value	Description
15:0	BIST_DA_0	RW	0h	[15:0] of DA .

7.4.1.202 BIST_DA_1 register

Table 301. BIST_DA_1 register - middle 16 bits of DA value[31:16] (address A889h)

Bit	Symbol	Access	Value	Description
15:0	BIST_DA_1	RW	0h	[31:16] of DA .

7.4.1.203 BIST_DA_2 register

Table 302. BIST_DA_2 register - higher 16 bits of DA value[47:32] (address A88Ah)

Bit	Symbol	Access	Value	Description
15:0	BIST_DA_2	RW	0h	[47:32] of DA .

7.4.1.204 BIST_SA_0 register

Table 303. BIST_SA_0 register - lower 16 bits of SA value[15:0] (address A88Bh)

Bit	Symbol	Access	Value	Description
15:0	BIST_SA_0	RW	0h	[15:0] of SA .

7.4.1.205 BIST_SA_1 register

Table 304. BIST_SA_1 register - middle 16 bits of SA value[31:16] (address A88Ch)

Bit	Symbol	Access	Value	Description
15:0	BIST_SA_1	RW	0h	[31:16] of SA

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7.4.1.206 BIST_SA_2 register

Table 305. BIST_SA_2 register - higher 16 bits of SA value[47:32] (address A88Dh)

Bit	Symbol	Access	Value	Description
15:0	BIST_SA_2	RW	0h	[47:32] of SA .

7.4.1.207 BIST_PTP_CONFIG register

Table 306. BIST_PTP_CONFIG register - PTP frame generation configuration (address A88Fh)

Bit	Symbol	Access	Value	Description
15:8	PTP_MSG_ INTERVAL	RW	0h	Configures the interval of ptp event message to be transmitted. For every programmed N frames, one PTP event frame is generated. N = PTP_MSG_INTERVAL. For ex: for every 10 frames, 9 normal frames are sent and 1 ptp frame is sent. 16'h88F7 is sent as ether type during ptp frame.
7:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3:0	PTP_MSG_TYPE	RW	0h	PTP message type as per IEEE 1588 PTP header. This configuration enables flexibility of selecting ptp message type. When all condition is met to send PTP frame over ethernet frame, PTP_MSG_TYPE replaces the first nibble of selected dataconstant data/incremental pattern/prbs data after ether type is sent. PTP frame generation enables basic functional checks of PTP IP and to exercise PTP IP esp. for HTOL testing.

7.4.1.208 BIST_ETHER_TYPE register

Table 307. BIST_ETHER_TYPE register - Ethernet type T/L (address A890h)

Bit	Symbol	Access	Value	Description
15:0	ETHER_TYPE	RW	0h	Ethernet type T/L.

7.4.1.209 PAYLOAD_CONFIG register

Table 308. PAYLOAD_CONFIG register - Payload data controls (address A891h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:12	PAYLOD_DATA_ TYPE	RW		Payload data type
			0	Programmable fixed data is sent
			1	Ramp data is sent increment by 1
			2*	PRBS output is sent see BIST_PRBS_SELECT
			3	Not used
11:10	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
9:8	PAYLOD_SIZE_ TYPE	RW		Indicates the increment in payload size.

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Table 308. PAYLOAD_CONFIG register - Payload data controls (address A891h)*...continued*

Bit	Symbol	Access	Value	Description
			0*	No increment in payload size Always PAYLOAD_SIZE
			1	Increment always from PAYLOAD_SIZE[7:0] until {PAYLOAD_SIZE[13:8], 8'hFF} and rolls-over to PAYLOAD_SIZE[7:0] again. The limit is between PAYLOAD_SIZE[7:0] & {PAYLOAD_SIZE[13:8], 8'hFF} – where PAYLOAD_SIZE is programmable
			2	For PAYLOAD_SIZE_TYPE= '10'=RANDOM, the payload data length is determined by a 14-bit LFSR x^14+x^13+x^12+x^2+1. PAYLOAD_SIZE is used as seed. PAYLOAD_SIZE[7:0] sets the minimum payload size. PAYLOAD[13:8] determined the maximum payload size. If LFSR[13:8} =< PAYLOAD_SIZE[13:8], the LFSR[13:0] value determines the payload size. If LFSR[13:8}> PAYLOAD_SIZE[13:8], the payload size is determined by LFSR[9:0]. For PAYLOAD_SIZE[13:8] =05h, the max payload size is 1535 bytes 5FFh. The number of payload data bytes equals PAYLOAD_SIZE[7:0] if LFSR[13:0]<={00h & PAYLOAD_SIZE[7:0]}. This generates frames with a pseudo-random payload size ranging from PAYLOAD_SIZE[7:0] to {PAYLOAD_SIZE[13:8] & FFh} bytes. If PAYLOAD_SIZE[13:8]>05h, jumbo frames will be generated. Jumbo frames up to the max size of 16383 bytes will be generated if PAYLOAD_SIZE[13:8]=3Fh. PAYLOAD_SIZE is used as seed for 14 bit LFSR which is used to derive random payload length.
			3	Not used
7:0	FIXED_PAYLD_DATA	RW	A5h	Byte value that will repeatedly send as frame payload.

7.4.1.210 PAYLOAD_SIZE register

Table 309. PAYLOAD_SIZE register - Length of frame payload data in bytes (address A892h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	PAYLOAD_SIZE	RW	2Eh	Maximum of 1500d=5DCh according to Ethernet standard and 9216d=2400h for jumbo-frames. This 14-bit field allows to send up to 3FFFh=16383 payload bytes in a frame. The minimum number of payload data bytes for Ethernet is 46d=2Eh. If a value smaller than 2Eh is programmed the generator will work, but runt frames will be generated. The usage of this field depends on PAYLOAD_SIZE_TYPE: PAYLOAD_SIZE_TYPE=FIXED: this field sets a fixed payload size. PAYLOAD_SIZE_TYPE=INCREMENT or RANDOM: bits[7:0] determine the minimum payload size, and bits[13:8] determine the maximum payload size.

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7.4.1.211 PRBS_DATA_CONFIG register

Table 310. PRBS_DATA_CONFIG register - PRBS data configuration (address A893h)

Bit	Symbol	Access	Value	Description
15:13	BIST_PRBS_SELECT	RW		Different type of PRBS is selected
			0*	PRBS7
			1	PRBS9
			2	PRBS11
			3	PRBS15
			4	Not used all other conditions
12	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
11	LFSR_SEED_USAGE	RW		This bit selects the seed value of the LFSR.
			0*	The seed of the LFSR returns to the programmed value for every packet. Seed value is taken from BIST_LFSR_SEED register
			1	First generated ethernet frame uses the programmed seed, and the seed keeps progressing for next frames.
10:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.212 BIST_LFSR_SEED register

Table 311. BIST_LFSR_SEED register - LFSR Seed value (address A894h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14:0	BIST_LFSR_SEED	RW	7FFFh	Initial value to be loaded into the LSFR for PRBS data.

7.4.1.213 GOOD_FRAMES_PLAN register

Table 312. GOOD_FRAMES_PLAN register - Number of good frames to be generated by BIST generator (address A8A2h)

Bit	Symbol	Access	Value	Description
15:0	GOOD_FRAMES_PLAN	RW	64h	Number of good frames to be generated by BIST generator.

7.4.1.214 G_GOOD_FRAME_CNT register

Table 313. G_GOOD_FRAME_CNT register - Generated good frames in continuous mode. (address A8A4h)

Bit	Symbol	Access	Value	Description
15:0	G_GOOD_FRAME_CNT	R	0h	Number of generated 'good frames' with a correct FCS value. This register stores the number of generated good frames when the packet generator is stopped by STOP bit. This register saturates at max value FFFFh.

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7.4.1.215 BAD_FRAMES_PLAN register

Table 314. BAD_FRAMES_PLAN register - Number of bad frames to be generated by BIST generator (address A8A6h)

Bit	Symbol	Access	Value	Description
15:0	BAD_FRAMES_PLAN	RW	0h	Number of bad frames to be generated by BIST generator

7.4.1.216 BIST_CHECK_CTRL register

Table 315. BIST_CHECK_CTRL register - BIST checker control register (address A8C0h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	BIST_CHECKER_EN	RW		BIST checker enable
			0*	BIST checker is disabled
			1	BIST checker is enabled
13	STATISTC_CNT_RST	RW	0h	If this bit is set, then resets statistic counters - good frame counter and bad frame counter R_GOOD_FRAME_CNT, R_BAD_FRAME_CNT . After reset, these counters hold fresh values. This bit is self-clearing bit. This bit is used only when BIST is in continuous mode.
12:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	BIST_CHECK_MODE	RW		Controls whether BIST is operating in continuous mode or production mode.
			0*	Production mode. BIST checker checks for BIST done and BIST fail status.
			1	Continuous mode. BIST checker keep checking the good frames, bad frames and rx_er when rx_dv is set. Results are available in statistical registers.

7.4.1.217 BIST_PROD_STATUS register

Table 316. BIST_PROD_STATUS register - BIST status used when BIST is in production mode and NOT valid for continuous mode (address A8C1h)

Bit	Symbol	Access	Value	Description
15	BIST_CHECK_DONE	R	0h	This bit indicates BIST status is ready for reading. This bit is set when expected good number of frames and bad number of frames is received or timed out.
14:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	BIST_CHECK_FAIL	R	0h	Indicates at least one of the following error status is set: 1. Good frame error: If the expected number of good frames are not received, then bit flag is set.

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Table 316. BIST_PROD_STATUS register - BIST status used when BIST is in production mode and NOT valid for continuous mode (address A8C1h)*...continued*

Bit	Symbol	Access	Value	Description
				2. Bad frame error: If the expected number of bad frames are not received, then bit flag is set Error is injected, but not received 3. rx_dv is not detected: If RX_DV is not received by the BIST checker, then this status is set. 4. RX_ER detected: This status is set when rx_er is set with rx_dv is high If BIST_CHECK_DONE is set and BIST_CHECK_FAIL is logic zero, then it is said to be PASSED. If BIST_CHECK_DONE is set and BIST_CHECK_FAIL is logic high, then it is said to be FAIL.
3	RX_ER_DETECT_ER	R	0h	This status is set when rx_er is set with rx_dv is high
2	RXDV_DETECT_ER	R	0h	If RX_DV is not received by the BIST checker, then this status is set.
1	BAD_FRAME_ERROR	R	0h	If the expected number of bad frames are not received, then bit flag is set.
0	GOOD_FRAME_ERROR	R	0h	If the expected number of good frames are not received, then bit flag is set

7.4.1.218 BIST_WAIT_TIMER register

Table 317. BIST_WAIT_TIMER register - Timer timeout value for checking all status in production mode. This timeout is useful when expected number of generated frames is never received. (address A8C4h)

Bit	Symbol	Access	Value	Description
15:0	BIST_WAIT_TIMER	RW	FFh	Timer timeout value for checking all status in production mode. This count begins when BIST checker is enabled. Hence this register should be programed before BIST checker is enabled. Ideally this value should be- time taken to send all frames by the BIST generator plus latency from BIST generator to BIST checker plus additional time to program BIST generator and checker enable bit. This is time out register and is useful when expected good number of frames and bad number of frames is not received. In this condition, time out asserts BIST_CHECK_DONE status. This count is in terms of microseconds. Note: This is 16 bit value hence can support upto 65ms only.

7.4.1.219 R_GOOD_FRAME_CNT register

Table 318. R_GOOD_FRAME_CNT register - Received good frame count (address A8D0h)

Bit	Symbol	Access	Value	Description
15:0	R_GOOD_FRAME_CNT	R	0h	Counts the number received frames for which CRC magic number is matched.

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Table 318. R_GOOD_FRAME_CNT register - Received good frame count (address A8D0h)

Bit	Symbol	Access	Value	Description
				This can be used in both production mode and continuous mode. This counter can be cleared by STATISTC_CNT_RST bit to count fresh good frame in continuous mode.

7.4.1.220 R_BAD_FRAME_CNT register

Table 319. R_BAD_FRAME_CNT register - Received bad frame count (address A8D2h)

Bit	Symbol	Access	Value	Description
15:0	R_BAD_FRAME_CNT	R	0h	Counts the number received frames for which CRC magic number is NOT matched. This can be used in both production mode and continuous mode. This counter can be cleared by STATISTC_CNT_RST bit to count fresh bad frame in continuous mode

7.4.1.221 R_RXER_FRAME_CNT register

Table 320. R_RXER_FRAME_CNT register - RX_ER detected when rx_dv is set. (address A8D4h)

Bit	Symbol	Access	Value	Description
15:0	R_RXER_FRAME_CNT	R	0h	Counts the number received frames for which rx_er is detected when rx_dv is high. This can be used in both production mode and continuous mode. This counter can be reset by toggling BIST_CHECKER_EN bit.

7.4.1.222 PORT_INFRA_CONTROL register

Table 321. PORT_INFRA_CONTROL register - Infrastructure control (address AC00h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	
14	CONFIG_ENABLE	RW	0h	1 = All infrastructure configuration registers can be changed 0 = Config-protected infrastructure configuration registers are write-protected Register list 30.AC06h port_infra_config 30.AFC6h-30.AFD1h xmii register group
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.223 INFRA_ABILITY register

Table 322. INFRA_ABILITY register - Infrastructure abilities (address AC04h)

Bit	Symbol	Access	Value	Description
15	XMII_ABILITY	R	1h	
14:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.224 INFRA_CONFIG register

Table 323. INFRA_CONFIG register - Infrastructure configuration (address AC06h)

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	TEST_ENABLE	RW	0h	1 = Test modes of INFRA sector can be used 0 = Test modes of INFRA sector are blocked

7.4.1.225 INFRA_IRQ_SOURCES register

Table 324. INFRA_IRQ_SOURCES register - Infrastructure interrupt sources (address AC08h)

Bit	Symbol	Access	Value	Description
15	GLOBAL_RESET_IRQ	RW	0h	Top-down propagated port-specific infrastructure reset due to a higher-level reset event. All port-specific infrastructural configuration settings got back to defaults. Write one to clear.
14	FIFO_ERROR	RW	0h	1 = FIFO underflow or overflow. Write one to clear.
13	RGMII_ID_LOCK_ERROR	RW	0h	1 = Lock-error of internal delays of RGMII. Write one to clear.
12:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	FRAME_WARN	R	0h	1 = Warning for Frame Monitors
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.226 INFRA_IRQ_ENABLES register

Table 325. INFRA_IRQ_ENABLES register - Infrastructure interrupt enables (address AC0Ah)

Bit	Symbol	Access	Value	Description
15	GLOBAL_RESET_IRQ	RW	0h	Enable for TOPDOWN_RESET interrupt
14	FIFO_ERROR	RW	0h	Enable for FIFO under/overflow interrupt
13	RGMII_ID_LOCK_ERROR	RW	0h	Enable for RGMII_ID_LOCK_ERROR interrupt
12:2	*reserved*	R	0h	Always write 0, ignore on read.

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Table 325. INFRA_IRQ_ENABLES register - Infrastructure interrupt enables (address AC0Ah)*...continued*

Bit	Symbol	Access	Value	Description
				Always write 0, ignore on read.
1	FRAME_WARN	RW	0h	Enable for frame monitor
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.227 INFRA_IRQ_MSTATUS register

Table 326. INFRA_IRQ_MSTATUS register - Infrastructure interrupt masked-status (address AC0Ch)

Bit	Symbol	Access	Value	Description
15	GLOBAL_RESET_IRQ	R	0h	Masked-status of TOPDOWN_RESET interrupt
14	FIFO_ERROR	R	0h	Masked-status of FIFO under/overflow IRQ
13	RGMII_ID_LOCK_ERROR	R	0h	Masked-status of RGMII_ID_LOCK_ERROR IRQ
12:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	FRAME_WARN	R	0h	Maked status for frame monitors
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.228 MONITOR_CONTROL register

Table 327. MONITOR_CONTROL register - Monitor control (address AC80h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	Reset
14:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.229 MONITOR_STATUS register

Table 328. MONITOR_STATUS register - Monitor status (address AC82h)

Bit	Symbol	Access	Value	Description
15	RX_PREMBL_WARN	RW	0h	1 = Minimum detected received pre-amble length shorter than configured limit Write one to clear.
14	TX_PREMBL_WARN	RW	0h	1 = Minimum detected transmitted pre-amble length shorter than configured limit Write one to clear.
13	RX_IPG_WARN	RW	0h	1 = Minimum detected transmitted IPG length shorter than configured limit Write one to clear.
12	TX_IPG_WARN	RW	0h	1 = Minimum detected transmitted IPG length shorter than configured limit

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Table 328. MONITOR_STATUS register - Monitor status (address AC82h)*...continued*

Bit	Symbol	Access	Value	Description
				Write one to clear.
11:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.230 MONITOR_CONFIG register

Table 329. MONITOR_CONFIG register - Monitor configuration (address AC86h)

Bit	Symbol	Access	Value	Description
15:10	*reserved*	R	0h	Always write 0, ignore on read.
9	LOST_FRAMES_ ENABLE	RW	0h	TX or RX lost frames are frames that are received on XMII or EPHY but could not be transmitted by the EPHY or XMII. Frames are received by XMII but could not transmitted by EPHY. Similarly, frames are received by EPHY but could not transmitted by XMII. If disabled, the counter preserved the last count value, until the counters are enabled again which implicitly resets their values.
8	ALL_FRAMES_ ENABLE	RW	0h	Enable bit to count all frames transmitted on ephy and on xmii here sgmii. If disabled, the counter preserved the last count value, until the counters are enabled again which implicitly resets their values.
7:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	PREAMBLE_ ENABLE	RW	0h	Enable bit to activate monitoring of pre-amble length of transmitted and received frames. If disabled, the corresponding counter minima shows as zero, but the configured limits are preserved.
0	IPG_ENABLE	RW	0h	Enable bit to activate monitoring of IPG length on INGRESS and EGRESS traffic of this port. If disabled, the corresponding counter minima shows as zero, but the configured limits are preserved.

7.4.1.231 TX_IPG_LENGTH register

Table 330. TX_IPG_LENGTH register - Egress Inter-Packet Gap Length (address AC90h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:8	LIMIT	RW	0h	Limit for the minimum allowed IPG length in bytes for the transmitted path. If an IPG length below this linit occurs the TX_IPG_WARN status bit is asserted, implying a FRAME_ WARN_IRQ
7:0	LENGTH	RW	FFh	Minimum detected Inter-Packet Gap length in equivalent bytes in the transmit path.

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7.4.1.232 RX_IPG_LENGTH register

Table 331. RX_IPG_LENGTH register - Ingress Inter-Packet Gap Length (address AC91h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:8	LIMIT	RW	0h	Limit for the minimum allowed IPG length in bytes for the transmitted path. If an IPG length below this linit occurs the TX_IPG_WARN status bit is asserted, implying a FRAME_WARN_IRQ
7:0	LENGTH	RW	FFh	Minimum detected Inter-Packet Gap length in equivalent bytes in the receive path. Value FFh=255d means ≥255 bytes.

7.4.1.233 TX_PREMBEL_COUNT register

Table 332. TX_PREMBEL_COUNT register - Egress Preamble Length (address AC92h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:4	LIMIT	RW	0h	Limit for minimum allowed pre-amble length in bytes for the transmit path. If a pre-amble length below this limit is detected, the TX_PREMBL_WARN bit is asserted, implying a FRAME_WARN_IRQ.
3:0	LENGTH	RW	Fh	Minimum detected pre-amble length in bytes for the transmit path. excluding SFD

7.4.1.234 RX_PREMBEL_COUNT register

Table 333. RX_PREMBEL_COUNT register - Ingress Preamble Length (address AC93h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7:4	LIMIT	RW	0h	Limit for minimum allowed pre-amble length in bytes for the receive path. If a pre-amble length below this limit is detected, the TX_PREMBL_WARN bit is asserted, implying a FRAME_WARN_IRQ.
3:0	LENGTH	RW	Fh	Minimum detected pre-amble length in bytes for the receive path. excluding SFD

7.4.1.235 TX_FRAME_COUNT register

Table 334. TX_FRAME_COUNT register - Egress frame count (address ACA0h)

Bit	Symbol	Access	Value	Description
15:0	COUNT_LSW	R	0h	Bit[15:0] of a 32-bit value for the number of transmitter frames over ephy since this monitoring feature was enabled.

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7.4.1.236 TX_FRAME_COUNT_XTD register

Table 335. TX_FRAME_COUNT_XTD register - Egress frame count extended (address ACA1h)

Bit	Symbol	Access	Value	Description
15:0	COUNT_MSW	R	0h	Bit[31:16] of a 32-bit value for the number of transmitter frames over ephysince this monitoring feature was enabled.

7.4.1.237 RX_FRAME_COUNT register

Table 336. RX_FRAME_COUNT register - Ingress frame count (address ACA2h)

Bit	Symbol	Access	Value	Description
15:0	COUNT_LSW	R	0h	Bit[15:0] of a 32-bit value for the number of transmitter frames over ephy since this monitoring feature was enabled.

7.4.1.238 RX_FRAME_COUNT_XTD register

Table 337. RX_FRAME_COUNT_XTD register - Ingress frame count extended (address ACA3h)

Bit	Symbol	Access	Value	Description
15:0	COUNT_MSW	R	0h	Bit[31:16] of a 32-bit value for the number of transmitter frames over ephy since this monitoring feature was enabled.

7.4.1.239 TX_LOST_FRAMES register

Table 338. TX_LOST_FRAMES register - Egress lost frames (address ACA4h)

Bit	Symbol	Access	Value	Description
15:0	COUNT_LSW	R	0h	Bit[15:0] of a 24-bit value for the number of transmitter lost frames since this monitoring feature was enabled.

7.4.1.240 TX_LOST_FRAMES_XTD register

Table 339. TX_LOST_FRAMES_XTD register - Egress lost frames extended (address ACA5h)

Bit	Symbol	Access	Value	Description
7:0	COUNT_MSW	R	0h	Bit[23:16] of a 24-bit value for the number of transmitter lost frames since this monitoring feature was enabled.

7.4.1.241 RX_LOST_FRAMES register

Table 340. RX_LOST_FRAMES register - Ingress lost frames (address ACA6h)

Bit	Symbol	Access	Value	Description
15:0	COUNT_LSW	R	0h	Bit[15:0] of a 24-bit value for the number of received frames which are lost since this monitoring feature was enabled.

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7.4.1.242 RX_LOST_FRAMES_XTD register

Table 341. RX_LOST_FRAMES_XTD register - Ingress lost frames extended (address ACA7h)

Bit	Symbol	Access	Value	Description
7:0	COUNT_MSW	R	0h	Bit[23:16] of a 24-bit value for the number of received frames which are lost since this monitoring feature was enabled.

7.4.1.243 PORT_MIN_TX_DELAY register

Table 342. PORT_MIN_TX_DELAY register - Minimum transmit path delay (address AD10h)

Bit	Symbol	Access	Value	Description
15:0	NSEC	R	0h	Minimum transmit path delay of this PORT from XMII to MDI in nsec.

7.4.1.244 PORT_MAX_TX_DELAY register

Table 343. PORT_MAX_TX_DELAY register - Maximum transmit path delay (address AD12h)

Bit	Symbol	Access	Value	Description
15:0	NSEC	R	0h	Maximum transmit path delay of this PORT from XMII to MDI in nsec.

7.4.1.245 PORT_MIN_RX_DELAY register

Table 344. PORT_MIN_RX_DELAY register - Minimum receive path delay (address AD14h)

Bit	Symbol	Access	Value	Description
15:0	NSEC	R	0h	Minimum receive path delay of this PORT from XMII to MDI in nsec.

7.4.1.246 PORT_MAX_RX_DELAY register

Table 345. PORT_MAX_RX_DELAY register - Maximum receive path delay (address AD16h)

Bit	Symbol	Access	Value	Description
15:0	NSEC	R	0h	Maximum receive path delay of this PORT from XMII to MDI in nsec.

7.4.1.247 XMII_CONTROL register

Table 346. XMII_CONTROL register - XMII control (address AFC0h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	Reset
14:4	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
3	LOGICAL_ISOLATE	RW	0h	1 = XMII data and control signals become inactive: makes XMII inputs numb and pulls outputs low, except for clock outputs which keep running.

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Table 346. XMII_CONTROL register - XMII control (address AFC0h)*...continued*

Bit	Symbol	Access	Value	Description
2	ELECTRICAL_ ISOLATE	RW	0h	1 = XMII port does not influence the XMII bus/wires states: makes XMII inputs numb for this PORT numb and outputs tri-stated. ***Note:*** *Only applicable for variant A.*
1	MAC_LOOPBACK	RW	0h	Data received on external xMII port is looped on internal GMII, and re-transmitted on external xMII port. 1 = Loopback enabled, 0 = Loopback disabled
0	PHY_LOOPBACK	RW	0h	Data received on internal GMII is looped on xMII, and re-transmitted on internal GMII. 1 = Loopback enabled, 0 = Loopback disabled

7.4.1.248 XMII_STATUS register

Table 347. XMII_STATUS register - XMII status (address AFC2h)

Bit	Symbol	Access	Value	Description
15	TXD_FIFO_ OVERFLOW	RW	0h	1 = Underflow of transmit path FIFO Write one to clear.
14	TXD_FIFO_ UNDERFLOW	RW	0h	1 = Overflow of transmit path FIFO Write one to clear.
13	RXD_FIFO_ OVERFLOW	RW	0h	1 = Underflow of receive path FIFO Write one to clear.
12	RXD_FIFO_ UNDERFLOW	RW	0h	1 = Overflow of receive path FIFO Write one to clear.
11	RGMII_TXID_ LOCK_ERROR	RW	0h	1 = Lock-error of integrated delay in TXC Write one to clear.
10	RGMII_RXID_ LOCK_ERROR	RW	0h	1 = Lock-error of integrated delay in RXC Write one to clear.
9:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.249 XMII_ABILITIES register

Table 348. XMII_ABILITIES register - XMII abilities (address AFC4h)

Bit	Symbol	Access	Value	Description
15	RGMII_ID_ABILITY	R	A: 1h B: 0h	1 = Ability to add Internal Delay in RXC/TXC clock signals for RGMII
14	RGMII_ABILITY	R	A: 1h B: 0h	1 = Ability to support RGMII for this PORT
13:1	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
0	SGMII_ABILITY	R	A: 0h B: 1h	1 = Ability to support SERDES-MII on this PORT. Specific SERDES abilities in AFC5h.

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7.4.1.250 SERDES_MII_ABILITIES register

Table 349. SERDES_MII_ABILITIES register - SERDES MII abilities (address AFC5h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	SGMII_ABILITY	R	A: 0h B: 1h	1 = SGMII is available on this PORT
0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

7.4.1.251 MII_BASIC_CONFIG register

Table 350. MII_BASIC_CONFIG register - XMII basic configuration (address AFC6h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7	ISOLATE_ON_LINKDROP	RW	0h	1= Logical isolate XMII on link drop with bit AFC0h.3 gets set automatically.
6:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4	ROLE	RW	0h	1 = MAC 0 = PHY
3:0	XMII_MODE	R	A: 7h B: 9h	3:0 = CMOS / SERDES 0000 = Reserved, 1000 = Reserved 0001 = Reserved, 1001 = SGMII 0010 = Reserved, 1010 = Reserved 0011 = Reserved, 1010 = Reserved 0100 = MII, 1100 = Reserved 0101 = RMII, 1101 = Reserved 0110 = Reserved, 1110 = Reserved 0111 = RGMII, 1111 = Reserved

7.4.1.252 XMII_CLK_CONFIG register

Table 351. XMII_CLK_CONFIG register - XMII Clock Configuration (address AFC8h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	CLOCK_DISABLE	RW	0h	1 = XMII clock outputs disabled 0 = XMII clock outputs enabled
13:0	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.

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7.4.1.253 XMII_CLK_IO_CONFIG register

Table 352. XMII_CLK_IO_CONFIG register - XMII Clock IO configuration (address AFC9h)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of of XMII DATA IO cells 0 = Auto-configration of XMII DATA IO cells: values in this register are ignored.
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Manual drive-strength setting of IO-cells used as CLOCK output for selected CMOS XMII mode 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Select slew rate of CLOCK output IO cells used for the configured CMOS xMII mode. 0 = slowest 3 = fastest

7.4.1.254 XMII_DATA_CONFIG register

Table 353. XMII_DATA_CONFIG register - XMII DATA configuration (address AFCAh)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	TXD30_SWAP	RW	0h	1 = TXD[0:3] TXD Bus swap 0 = TXD[3:0] default
13:12	TXCTL_SELECT	RW	0h	00 = TXCTL next to TXC default 01 = TXCTL between TXD[3:2] and TXD[1:0] 11 = TXCTL at opposite side of TXD[3:0] 10 = Reserved
11:8	*reserved*	R	0h	Always write 0, ignore on read.
7	RXD_DISABLE	RW	0h	1 = All XMII RXD and XMII RX_CTK outputs drive a low value Except RXC 0 = All XMII RXD and XMII RX_CTK outputs in normal operation
6	RXD30_SWAP	RW	0h	1 = RXD[3:0] becomes RXD[0:3] 0 = RXD[0] closest to RXC
5:4	RXCTL_SELECT	RW	0h	00 = RXCTL next to RXC default 01 = RXCTL between RX[3:2] and RXD[1:0] 11 = RXCTL at other side of RXD[3:0] 10 = Reserved
3:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.4.1.255 XMII_DATA_IO_CONFIG register

Table 354. XMII_DATA_IO_CONFIG register - XMII DATA IO configuration (address AFCBh)

Bit	Symbol	Access	Value	Description
15	MANUAL_CONFIG	RW	0h	1 = Manual config of of XMII DATA IO cells 0 = Auto-configration of XMII DATA IO cells: values in this register are ignored.
14:7	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
6:4	DRIVE_STRENGTH	RW	0h	Select drive strength of DATA output IO cells used for the configured CMOS xMII mode. 0 = weakest ~400 Ohm slow-corner 7 = strongest ~50 Ohm slow-corner
3:0	SLEW_RATE	RW	0h	Select slew rate of CLOCK output IO cells used for the configured CMOS xMII mode. 0 = slowest 3 = fastest

7.4.1.256 RGMII_TXC_DELAY_CONFIG register

Table 355. RGMII_TXC_DELAY_CONFIG register - RGMII TXC delay configuration (address AFCCh)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = Internal TXC delay enabled and applied, 0 = Internal TXC delay disabled and bypassed
14:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4:0	PHASE_SHIFT	RW	12h	The effective Data to Clock Skew ID as per below equation Phase-shift = 73.8 + 0.9*PHASE_SHIFT[4:0] degrees

7.4.1.257 RGMII_RXC_DELAY_CONFIG register

Table 356. RGMII_RXC_DELAY_CONFIG register - XMII RXC delay configuration (address AFCDh)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1 = Internal RXC delay enabled and applied, 0 = Internal RXC delay disabled and bypassed
14:5	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
4:0	PHASE_SHIFT	RW	12h	The effective Data to Clock Skew ID as per below equation Phase-shift = 73.8 + 0.9*PHASE_SHIFT[4:0] degrees

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7.4.1.258 SGMII_BASIC_CONTROL register

Table 357. SGMII_BASIC_CONTROL register - SGMII PCS register: SR MII MMD Control Register. (address B000h)

Bit	Symbol	Access	Value	Description
15	RST	RW	0h	Soft Reset. When the host sets this bit, this triggers the software reset process in which all internal blocks are reset, except the APB interface block. The registers are reset to their default values. When this bit is set, it also resets the PMA & PMD. This bit is self-cleared after 32 clock cycles of the APB bus clock. Type: Self Clearing.
14	LBE	RW	0h	Not used
13	SS13	RW	0h	Speed Selection (LSB). This bit, along with the SS6 bit of this register, indicates the speed. • SS6 SS13 -> Speed • 1 0 -> 1000 Mbps • 0 1 -> 100 Mbps • 0 0 -> 10 Mbps
12	AN_ENABLE	RW	1h	Enable Auto-Negotiation. When set to 1, this bit enables the Clause 37 auto-negotiation process.
11	LPM	RW	0h	Power-Down Mode. This bit controls the powerdown mode of the PCS. • 0: Normal operation • 1: The PCS goes to the powerdown mode When the host clears this bit, the PCS resumes the normal operation.
10	*reserved*	R	0h	Always write 0, ignore on read.
9	RESTART_AN	RW	0h	Restart Auto-Negotiation. When the host writes this bit, the PCS initiates the auto-negotiation process. This bit is used to restart the auto-negotiation which is already initiated by setting AN_ENABLE. The PCS clears this bit after restarting the autonegotiation. Type: Self Clearing.
8	DUPLEX_MODE	RW	1h[1] 0h[2]	Duplex Mode. This bit specifies the duplex mode of the PCS. • 0: Half duplex • 1: Full duplex If AN_ENABLE is set to 0, this bit determines the PHY link duplex mode. If AN_ENABLE is set to 1, then the PHY link duplex mode is independent of this bit (although the host can write any value) and is determined by the outcome of the Clause 37 auto-negotiation process. Note that only full duplex mode is supported.
7	*reserved*	R	0h	Always write 0, ignore on read.
6	SS6	RW	1h	Speed Selection (MSB).

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Table 357. SGMII_BASIC_CONTROL register - SGMII PCS register: SR MII MMD Control Register. (address B000h)*...continued*

Bit	Symbol	Access	Value	Description
				This bit, along with the SS13 bit of this register, indicates the speed. For more information, see description of the SS13 bit.
5:0	*reserved*	R	0h	Always write 0, ignore on read.

[1] Default value in managed operation
[2] Default value in autonomous operation and in managed operation once SGMII_ADVANCED_CONTROL.START_OPERATION is set.

7.4.1.259 SGMII_BASIC_STATUS register

Table 358. SGMII_BASIC_STATUS register - SGMII PCS register: SR MII MMD Status Register. (address B001h)

Bit	Symbol	Access	Value	Description
15	ABL100T4	R	0h	100BASE-T4 Ability. Always returns 0 because this functionality is not supported.
14	FD100ABL	R	0h	100BASE-X Full-Duplex Ability. Always returns 0 because this functionality is not supported.
13	HD100ABL	R	0h	100BASE-X Half-Duplex Ability. Always returns 0 because this functionality is not supported.
12	FD10ABL	R	0h	10 Mbps Full-Duplex Ability. Always returns 0 because this functionality is not supported.
11	HD10ABL	R	0h	10 Mbps Half-Duplex Ability. Always returns 0 because this functionality is not supported.
10	FD100T	R	0h	100BASE-T2 Full-Duplex Ability. Always returns 0 because this functionality is not supported.
9	HD100T	R	0h	100BASE-T2 Half-Duplex Ability. Always returns 0 because this functionality is not supported.
8	EXT_STS_ABL	R	1h	Extended Status Information. • 0: No Extended Status information is present at register address 0x000F of this MMD device • 1: Extended Status information is present at register address 0x000F of this MMD device Always returns 1.
7	UN_DIR_ABL	R	0h	Unidirectional Ability. • 0: The PCS is able to transmit frames only when the device has determined the valid link • 1: The PCS is able to transmit frames irrespective of whether device has determined the valid link or not
6	*reserved*	R	0h	Ignore on read.
5	AN_CMPL	R	0h	Auto-negotiation Complete. • 0: The AN process is not completed • 1: The AN process is completed When this bit is set to 1, the contents of the AN MMD Advertizement, AN MMD Link partner Ability, and AN MMD

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Table 358. SGMII_BASIC_STATUS register - SGMII PCS register: SR MII MMD Status Register. (address B001h)*...continued*

Bit	Symbol	Access	Value	Description
				Expansion registers are valid. This bit returns 0 if AN_ENABLE is set to 0.
4	RF	R	0h	Remote Fault. When set to 1, this bit indicates that the receive link of the link partner is down. This bit is set based on the auto-negotiated (1000BASE-X auto-negotiation) information from the link partner. • 0: The PCS did not detect a remote fault • 1: The PCS detected a remote fault Note that this bit is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0) because remote fault information is part of the exchanged base page. When using SGMII autonegotiation (PCS_MODE=0x2) this bit must be ignored since remote fault information is not exchanged. Type: Latch High.
3	AN_ABL	R	1h	Auto-negotiation Ability. Always returns 1 because the PCS is able to perform auto-negotiation.
2	LINK_STS	R	0h	Link Status. When set to 1, it indicates that the Rx link is up. If the link goes down, it is latched until the host performs the read operation to this register. • 0: Link Down • 1: Link Up Type: Latch Low. ***Note:*** *Read this bit twice to get the current link status.*
1	*reserved*	R	0h	Ignore on read.
0	EXT_REG_CAP	R	1h	Extended Register Capability. • 0: Extended Register capability does not exist • 1: Extended Register capability exists Always returns 1.

7.4.1.260 SGMII_SR_MII_DEV_ID1 register

Table 359. SGMII_SR_MII_DEV_ID1 register - SGMII PCS register (address B002h)

Bit	Symbol	Access	Value	Description
15:0	VS_MII_DEV_OUI_3_18	R	0h	Organizationally Unique Identifier[3:18]. This field contains Bits[18:3] of 24-bit OUI of device manufacturer.

7.4.1.261 SGMII_SR_MII_DEV_ID2 register

Table 360. SGMII_SR_MII_DEV_ID2 register - SGMII PCS register (address B003h)

Bit	Symbol	Access	Value	Description
15:10	VS_MMD_DEV_OUI_19_24	R	0h	Organizationally Unique Identifier[19:24].

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Table 360. SGMII_SR_MII_DEV_ID2 register - SGMII PCS register (address B003h)*...continued*

Bit	Symbol	Access	Value	Description
				This field contains Bits[24:19] of 24-bit OUI of device manufacturer.
9:4	VS_MMD_DEV_MMN_5_0	R	0h	Model Number. This field contains the 6-bit Model number of the device.
3:0	VS_MMD_DEV_RN_3_0	R	0h	Revision Number. This field contains the 4-bit Revision number of the device.

7.4.1.262 SGMII_AUTONEG_ADVERTISE register

Table 361. SGMII_AUTONEG_ADVERTISE register - SGMII PCS register: SR MII MMD AN Advertisement Register. (address B004h)

Bit	Symbol	Access	Value	Description
15	NP	R	0h	Next Page / SGMII Link Status. Next Page information is not exchanged. Instead, in SGMII PHY mode (TX_CONFIG=1), this field is read-only and indicates the PHY's Link Status that is communicated (advertised) to the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification. The value that is advertised depends on the setting of PHY_MODE_CTRL and bit 4 (SGMII_LINK_STS) of VR_MII_AN_CTRL.
14	*reserved*	R	0h	Always write 0, ignore on read.
13:12	RF	RW	0h	Remote Fault / SGMII Duplex Mode. This field indicates the fault signaling of the local device to be communicated (advertised) to the link partner. • 00: No Error • 01: Offline • 10: Link Failure • 11: Auto-negotiation Error Note that this field is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) remote fault information is not exchanged. Instead, in SGMII PHY mode (TX_CONFIG=1), bit 12 is read-only and indicates the duplex mode of operation that is communicated (advertised) to the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification. The value that is advertised depends on the setting of PHY_MODE_CTRL and bit 5 (FD) of SR_MII_AN_ADV. • 0: Half Duplex • 1: Full Duplex
11:9	*reserved*	R	0h	Always write 0, ignore on read.
8:7	PAUSE	RW	0h	Pause Ability. This field indicates the Pause ability of the local device to be communicated (advertised) to the link partner. Software can program suitable values based on the capability of the MAC. • 00: No Pause • 01: Asymmetric Pause towards the link partner

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Table 361. SGMII_AUTONEG_ADVERTISE register - SGMII PCS register: SR MII MMD AN Advertisement Register. (address B004h)*...continued*

Bit	Symbol	Access	Value	Description
				• 10: Symmetric Pause • 11: Symmetric Pause and Asymmetric Pause towards the local device Note that this field is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) Pause Ability information is not exchanged.
6	HD	RW	0h	Half Duplex. When this bit is set, it indicates that the device can operate in the half-duplex mode. Note: Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored. Half-duplex operation is not supported at chip level.
5	FD	RW	1h	Full Duplex. When this bit is set, it indicates that the device can operate in the full-duplex mode. Note: Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored.
4:0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.263 SGMII_AUTONEG_LP_ABILITY register

Table 362. SGMII_AUTONEG_LP_ABILITY register - SGMII PCS register: SR MII MMD AN Link Partner Base Ability Register. (address B005h)

Bit	Symbol	Access	Value	Description
15	LP_NP	R	0h	Link Partner Next Page / Link Partner Link Status. This bit indicates that the link partner can handle Next Page. To exchange information through Next Page, both devices (local and remote) should have the capability to handle Next Page. The PCS does not support Next Page (see SR_MII_AN_ADV.NP) therefore, the Next Page exchange does not happen. Note: Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) Next Page information is not exchanged and this field must be ignored. In SGMII auto-negotiation this field is used to indicate the Link Status of the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification.
14	LP_ACK	R	0h	Link Partner Acknowledge. ACK bit from the Link Partner. This bit indicates that the link partner has successfully received the page sent by the local device.
13:12	LP_RF	R	0h	Link Partner Remote Fault / Link Partner Duplex Mode.

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Table 362. SGMII_AUTONEG_LP_ABILITY register - SGMII PCS register: SR MII MMD AN Link Partner Base Ability Register. (address B005h)*...continued*

Bit	Symbol	Access	Value	Description
				This field indicates the fault signaling of the link partner. • 00: No Error • 01: Offline • 10: Link Failure • 11: Auto-negotiation Error Note: Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) remote fault information is not exchanged and this field must be ignored. In SGMII auto-negotiation bit 12 of this field is used to indicate the duplex mode of operation of the SGMII link partner, in line with the definition of the tx_config_Reg in the SGMII specification.
11:9	*reserved*	R	0h	Ignore on read.
8:7	LP_PAUSE	R	0h	Link Partner Pause Ability. This field indicates the Pause ability of the link partner. • 00: No Pause • 01: Asymmetric Pause towards the link partner • 10: Symmetric Pause • 11: Both Symmetric Pause and Asymmetric Pause towards the local device Note that this field is only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) Pause Ability information is not exchanged.
6	LP_FD	R	0h	Link Partner Half Duplex. When this bit is set, it indicates that the link partner is capable of operating in the halfduplex mode. Note: Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored.
5	LP_HD	R	0h	Link Partner Full Duplex. When this bit is set, it indicates that the link partner is capable of operating in the full-duplex mode. Note: Only valid when using 1000BASE-X auto-negotiation (PCS_MODE=0x0). When using SGMII autonegotiation (PCS_MODE=0x2) duplex information is not exchanged and this field must be ignored.
4:0	*reserved*	R	0h	Ignore on read.

7.4.1.264 SGMII_AUTONEG_EXPANSION register

Table 363. SGMII_AUTONEG_EXPANSION register - SGMII PCS register: SR MII MMD AN Expansion Register. (address B006h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Ignore on read.
2	LD_NP_ABL	R	0h	Local Device Next Page Able.

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Table 363. SGMII_AUTONEG_EXPANSION register - SGMII PCS register: SR MII MMD AN Expansion Register. (address B006h)*...continued*

Bit	Symbol	Access	Value	Description
				• 0: The local device does not have the next page ability • 1: The local device has the next page ability The PCS always returns this bit as 0 because it does not support Next Page.
1	PG_RCVD	R	0h	Page Received. This bit indicates that the local device received a page from the link partner. • 0: The local device did not receive a new page • 1: The local device received a new page Type: Latch High.
0	*reserved*	R	0h	Ignore on read.

7.4.1.265 SGMII_EXTENDED_STATUS register

Table 364. SGMII_EXTENDED_STATUS register - SGMII PCS register: SR MII MMD Extended Status Register. (address B00Fh)

Bit	Symbol	Access	Value	Description
15	CAP_1G_X_FD	R	1h	1000BASE-X Full-Duplex Capable. • 0: Not 1000BASE-X full-duplex capable • 1: 1000BASE-X full-duplex capable The PCS always returns this bit as 1 because it does support this feature.
14	CAP_1G_X_HD	R	1h	1000BASE-X Half-Duplex Capable. • 0: Not 1000BASE-X half-duplex capable • 1: 1000BASE-X half-duplex capable The PCS always returns this bit as 1 because in principle it does have the capability to support a half-duplex MAC. However, it does not indicate actual half-duplex operation, this is not supported at chip level.
13	CAP_1G_T_FD	R	0h	1000BASE-T Full-Duplex Capable. • 0: Not 1000BASE-T full-duplex capable • 1: 1000BASE-T full-duplex capable The PCS always returns this bit as 0 because it does not support this feature.
12	CAP_1G_T_HD	R	0h	1000BASE-T Half-Duplex Capable. • 0: Not 1000BASE-T half-duplex capable • 1: 1000BASE-T half-duplex capable The PCS always returns this bit as 0 because it does not support this feature.
11:0	*reserved*	R	0h	Ignore on read.

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7.4.1.266 SGMII_ADVANCED_CONTROL register

Table 365. SGMII_ADVANCED_CONTROL register - Advanced control (address B010h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	CONFIG_ENABLE	RW	0h	SGMII PHY configuration register write protection 0* = SGMII PHY configuration registers cannot be changed 1 = SGMII PHY configuration registers can be changed
13:2	*reserved*	R	0h	Always write 0, ignore on read.
1	GOTO_STANDBY	RW	0h	Goto standby mode The transition to STANDBY mode needs 100 µs delay and no SMI transaction should be issued toward SGMII 0* = no action 1 = go to standby
0	START_ OPERATION	RW	0h	Start operation by exiting standby 0* = no action 1 = exit standby

7.4.1.267 SGMII_ADVANCED_STATUS register

Table 366. SGMII_ADVANCED_STATUS register - Advanced status (address B011h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Ignore on read.
2	LINK_STATUS	R	0h	This bit represents the actual link_status value of the SGMII PHY.
1:0	*reserved*	R	0h	Ignore on read.

7.4.1.268 SGMII_ADVANCED_CONFIG register

Table 367. SGMII_ADVANCED_CONFIG register - Advanced configuration (address B013h)

*Legend: *reset value*

Bit	Symbol	Access	Value	Description
15:1	*reserved*	R	0h	Always write 0, ignore on read.
0	AUTO	RW	0h[1] 1h[2]	Advanced configuration 0* = Phy pauses in STANDBY at startup until the START_ OPERATE bit is set. 1 = PHy autonomously prodeeds to Normal mode for both startup and wake-up.

[1] Default value in managed operation
[2] Default value in autonomous operation

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7.4.1.269 SGMII_IRQ_SOURCES register

Table 368. SGMII_IRQ_SOURCES register - SGMII interrupt source (address B014h)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET	RW	0h	Top-down SGMII reset due to global reset event. All configuration settings return to defaults. This bit is W1C (Cleared on W1). • 0* = no topdown reset since last clear • 1 = topdown reset since last clear
14:5	*reserved*	R	0h	Always write 0, ignore on read.
4	FIFO_OVF	RW	0h	SGMII PCS RX FIFO overflow. This bit is W1C (self-clearing when W1). • 0* = no RX FIFO overflow since last clear • 1 = RX FIFO overflow since last clear
3	FIFO_UNF	RW	0h	SGMII PCS RX FIFO underflow. This bit is W1C (self-clearing when W1). • 0* = no RX FIFO underflow since last clear • 1 = RX FIFO underflow since last clear
2	AUTONEG_EVENT	R	0h	First read B001h 5 to check Auto-Negotiation has completed. If so, the contents of the AN MMD Advertisement,AN MMD Link partner Ability, and AN MMD Expansion registers are valid. This bit is W1C (Cleared on W1) • 0* = no Auto-Negotiation process (re)start or completion event • 1 = Auto-Negotiation process has been (re)started or completed
1	RX_LINK_STATUS_EVENT	RW	0h	The actual SGMII link status is available in register B011h 2. This bit is W1C (self-clearing when W1). • 0* = This link did not change since last clear • 1 = This link status changed since last clear
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.270 SGMII_IRQ_ENABLES register

Table 369. SGMII_IRQ_ENABLES register - SGMII interrupt enables (address B015h)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_RESET_ENABLE	RW	0h	If set, the respective IRQ is enabled. This setting is preserved during deep sleep.
14 to 5	*reserved*	RW	0h	Always write 0, ignore on read.
4	FIFO_OVF_ENABLE	RW	0h	If set, the respective IRQ is enabled. This setting is preserved during deep sleep.
3	FIFO_UNF_ENABLE	RW	0h	If set, the respective IRQ is enabled. This setting is preserved during deep sleep.
2	AUTONEG_EVENT_ENABLE	RW	0h	If set, the respective IRQ is enabled. This setting is preserved during deep sleep.
1	RX_LINK_STATUS_EVENT_ENABLE	RW	0h	If set, the respective IRQ is enabled. This setting is preserved during deep sleep.

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Table 369. SGMII_IRQ_ENABLES register - SGMII interrupt enables (address B015h)*...continued*

Bit	Symbol	Access	Value	Description
0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.271 SGMII_IRQ_MSTATUS register

Table 370. SGMII_IRQ_MSTATUS register - SGMII interrupt mask status. This is AND of IRQ_SOURCE and IRQ_ENABLE. (address B016h)

Bit	Symbol	Access	Value	Description
15	TOPDOWN_ RESET_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to TOPDOWN_RESET_IRQ.
14:5	*reserved*	R	0h	Ignore on read.
4	FIFO_OVF_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to FIFO_OVF_IRQ.
3	FIFO_UNF_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to FIFO_UNF_IRQ.
2	AUTONEG_ EVENT_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to AUTONEG_EVENT_IRQ.
1	RX_LINK_STATUS_ EVENT_MSTS	R	0h	This is AND of IRQ_SOURCE and IRQ_ENABLE related to RX_LINK_STATUS_EVENT_IRQ.
0	*reserved*	R	0h	Ignore on read.

7.4.1.272 SGMII_LED_TRIGGER0 register

Table 371. SGMII_LED_TRIGGER0 register - SGMII LED trigger 0 (address B040h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	If set, the trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Reserved 1 = Link_status 2 = Role: MAC(0) or PHY(1) 3 = PHY active (AFE enabled) 4 = Frame reception (RX_DV = true) 5 = Symbol error 6 = Frame transmission (TX_EN=true) 7 = Frame activity (RX_DV=true | TX_EN=true)

7.4.1.273 SGMII_LED_TRIGGER1 register

Table 372. SGMII_LED_TRIGGER1 register - see SGMII_LED_TRIGGER0 (address B041h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	If set, the trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.

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Table 372. SGMII_LED_TRIGGER1 register - see SGMII_LED_TRIGGER0 (address B041h)*...continued*

Bit	Symbol	Access	Value	Description
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Reserved 1 = Link_status 2 = Role: MAC(0) or PHY(1) 3 = PHY active (AFE enabled) 4 = Frame reception (RX_DV = true) 5 = Symbol error 6 = Frame transmission (TX_EN=true) 7 = Frame activity (RX_DV=true | TX_EN=true)

7.4.1.274 SGMII_LED_TRIGGER2 register

Table 373. SGMII_LED_TRIGGER2 register - see SGMII_LED_TRIGGER0 (address B042h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	If set, the trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Reserved 1 = Link_status 2 = Role: MAC(0) or PHY(1) 3 = PHY active (AFE enabled) 4 = Frame reception (RX_DV = true) 5 = Symbol error 6 = Frame transmission (TX_EN=true) 7 = Frame activity (RX_DV=true | TX_EN=true)

7.4.1.275 SGMII_LED_TRIGGER3 register

Table 374. SGMII_LED_TRIGGER3 register - see SGMII_LED_TRIGGER0 (address B043h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	If set, the trigger is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0 = Reserved 1 = Link_status 2 = Role: MAC(0) or PHY(1) 3 = PHY active (AFE enabled) 4 = Frame reception (RX_DV = true) 5 = Symbol error 6 = Frame transmission (TX_EN=true) 7 = Frame activity (RX_DV=true | TX_EN=true)

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7.4.1.276 SGMII_PCS_CONTROL register

Table 375. SGMII_PCS_CONTROL register - SGMII PCS register (address B070h)

Bit	Symbol	Access	Value	Description
15	VR_RST	RW	0h	Vendor-Specific Soft Reset. When the host sets this bit, this triggers the vendor-specific software reset process in which all internal blocks are reset, except the APB interface block and PCS registers. When this bit is set, it does reset the PMA & PMD registers. This bit is self-cleared after 32 clock cycles of the APB bus clock. Type: Self Clearing.
14	R2TLBE	RW	0h	PCS Rx to Tx Loopback Enable. This bit controls the PCS loopback path from the GMII Rx to the GMII Tx at the GMII interface. • 0 = Loopback path is disabled • 1 = Loopback path is enabled
13	*reserved*	R	1h[1] 0h[2]	Always write 0, ignore on read.
12	CL37_BP	R	0h	Enable Clause 37 AN in Backplane Configuration. The PCS always returns this bit as 0 because it does not support this feature.
11:10	*reserved*	R	0h	Always write 0, ignore on read.
9	MAC_AUTO_SW	RW	0h[1] 1h[2]	Automatic Speed Mode Change after SGMII CL37 AN Completion. This bit controls the automatic change of the PCS speed and duplex mode settings after completion of SGMII Clause 37 auto-negotiation. • 0 = The PCS will not automatically change the speed/ duplex mode and operate at the speed/duplex mode according to the values programmed to SR MII MMD Control Register (SGMII_BASIC_CONTROL). After the completion of CL37 AN, the application has to read the negotiated Speed/Duplex Mode from the VR MII MMD AN Interrupt and Status Register (SGMII_AUTONEG_STATUS) and then program the SR MII MMD Control Register (SGMII_BASIC_CONTROL) appropriately. • 1 = The PCS automatically switches to the negotiated SGMII speed/duplex mode settings, after the completion of Clause 37 auto-negotiation. This mode is valid only when PCS is configured as MAC-side SGMII (TX_CONFIG=0 and PCS_MODE set to the SGMII) and should be set only when auto-negotiation is enabled (AN_ENABLE=1). ***Note:*** *Only PCS speed/duplex mode settings are impacted and the application must still adjust other chip level speed/ duplex mode settings appropriately.*
8	INIT	RW	0h	Datapath Initialization Control. This bit can be set to flush/initialize the various FIFOs implemented in the PCS. This is a self-clearing bit. After writing a 1 to this bit, the application should poll this bit continuously and proceed to any other operation only after reading this bit as 0. ***Note:*** *When this bit is programmed to 1, the RXFIFO_OVF/ RXFIFO_UNF bits of VR MII MMD Digital Status Register*

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Table 375. SGMII_PCS_CONTROL register - SGMII PCS register (address B070h)*...continued*

Bit	Symbol	Access	Value	Description
				(SGMII_PCS_STATUS) can get set incorrectly. Hence, these register bits must be read so that they get cleared. Type: Self Clearing.
7	MSK_RD_ERR	R	0h	Mask Running Disparity Error. The PCS always returns this bit as 0 because it does not support this feature.
6	PRE_EMP	RW	0h	preemption Packet Enable. This bit must be set to 1 to allow the PCS to properly receive/ transmit preemption packets in SGMII 10M/100M Modes.
5	*reserved*	R	0h	Always write 0, ignore on read.
4	DTXLANED_0	RW	0h	Disable Tx Lane of the PMD. This bit should always be set to 0 as the PCS does not support this feature.
3	*reserved*	RW	0h	Always write 0, ignore on read.
2	EN_2_5G_MODE	RW	0h	Enable 2.5G SGMII Mode. This bit should be set to 1 to enable 2.5G SGMII mode of operation. • 0 = 2.5G SGMII mode is disabled and 1000M/100M/10M operation is supported • 1 = 2.5G SGMII mode is enabled and only 2500M operation is supported ***Note:*** *This bit must be set to 0 to allow SGMII to operate in 10/100/1000M.*
1	BYP_PWRUP	R	0h	Reserved
0	PHY_MODE_CTRL	RW	0h[1] 1h[2]	PHY mode AN advertisement control. This bit controls the Clause 37 auto-negotiation advertisement when operating in SGMII PHY mode. • 0 = During SGMII autonegotiation the PCS advertises the values programmed to SGMII_LINK_STS of VR MII MMD AN Control Register (SGMII_PCS_CONFIG2), SS13 and SS6 of SR MII MMD Control Register (SGMII_BASIC_ CONTROL) and FD of SR MII MMD AN Advertisement Register (SGMII_ AUTONEG_ADVERTISE). • 1 = During SGMII autonegotiation the PCS advertises the values of internal hardware input ports, controlled by chip level settings, in the SR MII MMD AN Advertisement Register (SGMII_AUTONEG_ADVERTISE). ***Note:*** *This bit should be set only when the PCS is configured as PHY-side SGMII i.e. TX_CONFIG=1.*

[1] Default value in managed operation
[2] Default value in autonomous operation and in managed operation once SGMII_ADVANCED_CONTROL.START_OPERATION is set.

7.4.1.277 SGMII_PCS_STATUS register

Table 376. SGMII_PCS_STATUS register - SGMII PCS register (address B071h)

Bit	Symbol	Access	Value	Description
15 to 7	*reserved*	R	0h	ignore on read.

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Table 376. SGMII_PCS_STATUS register - SGMII PCS register (address B071h)*...continued*

Bit	Symbol	Access	Value	Description
6	RXFIFO_OVF	R	0h	Rx FIFO Overflow. This bit indicates overflow of the clock rate compensation FIFO. • 0 = Normal operation • 1 = FIFO overflow Type: Latch High.
5	RXFIFO_UNDF	R	0h	Rx FIFO Underflow. This bit indicates underflow of the clock rate compensation FIFO. • 0 = Normal operation • 1 = FIFO overflow Type: Latch High.
4 to 0	*reserved*	R	0h	Always write 0, ignore on read.

7.4.1.278 SGMII_PCS_CONFIG1 register

Table 377. SGMII_PCS_CONFIG1 register - SGMII PCS register (address B073h)

Bit	Symbol	Access	Value	Description
15:5	*reserved*	R	0h	Always write 0, ignore on read.
4	TX_POL_INV_0	RW	0h[1] 1h[2]	Tx Polarity Invert When set, the data polarity on the Tx differential lines is reversed/inverted. Can be used to swap TXP and TXN. • 0 = Do not invert data polarity on Tx differential lines • 1 = Invert data polarity on Tx differential lines
3:1	*reserved*	R	0h	Always write 0, ignore on read.
0	RX_POL_INV_0	RW	0h	Rx Polarity Invert When set, the data received from Rx serial line is inverted. Can be used to swap RXP and RXN. • 0 = Do not invert data polarity of data received from Rx differintial lines. • 1 = Invert data polarity of data received from Rx differential lines.

[1] Default value in managed operation
[2] Default value in autonomous operation and in managed operation once SGMII_ADVANCED_CONTROL.START_OPERATION is set.

7.4.1.279 SGMII_PCS_CONFIG2 register

Table 378. SGMII_PCS_CONFIG2 register - SGMII PCS register (address B074h)

Bit	Symbol	Access	Value	Description
15:9	*reserved*	R	0h	Always write 0, ignore on read.
8	MII_CTRL	RW	0h[1] 1h[2]	SGMII MII Control. This bit controls the width of the internal MII interface when operating at SGMII speed modes of 10 Mbps or 100 Mbps. • 0 = 4-bit MII • 1 = 8-bit MII This bit must always be set to 1 for proper operation of the internal MII interface.
7:5	*reserved*	R	0h	Always write 0, ignore on read.

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Table 378. SGMII_PCS_CONFIG2 register - SGMII PCS register (address B074h)*...continued*

Bit	Symbol	Access	Value	Description
4	SGMII_LINK_STS	RW	0h	SGMII Link Status. This bit is used in Bit 15 of the tx_config_Reg during SGMII Clause 37 auto-negotiation when operating in SGMII PHYside. • 0 = Link Down • 1 = Link Up This mode is relevant only when PCS is configured as PHY-side SGMII (TX_CONFIG=1 and PCS_MODE set to the SGMII) and when auto-negotiation is enabled (AN_ ENABLE=1).
3	TX_CONFIG	RW	0h	SGMII Transmit Configuration Register select. This bit controls the tx_config_Reg values to be used during the Clause 37 auto-negotiation in SGMII mode. • 0 = Configures the PCS as MAC-side SGMII and uses MAC-to-PHY tx_config_Reg values for CL37 AN • 1 = Configures the PCS as PHY-side SGMII and uses PHY-to-MAC tx_config_Reg values for CL37 AN This mode is relevant only when PCS is configured as SGMII (PCS_MODE set to the SGMII) and when auto-negotiation is enabled (AN_ENABLE=1).
2:1	PCS_MODE	RW	0h[1] 2h[2]	PCS Mode. This field controls the PCS auto-negotiation (and operating) mode. • 00 = 1000BASE-X mode (Clause 37 autonegotiation with tx_ config_Reg according to IEEE802.3 Clause 37 and 1000Base-X) • 01 = Reserved • 10 = SGMII mode (Clause 37 auto-negotiation with tx_ config_ Reg according to the SGMII specification) • 11 = Reserved
0	*reserved*	R	0h[1] 1h[2]	Always write 1, ignore on read.

[1] Default value in managed operation
[2] Default value in autonomous operation and in managed operation once SGMII_ADVANCED_CONTROL.START_OPERATION is set.

7.4.1.280 SGMII_AUTONEG_STATUS register

Table 379. SGMII_AUTONEG_STATUS register - SGMII PCS register (address B075h)

Bit	Symbol	Access	Value	Description
15:7	*reserved*	R	0h	Ignore on read.
6	LP_CK_STP	R	0h	Link Partner EEE Clock Stop Capability. The PCS always returns this bit as 0 because it does not support this feature.
5	LP_EEE_CAP	R	0h	Link Partner EEE Capability. The PCS always returns this bit as 0 because it does not support this feature.
4:1	CL37_ANSGM_STS	R	0h	Clause 37 AN SGMII Status. This field indicates the status received from remote link after the SGMII autonegotiation is complete. • CL37_ANSGM_STS[0] - 0 = Half Duplex

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Table 379. SGMII_AUTONEG_STATUS register - SGMII PCS register (address B075h)*...continued*

Bit	Symbol	Access	Value	Description
				• CL37_ANSGM_STS[0] - 1 = Full Duplex • CL37_ANSGM_STS[2:1] - 00 = 10 Mbps speed link • CL37_ANSGM_STS[2:1] - 01 = 100 Mbps speed link • CL37_ANSGM_STS[2:1] - 10 = 1000 Mbps speed link • CL37_ANSGM_STS[2:1] - 11 = Reserved • CL37_ANSGM_STS[3] - 0 = Link is Down • CL37_ANSGM_STS[3] - 1 = Link is Up This field is valid only when PCS is configured as SGMII (PCS_MODE set to the SGMII) and the autonegotiation is complete.
0	CL37_ANCMPLT_EVENT	R	0h	Clause 37 AN Complete Event. The PCS sets this bit when Clause 37 autonegotiation is complete. The host must clear this bit by writing 0 to it. Type: Self Set / Write Clear.

7.5 Shared

7.5.1 Register Description

Table 380. Register overview: Shared

Address	Name	Access	Reset	Description
2h	PHY_IDENTIFIER1	R	1Bh	PHY identifier MMD device identifier MSW
3h	PHY_IDENTIFIER2	R	B030h	PHY identifier MMD device identifier LSW
5h	DEVICES_IN_PACKAGE1	R	Bh	Devices in package LSW
6h	DEVICES_IN_PACKAGE2	R	4000h	Devices in package MSW
Eh	PACKAGE_IDENTIFIER1	R	1Bh	PHY package identifier MSW
Fh	PACKAGE_IDENTIFIER2	R	B030h	PHY package identifier LSW
8021h	COMPOSITE_STATUS	RW	3000h	Composite SGMII and Ethernet PHY status. Only applicable for B-variant.
8040h	PORT_CONTROL	RW	0h	PORT control
8046h	PORT_ABILITIES	R	818h	PORT abilities
8048h	PORT_FUNC_ENABLES	RW	2004h	PORT function enables
8070h	PORT_LEVEL_IRQ_SOURCES	RW	0h	PORT configuration interrupt sources
8072h	PORT_LEVEL_IRQ_ENABLES	RW	0h	PORT top-level interrupt enables
8074h	PORT_LEVEL_IRQ_MSTATUS	R	0h	PORT top-level interrupt masked-status
8078h	PORT_FUNC_IRQ_STATUS	R	0h	PORT function interrupt status
807Ah	PORT_FUNC_IRQ_MSTATUS	RW	B01Eh	PORT function interrupt enables
807Ch	PORT_FUNC_IRQ_MSTATUS	R	0h	PORT function interrupt masked-status

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Table 380. Register overview: Shared...*continued*

Address	Name	Access	Reset	Description
8080h	PHY_CAT_IRQ_STATUS	R	0h	PHY interrupt categories group : Interrupt status
8081h	PHY_CAT_IRQ_ENABLES	RW	410h	PHY interrupt categories group : Interrupt enables
8082h	PHY_CAT_IRQ_MSTATUS	R	0h	PHY interrupt categories group : Masted interrupt status
80A0h	PHY_FUNCTIONAL_IRQ_SOURCES	R	0h	PHY functional baseline interrupt group : Interrupt sources
80A1h	PHY_FUNCTIONAL_IRQ_ENABLES	RW	0h	PHY functional baseline interrupt group : Interrupt enables
80A2h	PHY_FUNCTIONAL_IRQ_STATUS	R	0h	PHY functional baseline interrupt group : Masked interrupt status
80D0h	PHY_DIAGNOSTIC_IRQ_SOURCES	R	0h	PHY diagnosis & test interrupt group : Interrupt sources
80D1h	PHY_DIAGNOSTIC_IRQ_ENABLES	RW	0h	PHY diagnosis & test interrupt group : Interrupt enables
80D2h	PHY_DIAGNOSTIC_IRQ_MSTATUS	R	0h	PHY diagnosis & test interrupt group : Masked interrupt status
8100h	PHY_CONTROL	RW	0h	Top-level PHY functionality group : PHY control
8102h	PHY_STATUS	R	0h	Top-level PHY functionality group : PHY status
8104h	PHY_LATCHED_STATUS	R	0h	Top-level PHY functionality group : PHY latched status
8106h	PHY_EXTENDED_ABILITIES	R	6h	Top-level PHY functionality group : PHY non-IEEE abilities
8108h	PHY_CONFIG	RW	4h	Top-level PHY functionality group : PHY config
810Ch	PHY_STATE	R	1h	Top-level PHY functionality group : PHY state report
810Eh	PHY_PARAMETERS	RW	0h	Top-level PHY functionality group : PHY parameters
8180h	WAKE_SLEEP_CONTROL	RW	0h	Wake-Sleep group : Wake-Sleep control
8181h	WAKE_SLEEP_STATUS	RW	0h	Wake-Sleep group : Wake-Sleep status
8182h	WAKE_SLEEP_CONFIG	RW	8012h	Wake-Sleep group : Wake-Sleep config
8184h	WAKE_SLEEP_PARAMETERS	RW	6330h	Wake-Sleep group : Wake-Sleep parameters
8190h	LED_TRIGGER0	RW	8000h	LED trigger 0
8191h	LED_TRIGGER1	RW	8004h	LED trigger 1
8192h	LED_TRIGGER2	RW	8006h	LED trigger 2
8193h	LED_TRIGGER3	RW	0h	LED trigger 3

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Table 380. Register overview: Shared...*continued*

Address	Name	Access	Reset	Description
8320h	SIGNAL_QUALITY	RW	70h	Slicer signal quality group : Signal Quality Indicator SQI
8324h	MSE	RW	0h	Slicer signal quality group : Mean Square Error MSE
8325h	MAX_MSE	R	0h	Maximum MSE
8340h	LINK_TRAINING_TIMER	R	FF0h	Timing group : Link training timer
8341h	LOC_RCVR_STATUS_ TIMER	R	FF0h	Timing group : Local Receiver Status timer
8342h	REM_RCVR_STATUS_ TIMER	R	FF0h	Timing group : Remote Receiver Status timer
8343h	FOLLOWER_SILENT_ TIMER	R	FF0h	Timing group : Follower transmit timer
8344h	ENERGY_DETECT_TIMER	R	FF0h	Energy Detect Timer
8345h	LINK_SYNC_TIMER	R	FFF0h	Link Sync Timer. Shows the time spent in link sync state, when a transreceiver tries to discover if there a link partner present and ready to train.
8346h	LINK_UP_TIMER	R	FFF0h	Link up Timer. Shows the total time spent in link sync and training iterations until it gets link up.
8350h	SYMBOL_ERROR_ COUNTER	R	0h	Error & Fail counter group : Symbol error counter
8351h	SYMBOL_ERROR_ COUNTER_XTD	RW	0h	Symbol error counter extension
8353h	LINK_LOSSES_AND_ FAILURES	R	0h	Link failure counter
8354h	RS_FRAME_ERROR_ COUNTER	RW	0h	RS frame error counter
8355h	CORRECTED_RS_FRAME_ COUNTER	RW	0h	Corrected RS frame counter
8360h	TDR_CONTROL	RW	1000h	TDR control
8361h	TDR_STATUS	R	0h	TDR status
8362h	FAULT_DISTANCE	R	0h	Distance scale
8363h	DISTANCE_SCALE	RW	7B2Ah	Distance scale
8367h	WAVELET_CONFIG	RW	0h	Wavelet configuration
83F0h	PHY_COMPLAINCE_TEST_ CONFIG	RW	0h	PHY compliance test config .. It must be read as "PHY_COMPLIANCE_TEST_CONFIG"

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7.5.1.1 PHY_IDENTIFIER1 register

Table 381. PHY_IDENTIFIER1 register - PHY identifier MMD device identifier MSW (address 2h)

Bit	Symbol	Access	Value	Description
15:0	OUI_3_TO_18	R	1Bh	OUI = 00-60-37 OUI bits 3:18 = 0000.0000.0001.1011

7.5.1.2 PHY_IDENTIFIER2 register

Table 382. PHY_IDENTIFIER2 register - PHY identifier MMD device identifier LSW (address 3h)

Bit	Symbol	Access	Value	Description
15:10	OUI_19_TO_24	R	2Ch	OUI = 00-60-37 OUI bits 19:24 = 1011.00
9:4	MODEL	R	3h	6-bit PHY model number
3:0	REVISION	R	0h	4-bit PHY revision number

7.5.1.3 DEVICES_IN_PACKAGE1 register

Table 383. DEVICES_IN_PACKAGE1 register - Devices in package LSW (address 5h)

Bit	Symbol	Access	Value	Description
15:8	*reserved*	R	0h	Always write 0, ignore on read.
7	AN_PRESENT	R	0h	0=Auto-Negotiation not present
6	TC_PRESENT	R	0h	0=TC not present
5	DTE_XS_PRESENT	R	0h	0=DTE XS not present
4	PHY_XS_PRESENT	R	0h	0=PHY XS not present
3	PCS_PRESENT	R	1h	1=PCS present
2	WIS_PRESENT	R	0h	0=WIS not present
1	PMD_PMA_ PRESENT	R	1h	1=PMD/PMA present
0	CL22_PRESENT	R	1h	1=Clause 22 registers present

7.5.1.4 DEVICES_IN_PACKAGE2 register

Table 384. DEVICES_IN_PACKAGE2 register - Devices in package MSW (address 6h)

Bit	Symbol	Access	Value	Description
15	MMD31_PRESENT	R	0h	0=Vendor specific device 2 not present
14	MMD30_PRESENT	R	1h	1=Vendor specific device 1 present
13	CL22_EXT_ PRESENT	R	0h	0=Clause 22 extension not present
12:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.5 PACKAGE_IDENTIFIER1 register

Table 385. PACKAGE_IDENTIFIER1 register - PHY package identifier MSW (address Eh)

Bit	Symbol	Access	Value	Description
15:0	OUI_3_TO_18	R	1Bh	OUI = 00-60-37 OUI bits 3:18 = 0000.0000.0001.1011

7.5.1.6 PACKAGE_IDENTIFIER2 register

Table 386. PACKAGE_IDENTIFIER2 register - PHY package identifier LSW (address Fh)

Bit	Symbol	Access	Value	Description
15:10	OUI_19_TO_24	R	2Ch	OUI = 00-60-37 OUI bits 19:24 = 1011.00
9:4	MODEL	R	3h	6-bit PHY model number
3:0	REVISION	R	0h	4-bit PHY revision number

7.5.1.7 COMPOSITE_STATUS register

Table 387. COMPOSITE_STATUS register - Composite SGMII and Ethernet PHY status. Only applicable for B-variant. (address 8021h)

Bit	Symbol	Access	Value	Description
15	AN_LINK_STATUS	R	0h	1 = OKAY, 0= NOT_OKAY
14:12	LINK_STATUS_ SELECT	RW	3h	000 = Reserved 001 = SGMII RX Link Status 010 = ePHY Link Availability 011 = ePHY Link Available & SGMII RX Link Status 1xx = Reserved
11:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.8 PORT_CONTROL register

Table 388. PORT_CONTROL register - PORT control (address 8040h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	1 = Reset all functionality of this PORT.
14	CONFIG_ENABLE	RW	0h	1 = All configuration registers of this PORT (8048h PORT_ FUNCT_ENABLES) can be changed. 0 = Function-critical configuration registers of this PORT are write-protected.
13:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.9 PORT_ABILITIES register

Table 389. PORT_ABILITIES register - PORT abilities (address 8046h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.

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Table 389. PORT_ABILITIES register - PORT abilities (address 8046h)*...continued*

Bit	Symbol	Access	Value	Description
11	BIST_ABILITY	R	1h	1 = This PORT is able to support BIST 0 = This PORT is not able to support BIST
10:5	*reserved*	R	0h	Always write 0, ignore on read.
4	FUSA_ABILITY	R	1h	1 = This PORT is able to support FUSA 0 = This PORT is not able to support FUSA
3	PTP_ABILITY	R	1h	1 = This PORT is able to support PTP 0 = This PORT is not able to support PTP
2:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.10 PORT_FUNC_ENABLES register

Table 390. PORT_FUNC_ENABLES register - PORT function enables (address 8048h)

Bit	Symbol	Access	Value	Description
15:14	*reserved*	R	0h	Always write 0, ignore on read.
13	MII_ENABLE	RW	1h	1 = Enabled, 0 = Disabled
12	*reserved*	R	0h	Always write 0, ignore on read.
11	BIST_ENABLE	RW	0h	1 = BIST functionality is enabled for this PORT 0 = BIST functionality is disabled for this PORT
10:4	*reserved*	R	0h	Always write 0, ignore on read.
3	PTP_ENABLE	RW	0h	1 = PTP functionality is enabled for this PORT. 0 = PTP functionality is disabled for this PORT.
2	EPHY_ENABLE	RW	1h	1 = Ethernet PHY of this PORT is enabled. 0 = Ethernet PHY of this PORT is disabled.
1	*reserved*	R	0h	Always write 0, ignore on read.
0	TEST_ENABLE	RW	0h	1 = Test mode usage is enabled for this PORT 0 = Test mode usage is disabled for this PORT

7.5.1.11 PORT_LEVEL_IRQ_SOURCES register

Table 391. PORT_LEVEL_IRQ_SOURCES register - PORT configuration interrupt sources (address 8070h)

Bit	Symbol	Access	Value	Description
15	PORT_RESET_IRQ	RW	0h	Interrupt bit indicating a reset of the whole PORT by a propagated GLOBAL reset event. Write one to clear.
14:13	*reserved*	R	0h	Always write 0, ignore on read.
12	RECONFIG_DENIED	RW	0h	1 = Denied reconfig attempt to a register of this PORT 0 = No denied reconfig attempts to registers of this PORT Write one to clear.
11	INIT_ERROR	RW	0h	1 = PORT initialization at reboot or restart did not complete successfully Write one to clear.

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Table 391. PORT_LEVEL_IRQ_SOURCES register - PORT configuration interrupt sources (address 8070h)*...continued*

Bit	Symbol	Access	Value	Description
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.12 PORT_LEVEL_IRQ_ENABLES register

Table 392. PORT_LEVEL_IRQ_ENABLES register - PORT top-level interrupt enables (address 8072h)

Bit	Symbol	Access	Value	Description
15	PORT_RESET_IRQ	RW	0h	1= enables the respective source to influence the interrupt output
14:13	*reserved*	R	0h	Always write 0, ignore on read.
12	RECONFIG_ DENIED	RW	0h	1= enables the respective source to influence the interrupt output
11	INIT_ERROR	RW	0h	1= enables the respective source to influence the interrupt output
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.13 PORT_LEVEL_IRQ_MSTATUS register

Table 393. PORT_LEVEL_IRQ_MSTATUS register - PORT top-level interrupt masked-status (address 8074h)

Bit	Symbol	Access	Value	Description
15	PORT_RESET_IRQ	R	0h	1=respective interrupt signal AND respective enables are both '1'
14:13	*reserved*	R	0h	Always write 0, ignore on read.
12	RECONFIG_ DENIED	R	0h	1=respective interrupt signal AND respective enables are both '1'
11	INIT_ERROR	R	0h	1=respective interrupt signal AND respective enables are both '1'
10:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.14 PORT_FUNC_IRQ_STATUS register

Table 394. PORT_FUNC_IRQ_STATUS register - PORT function interrupt status (address 8078h)

Bit	Symbol	Access	Value	Description
15	PORT_LEVEL_IRQ	R	0h	Status bit indicating an interrupt at PORT-level from functionality that does not belong to one specific sub-function.
14	*reserved*	R	0h	Always write 0, ignore on read.
13	SERDES_MII_IRQ	R	0h	Status bit indicating an interrupt from the SERDES-MII SGMII function of this port when it is configured to use this MII type.
12	INFRA_IRQ	R	0h	Status bit indicating an interrupt from INFRAstructure functionality for this port. Note this includes XMII functionality.
11:5	*reserved*	R	0h	Always write 0, ignore on read.

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Table 394. PORT_FUNC_IRQ_STATUS register - PORT function interrupt status (address 8078h)*...continued*

Bit	Symbol	Access	Value	Description
4	FUSA_IRQ	R	0h	Status bit indicating an interrupt from the fusa function of this port.
3	PTP_IRQ	R	0h	Status bit indicating an interrupt from the PTP function of this port.
2	EPHY_IRQ	R	0h	Status bit indicating an interrupt from the ePHY of this port.
1	GLOBAL_IRQ	R	0h	Status bit indicating an interrupt from GLOBAL functions.
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.15 PORT_FUNC_IRQ_MSTATUS register

Table 395. PORT_FUNC_IRQ_MSTATUS register - PORT function interrupt masked-status (address 807Ch)

Bit	Symbol	Access	Value	Description
15	PORT_LEVEL_IRQ	R	0h	Masked-status of PORT interrupt
14	*reserved*	R	0h	Always write 0, ignore on read.
13	SERDES_MII_IRQ	R	0h	Masked-status of SERDES-MII interrupt
12	INFRA_IRQ	R	0h	Masked-status of INFRA interrupt
11:5	*reserved*	R	0h	Always write 0, ignore on read.
4	FUSA_IRQ	R	0h	Masked-status of the fusa interrupt
3	PTP_IRQ	R	0h	Masked-status of the PTP interrupt
2	EPHY_IRQ	R	0h	Masked-status of ePHY interrupt
1	GLOBAL_IRQ	R	0h	Masked-status of GLOBAL interrupt
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.16 PORT_FUNC_IRQ_MSTATUS register

Table 396. PORT_FUNC_IRQ_MSTATUS register - PORT function interrupt masked-status (address 807Ch)

Bit	Symbol	Access	Value	Description
15	PORT_LEVEL_IRQ	R	0h	Masked-status of PORT interrupt
14	*reserved*	R	0h	Always write 0, ignore on read.
13	SERDES_MII_IRQ	R	0h	Masked-status of SERDES-MII interrupt
12	INFRA_IRQ	R	0h	Masked-status of INFRA interrupt
11:5	*reserved*	R	0h	Always write 0, ignore on read.
4	FUSA_IRQ	R	0h	Masked-status of the fusa interrupt
3	PTP_IRQ	R	0h	Masked-status of the PTP interrupt
2	EPHY_IRQ	R	0h	Masked-status of ePHY interrupt
1	GLOBAL_IRQ	R	0h	Masked-status of GLOBAL interrupt
0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.17 PHY_CAT_IRQ_STATUS register

Table 397. PHY_CAT_IRQ_STATUS register - PHY interrupt categories group : Interrupt status (address 8080h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	PHY_ DIAGNOSTIC_IRQ	R	0h	1=Active interrupts present in 'Diagnosis & Test interrupts' source register
9:5	*reserved*	R	0h	Always write 0, ignore on read.
4	PHY_ FUNCTIONAL_IRQ	R	0h	1=Active interrupts present in 'Functional baseline' source register
3:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.18 PHY_CAT_IRQ_ENABLES register

Table 398. PHY_CAT_IRQ_ENABLES register - PHY interrupt categories group : Interrupt enables (address 8081h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	PHY_ DIAGNOSTIC_IRQ	RW	1h	1=Enable diagnosis & test category interrupt
9:5	*reserved*	R	0h	Always write 0, ignore on read.
4	PHY_ FUNCTIONAL_IRQ	RW	1h	1=Enable functional baseline category interrupt
3:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.19 PHY_CAT_IRQ_MSTATUS register

Table 399. PHY_CAT_IRQ_MSTATUS register - PHY interrupt categories group : Masted interrupt status (address 8082h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	PHY_ DIAGNOSTIC_IRQ	R	0h	1=Diagnosis & test category unmasked status
9:5	*reserved*	R	0h	Always write 0, ignore on read.
4	PHY_ FUNCTIONAL_IRQ	R	0h	1=Functional baseline category unmasked status
3:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.20 PHY_FUNCTIONAL_IRQ_SOURCES register

Table 400. PHY_FUNCTIONAL_IRQ_SOURCES register - PHY functional baseline interrupt group : Interrupt sources (address 80A0h)

Bit	Symbol	Access	Value	Description
15	POLARITY_ CONFLICT	R	0h	1=Polarity conflict latched status. Write one to clear.
14	TRAINING_ FAILURE	R	0h	1=Training failure latched status Write one to clear.
13	TRANSMIT_ FAILURE	R	0h	1=Transmit failure latched status Write one to clear.
12:7	*reserved*	R	0h	Always write 0, ignore on read.
6	OAM_TX_ MESSAGE_ACK	R	0h	1=1000Base-T1 OAM Message received latched status Write one to clear.
5	OAM_RX_ MESSAGE_VALID	R	0h	1=Link partner 1000BASE-T1 OAM message valid latched status Write one to clear.
4	FAILSAFE_EVENT	R	0h	1=Failsafe event latched status Write one to clear.
3	WAKE_SLEEP_ EVENT	R	0h	1=Sleep Status Change latched status Write one to clear.
2	LINK_AVAILABLE_ EVENT	R	0h	1=Link Availability Change latched status Write one to clear.
1	LINK_STATUS_ EVENT	R	0h	1=Link Status Change latched status Write one to clear.
0	RESERVED	R	0h	Reserved. Read value is undefined, only zero should be written.

7.5.1.21 PHY_FUNCTIONAL_IRQ_ENABLES register

Table 401. PHY_FUNCTIONAL_IRQ_ENABLES register - PHY functional baseline interrupt group : Interrupt enables (address 80A1h)

Bit	Symbol	Access	Value	Description
15	POLARITY_ CONFLICT	RW	0h	1=Polarity conflict interrupt enable
14	TRAINING_ FAILURE	RW	0h	1=Training failure interrupt enable
13	TRANSMIT_ FAILURE	RW	0h	1=Transmit failure interrupt enable
12:7	*reserved*	R	0h	Always write 0, ignore on read.
6	OAM_TX_ MESSAGE_ACK	RW	0h	1=OAM TX message ack interrupt enable
5	OAM_RX_ MESSAGE_VALID	RW	0h	1=OAM RX message valid interrupt enable
4	FAILSAFE_EVENT	RW	0h	1=Failsafe event interrupt enable

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Table 401. PHY_FUNCTIONAL_IRQ_ENABLES register - PHY functional baseline interrupt group : Interrupt enables (address 80A1h)*...continued*

Bit	Symbol	Access	Value	Description
3	WAKE_SLEEP_ EVENT	RW	0h	1=Sleep Status change interrupt enable
2	LINK_AVAILABLE_ EVENT	RW	0h	1=Link Availability change interrupt enable
1	LINK_STATUS_ EVENT	RW	0h	1=Link Status change interrupt enable
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.22 PHY_FUNCTIONAL_IRQ_STATUS register

Table 402. PHY_FUNCTIONAL_IRQ_STATUS register - PHY functional baseline interrupt group : Masked interrupt status (address 80A2h)

Bit	Symbol	Access	Value	Description
15	POLARITY_ CONFLICT	R	0h	1=Polarity conflict unmasked status
14	TRAINING_ FAILURE	R	0h	1=Training failure unmasked status
13	TRANSMIT_ FAILURE	R	0h	1=Transmit failure unmasked status
12:7	*reserved*	R	0h	Always write 0, ignore on read.
6	OAM_TX_ MESSAGE_ACK	R	0h	1=OAM TX message ack event unmasked status
5	OAM_RX_ MESSAGE_VALID	R	0h	1=OAM RX message valid unmasked status
4	FAILSAFE_EVENT	R	0h	1=Failsafe event unmasked status
3	WAKE_SLEEP_ EVENT	R	0h	1=Sleep Status Change unmasked status
2	LINK_AVAILABLE_ EVENT	R	0h	1=Availability Change unmasked status
1	LINK_STATUS_ EVENT	R	0h	1=Link Status Change unmasked status
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.23 PHY_DIAGNOSTIC_IRQ_SOURCES register

Table 403. PHY_DIAGNOSTIC_IRQ_SOURCES register - PHY diagnosis & test interrupt group : Interrupt sources (address 80D0h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	RS_FRAME_ ERROR	R	0h	1=Reed-Solomon Frame Error latched status Write one to clear.
1	SQI_WARN	R	0h	1=Signal Quality Warning latched status

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Table 403. PHY_DIAGNOSTIC_IRQ_SOURCES register - PHY diagnosis & test interrupt group : Interrupt sources (address 80D0h)*...continued*

Bit	Symbol	Access	Value	Description
				Write one to clear.
0	RESERVED	R	0h	Reserved. Read value is undefined, only zero should be written.

7.5.1.24 PHY_DIAGNOSTIC_IRQ_ENABLES register

Table 404. PHY_DIAGNOSTIC_IRQ_ENABLES register - PHY diagnosis & test interrupt group : Interrupt enables (address 80D1h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	RS_FRAME_ ERROR	RW	0h	1=RS-frame error interrupt enable
1	SQI_WARN	RW	0h	1=Signal Quality Warning interrupt enable
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.25 PHY_DIAGNOSTIC_IRQ_MSTATUS register

Table 405. PHY_DIAGNOSTIC_IRQ_MSTATUS register - PHY diagnosis & test interrupt group : Masked interrupt status (address 80D2h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	RS_FRAME_ ERROR	R	0h	1=RS_FRAME_ERROR unmasked status
1	SQI_WARN	R	0h	1=Signal Quality Warning unmasked status
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.26 PHY_CONTROL register

Table 406. PHY_CONTROL register - Top-level PHY functionality group : PHY control (address 8100h)

Bit	Symbol	Access	Value	Description
15	RESET	RW	0h	1=PHY will be fully reset including AO registers.
14	CONFIG_ENABLE	RW	0h	1=All configuration registers of this EPHY can be changed
13:2	*reserved*	R	0h	Always write 0, ignore on read. Always write 0, ignore on read.
1	GOTO_STANDBY	RW	0h	1=Directs the PHY to STANDBY state for reconfiguration
0	START_ OPERATION	RW	0h	1=Exit standby state and start PHY operation

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7.5.1.27 PHY_STATUS register

Table 407. PHY_STATUS register - Top-level PHY functionality group : PHY status (address 8102h)

Bit	Symbol	Access	Value	Description
15:11	*reserved*	R	0h	Always write 0, ignore on read.
10	BLOCK_LOCK	R	0h	1=Block Lock assertion status
9	REM_PHY_READY	R	0h	1=Remote PHY ready status
8	LOC_PHY_READY	R	0h	1=Local PHY ready status
7	DETECTED_ POLARITY	R	0h	1=Inversed polarity detected
6	LINK_AVAILABLE	R	0h	1=Link availability LOC_PHY_READY and REM_PHY_ READY and link_status
5	SENDN_OR_DATA	R	0h	1= PHY control FSM is in Send Idle or Data state
4	REM_RCVR_ STATUS	R	0h	1=Remote receiver status
3	SCRAMBLER_ STATUS	R	0h	1=Scrambler status
2	LINK_STATUS	R	0h	1=Link status
1	LOC_RCVR_ STATUS	R	0h	1=Local receiver status
0	RESERVED	R	0h	Reserved. Read value is undefined, only zero should be written.

7.5.1.28 PHY_LATCHED_STATUS register

Table 408. PHY_LATCHED_STATUS register - Top-level PHY functionality group : PHY latched status (address 8104h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	REMOTE_FAILURE	R	0h	1=Remote failure latch high status Write one to clear.
1	EXPIRED_MAX_ WAIT	R	0h	1=Expired max wait timer latch high status Write one to clear.
0	RESERVED	R	0h	Reserved. Read value is undefined, only zero should be written.

7.5.1.29 PHY_EXTENDED_ABILITIES register

Table 409. PHY_EXTENDED_ABILITIES register - Top-level PHY functionality group : PHY non-IEEE abilities (address 8106h)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2	TC10_ABILTY	R	1h	1=PHY is able to support TC10 Wake-Sleep

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Table 409. PHY_EXTENDED_ABILITIES register - Top-level PHY functionality group : PHY non-IEEE abilities (address 8106h) *...continued*

Bit	Symbol	Access	Value	Description
1	BASIC_SLEEP_ ABILITY	R	1h	1=PHY is able to support BASIC SLEEP state
0	RESERVED	R	0h	Reserved. Read value is undefined, only zero should be written.

7.5.1.30 PHY_CONFIG register

Table 410. PHY_CONFIG register - Top-level PHY functionality group : PHY config (address 8108h)

Bit	Symbol	Access	Value	Description
15:6	*reserved*	R	0h	Always write 0, ignore on read.
5	STATUS_FILTER_ DISABLE	RW	0h	1=Disable hi_rfer status and associated effect. Enabling of this bit is strongly discouraged for real applications.
4	*reserved*	R	0h	Always write 0, ignore on read.
3	POLARITY_SWAP	RW	0h	1=MDI pin polarity swap Enforce an additional polarity swap in order to virtually swap the DP/DN pins The paired behavior of bits 3:2 becomes: 00 : Polarity correction, default pin polarity 01 : No polarity correction, default pin polarity 10 : Polarity correction plus forced polarity swap 11 : Forced polarity swap, no polarity correction
2	POLARITY_ CORRECT_ DISABLE	RW	1h	1=Polarity correction disable. The default value on reset removal further depends on strap pin "polarity_correction" control. The strap pin expected to be set '1' in Normal operation.
1	TEST_MODE_ ENABLE	RW	0h	1=Selecting a valid test mode brings the PHY to test mode
0	AUTO_OPERATION	RW	0h	1=PHY autonomously proceeds to Normal mode for both start-up and wake-up.

7.5.1.31 PHY_STATE register

Table 411. PHY_STATE register - Top-level PHY functionality group : PHY state report (address 810Ch)

Bit	Symbol	Access	Value	Description
15	BUSY	R	0h	1 = BUSY state indication status. During state transitions, the busy bit is asserted and the previous state indication is maintained until the next state is fully reached. ***Note:*** *When the PHY is configured in TC10 and is in SLEEP mode, this bit indicates "busy" even though the PHY is not in a transitions state.*
14	TEST_OR_ DIAGNOSIS	R	0h	1 = Test & Diagnosis indication status
13:12	*reserved*	R	0h	Always write 0, ignore on read.
11:8	TRANSMIT_STATE	R	0h	Transmit state

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Table 411. PHY_STATE register - Top-level PHY functionality group : PHY state report (address 810Ch)*...continued*

Bit	Symbol	Access	Value	Description
				000 = Trasmit state is in Disabled 001 = Transmit state is in SEND_Z 010 = Transmit state is in SEND_S 011 = Transmit state is in SEND_T 100 = Transmit state is in SEND_IDLE state 101 = Transmit state is in SEND_DATA state 110 = Transmit state is in SEND_WUR state 111 = Transmit state is in SEND_LPS state
7:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	BASIC_STATE	R	1h	Basic states 0000 = PHY control is in Disable Transmitter sttae 0001 = PHY in forced standby stte 0010 = Reserved 0011 = PHY TC10 in sleep state 1010 = PHY in link sync state 1001 = PHY control is in SILENT state 1000 = PHY control is in Training state 1100 = PHY control is in SEND_IDLE1 or SEND_IDLE2 state 1101 = PHY control is in SEND_DATA state 1110 = Reserved 1111 = TC 10 state Entering Send LPS state with TC10 enabled after Link UP state 1011 = TC10 in Sleep Silent state 0110 = Not valid for 1000BaseT1. Reserved

7.5.1.32 PHY_PARAMETERS register

Table 412. PHY_PARAMETERS register - Top-level PHY functionality group : PHY parameters (address 810Eh)

Bit	Symbol	Access	Value	Description
15:3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	LINK_AVAILABLE_HYST_TIME	RW	0h	Link availability hysteresis timer 000=0MS 001=250US 010=500US 011=1MS 100=2MS 101=4MS 110=8MS 111=16MS

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7.5.1.33 WAKE_SLEEP_CONTROL register

Table 413. WAKE_SLEEP_CONTROL register - Wake-Sleep group : Wake-Sleep control (address 8180h)

Bit	Symbol	Access	Value	Description
15	WAKEUP_PHY_REQUEST	RW	0h	1=Wake-up request to this PHY is enabled
14:3	*reserved*	R	0h	Always write 0, ignore on read.
2	SLEEP_REJECT	RW	0h	1=Reject sleep request received on MDI
1	SLEEP_ACCEPT	RW	0h	1=Accept sleep request received on MDI
0	SLEEP_REQUEST	RW	0h	1=Start the procedure to bring this PHY and its link partner into SLEEP state

7.5.1.34 WAKE_SLEEP_STATUS register

Table 414. WAKE_SLEEP_STATUS register - Wake-Sleep group : Wake-Sleep status (address 8181h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	WU_IO_RECEIVED	RW	0h	1=Wake-up request on WU_IO pin received Write one to clear.
13	WU_SMI_RECEIVED	RW	0h	1=Wake-up request received by WU_SMI Write one to clear.
12	WU_PHY_RECEIVED	RW	0h	1=Wake-up request received by WU_PHY bit Write one to clear.
11	WUP_RECEIVED	RW	0h	1=Wake-Up Pulse received Write one to clear.
10	WUR_RECEIVED	RW	0h	1=Wake-Up Transfer from another PHY Write one to clear.
9:4	*reserved*	R	0h	Always write 0, ignore on read.
3	AUTO_SLEEP_EVENT	RW	0h	1=Auto sleep has been triggered Write one to clear.
2	SLEEP_FAILED	RW	0h	1=Failed to enter Sleep mode Write one to clear.
1	LPS_RECEIVED	RW	0h	1=Sleep request received Write one to clear.
0	SLEEP_STATUS	R	0h	1=PHY has entered Sleep state

7.5.1.35 WAKE_SLEEP_CONFIG register

Table 415. WAKE_SLEEP_CONFIG register - Wake-Sleep group : Wake-Sleep config (address 8182h)

Bit	Symbol	Access	Value	Description
15:14	FUNCTION_SELECT	RW	2h	Wake-Sleep function select 000=No Wake-Sleep functionality 001=Basic Wake-Sleep functionality 010=TC10 Wake-Sleep functionality

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Table 415. WAKE_SLEEP_CONFIG register - Wake-Sleep group : Wake-Sleep config (address 8182h)*...continued*

Bit	Symbol	Access	Value	Description
				011=reserved
13	*reserved*	R	0h	Always write 0, ignore on read.
12	RESERVED	RW	0h	Reserved. Read value is undefined, only zero should be written.
11	WU_IO_ENABLE	RW	0h	1=Wake-up PHY on received WU_IO assertion
10	WU_SMI_ENABLE	RW	0h	1=Wake-Up by SMI enabled
9	FWD_WU_PHY_TO_WU_IO	RW	0h	1=Forward wake request by WU_PHY control bit to WU_IO
8	FWD_WUPWUR_TO_WU_IO	RW	0h	1=Forward WUP/WUR requests received on MDI by this PHY to the device WU_IO pin
7:5	*reserved*	R	0h	Always write 0, ignore on read.
4	WUR_IN_SENDLPS	RW	1h	1=PHY can launch a WUR during SEND_LPS
3	AUTO_REJECT	RW	0h	1=Sleep requests are auto-rejected by the responder on sleep_ack_timer expiration
2	AUTO_SLP_ON_IDLE	RW	0h	1=Long idle autonomous sleep request enabled
1	AUTO_SLEEP_ON_SILENCE	RW	1h	1=PHY automatically returns to SLEEP after wake-up from remote end if no signal is received from the leader.It is of false remote wake up condition.
0	NO_AUTO_ON_WAKE	RW	0h	1=PHY pauses in STANDBY after wake-up if auto_operate=0, until start_operation bit is set.

7.5.1.36 WAKE_SLEEP_PARAMETERS register

Table 416. WAKE_SLEEP_PARAMETERS register - Wake-Sleep group : Wake-Sleep parameters (address 8184h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14:12	SLEEP_ACK_TIMER	RW	6h	Sleep ack timer 000=0US 001=250US 010=500US 011=1MS 100=2MS 101=4MS 110=8MS 111=12MS
11	*reserved*	R	0h	Always write 0, ignore on read.
10:8	AUTO_SLEEP_IDLE_TIMER	RW	3h	Auto sleep idle timer 000=125MS 001=250MS 010=500MS 011=1S 100=2S

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Table 416. WAKE_SLEEP_PARAMETERS register - Wake-Sleep group : Wake-Sleep parameters (address 8184h)*...continued*

Bit	Symbol	Access	Value	Description
				101=4S 110=8S 111=16S
7	*reserved*	R	0h	Always write 0, ignore on read.
6:4	AUTO_SLEEP_ SILENCE_TIMER	RW	3h	Auto sleep silence timer 000=125MS 001=250MS 010=500MS 011=1S 100=2S 101=4S 110=8S 111=16S
3:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.37 LED_TRIGGER0 register

Table 417. LED_TRIGGER0 register - LED trigger 0 (address 8190h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1=LED trigger-o is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0000=Link_available 0001=Link_status 0010=Role: Leader1 or Follower0 0011=PHY active AFE enabled 0100=Frame reception RX_DV = true 0101=Symbol error 0110=Frame transmission TX_EN=true 0111=Frame activity RX_DV=true | TX_EN=true 1000-1111=reserved

7.5.1.38 LED_TRIGGER1 register

Table 418. LED_TRIGGER1 register - LED trigger 1 (address 8191h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	1=LED trigger-1 is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	4h	Selectable LED functions: 0000=Link_available 0001=Link_status 0010=Role: Leader1 or Follower0 0011=PHY active AFE enabled

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Table 418. LED_TRIGGER1 register - LED trigger 1 (address 8191h)*...continued*

Bit	Symbol	Access	Value	Description
				0100=Frame reception RX_DV = true 0101=Symbol error 0110=Frame transmission TX_EN=true 0111=Frame activity RX_DV=true | TX_EN=true

7.5.1.39 LED_TRIGGER2 register

Table 419. LED_TRIGGER2 register - LED trigger 2 (address 8192h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	1h	
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	6h	Selectable LED functions: 0000=Link_available 0001=Link_status 0010=Role: Leader1 or Follower0 0011=PHY active AFE enabled 0100=Frame reception RX_DV = true 0101=Symbol error 0110=Frame transmission TX_EN=true 0111=Frame activity RX_DV=true | TX_EN=true

7.5.1.40 LED_TRIGGER3 register

Table 420. LED_TRIGGER3 register - LED trigger 3 (address 8193h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1=LED trigger-3 is enabled
14:4	*reserved*	R	0h	Always write 0, ignore on read.
3:0	FUNCTION	RW	0h	Selectable LED functions: 0000=Link_available 0001=Link_status 0010=Role: Leader1 or Follower0 0011=PHY active AFE enabled 0100=Frame reception RX_DV = true 0101=Symbol error 0110=Frame transmission TX_EN=true 0111=Frame activity RX_DV=true | TX_EN=true

7.5.1.41 SIGNAL_QUALITY register

Table 421. SIGNAL_QUALITY register - Slicer signal quality group : Signal Quality Indicator SQI (address 8320h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.
14	VALID	R	0h	1=SQI is available Valid

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Table 421. SIGNAL_QUALITY register - Slicer signal quality group : Signal Quality Indicator SQI (address 8320h)*...continued*

Bit	Symbol	Access	Value	Description
13:11	*reserved*	R	0h	Always write 0, ignore on read.
10:8	SQI_WARN_LIMIT	RW	0h	SQI warning limit. Threshold level for creation of an SQI warning interrupt
7	*reserved*	R	0h	Always write 0, ignore on read.
6:4	SQI_WORST	R	7h	SQI minimum Lowest occured SQI level. Type :Latch Low
3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	SQI	R	0h	Actual SQI level

7.5.1.42 MSE register

Table 422. MSE register - Slicer signal quality group : Mean Square Error MSE (address 8324h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1=MSE enable
14	VALID	R	0h	1= MSE valid
13:12	*reserved*	R	0h	Always write 0, ignore on read.
11:2	MSE	R	0h	Mean-Square Error on slicer Mean-square value of slicer error N=8
1:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.43 MAX_MSE register

Table 423. MAX_MSE register - Maximum MSE (address 8325h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:2	MAX_MSE	R	0h	Peak MSE on slicer. Type : Latch High Maximum value of mean-square slicer error N=8
1:0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.44 LINK_TRAINING_TIMER register

Table 424. LINK_TRAINING_TIMER register - Timing group : Link training timer (address 8340h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:4	TIME_MS	R	FFh	Training time. 8-bit time duration in ms units.
3:1	TIME_SUBMS	R	0h	Training time fraction. Fractional part for 125us timing accuracy
0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.45 LOC_RCVR_STATUS_TIMER register

Table 425. LOC_RCVR_STATUS_TIMER register - Timing group : Local Receiver Status timer (address 8341h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:4	TIME_MS	R	FFh	time. 8-bit time duration in ms units.
3:1	TIME_SUBMS	R	0h	Time fraction
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.46 REM_RCVR_STATUS_TIMER register

Table 426. REM_RCVR_STATUS_TIMER register - Timing group : Remote Receiver Status timer (address 8342h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:4	TIME_MS	R	FFh	time. 8-bit time duration in ms units.
3:1	TIME_SUBMS	R	0h	Time fraction
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.47 FOLLOWER_SILENT_TIMER register

Table 427. FOLLOWER_SILENT_TIMER register - Timing group : Follower transmit timer (address 8343h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:4	TIME_MS	R	FFh	time. 8-bit time duration in ms units.
3:1	TIME_SUBMS	R	0h	Time fraction
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.48 ENERGY_DETECT_TIMER register

Table 428. ENERGY_DETECT_TIMER register - Energy Detect Timer (address 8344h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:4	TIME_MS	R	FFh	time. 8-bit time duration in ms units.
3:1	TIME_SUBMS	R	0h	Fractional part for 125us timing accuracy
0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.49 LINK_SYNC_TIMER register

Table 429. LINK_SYNC_TIMER register - Link Sync Timer. Shows the time spent in link sync state, when a transreceiver tries to discover if there a link partner present and ready to train. (address 8345h)

Bit	Symbol	Access	Value	Description
15:4	TIME_MS	R	FFFh	time. 12-bit time duration in ms units. 0d - 4000d : 0 - 4000msec 4001d = more than 4sec 4095d = No value
3:1	TIME_SUBMS	R	0h	Fractional part for 125us timing accuracy
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.50 LINK_UP_TIMER register

Table 430. LINK_UP_TIMER register - Link up Timer. Shows the total time spent in link sync and training iterations until it gets link up. (address 8346h)

Bit	Symbol	Access	Value	Description
15:4	TIME_MS	R	FFFh	time. 12-bit time duration in ms units. 0d - 4000d : 0 - 4000msec 4001d = more than 4sec 4095d = No value
3:1	TIME_SUBMS	R	0h	Fractional part for 125us timing accuracy
0	*reserved*	R	0h	Always write 0, ignore on read.

7.5.1.51 SYMBOL_ERROR_COUNTER register

Table 431. SYMBOL_ERROR_COUNTER register - Error & Fail counter group : Symbol error counter (address 8350h)

Bit	Symbol	Access	Value	Description
15:0	SYMBOL_ERRORS	R	0h	Symbol error counter Accumulative total number of symbol errors

7.5.1.52 SYMBOL_ERROR_COUNTER_XTD register

Table 432. SYMBOL_ERROR_COUNTER_XTD register - Symbol error counter extension (address 8351h)

Bit	Symbol	Access	Value	Description
15	EXTENDED_ COUNTER_ LENGTH	RW	0h	Extended counter length 0=16-bit counter 1=30-bit counter
14	*reserved*	R	0h	Always write 0, ignore on read.
13:0	SYMBOL_ ERRORS_MSW	R	0h	Symbol error counter extension

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7.5.1.53 LINK_LOSSES_AND_FAILURES register

Table 433. LINK_LOSSES_AND_FAILURES register - Link failure counter (address 8353h)

Bit	Symbol	Access	Value	Description
15:10	LINK_LOSSES	R	0h	A link loss is counted when the loc_rcvr_status becomes not_okay and this PHY will re-train and capture a new link_training_time.
9:0	LINK_FAILURES	R	0h	Number of link failure events. Hi_RFER status event which removes loc_phy_ready which is the source of link failure. 0-1022: number of link failures 1023: 1023 or more link failures

7.5.1.54 RS_FRAME_ERROR_COUNTER register

Table 434. RS_FRAME_ERROR_COUNTER register - RS frame error counter (address 8354h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1=RX frame error counter enable
14:10	*reserved*	R	0h	Always write 0, ignore on read.
9:0	RS_FRAME_ERRORS	R	0h	Number of received uncorrectable Reed-Solomon frames.

7.5.1.55 CORRECTED_RS_FRAME_COUNTER register

Table 435. CORRECTED_RS_FRAME_COUNTER register - Corrected RS frame counter (address 8355h)

Bit	Symbol	Access	Value	Description
15:0	CORRECTED_RS_FRAMES	RW	0h	corrected RS frame status counter – WC Write Clear

7.5.1.56 TDR_CONTROL register

Table 436. TDR_CONTROL register - TDR control (address 8360h)

Bit	Symbol	Access	Value	Description
15	ENABLE	RW	0h	1=Enable
14	START	RW	0h	1=Start test .
13	DISABLE_CALIBRATION	RW	0h	Disable calibration
12	INCLUDE_LINK_TEST	RW	1h	1=Link-up test before starting TDR
11	PEAK_INTERVAL_CHECK	RW	0h	1=Enabled. Distance between positive and negative pulse reflections is checked and no fail is reported if this does not match the expected distance within reasonable margins.
10:0	*reserved*	R	0h	Always write 0, ignore on read.

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7.5.1.57 TDR_STATUS register

Table 437. TDR_STATUS register - TDR status (address 8361h)

Bit	Symbol	Access	Value	Description
15	VALID	R	0h	1=TDR valid
14:12	TDR_STATE	R	0h	TDR state information 000=NO_TST_ONGOING 001=LINK_TEST 010=TDR_CALIBRATION 011=TDR_CABLE_TEST 100=Reserved 101=Reserved 110=CALCULATING_RESULT 111=Reserved
11:3	*reserved*	R	0h	Always write 0, ignore on read.
2:0	FAULT_TYPE	R	0h	000=NO_CABLE_FAULT_DETECTED 001=SHORT 010=OPEN 011=Reserved 100=Reserved 101=Reserved 110=Reserved 111=NOT_ABLE_TO_DETECT

7.5.1.58 FAULT_DISTANCE register

Table 438. FAULT_DISTANCE register - Distance scale (address 8362h)

Bit	Symbol	Access	Value	Description
15:12	*reserved*	R	0h	Always write 0, ignore on read.
11:0	FAULT_DISTANCE	R	0h	Distance to fault location in units as configured by the DISTANCE_SCALE register

7.5.1.59 DISTANCE_SCALE register

Table 439. DISTANCE_SCALE register - Distance scale (address 8363h)

Bit	Symbol	Access	Value	Description
15:0	DISTANCE_SCALE	RW	7B2Ah	Distance scale factor = 2*256*16/0.65*C*1.33333e-9 ; C- velocity of light Number of sub phases – 256 Fractional bits – 4 multiplied by 16

7.5.1.60 WAVELET_CONFIG register

Table 440. WAVELET_CONFIG register - Wavelet configuration (address 8367h)

Bit	Symbol	Access	Value	Description
15	*reserved*	R	0h	Always write 0, ignore on read.

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Table 440. WAVELET_CONFIG register - Wavelet configuration (address 8367h)*...continued*

Bit	Symbol	Access	Value	Description
14	TYPE	RW	0h	Type 0=TYPE-I 1=TYPE-II
13	SPLIT	RW	0h	1=Split enabled
12:6	*reserved*	R	0h	Always write 0, ignore on read.
5:3	LOG2_T1	RW	0h	Length of +1/-1 intervals in 2^N UI 001=UI_2 010=UI_4 011=UI_8 100=Reserved 101=Reserved 110=Reserved 111=Reserved
2:0	LOG2_T0	RW	0h	Length of zero mid-interval in 2^N UI. 000=UI_1 001=UI_2 010=UI_4 011=UI_8 100=Reserved 101=Reserved 110=Reserved 111=Reserved

7.5.1.61 PHY_COMPLAINCE_TEST_CONFIG register

Table 441. PHY_COMPLAINCE_TEST_CONFIG register - PHY compliance test config .. It must be read as "PHY_COMPLIANCE_TEST_CONFIG" (address 83F0h)

Bit	Symbol	Access	Value	Description
15:2	*reserved*	R	0h	Always write 0, ignore on read.
1	TX_SEND_Z	RW	0h	1=Transmit SEND_Z enable control
0	TX_CLK_OUT_ENABLE	RW	0h	1=Transmit clock output enable

8 Limiting values

Table 442. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134); all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_x	Voltage on pin x	on pins MDC, INT_N, RST_N, INH, WAKE_IN_OUT, RX_CTL, RXDx, RXC, TXC, TX_CTL, TXDx, MDIO, GPIOx	–0.5	—	4	V

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Table 442. Limiting values...*continued*

In accordance with the Absolute Maximum Rating System (IEC 60134); all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		on pins XI, XO; differential mode or XTAL mode	–0.5	—	1.6	V
		on pins XI, XO; single ended IO mode	–0.5	—	4	V
		on pins TRX_P, TRX_M	–0.5	—	4	V
		on pins SIN, SIP	–0.5	—	1.6	V
		on pins SOP, SON	–0.5	—	1.6	V
		on pin VDDIO	–0.5	—	4	V
		on pin VDD_CORE	–0.5	—	1.6	V
		on pin VDDA_AO	–0.5	—	4	V
		on pin VREGA_IN	–0.5	—	4	V
		on pins VDDA_TRX	–0.5	—	4	V
		on pins VDDA_SI, VDDA_SO	–0.5	—	4	V
		on pins VDDA_RX	–0.5	—	4	V
V_{trt}	transient voltage	on on pins TRX_P, TRX_M [1]				
		pulse 1	–100	—	—	V
		pulse 2a	—	—	+75	V
		pulse 3a	–150	—	—	V
		pulse 3b	—	—	100	V
V_{ESD}	electrostatic discharge voltage	IEC 61000-4-2; 150 pF, 330 Ω [2]				
		on pins TRX_P, TRX_M to GND	–6	—	+6	kV
V_{ESD}	electrostatic discharge voltage	Human Body Model (HBM)				
		on any pin [3]	–2	—	+2	kV
		on pins TRX_P, TRX_M to GND [4]	–4	—	+4	kV
V_{ESD}	electrostatic discharge voltage	Charge Device Model (CDM) [5]				
		on all pins	–500	—	+500	V
T_{vj}	virtual junction temperature	[6]	–40	—	+150	°C
T_{stg}	storage temperature	[7]	–55	—	+150	°C

[1] Verified by an external test house according to IEC TS62228 Section 4.2.4; parameters for standardisation defined in ISO7637 Part 2 [1][2]
[2] Verified by an external test house according to IEC TS62228 Section 4.3 [1][2]
[3] According to AEC-Q100-002
[4] Pins stressed to reference group containing all ground and supply pins, emulating the application circuits. HBM pulse as specified in AEC-Q100-002 used.
[5] According to AEC-Q100-011
[6] In accordance with IEC 60747-1. An alternative definition of virtual junction temperature is: $T_{vj} = T_{amb} + P \times R_{th(j-a)}$, where $R_{th(j-a)}$ is a fixed value used in the calculation of T_{vj}. The rating for T_{vj} limits the allowable combinations of power dissipation (P) and ambient temperature (T_{amb}).
[7] T_{stg} in application according to IEC61360-4. For component transport and storage conditions, see instead IEC61760-2.

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9 Thermal characteristics

Table 443. Thermal characteristics

Value determined for free convection conditions on a JEDEC 2S2P board[1].

Symbol	Parameter[2]	Conditions	Typ	Unit
$R_{th(j-a)}$	thermal resistance from junction to ambient	HVQFN36	28	K/W
$R_{th(j-c)}$	thermal resistance from junction to case	HVQFN36	3	K/W
Ψ_{j-top}	thermal characterization parameter from junction to top of package	HVQFN36	8	K/W

[1] According to JEDEC JESD51-2, JESD51-5 and JESD51-7 at natural convection on 2s2p board. Board with two inner copper layers (thickness: 35 µm) and thermal via array under the exposed pad connected to the first inner copper layer (thickness: 70 µm)

[2] Case temperature refers to the center of the heat sink at the bottom of the package.

10 Static characteristics

Table 444. Static characteristics

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Power supply						
V_{DD}	supply voltage	on pin VDDIO, 1.8 V signaling	1.65	1.8	1.95	V
		on pin VDDIO, 2.5 V signaling	2.25	2.5	2.75	V
		on pin VDDIO, 3.3 V signaling	2.97	3.30	3.63	V
		on pin VDD_CORE	1.045	1.1	1.155	V
		on pin VDDA_AO, 3.3 V operation [2]	3.135	3.3	3.465	V
		on pin VDDA_AO, 2.5 V operation	2.375	2.5	2.625	V
		on pin VREGA_IN, 3.3 V operation	3.135	3.3	3.465	V
		on pin VREGA_IN, 1.8 V operation	1.71	1.8	1.89	V
		on pins VDDA_TRX	1.71	1.8	1.89	V
		on pins VDDA_RX	1.71	1.8	1.89	V
		on pins VDDA_SI, VDDA_SO	1.045	1.1	1.155	V
Current consumption (core)[3][4]						
$I_{DD(RMS)}$	RMS supply current	on pin VDD_CORE mode DEVICE_OPERATIONAL 1000BASE-T1 link-up MACsec disabled 100% load RX/TX	—	245	330	mA
		on pin VDD_CORE mode DEVICE_OPERATIONAL during 1000BASE-T1 link-up phase MACsec disabled	—	275	350	mA

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Table 444. Static characteristics...*continued*

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		on pin VDD_CORE mode DEVICE_STANDBY DEVICE_SLEEP; variant A MACsec disabled	—	10	49	mA
		on pin VDD_CORE mode DEVICE_STANDBY DEVICE_SLEEP; variant B MACsec disabled	—	8	38	mA
		on pin VDD_CORE mode DEVICE_OPERATIONAL 1000BASE-T1 link-up 100% load RX/TX MACsec enabled	—	290	380	mA
		on pin VDD_CORE mode DEVICE_OPERATIONAL during 1000BASE-T1 link-up phase MACsec enabled	—	310	380	mA
		on pin VDD_CORE mode DEVICE_STANDBY DEVICE_SLEEP; variant A MACsec enabled	—	18	60	mA
		on pin VDD_CORE mode DEVICE_STANDBY DEVICE_SLEEP; variant B MACsec enabled	—	18	60	mA
Current consumption (IO)[3]						
$I_{DD(RMS)}$	RMS supply current	on pins VDDIO				
		RGMII mode (variant A only); C_L= 5 pF (far end with 15 cm PCB trace); frame size 1500 bytes; worst-case pattern (0xA5A5); default IO settings; no external pull-up, pull_down resistors VDDIO = 1.8 V	—	—	60	mA
		RGMII mode (variant A only); C_L=5 pF (far end with 8cm PCB trace ; frame size 128 bytes; PRBS pattern; default IO settings; 25 °C ambient; no external pull-up, pull_down resistors VDDIO = 1.8 V	—	16	—	mA
		RGMII mode (variant A only); C_L= 5 pF (far end with 15 cm PCB trace); frame size 1500 bytes; worst-case pattern (0xA5A5); default IO settings;	—	—	95	mA

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Table 444. Static characteristics...*continued*

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		VDDIO = 3.3 V				
		RGMII mode (variant A only); C_L=5 pF (far end with 8cm PCB trace ; frame size 128 bytes; PRBS pattern; default IO settings; 25 °C ambient VDDIO = 3.3 V	—	38	—	mA
		SGMII mode	—	2	5	mA
Current consumption (other pins)[3]						
$I_{DD(RMS)}$	RMS supply current	on pin VREGA_IN, variant A	—	1.5	2.5	mA
		on pin VREGA_IN, variant B	—	7	8.5	mA
		on pins VDDA_TRX	—	30	35	mA
		on pins VDDA_RX	—	95	110	mA
		on pin VDDA_SI; SGMII link-up; (variant B only)	—	20	24	mA
		on pin VDDA_SO; SGMII link-up; (variant B only)	—	13	15	mA
		on pin VDDA_AO	—	1	2	mA
I_{sleep}	sleep current	on pin VDDA_AO; device mode DEVICE_DEEP_SLEEP [5]	—	—	35	µA
Digital I/O pins[6]						
V_{IH}	HIGH-level input voltage	3.3, 2.5, 1.8 V signaling	0.7 × V_{DDIO}	—	—	V
V_{IL}	LOW-level input voltage	3.3, 2.5, 1.8 V signaling	—	—	0.3 × V_{DDIO}	V
V_{OH}	HIGH-level output voltage	3.3 V signaling I_{OH} = –2 mA	V_{DDIO} - 0.4	—	—	V
		2.5 V signaling I_{OH} = –2 mA	V_{DDIO} - 0.4	—	—	V
		1.8 V signaling I_{OH} = –2 mA	V_{DDIO} - 0.4	—	—	V
V_{OL}	LOW-level output voltage	3.3 V signaling I_{OL} = 2 mA	—	—	0.4	V
		2.5 V signaling I_{OL} = 2 mA	—	—	0.4	V
		1.8 V signaling I_{OL} = 2 mA	—	—	0.4	V
C_i	input capacitance	[3]	—	—	5.0	pF
V_{hys}	input hysteresis voltage	[3]	0.1 × V_{DDIO}	—	—	V

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Table 444. Static characteristics...*continued*

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
I_{OSH}	HIGH-level short-circuit output current	[3]	—	—	144	mA
I_{OSL}	LOW-level short-circuit output current	[3]	—	—	144	mA
I_{IH}	HIGH-level input current	V_{IH}= $V_{DD(IO)}$ internal pull-up or pull-down disabled	—	—	10	µA
I_{IL}	LOW-level input current	V_{IL}=0 V internal pull-up or pull-down disabled	–10	—	—	µA
R_{pd}	pull-down resistance		40	50	62	kΩ
R_{pu}	pull-up resistance		40	50	62	kΩ
R	output impedance	Default register settings. VDDIO = 3.3 V [3]	—	50	—	Ω
		Default register settings. VDDIO = 2.5 V [3]	—	52	—	Ω
		Default register settings. VDDIO = 1.8 V [3]	—	48	—	Ω
SGMII[3]						
$\|V_{o(dif)}\|$	differential output voltage (absolute value)	R_L = 100 Ω	150	—	400	mV
$R_{term(se)}$	single-ended termination resistance		—	50	—	Ω
$V_{ring(o)}$	ringing voltage		—	—	0.1 × Vo(dif)	V
$Z_{o(se)}$	single-ended output impedance		40	—	140	Ω
$\Delta Z_{o(se)}$	single-ended output impedance mismatch		—	—	10	%
$\|\Delta V_{o(dif)}\|$	differential output voltage difference	change in differential output voltage between complementary output states R_L = 100 Ω	—	—	25	mV
ΔV_{cm}	common-mode voltage difference	change in output offset voltage between complementary output states R_L = 100 Ω	—	—	25	mV
$V_{th(i)dif}$	differential input threshold voltage		–50	—	50	mV
$Z_{i(dif)RX}$	receiver differential input impedance		80	100	120	Ω
1000BASE-T1[3]						
$\|V_{o(dif)}\|$	differential output voltage (absolute value)	load 100 Ω measured at MDI	—	1.2	2.2	V
R_{term}	termination resistance		—	100	—	Ω

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Table 444. Static characteristics...*continued*

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified; positive currents flow into the IC. [1]

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Oscillator[3]						
C_i	input capacitance	on pin XI	—	2	—	pF
C_L	load capacitance	[7]	—	8	—	pF
Voltage monitors						
V_{uvd}	undervoltage detection voltage	on pin VDD_CORE	0.99	—	—	V
		on pin VDDA_RX [8]	1.62	—	—	V
		on pin VREGA_IN, variant A	1.62	—	—	V
		on pin VREGA_IN, variant B	2.25	—	—	V
V_{uvr}	undervoltage recovery voltage	on pin VDD_CORE	—	—	1.045	V
		on pin VDDA_RX [8]	—	—	1.71	V
		on pin VREGA_IN, variant A	—	—	1.71	V
		on pin VREGA_IN, variant B	—	—	2.38	V
V_{ovd}	overvoltage detection voltage	on pin VDD_CORE	—	—	1.21	V
		on pin VDDA_RX [8]	—	—	1.98	V
		on pin VREGA_IN	—	—	3.63	V
V_{ovr}	overvoltage recovery voltage	on pin VDDA_RX [8]	1.89	—	—	V
		on pin VREGA_IN	3.47	—	—	V

[1] All parameters are guaranteed over the junction temperature range by design. Factory testing uses correlated test conditions to cover the specified temperature and power supply voltage ranges.
[2] Note that INH, RST_N, and WAKE_IN_OUT logic levels are according to VDDA_AO
[3] Not measured in production, guaranteed by design.
[4] typical values for 25 °C and typical operating voltages
[5] At an ambient temperature of 85 °C; supply on pins other than VDDA_AO is expected to be removed in DEVICE_DEEP_SLEEP.
[6] Some pins are input-only or output-only. In this case only the relevant information applies. V_{DDIO} in the relative parameters applies according to their reference voltage specified in the pinning table.
[7] Value of external load capacitors depend on the selected crystal. Consult the Appnote for details.
[8] VDD_RX and VDD_TRX are expected to be connected on the PCB

11 Dynamic characteristics

Table 445. Dynamic characteristics

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Digital I/Os (JTAG, GPIOs, MDIO)[1][2]						
t_r	rise time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load. 20% - 80%				
		1.8 V, default IO settings	0.4	—	3.9	ns
		2.5 V, default IO settings	0.9	—	4.3	ns
		3.3 V, default IO settings	1.2	—	4.7	ns

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Table 445. Dynamic characteristics...*continued*

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
t_f	fall time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load. 20% - 80%				
		1.8 V, default IO settings	0.4	—	3.9	ns
		2.5 V, default IO settings	0.9	—	4.3	ns
		3.3 V, default IO settings	1.2	—	4.7	ns
Digital I/Os (RGMII)[1][2]						
t_r	rise time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load. 20 % - 80 % default IO settings	0.24	—	1.5	ns
		5 pF load. 20% - 80%, default IO settings	0.24	—	1	ns
t_f	fall time	15 cm PCB transmission line: 50 Ω; 5 pF far-end load. 20% - 80% default IO settings	0.24	—	1.5	ns
		5 pF load. 20% - 80% default IO settings	0.24	—	1	ns
RGMII transmit/receive timing[1]						
T_{clk}	clock period	on pin TXC	—	8	—	ns
δ	duty cycle	on pin TXC	45	50	55	%
$t_{sk(o)TX}$	transmitter output skew time	RGMII mode, from pin RXC to RXD[3:0], RX_CTL[3] 25 pF load default IO settings 15 pF load	–500	—	500	ps
$t_{sk(i)RX}$	receiver input skew time	RGMII mode, from pin TXC to TXD[3:0], TX_CTL	1	—	2.6	ns
$t_{su(o)TX}$	transmitter output setup time	RGMII-ID mode, from pins RXD[3:0], RX_CTL to RXC[4] 25 pF load default IO settings 15 pF load	1.2	—	—	ns
$t_{h(o)TX}$	transmitter output hold time	RGMII-ID mode, from pin RXC to RXD[3:0], RX_CTL 25 pF load default IO settings 15 pF load	1.2	—	—	ns
$t_{su(o)RX}$	receiver output setup time	RGMII-ID mode, from pins TXD[3:0], TX_CTL to TXC	1	—	—	ns
$t_{h(o)RX}$	receiver output hold time	RGMII-ID mode, from pin TXC to TXD[3:0], TX_CTL	1	—	—	ns

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Table 445. Dynamic characteristics...*continued*

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
SGMII[1]						
t_r	rise time	20% - 80%	40	—	200	ps
t_f	fall time	20% - 80%	40	—	200	ps
t_{jit}	jitter time	TIE (Time Interval Error): RMS value	—	—	10	ps
MDIO timing[1]						
T_{clk}	clock period	on pin MDC	40	—	1000	ns
δ	duty cycle	on pin MDC	—	50	—	%
t_{su}	set-up time	from pin MDIO to MDC	10	—	—	ns
t_h	hold time	from pin MDC to MDIO	10	—	—	ns
t_d	delay time	from pin MDC to MDIO	0	—	15	ns
JTAG[1]						
f_{clk}	clock frequency	on pin TCK	0.1	—	25	MHz
δ	duty cycle	on pin TCK	40	50	60	%
t_{su}	set-up time		4	—	—	ns
t_h	hold time		25	—	—	ns
t_{ov}	output valid time		—	—	20	ns
CLK_OUT[1]						
f_{clk}	clock frequency	on pins CLK_OUT	—	25	—	MHz
Δf	frequency deviation	on pins CLK_OUT		—	100	ppm
MDI[1]						
$t_{d,tx}$	transmit latency	RGMII from TX_EN to SSD on MDI MACsec enabled	879	—	1817	ns
		SGMII /S/ to SSD on MDI MACsec enabled	1407	—	2002	ns
$t_{d,rx}$	receive latency	SSD to RX_DV on RGMII MACsec enabled	5405	—	5790	ns
		SSD to /S/ on SGMII MACsec enabled	5154	—	5891	ns
Voltage monitor[1]						
$t_{uvd(det)}$	undervoltage detection time	from crossing detection level to IRQ generation	—	—	5	ms
$t_{ovd(det)}$	overvoltage detection time	from crossing detection level to IRQ generation	—	—	5	ms
Power-up timing[1]						
dV_{ru}/dt	rate of change of ramp-up voltage	supply ramp up rate; 20 % - 80 %				
		on pin VDDA_AO	1	—	35	V/ms

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Table 445. Dynamic characteristics...*continued*

T_{vj} = –40 °C to + 150 °C; all supply voltages within the operating range as defined in V_{DD}, unless otherwise specified; all voltages are defined with respect to ground unless otherwise specified. Not measured in production, guaranteed by design.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
		on pin VDD_CORE	1	—	35	V/ms
		on pin VDDIO	1	—	35	V/ms
		on pin VDDA_SI, VDDA_SO	1	—	35	V/ms
		on pin VREGA_IN	1	—	35	V/ms
$t_{init(ps)}$	pin strapping initialization time	from deasserting RST_N pin to completion of pin strapping	—	—	2	ms
Oscillator[1]						
f_{osc}	oscillator frequency		—	25	—	MHz
Δf	frequency deviation	over lifetime and temperature range [5]				
		SGMII	–100	—	100	ppm
		RGMII [6]	–50	—	50	ppm
Oscillator Follower mode (external clock)[1]						
δ	duty cycle	on pin XI	45	50	55	%
$t_{jit(i)rms}$	RMS input jitter time	on pin XI	—	—	2	ps
Reset[1]						
$t_{det(rst)}$	reset detection time	on pin RST_N	5	—	—	µs

[1] Not measured in production, guaranteed by design.
[2] For reference only. Run simulations with the IBIS model and your custom board for accurate results.
[3] Device configured to operate in RGMII mode.
[4] Device configured to operate in RGMII-ID mode, delay is in the transmitter.
[5] The frequency deviation depends on the selected crystal
[6] Device can tolerate a 100 ppm crystal in RGMII mode, but clock output ppm specifications are not met. See AN14517 [6] for further information.

12 Timing diagrams

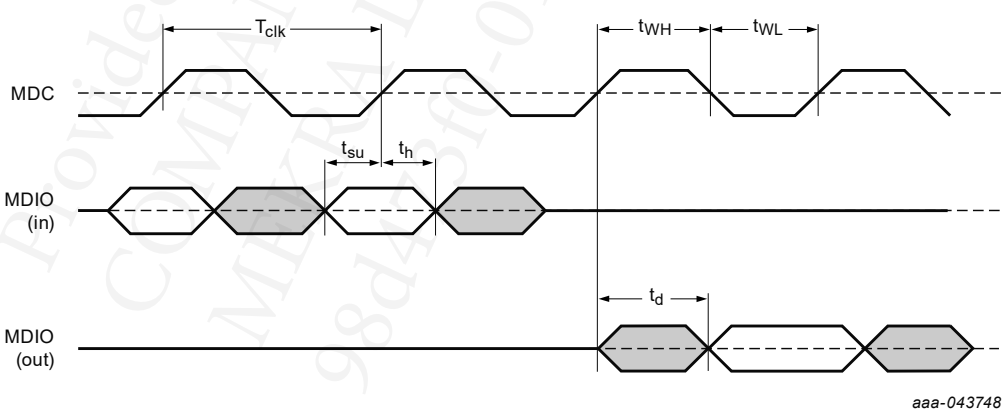


Figure 41. MDIO timing diagram

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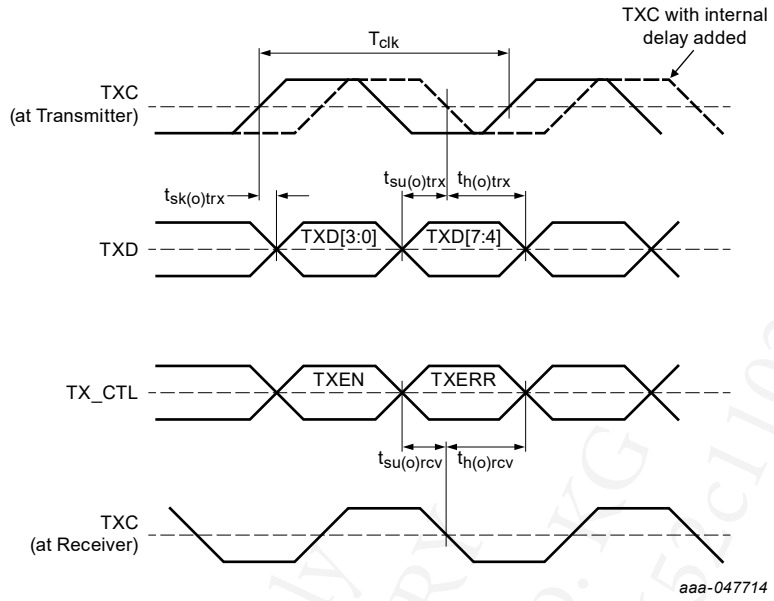


Figure 42. RGMII transmit timing diagram

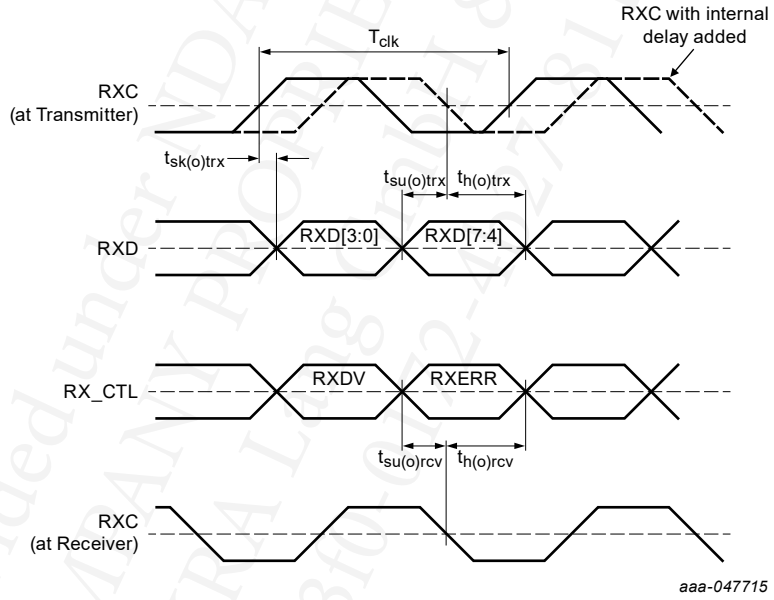


Figure 43. RGMII receive timing diagram

13 Application diagrams

The schematics in Figure 1 to Figure 2 show application examples. See AN14517 [6] for further information.

13.1 Supply

The diagrams in Figure 44 and Figure 45 show the power supply circuit diagrams for TJA1121A resp. TJA1121B.

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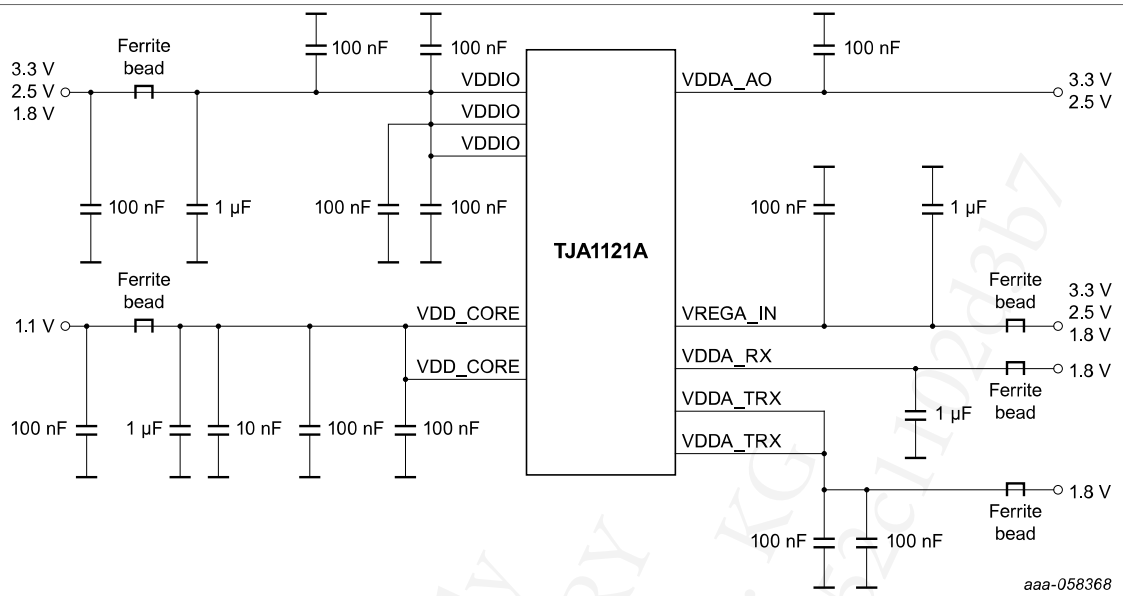


Figure 44. Supply circuit diagram TJA1121A

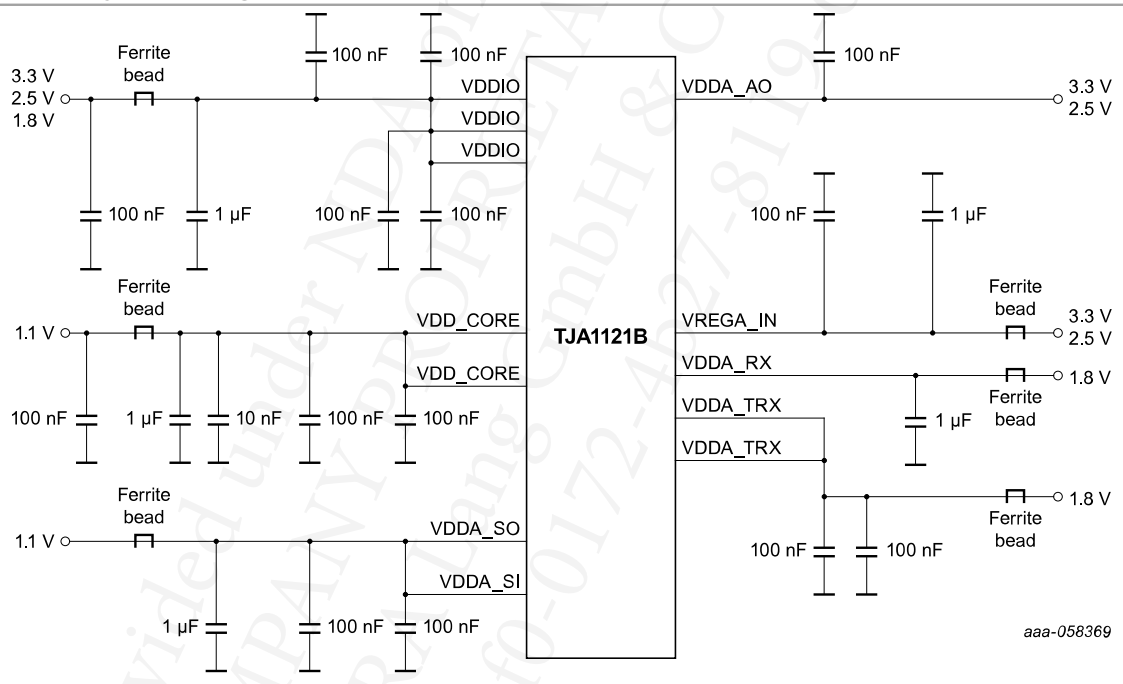


Figure 45. Supply circuit diagram TJA1121B

13.2 Wakeup-Sleep

In case the wakeup/sleep feature is used, VDDA_AO must be supplied by a dedicated standby supply. The VDDA_AO supply of the device can operate on 2.5 V or 3.3 V. See AN14517 [6] for further information. The pins INH, WAKE_IN_OUT and RST_N are in the VDDA_AO domain. Thus any interfacing to these pins must be compatible with the level of VDDA_AO.

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13.3 Reset

The RST_N pin is part of the VDDA_AO voltage domain. This makes sure that the device is not automatically going through a reset sequence after a deep sleep, during which only VDDA_AO is present. In case VDDA_AO and VDDIO are different supplies it must be ensured that RST_N is driven with voltage levels according to VDDA_AO.

For example, if VDDIO is 1.8 V then all microcontroller pins operate in 1.8 V mode. In this case, RST_N cannot be connected directly to a microcontroller pin, because RST_N has thresholds relative to VDDA_AO and not VDDIO. See AN14517 [6] for examples.

Figure 46 shows two possibilities how a microcontroller and TJA1121 can be connected. Additional components such as pull-up resistors may be required. RST_N is active in all device states, including system startup, sleep and wakeup, and asserting its low will result in a device reset.

Note: *Never assert RST_N during wake/sleep cycles. A device reset clears all wakeup/sleep related state. Attention when the microcontroller controls RST_N. Check that the GPIO behavior of the microcontroller during startup/reset is well defined. A glitch on the GPIO outputs can lead to a TJA1121 reset.*

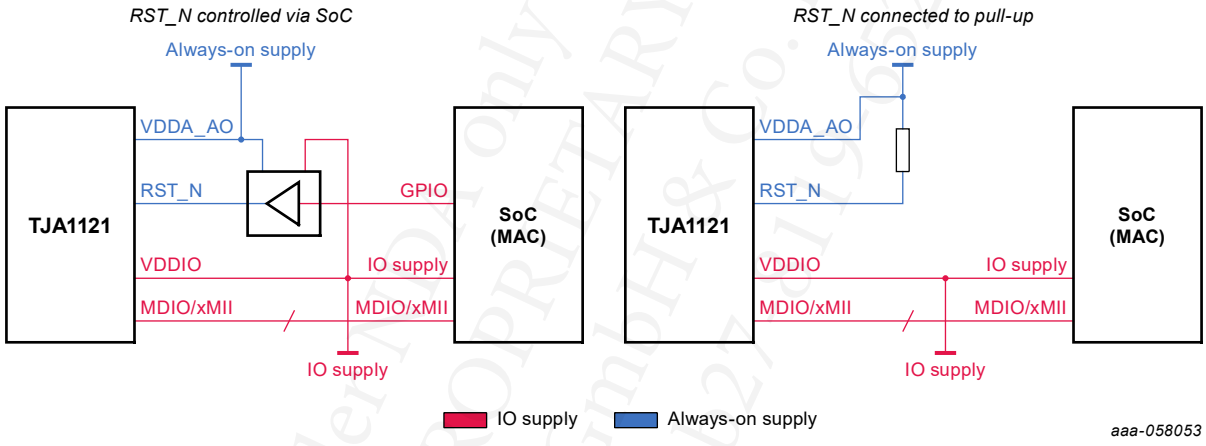


Figure 46. Driving the RST_N pin

14 Test information

This product has been qualified in accordance with the Automotive Electronics Council (AEC) standard Q100 Rev-H - *Failure mechanism based stress test qualification for integrated circuits*, and is suitable for use in automotive applications.

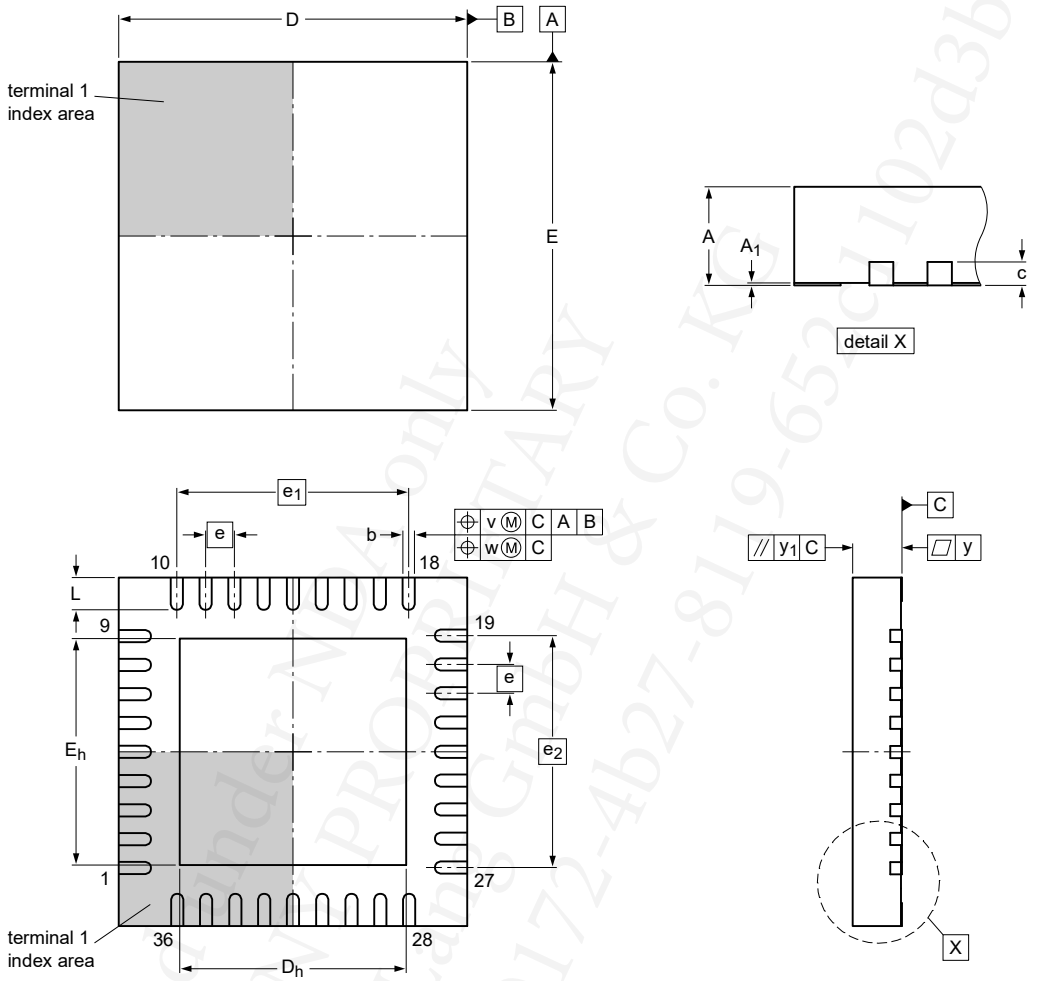
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15 Package outline

HVQFN36: plastic thermal enhanced very thin quad flat package; no leads; 36 terminals; body 6 x 6 x 0.85 mm **SOT1092-2**



Dimensions

Unit		$A^{(1)}$	A_1	b	c	$D^{(1)}$	D_h	$E^{(1)}$	E_h	e	e_1	e_2	L	v	w	y	y_1
mm	max	1.00	0.05	0.30		6.1	4.05	6.1	4.05				0.65				
	nom	0.85	0.02	0.21	0.2	6.0	3.90	6.0	3.90	0.5	4	4	0.55	0.1	0.05	0.05	0.1
	min	0.80	0.00	0.18		5.9	3.75	5.9	3.75				0.45				

Note
1. Plastic or metal protrusions of 0.075 mm maximum per side are not included.

sot1092-2_po

Outline version	References				European projection	Issue date
	IEC	JEDEC	JEITA			
SOT1092-2		---	---			~~09-02-23~~ 09-02-24

Figure 47. Package outline TJA1121 HVQFN36

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16 References

[1] DIN EN 61000-4-2:2009-12, Electromagnetic compatibility (EMC) - Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test (IEC 61000-4-2:2008)
[2] IEC 62228-5:2021, Integrated circuits - EMC evaluation of transceivers - Part 5: Ethernet transceivers, 2021-04-26
[3] IEEE Std 802.3-2022, 29 July 2022
[4] RGMII Specification 2.0, Broadcom, HP, Marvell, 1 April 2002
[5] Serial-GMII Specification 1.8, Cisco Systems, 27 April 2005
[6] AN14517, TJA1121 Application Notes, NXP, http://www.nxp.com/TJA1121
[7] Advanced diagnostic features for 1000BASE-T1 automotive Ethernet PHYs v2.0, OPEN Alliance, May 22, 2020
[8] OPEN Sleep/Wake-up Specification, v2.0, OPEN Alliance, Feb 21, 2017
[9] MACsec core user manual, UM12059, NXP, http://www.nxp.com/TJA1121
[10] ISO 21111-2:2020 Road vehicles — In-vehicle Ethernet — Part 2: Common physical entity requirements

17 Revision history

Table 446. Revision history

Document ID	Release date	Description
TJA1121 v.1.0	03 December 2024	• Initial version

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Document status[1][2]	Product status[3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
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Product [short] data sheet	Production	This document contains the product specification.

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[2] The term 'short data sheet' is explained in section "Definitions".
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